RV College of Engineering

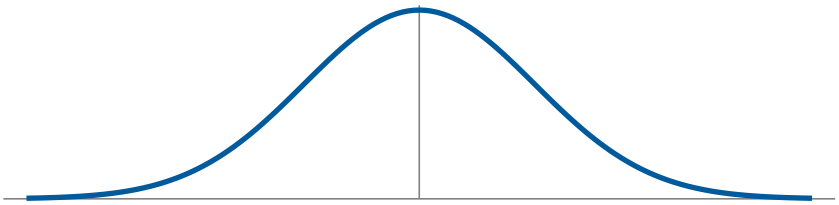
Department of Industrial Engineering and Management

Statistics for Data Analytics

Course Code: IM241AT

Semester IV • Professional Core Course

Master Compilation of Course Notes — All Five Units



A Student Compilation for Study and Reference

Semester: IV						
STATISTICS FOR DATA ANALYTICS **Category: Professional Core Course** **(Theory)**						
Course Code	**:**	**IM241AT**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3.00 Hours**

Unit-I	**09 Hrs**
Data Summary and Presentation: Data types, tabular and graphical displays: Stem and Leaf diagrams, Histograms, Box plots, Radar diagrams. **Concepts of Probability:** Sample spaces and Events, Interpretations of probability, Addition rules, Conditional probability, Multiplication and Total probability rules, Independence, Bayes Theorem. Numerical Problems	
Unit-II	**09 Hrs**
Concepts of Random Variables: Random Variables, Discrete and continuous random variables. Probability distributions and mass functions, Numerical Problems **Discrete Probability Distributions:** Discrete uniform, Binominal, Poisson, Geometric, Negative binomial, Applications, Numerical Problems.	
Unit-III	**09 Hrs**
Continuous Probability Distributions: Continuous uniform, Normal, Normal approximations, Exponential, Applications, Numerical Problems. **Sampling Distribution:** Sampling distribution, Central Limit Theorem, Sampling distribution of means and derived quantities, Numerical Problems.	
Unit-IV	**09 Hrs**
Simple Linear Regression and Correlation: Empirical models, Simple Linear Regression, Properties of Least square Estimators and Estimation of variances, Common abuses of regression, Prediction of new observations, Correlation, Numerical Problems. Interpretation of graphical output from software packages such as Minitab	
Unit-V	**09 Hrs**
Statistical Inference for a single sample: Hypothesis testing, Inference on the mean of a normal population (variance known and unknown), Inference on the variance of a Normal population, Testing for Goodness of Fit, Tests of association, Numerical Problems **Interval Estimation:** Confidence Intervals on mean (variance known and unknown), and variance of a normal population	

Course Outcomes: After completing the course, the students will be able to	
CO1	Describe and report data set using data analysis, presentation and interpretation techniques to understand various phenomena in the fields of science and engineering.
CO2	Apply various statistical processing techniques to handle a set of data to estimate probabilities.
CO3	Apply an appropriate statistical tool and analyze a specific set of data to estimate and draw conclusions about population parameters
CO4	Draw inferences about population parameters and relations between variables based on analysis of sample data

References Books	
1.	Engineering Statistics, Douglas C. Montgomery, George C. Runger, Norma FarisHubele, 5th Edition, 2011, John Wiley & Sons, Inc., ISBN-13: 978- 0-470-63147-8
2.	Applied statistics and Probability for Engineers, Douglas C Montgomery, George C Runger, Wiley, 4th Edition, 2007, Asia Student Edition, ISBN: 978-81-265-2315-3.
3.	Statistics for Management, Richard I Levin, David S Rubin, 7th Edition, 1997, Prentice Hall India, ISBN: 9780134762920.
4.	Probability and Statistics for Engineers and Scientists, Walpole, Myers, Myers, Ye, 8th Edition, 2007, Pearson Education Inc., ISBN: 978-81-317-1552-9.
5.	Softwares : Microsoft Excel / Minitab / Matlab / R

6.	Online resources: a) http://172.16.44.44/nnptel.html - choose NOC:Introduction to Data Analytics(Course sponsored by Aricent) b) https://onlinecourses.nptel.ac.in/noc22_mg31/preview -Introduction to probability and Statistics c) https://newonlinecourses.science.psu.edu/statprogram/undergraduate-studies d) https://www.khanacademy.org/math/statistics-probability

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40**	**40**
MAXIMUM MARKS FOR THE CIE THEORY		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: (Internal Choice)	16
5 & 6	Unit 3: (Internal Choice)	16
7 & 8	Unit 4: (Internal Choice)	16
9 & 10	Unit 5: (Internal Choice)	16
	TOTAL	**100**

A Note on This Compilation

This book is **not** the official RV College of Engineering notes. It is a compilation of student notes together with the handwritten notes provided by **Dr. Ramaa A Nadig**, Department of Industrial Engineering and Management.

All information contained in this book has been drawn from her notes and lectures. The material has been organised, typeset, and arranged to follow the prescribed course syllabus, but it should be treated as a supplementary study aid rather than an official course document.

Every effort has been made to reproduce the source notes faithfully and to preserve their content without addition or deletion. Where the original handwritten working was unclear, the final answers recorded in the notes were treated as authoritative. Readers should always cross-check against the prescribed textbooks and their instructor's guidance.

Compiled from the notes and lectures of Dr. Ramaa A Nadig, Department of Industrial Engineering and Management.

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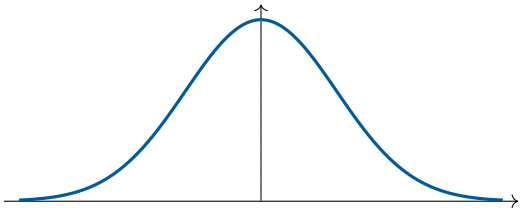
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Unit 1



Chapter 1

Introduction to Engineering Statistics

1.1 What Is Statistics?

At its core, statistics may be defined as *the measurement of uncertainty*. It is the discipline that allows researchers to evaluate conclusions derived from sample data and to quantify how much confidence those conclusions deserve.

Statistics

Statistics is the art and science of collecting, classifying, presenting, interpreting, and analysing numerical data, as well as drawing conclusions about the system from which the data were obtained.

In practice, statistics refers to a scientific approach used to:

- Collect data.
- Interpret and analyse data.
- Assess the reliability of conclusions based on sample data.

A closely related idea is the notion of a single statistic.

Statistic

A **statistic** is a characteristic of a sample. Generally, a statistic is used to estimate the value of a population parameter.

Viewed more broadly, statistics is the collection, analysis, and use of data to understand and predict factors of interest in a scientific way, so as to make decisions, solve problems, and design products or processes. The field of statistics can be broken into two major areas: **descriptive statistics** and **inferential statistics**, both of which are developed in detail later in this unit.

1.2 Why Learn Statistics?

Statistics earns its place in engineering and science for two principal reasons:

- The potential outcome of events can be predicted with the help of statistics in situations where other techniques are not adequate.
- It supports decision making based on what is already known. Familiar examples include the analysis of climate disasters and the forecasting of agricultural production.

1.3 The Role of Statistics in Engineering

Problems of interest are solved by engineers using the scientific method. Statistics provides the formal machinery for the data-driven steps of that method. A typical engineering problem-solving cycle proceeds as follows:

1. Describe the problem.
2. Identify the factors that affect the problem.
3. Propose a model.
4. Collect data from experiments to validate the model.
5. Refine the model based on the data.
6. Develop a solution.
7. Collect data to confirm the solution's effectiveness and efficiency.
8. Conclude and give recommendations.

More specifically, the role of statistics within this cycle includes the following:

- Collection, presentation, analysis, and use of data to make decisions.
- Helping to describe and understand *variability*.
- Making conclusions about a population based on sample observations — the process of *statistical inference*. For example: if sample data exist, are they sufficient to infer something about the population?
- Distinguishing between inherent variability and changes due to deliberate modification of the inputs.

1.4 Applications of Statistics

Statistics is applied across virtually every branch of engineering and the sciences. Representative application areas include:

- **Process improvement** — for example, studying the mean productivity of processes.
- **Product design** — assessing the effect of design parameters.
- **Health care** — for example, evaluating whether a new vaccination programme is going to be effective.
- **Strategic decision making** — decisions based on surveys.
- **Prediction of parameters** — such as financial returns, inflation, and weather.

Numerical data arise in essentially every sphere of human experience. A non-exhaustive catalogue of such data includes: birth and death data; crime data and food-product consumption; agricultural production of different crops; industrial production across heavy, medium, and small-scale industry; earnings through export of various commodities; sports data such as runs scored, wickets taken, goals scored, matches won, and running times; social data such as age at marriage, age of a parent at the birth of a child, and infant-mortality rate; medical data such as the incidence of certain diseases, control and survival rates, the effects and side effects of medicines, and

the relationship between certain diseases and habits; accident data for road, rail, air, and ship, together with fatality rates; counts of vehicles, lengths of roads and rails, and numbers of valid licence holders; numbers of voters and constituencies; astronomical and cosmological data such as planets, the solar system, stars, galaxies, comets, solar and lunar eclipses, and the positions of certain stars and planets; chemical compositions and the study of electrons and neutrons in metallurgical work; genetic studies of human behaviour; biometric studies; the percentage of defective items and the number of items produced; data on consumers of certain products, their capacity to spend, their income levels, and profession; weather data; and placement data.

1.5 A Brief History of Statistics

1.5.1 Western Developments

The development of statistics as a discipline can be traced through a sequence of milestones:

- The word *statistics* was coined by the German scholar Gottfried Achenwall around the middle of the eighteenth century. It is derived from the word *state*.
- It was used for the first time in Britain by John Sinclair (1791–1799).
- The term appeared in the *Encyclopaedia Britannica* in 1797.
- The earliest work on statistical methodology was related to the normal distribution (De Moivre, 1733; the paper was later discovered by Karl Pearson in 1924).
- The French mathematician Laplace, in his manuscript of 1783, stated that the normal curve can be used for describing the probability distribution of errors.
- In 1809, Gauss developed the normal distribution in a study of the theory of the motions of heavenly bodies; hence the alternative name *Gaussian distribution*.
- The Belgian mathematician Quetelet (1796–1874) applied statistics to such quantities as the heights of a large number of people.
- Gregor Mendel performed various genetic experiments, mostly on peas, and concluded that the laws of heredity are statistical in nature.
- Francis Galton, in *Hereditary Genius* (1869) and *Natural Inheritance* (1889), argued that hereditary laws are statistical in nature, and so introduced the concepts of correlation and regression.
- Karl Pearson introduced the concept of statistics in a systematic way. In 1901 he started the famous journal *Biometrika*, one of the top journals for the study of statistics. He was the first to discuss the chi-square test statistic, used for testing data for the normal distribution or a gamma distribution — the test of goodness of fit.
- G. U. Yule, in 1910, gave the distinction between a population and a random sample; later, R. A. Fisher also gave a clear-cut distinction between population and sample.
- The definition of random sampling was given by John Venn and C. S. Peirce.
- The first table of random digits was printed by Tippett in 1927.
- Gosset ("Student") contributed the t-distribution.

- R. A. Fisher is regarded as the father of modern statistics. He developed the theory and foundations of statistical inference, maximum-likelihood estimation, unbiased estimation, and the analysis of variance.
- Neyman and E. S. Pearson started developing the testing of hypotheses.
- Fisher and Neyman are credited with giving confidence intervals; Neyman further developed stratified random sampling.
- In Russia, around 1939, Kolmogorov developed a rigorous theory of probability and stochastic processes.

1.5.2 Indian Developments

- Kautilya's *Arthashastra*, written in 321 BC, gives a detailed description of how to collect data on various aspects of the state.
- Modern statistical methodology in India was initiated by P. C. Mahalanobis, who studied in England and came back to India to use statistical tools regularly. With his efforts, the Indian Statistical Institute was started in 1931.
- R. C. Bose contributed to the subject of the design of experiments.
- S. N. Roy and C. R. Rao contributed greatly to multivariate analysis, the design of experiments, and linear models.
- With the efforts of Mahalanobis, the Central Statistical Organisation was established in 1951 by the Government of India to collect data on various aspects of the state. In 1950, the National Sample Survey Organisation was established to collect data on various socio-economic aspects of life.

1.6 Collecting Engineering Data

Engineering data are collected in three principal ways:

- A retrospective study using historical data.
- An observational study.
- A designed experiment.

1.6.1 Retrospective Study

A retrospective study would use either all or a sample of the historical process data archived over some period of time. This study may involve a large amount of data, but that data may contain relatively little useful information about the problem. Furthermore, some of the relevant data may be missing, and there may be transcription or recording errors resulting in outliers. As a result of these types of issues, statistical analysis of historical data sometimes identifies interesting phenomena, but solid and reliable explanations of these phenomena are often difficult to obtain.

1.6.2 Observational Study

In an observational study, the engineer observes the process or population, disturbing it as little as possible, and records the quantities of interest.

1.6.3 Designed Experiment

In a designed experiment, the engineer makes deliberate or purposeful changes in the controllable variables of the system or process, observes the resulting system output data, and then makes an inference or decision about which variables are responsible for the observed changes in output performance.

1.7 Models

Mathematical Model

A **mathematical model** can be defined as a description of a system using mathematical concepts and languages, in order to facilitate proper explanation of a system, to study the effects of different components, and to make predictions on patterns of behaviour.

Three useful categories of model are distinguished below.

Mechanistic model. A mechanistic model is built from underlying knowledge of the basic physical mechanism that relates the variables of interest. It is a description of a complex system based on a description of its components and their interconnections; such models mostly have a tangible, physical aspect.

Empirical model. An empirical model uses our engineering and scientific knowledge of the phenomenon, but it is not directly developed from theoretical or first principles of the underlying mechanism. In general, one example of an empirical model is regression. To estimate the parameters, the method of least squares is used, which traces its origin to work by Karl Gauss.

Probability model. Probability models help quantify the risks involved in statistical inference — that is, the risks involved in decisions made every day.

Example 1.1 (Two ways to model the tides)**.** Consider the prediction of tides. There are two contrasting modelling philosophies:

- **Empirical** — based on past data, without explicitly knowing the mechanism.
- **Mechanistic** — based on the physical interaction of the Earth, Sun, and Moon.

1.8 Raw and Processed Data

Data may be classified by the degree of processing applied to them.

- **Raw data** — data as collected, without further processing.
- **Processed data** — data processed after collection.

The distinction is best appreciated through paired examples.

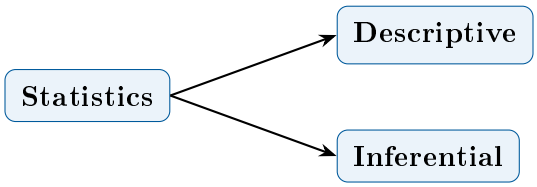
Table 1.1: Raw data versus processed data

Raw	**Processed**
Height of all students in a particular class.	Average height of each class of the IV semester.
Online measurement of the weight of each pack of a food item.	Report containing the average weight of a pack for a food item per day.
Weight of a casting part measured at different manufacturing looms in different units of measurement.	Weight of components converted to a single unit for comparison purposes.
Downtime in days of a machine in a factory or shop floor.	Inflation rates for the past 10 years; yearly sales figures of automobile manufacturers in India.
Length of phone calls for 200 persons calling through a particular exchange.	—

Chapter 2

Descriptive and Inferential Statistics

The field of statistics divides naturally into two complementary branches. Descriptive statistics is concerned with *summarising* data that are in hand, whereas inferential statistics is concerned with *generalising* from a sample to a larger population. Probability is the conceptual bridge that links the two.



2.1 Descriptive Statistics

Descriptive Statistics

Descriptive statistics is the branch of statistics which characterises and summarises the most prominent features of a given set of data.

Some of the commonly used descriptive measures include means, medians, standard deviations, percentiles, graphs, tables, and charts. Descriptive statistics can be used to describe the elements of a population as a whole, or they can be used to describe data that represent just a sample of elements from the entire population.

2.2 Inferential Statistics

Inferential Statistics

Inferential statistics is the branch of statistics that deals with drawing conclusions about a population based on information obtained from a sample drawn from that population.

2.3 Probability: The Bridge Between the Two Branches

Probability is the bridge between descriptive statistics and inferential statistics. It is also a link between a population and a sample. When certain properties of a population are known or can be assumed, the probability principles involved allow one to deduce the expected properties of a sample. Conversely, probability together with sample data allows one to evaluate the amount of uncertainty when making conclusions about a population. Hence probability is the heart of both the deductive and inferential processes involved in statistical analysis.

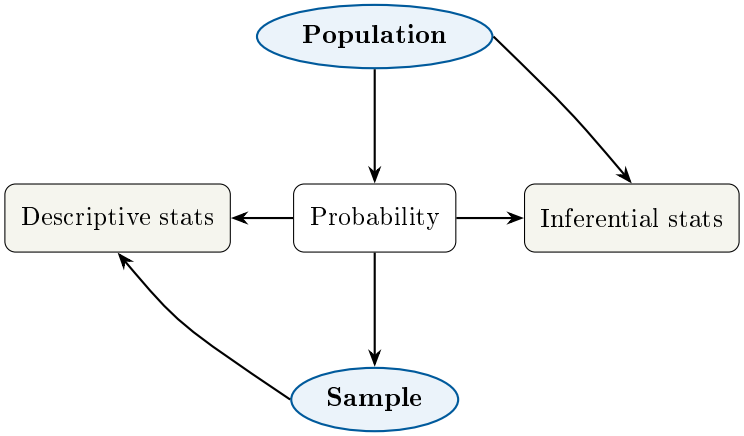


Figure 2.1: Probability links a population to its samples and connects descriptive and inferential statistics. Knowledge of the population, processed through probability, yields expected properties of a sample; sample data, processed through probability, support inferences about the population.

Chapter 3

Types of Data

Before any data can be summarised or analysed, they must be classified by type, because the type of data determines which summaries, visualisations, and operations are meaningful. Data are first divided into **numerical** and **categorical** types, each of which is further subdivided.

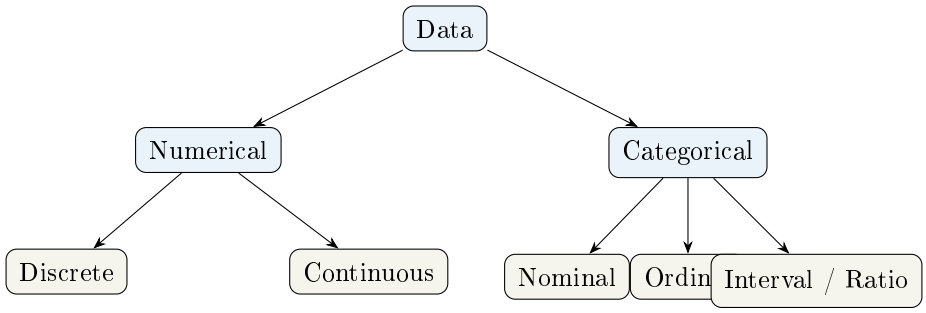


Figure 3.1: Classification of data into numerical and categorical types and their subtypes.

The four measurement scales — nominal, ordinal, interval, and ratio — form a hierarchy of increasing mathematical structure. Nominal and ordinal scales describe categorical data; interval and ratio scales describe numerical data.

3.1 Categorical Data

Categorical data represent characteristics. They can therefore represent things such as a person's gender or the language they speak. Categorical data can also take on numerical values — for example, 1 for female and 0 for male — but it is essential to note that those numbers do not have any mathematical meaning.

3.1.1 Nominal Data

Nominal Data

Nominal data represent discrete units and are used to label variables that have no quantitative value. Nominal data have no order.

Example 3.1 (Nominal variables)**.** • **Gender:** Male, Female. (A variable with only two categories, such as gender, is called *dichotomous*.)

• **Language spoken:** French, English, German, Spanish.

3.1.2 Ordinal Data

Ordinal Data

Ordinal values represent discrete and ordered units. Ordinal data are therefore the same as nominal data except that the ordering matters.

Example 3.2 (Educational background)**.** The categories

1. Elementary 2. High school 3. Undergraduate (UG) 4. Postgraduate (PG)

are ordinal: the order is important, but the difference between the values is not really known.

3.2 Numerical Data

3.2.1 Discrete Data

Discrete Data

Discrete numerical data cannot be measured but can be counted.

Example 3.3. The number of heads in 100 coin flips is discrete data.

3.2.2 Continuous Data

Continuous Data

Continuous data represent measurements; therefore their values cannot be counted but can be measured.

Example 3.4. The height of a person is continuous data.

3.2.3 Interval Data

Interval Data

Interval data represent ordered units that have the same difference between successive values.

Example 3.5. Temperature measured on a scale such as $-10,\ -5,\ 0,\ +5,\ +10$ is interval data.

The problem with interval values is that they do not have a true zero. As a consequence, with interval data we can add and subtract, but we cannot multiply, divide, or calculate ratios.

3.2.4 Ratio Data

Ratio Data

Ratio data are also ordered units that have the same difference between successive values. Ratio data are the same as interval values, with the crucial difference that they *do* have an absolute zero.

Example 3.6. Length in inches, such as $0,\ 5,\ 10,\ 15$, is ratio data. Because length has an absolute zero, ratios between lengths are meaningful.

3.3 Data Collection and Visualisation by Type

The admissible summaries and visualisations depend on the data type.

- **Nominal data** — summarised by frequency, proportion, and percentage; visualised with the pie chart and the bar chart.
- **Ordinal data** — visualised with the pie chart and bar chart, and additionally summarised using percentiles, median, mode, and the interquartile range.

Chapter 4

Graphical Presentation of Data

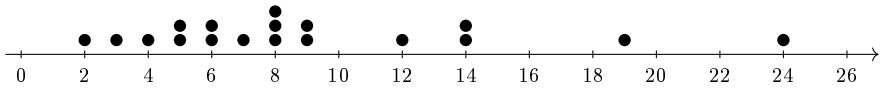
Graphical methods reveal features of a data set — its centre, spread, extremes, gaps, and possible associations — that are difficult to perceive from a raw list of numbers. This chapter develops the dot plot, the scatter plot, the frequency distribution and histogram, the box plot, the stem-and-leaf plot, and the radar chart, and illustrates each with worked problems.

4.1 The Dot Plot (Univariate)

When only one variable is considered and plotted, dot plots are used. They are a simple, attractive display that reveal the extremes, the centering, and the voids or gaps in the data.
Example 4.1 (Constructing a dot plot)**.** Suppose the data points are

$$12,\ 14,\ 9,\ 8,\ 7,\ 8,\ 4,\ 5,\ 14,\ 19,\ 2,\ 3,\ 6,\ 6,\ 5,\ 8,\ 9,\ 24.$$

The corresponding dot plot is shown below.



Dot plots are used when the number of units is relatively small — say $n < 30$ — or where there are relatively few distinct values represented in the data set. In such cases, dot plots are effective displays.

4.2 The Scatter Plot (Bivariate)

When we wish to jointly display results for two variables, the bivariate equivalent of the dot plot is called the **scatter plot**. Similarly, a three-dimensional scatter plot is used with three variables.

4.3 Frequency Distribution and Histogram

A frequency distribution is a more useful summary of data. To construct a frequency distribution, we must divide the range of the data into intervals, which are usually called **class intervals**. If possible, the class intervals should be of equal width, to enhance the visualisation.

The number of class intervals used depends on the number of observations and the amount of scatter or dispersion in the data. Generally, 5 to 20 class intervals are found satisfactory in most cases, and the number of class intervals should increase with n. Choosing a number of class intervals approximately equal to the square root of the number of observations often works well.

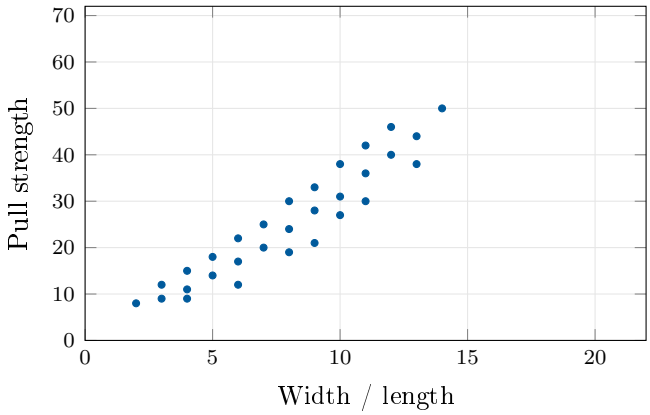


Figure 4.1: A scatter plot. Scatter plots show the region of the plane where data points fall, as well as the data density associated with that region. They also suggest possible association between the variables. Unlike dot plots, scatter plots can be used for large data sets as well.

Choosing the number of classes

Let n be the number of observations and let k denote the number of classes. A practical rule is

$$k \approx \sqrt{n} \quad (n \leq 100), \qquad k \approx 1 + 3.322 \log_{10} n \quad (n > 100),$$

where the second expression is **Sturges' formula**. More generally, $6 \leq k \leq 15$ classes is recommended.

Remark 4.1 (Requirements on histogram classes)**.** For a histogram, the classes should be *all-inclusive* and *mutually exclusive*: every observation falls in exactly one class.

A histogram of cumulative frequency or cumulative relative frequency is called an **ogive curve**.

Worked Problem 1: Shelf Life of Photographic Film

Example 4.2 (Histogram and summary statistics)**.** The shelf life of a high-speed photographic film is being investigated by the manufacturer. The following data (life in days) are available:

126	131	116	125	134	120	125	150	130	149
129	132	128	126	127	122	111	148	120	117
134	136	130	134	120	129	147	126	117	143
141	145	162	129	127	133	129	140	131	133

Here the minimum is 111 and the maximum is 162, so the data span roughly 110 to 170. Construct the histogram and comment on the properties of the data.

The frequency table, using class intervals of width 10, is as follows.

The summary statistics for these data are:

$$\text{Mean } \bar{x} = \frac{\sum x_i}{n} = 131.30, \qquad \text{Median} = 129.5, \qquad \text{Mode} = 129,$$

$$s^2 = 113.85, \qquad s = 10.67.$$

The sample variance is computed from

$$s^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1} = \frac{S_{xx}}{n-1}.$$

Table 4.1: Frequency distribution for the shelf-life data

Class interval	Frequency	Relative frequency	Cumulative rel. freq.
$110 \le x < 120$	4	0.100	0.100
$120 \le x < 130$	16	0.400	0.500
$130 \le x < 140$	11	0.275	0.775
$140 \le x < 150$	8	0.200	0.975
$150 \le x < 160$	0	0.000	0.975
$160 \le x < 170$	1	0.025	1.000
Total	**40**	**1.000**	

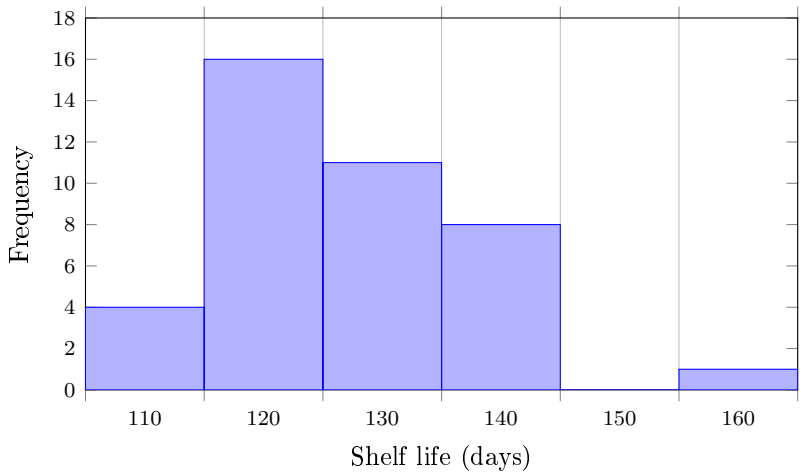


Figure 4.2: Histogram of the shelf-life data. The distribution is unimodal, slightly right-skewed, centred near 130 days, with a single isolated large value near 162.

Worked Problem 2: Number of Defects in Parts

Example 4.3 (Histogram, box plot, and measures of centre)**.** 62 parts were analysed for the number of defects, with the following distribution:

Number of defects i	2	3	4	5	6	7	10
Number of parts with i defects	30	15	9	4	2	1	1

(i) Plot a histogram for the number of defects using 5 classes. (ii) Construct a box plot for the number of defects in a part. (iii) Compute the mean, median, and mode for the number of defects in a part.

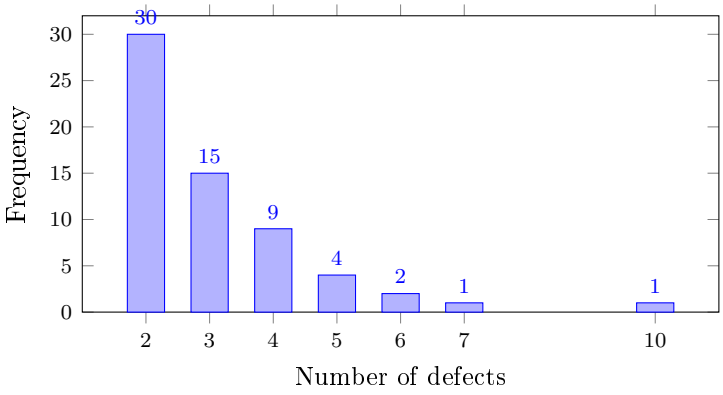


Figure 4.3: Histogram for the number of defects per part.

Histogram.

Measures of centre. The mean is

$$\bar{x} = \frac{2(30) + 3(15) + 4(9) + 5(4) + 6(2) + 7(1) + 10(1)}{62} = \frac{190}{62} = 3.06.$$

The mode is 2, the most frequently occurring value.

Quartiles and the box plot. The cumulative frequencies are

$$30,\ 45,\ 54,\ 58,\ 60,\ 61,\ 62.$$

Using $\dfrac{n+1}{4} = \dfrac{63}{4} = 15.75$, the quartile positions are

$$Q_1 = 15.75^{\text{th}}, \quad Q_2 = 31.5^{\text{th}}, \quad Q_3 = 47.25^{\text{th}} \text{ position},$$

giving

$$Q_1 = 2, \quad Q_2 = 3, \quad Q_3 = 4, \qquad \text{IQR} = 4 - 2 = 2.$$

The fences are

$$\text{lower fence} = 2 - 1.5(2) = -1, \qquad \text{upper fence} = 4 + 1.5(2) = 7.$$

Hence values beyond 7 (here, the value 10) are flagged as outliers.

4.4 Percentiles, Quartiles, and the Box Plot

The box plot rests on the notions of percentiles and quartiles, which locate positions within an ordered data set.

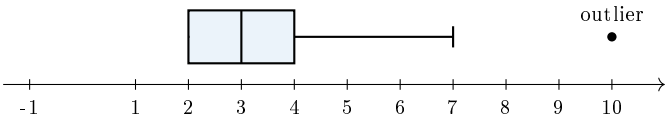


Figure 4.4: Box plot for the number of defects. The box spans $Q_1 = 2$ to $Q_3 = 4$ with the median at 3; the value 10 lies beyond the upper fence and is an outlier.

Quartiles and interquartile range

The **first**, **second**, and **third quartiles** Q_1, Q_2, Q_3 divide ordered data into four equal parts. The second quartile Q_2 is the median. The **interquartile range** is

$$\mathrm{IQR} = Q_3 - Q_1.$$

To locate quartiles by rank, order the data and use positions such as $(n+1)/4$, $(n+1)/2$, and $3(n+1)/4$. For instance, with 100 numbers, the median is the average of the two observations with ranks 50 and 51; the tenth percentile is the observation with rank $(10+11)/2 = 10.5$; the first quartile lies at rank $(25+26)/2 = 25.5$; and the third quartile lies at rank 75.5.

The box plot

The **box plot** displays the three quartiles together with the minimum and maximum of the data on a rectangular box, aligned either horizontally or vertically. The box encloses the interquartile range, with the left line at the first quartile Q_1 and the right line at the third quartile Q_3. A line is drawn through the box at the second quartile (the 50th percentile, or median) $Q_2 = \tilde{x}$. Lines called **whiskers** extend from the ends of the box to the extreme values. In large data sets the whiskers may extend only up to the 10th and 90th percentiles, or the 5th and 95th percentiles.

The box plot is especially useful in comparing two or more samples. Outliers are conventionally identified using fences placed at a distance of 1.5 IQR beyond each quartile.

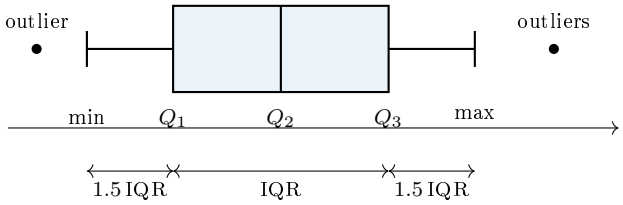


Figure 4.5: Anatomy of a box plot. The box spans the interquartile range from Q_1 to Q_3, the central line marks the median Q_2, and whiskers reach the most extreme non-outlier values. Points beyond 1.5 IQR of the quartiles are outliers.

Worked Problem 3: Furnace Temperatures

Example 4.4 (Mean, variance, median, and box plot)**.** The nine measurements that follow are furnace temperatures recorded on successive batches in a semiconductor manufacturing process (units are in °F):

$$953,\ 950,\ 948,\ 955,\ 951,\ 949,\ 957,\ 954,\ 955.$$

(a) Calculate the sample mean, variance, and standard deviation. (b) Find the median. How much could the largest temperature measurement increase without changing the median value?

(c) Construct a box plot of the data.

(a) Mean, variance, standard deviation.

$$\bar{x} = \frac{953 + 950 + 948 + 955 + 951 + 949 + 957 + 954 + 955}{9} = 952.444.$$

$$s^2 = \frac{(953 - 952.44)^2 + \cdots + (955 - 952.44)^2}{9 - 1} = 9.53, \qquad s = \sqrt{9.53} = 3.0867.$$

(b) Median. Sorting in ascending order:

$$948,\ 949,\ 950,\ 951,\ 953,\ 954,\ 955,\ 955,\ 957.$$

The number of measurements is odd, so the median is the middle value of the sorted data set:

$$M = Q_2 = 953.$$

There is no limit to which the highest temperature measurement can increase without changing the median value, because the middle value is not affected by the largest value in the data set.

Quartiles.

$$Q_1 = \frac{n+1}{4} = \frac{10}{4} = 2.5^{\text{th}} \text{ value} \Rightarrow Q_1 = 949.5,$$

$$Q_2 = 5^{\text{th}} \text{ value} = 953, \qquad Q_3 = 7.5^{\text{th}} \text{ position} = 955.$$

$$\text{Range} = 957 - 948 = 9, \qquad \text{IQR} = Q_3 - Q_1 = 955 - 949.5 = 5.5.$$

$$\text{Lower fence} = 949.5 - 1.5(5.5) = 941.25, \qquad \text{Upper fence} = 955 + 1.5(5.5) = 963.25.$$

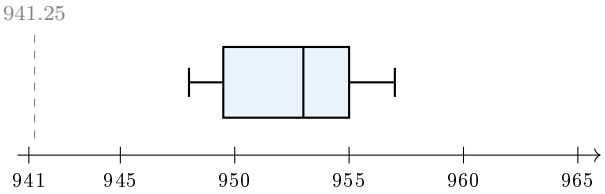


Figure 4.6: Box plot of the furnace-temperature data. No points fall beyond the fences, so there are no outliers.

(c) Box plot.

Worked Problem 4: A Ten-Point Data Set

Example 4.5. For the data

$$38,\ 39,\ 41,\ 45,\ 46,\ 47,\ 48,\ 48,\ 51,\ 52 \qquad (n = 10),$$

the quartiles are located as follows:

$$Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}} = \left(\frac{11}{4}\right)^{\text{th}} = 2.75^{\text{th}} \text{ value} = 40.5,$$

$$Q_2 = \left(\frac{n+1}{2}\right)^{\text{th}} = 5.5^{\text{th}} \text{ value} = 46.5,$$

$$Q_3 = \frac{3(n+1)^{\text{th}}}{4} = \frac{11 \times 3}{4} = 8.25^{\text{th}} \text{ value} = 48.75.$$

$$\text{IQR} = Q_3 - Q_1 = 48.75 - 40.5 = 8.25.$$

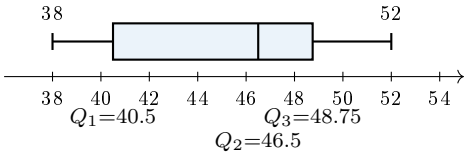


Figure 4.7: Box plot for the ten-point data set of Problem 4.

4.5 The Stem-and-Leaf Plot

Stem-and-leaf plot

Suppose that the data are represented by $x_1, x_2, \ldots, x_n$ and that each number x_i consists of at least two digits. To construct a **stem-and-leaf plot**, we divide each number x_i into two parts: a **stem**, consisting of one or more of the leading digits, and a **leaf**, consisting of the remaining digit. For example, if the number is 76, then we could divide it into stem 7 and leaf 6.

In general, choose few stems compared to the number of observations. It is best to choose between 5 and 20 stems. Once a set of stems has been chosen, they are listed along the left-hand margin of the display, and beside each stem all leaves corresponding to the observed data values are listed in the order in which they are encountered in the data set.

In order to assist in finding percentiles, we order the leaves by magnitude, producing an **ordered stem-and-leaf plot**. The first and third quartiles are occasionally denoted by the symbols Q_1 and Q_3 respectively, and the interquartile range is $\text{IQR} = Q_3 - Q_1$.

Worked Problem 5: Defects in Twenty Parts

Example 4.6. The number of defects when 20 parts were analysed were recorded as

35, 18, 20, 29, 21, 8, 16, 14, 13, 25, 49, 21, 38, 18, 92, 19, 19, 15, 22, 29.

(a) Draw the box plot. (b) Draw the stem-and-leaf diagram. What inferences can be drawn from these plots?

Ordered stem-and-leaf plot.

Stem	Leaf	Frequency
0	8	1
1	3 4 5 6 8 8 9 9	8
2	0 1 1 2 5 9 9	7
3	5 8	2
4	9	1
5		0
6		0
7		0
8		0
9	2	1

Inference: there are large gaps in the data, with minimum 8 and maximum 92.

Box plot. Arranging in ascending order:

$$8,\ 13,\ 14,\ 15,\ 16,\ 18,\ 18,\ 19,\ 19,\ 20,\ 21,\ 21,\ 22,\ 25,\ 29,\ 29,\ 35,\ 38,\ 49,\ 92.$$

$$Q_2 = \left(\frac{n+1}{2}\right)^{\text{th}} = 10.5^{\text{th}} \text{ position} = 20.5,$$

$$Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}} = 5.25^{\text{th}} \text{ position} = 16.5,$$

$$Q_3 = \frac{3(n+1)}{4}^{\text{th}} = 15.75^{\text{th}} \text{ position} = 29.$$

$$\text{IQR} = 29 - 16.5 = 12.5.$$

$$\text{Lower fence} = 16.5 - 12.5(1.5) = -2.25, \qquad \text{Upper fence} = 29 + 12.5(1.5) = 47.75.$$

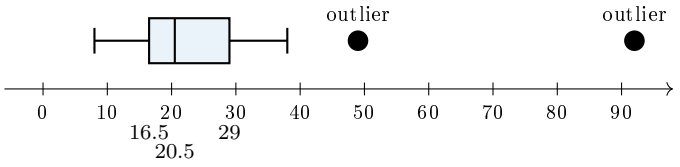


Figure 4.8: Box plot for Problem 5. **Interpretation:** the data set is unsymmetric; the middle 50% of the data lie between 16.5 and 29. The value 49 is a suspected outlier, and 92 is an outlier.

Worked Problem 6: Twenty Weight Measurements

Example 4.7. 20 weight measurements in a process were recorded as

$$2,\ 8,\ 11,\ 10,\ 8,\ 4,\ 2,\ 2,\ 9,\ 8,\ 4,\ 11,\ 12,\ 2,\ 4,\ 4,\ 20,\ 1,\ 10,\ 15.$$

(i) Draw a box plot for the data, showing the fences, quartiles, max, and min clearly. (ii) What are the range and the IQR? (iii) Are there any outliers?

Sorted data.

$$1,\ 2,\ 2,\ 2,\ 2,\ 4,\ 4,\ 4,\ 4,\ 8,\ 8,\ 8,\ 9,\ 10,\ 10,\ 11,\ 11,\ 12,\ 15,\ 20.$$

$$Q_1 = \frac{21}{4} = 5.25^{\text{th}} \text{ position} = 2.5, \quad Q_2 = \frac{21}{2} = 10.5^{\text{th}} = 8, \quad Q_3 = \frac{3(21)}{4} = 15.75^{\text{th}} = 10.75.$$

$$\text{Range} = 20 - 1 = 19, \qquad \text{IQR} = 10.75 - 2.5 = 8.25.$$

$$\text{Lower fence} = 2.5 - 1.5(8.25) = -9.875, \qquad \text{Upper fence} = 10.75 + 1.5(8.25) = 23.125.$$

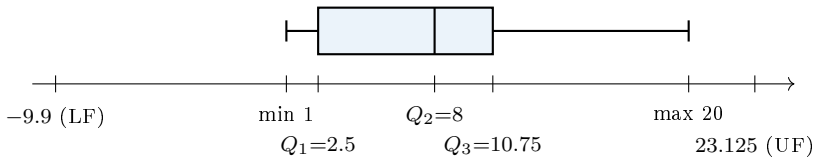


Figure 4.9: Box plot for Problem 6. Since both the minimum and maximum lie within the fences, there are **no outliers**.

Worked Problem 7: Motor-Fuel Octane Ratings

Example 4.8. The data on the motor-fuel octane ratings of several blends of gasoline are given in the table below. Construct a stem-and-leaf diagram and highlight the important feature that can be noticed from it.

93	90.2	91	92.4	90.5	88.7	89.8	89.5
89.8	84.6	91	87.3	92.1	94.8	90.2	89.8
88.8	96.7	88.7	91.3	85.9	85.6	84.7	89.3
89.0	89.4	92.6	87.7	92.3	92.3	86.5	
89.5	87.3	88.7	94.1	87.2	85.4	90.7	

Here the minimum is 84 and the maximum is 96.

Ordered stem-and-leaf plot (stems are integer ranges, leaves are tenths).

Stem	Ordered leaf	Cum. freq.
84	6 7	2
85	4 6 9	5
86	5	6
87	2 3 3 7	10
88	7 7 7 8	14
89	0 3 4 5 5 7 8 8	(8)
90	2 2 5 7	16
91	0 0 3	12
92	1 3 3 4 6	9
93	0	4
94	1 8	3
95		
96	7	1

The important feature is that the bulk of the octane ratings concentrate in the high-80s to low-90s, with the modal stem at 89.

Worked Problem 8: Furnace Temperatures (Revisited)

This problem repeats the nine furnace-temperature measurements of Problem 3, with identical data

$$953,\ 950,\ 948,\ 955,\ 951,\ 949,\ 957,\ 954,\ 955,$$

and asks again for (a) the sample mean, variance, and standard deviation; (b) the median and the amount by which the largest measurement may increase without changing the median; and (c) a box plot. The full solution is given in Problem 3; the principal results are repeated for convenience:

$$\bar{x} = 952.444, \quad s^2 = 9.53, \quad s = 3.0867, \quad M = Q_2 = 953,$$

$$Q_1 = 949.5,\ Q_3 = 955,\ \text{Range} = 9,\ \text{IQR} = 5.5,$$

$$\text{Lower fence} = 941.25, \quad \text{Upper fence} = 963.25.$$

As before, the median is unaffected by increases in the largest value, so there is no upper limit on how much the maximum may grow without changing the median.

Worked Problem 9: History Test Scores

Example 4.9. Observe the stem-and-leaf diagram of history test scores (July 2017) and answer the following.

Stem	Leaf
6	1 1 4 6 7 8
7	2 3 5 7 9
8	1 3 5 6 6 7 7 8 9
9	0 0 3 4 6 8 9 9
10	0 0

(i) **What is the best score?** 100.

(ii) **How many students took the test?** 30.

(iii) **How many students scored 90?** 2.

(iv) **What is the lowest score?** 61.

(v) **Find the range for the scores.** $100 - 61 = 39$.

Worked Problem 10: Process Times in a Chemical Industry

Example 4.10. Process times in a chemical industry are recorded in hours as

4.2, 4.7, 4.7, 5.0, 3.8, 3.6, 3.0, 5.1, 3.1, 3.8, 4.8, 4.0, 5.2, 4.3, 2.8, 2.0, 2.8, 3.3, 4.8, 5.0.

(i) Calculate the sample mean and standard deviation. (ii) Construct a box plot of the data. (iii) Construct a stem-and-leaf plot and comment on the data.

Mean and standard deviation.

$$\bar{x} = \frac{80}{20} = 4.0, \qquad s^2 = \frac{16.46}{19} = 0.866, \qquad s = 0.9307.$$

Ordered stem-and-leaf plot (stem = units, leaf = tenths).

Stem	Ordered leaf	Freq.
2	0 8 8	3
3	0 1 3 6 8 8	6
4	0 2 3 7 7 8 8	(7)
5	0 0 1 2	4

Quartiles.

$$Q_1 = \left(\frac{1}{4}\right) \times 21 = 5.25^{\text{th}} \text{ position} = 3.15,$$

$$Q_2 = 10.5^{\text{th}} \text{ position} = 4.1, \qquad Q_3 = 15.75^{\text{th}} \text{ position} = 4.8.$$

$$\text{IQR} = 4.8 - 3.15 = 1.65, \quad \text{LF} = 3.15 - 1.5(1.65), \quad \text{UF} = 4.8 + 1.5(1.65).$$

Worked Problem 11: Body Parameters of Asian Male Samples

Example 4.11. The following data represent variations in body parameters of 50 Asian male samples. Construct a stem-and-leaf plot and compute the measures of central tendency.

28	56	68	58	34	74	52	37	62	98
33	40	48	08	34	65	37	19	46	76
16	70	99	35	67	61	07	10	29	94
72	66	26	28	85	17	14	39	66	03
16	79	23	92	09	30	73	55	23	46

Ordered stem-and-leaf plot.

Stem	Ordered leaf
0	3 7 8 9
1	0 4 6 6 7 9
2	3 3 6 8 8 9
3	0 3 4 4 5 7 7 9
4	0 6 6 8
5	2 5 6 8
6	1 2 6 6 7 8
7	0 2 3 4 6 9
8	5
9	2 4 8 9

Measures of central tendency.

$$\text{Mean} = 45.46, \qquad \text{Median} = 39.5.$$

4.6 The Radar Chart

Radar chart

A **radar chart** is a graphical method of displaying multivariate data in the form of a two-dimensional chart of three or more quantitative variables represented on axes starting from the same point. Radar charts — also known as spider charts, polar charts, web charts, or star plots — are a way to visualise multivariate data. They are used to plot one or more groups of values over multiple common variables.

The radar chart consists of a sequence of equiangular spokes, called radii, with each spoke representing one of the variables. The data length of a spoke is proportional to the magnitude of the variable for the data point relative to the maximum magnitude of the variable across all data points. A line is drawn connecting the data values for each spoke. This gives the plot a star-like appearance, and the origin of one of its popular names.

A radar chart can be used to answer the following questions:

- Which observations are most similar; that is, are there clusters of observations? (Radar charts are used to examine the relative values for a single data point and to locate similar points or dissimilar points.)
- Are there outliers?

Developing radar charts.

1. Identify the alternatives to be compared.
2. Generate criteria to rate each alternative.
3. Rate each alternative based on the criteria.
4. Select a suitable structure and set relevant options; set and label the axes.
5. Label the alternative's ratings on the chart.

Guidelines.

1. No more than 15 alternatives should be chosen, or else the chart will look cramped.
2. Use at least 3 criteria, but more than 7 will become complex.
3. Standardise ratings to a uniform scale.

Example 4.12 (Manufacturing process ratings)**.** Three criteria are used to compare a process against a best-in-class benchmark.

Category	Our score	Best-in-class	Standardised
1. Safety	Red. 5 accidents	Red. 12 accidents	$5/12 = 42\%$
2. Training expense	Rs. 10,000	Rs. 20,000	50%
3. Standard operating procedures	74%	96%	$74/96 = 77\%$

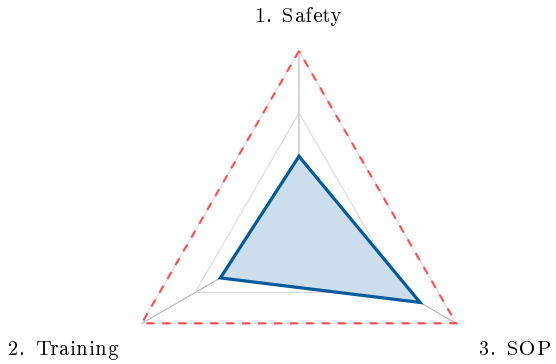


Figure 4.10: Radar chart comparing the process (solid) against the best-in-class benchmark (dashed), with each axis standardised to a common scale.

Chapter 5

Concepts of Probability and Random Variables

Probability provides the formal language for reasoning about uncertainty. This chapter introduces random experiments, sample spaces, events, the algebra of events, and the counting techniques that underlie the computation of probabilities for equally likely outcomes.

Engineering systems are usefully pictured as transforming inputs into outputs under the influence of two kinds of variables: **controlled variables**, which the engineer sets, and **noise variables**, which the engineer cannot control. It is the noise variables that introduce randomness into the output.

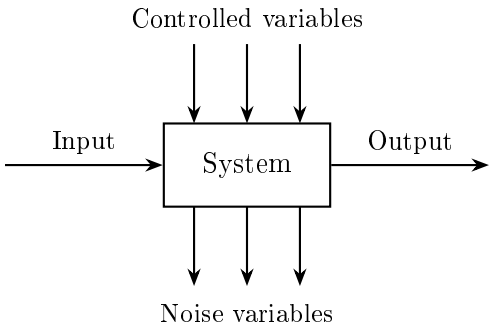


Figure 5.1: A system viewed as a transformation of inputs to outputs, subject to controlled and noise variables.

5.1 Random Experiments and Sample Spaces

Random experiment

A **random experiment** is an experiment that can result in different outcomes, even though it is repeated in the same manner every time.

Sample space

The set of all possible outcomes of a random experiment is called the **sample space** of the experiment. The sample space is denoted by S. A sample space is often defined based on the objectives of the analysis.

Example 5.1 (Thickness of a molded part)**.** Consider an experiment in which you select a molded plastic part, such as a connector, and measure its thickness. The possible values for thickness depend on the resolution of the measuring instrument, and also depend on upper and

lower bounds for thickness. It might be convenient to define the sample space as simply the positive real line, since a negative value for thickness cannot occur:

$$S = \mathbb{R}^+ = \{\, x \mid x > 0 \,\}.$$

If it is known that all connectors will be between 10 and 11 millimetres thick, the sample space could be

$$S = \{\, x \mid 10 < x < 11 \,\}.$$

If the objective of the analysis is to consider only whether a particular part is low, medium, or high for thickness, the sample space will be the set of three outcomes:

$$S = \{\text{low},\ \text{medium},\ \text{high}\}.$$

If the objective is to consider only whether or not a particular part conforms to the manufacturing specifications, the sample space is

$$S = \{\text{yes},\ \text{no}\},$$

indicating whether or not the part conforms.

Discrete and continuous sample spaces

A sample space is **discrete** if it consists of a finite or countably infinite set of outcomes. A sample space is **continuous** if it contains an interval (either finite or infinite) of real numbers.

In the previous examples, the choice $S = \mathbb{R}^+$ is an example of a continuous sample space, whereas $S = \{\text{yes}, \text{no}\}$ is a discrete sample space.

Example 5.2 (Sample spaces for various experiments)**.**

Experiment	Sample space
Throw a die	$\{1, 2, \ldots, 6\}$
Record claim amounts in all insurance policies	$\{0, 1, \ldots, \infty\}$
Record height of students (cm)	$\{\, x \mid 100 \leq x \leq 250 \,\}$
Record number of accidents per day in front of a gate	$\{0, 1, 2, \ldots, 100\}$

The sample spaces that contain the outcomes of dice rolls, drawings from a bag of mixed-colour balls, and dealings from a regular 52-card deck are examples of discrete sample spaces. Another example is the number of roulette-wheel spins made before the ball lands on 25; the number can range from $1, 2, 3, \ldots$ up to infinity, but the number has to be an integer. Sample spaces that contain the outcomes of temperature readings, height measurements, and salaries are examples of continuous sample spaces.

5.1.1 Sampling With and Without Replacement

In random experiments in which items are selected from a batch, we indicate whether or not the selected item is replaced before the next one is selected.

Example 5.3 (With and without replacement)**.** If the batch consists of 3 items $\{a, b, c\}$ and our experiment is to select 2 items *without* replacement, the sample space of ordered outcomes can be represented as

$$S_{\text{without}} = \{ab, ac, ba, bc, ca, cb\}, \quad \text{or unordered as } \{\{a, b\}, \{b, c\}, \{c, a\}\}.$$

If items are *replaced* before the next one is selected, the sampling is referred to as *with replacement*. Then the possible ordered outcomes are

$$S_{\text{with}} = \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}.$$

Example 5.4 (Good and defective parts)**.** Suppose that there are 5 defective parts and 95 good parts in a batch. To study the quality of the batch, 2 are selected without replacement. Let g denote a good part and d denote a defective part. The ordered sample space in this case is

$$S = \{gg, gd, dg, dd\}.$$

If there is only one defective part in the batch, then the sample space is

$$S = \{gg, gd, dg\}.$$

5.1.2 Tree Diagrams

Sample spaces can also be described with tree diagrams. Consider a digital communication system in which each message is classified as to whether it is received within the time specified by the system design — either "on time" or "late". If 3 messages are classified, the possible results of the 3 messages can be displayed by 8 branches in the tree diagram.

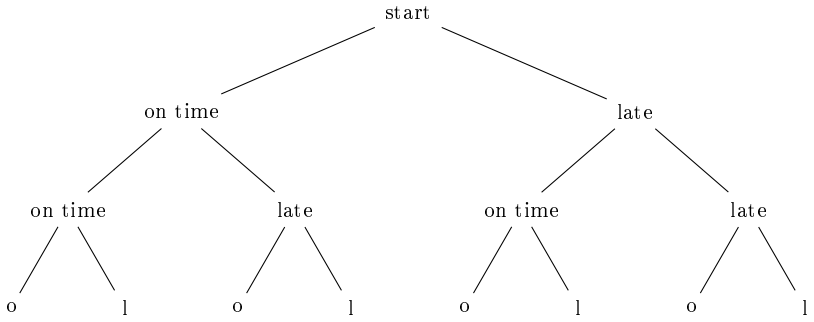


Figure 5.2: Tree diagram for the classification of 3 messages, each "on time" (o) or "late" (l), giving $2^3 = 8$ branches.

5.2 Events and the Algebra of Events

Event

An **event** is a subset of the sample space of a random experiment.

Union, intersection, complement

For two events E_1 and E_2:

- The **union** $E_1 \cup E_2$ is the event that consists of all outcomes that are contained in either of the two events.
- The **intersection** $E_1 \cap E_2$ is the event that consists of all outcomes that are contained in both of the two events.
- The **complement** of an event in the sample space is the set of outcomes in the sample space that are not in the event. It is denoted E' or E^c.

Example 5.5 (Events on a continuous sample space)**.** Measurements of the thickness of a plastic connector might be modelled with sample space $S = \mathbb{R}^+$, the set of positive real numbers. Let

$$E_1 = \{\, x \mid 10 \le x < 12 \,\}, \qquad E_2 = \{\, x \mid 11 < x < 15 \,\}.$$

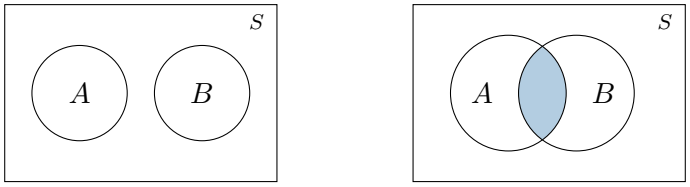
Then

$$E_1 \cup E_2 = \{\, x \mid 10 \le x < 15 \,\}, \qquad E_1 \cap E_2 = \{\, x \mid 11 < x < 12 \,\},$$
$$E_1' = \{\, x \mid x < 10 \text{ or } 12 \le x \,\}, \qquad E_1' \cap E_2 = \{\, x \mid 12 \le x < 15 \,\}.$$

Mutually exclusive events

Two events E_1 and E_2 such that $E_1 \cap E_2 = \varnothing$ are said to be **mutually exclusive**.

Venn diagrams are used to represent a sample space and events in the sample space.



Mutually exclusive events $A \cap B$

Figure 5.3: Venn diagrams: mutually exclusive events (left) and the intersection $A \cap B$ of two overlapping events (right).

5.2.1 Laws of Set Operations

Property 5.1 (Complement of a complement)**.**

$$(E')' = E.$$

Property 5.2 (Distributive laws)**.**

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C),$$
$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C).$$

Property 5.3 (De Morgan's laws)**.**

$$(A \cup B)' = A' \cap B', \qquad (A \cap B)' = A' \cup B'.$$

Property 5.4 (Commutativity)**.**

$$A \cap B = B \cap A, \qquad A \cup B = B \cup A.$$

5.3 Counting Techniques

Counting techniques determine the number of outcomes in each event, which is essential for computing probabilities when outcomes are equally likely.

Multiplication rule

If an operation can be described as a sequence of k steps, and the number of ways of completing the first step is n_1, and the number of ways of completing the second step is n_2 for each way of completing the first step, and so forth, then the total number of ways of completing the operation is

$$n_1 \times n_2 \times \cdots \times n_k.$$

5.3.1 Permutations

Permutation

A **permutation** is an ordered sequence of elements of a set. For a set $S = \{a, b, c\}$, the permutations are $abc, acb, bac, bca, cab, cba$. The number of permutations of n different elements is

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1.$$

Permutations of subsets

The number of permutations of subsets of r elements selected from a set of n different elements is

$$P_r^n = n \times (n-1) \times (n-2) \times \cdots \times (n-r+1) = \frac{n!}{(n-r)!}.$$

Example 5.6 (Circuit-board component placement)**.** A printed circuit board has 8 different locations in which a component can be placed. If four different components are to be placed on the board, how many different designs are possible? Using the permutation formula,

$$P_4^8 = 8 \times 7 \times 6 \times 5 = \frac{8!}{4!} = 1680 \text{ different designs.}$$

Permutations of similar objects

The number of permutations of $n = n_1 + n_2 + \cdots + n_r$ objects, of which n_1 are of one type, n_2 are of a second type, ..., and n_r are of an rth type, is

$$\frac{n!}{n_1!\, n_2! \cdots n_r!}.$$

Example 5.7 (Part labels from lines)**.** A part is labelled by printing with 4 thick lines, 3 medium lines, and 2 thin lines. If each ordering of 9 lines represents a different label, how many different labels can be generated by using this scheme? The number of possible part labels is

$$\frac{9!}{4!\,3!\,2!} = 1260.$$

5.3.2 Combinations

Combination

The number of **combinations** — subsets of size r that can be selected from a set of n elements — is denoted $\binom{n}{r}$ or C_r^n, and

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}.$$

Example 5.8 (Samples with exactly two defectives)**.** A bin of 50 parts manufactured contains 3 defective parts and 47 non-defective parts. A sample of six parts is selected from the 50 parts and not replaced. How many different samples are there of size six that contain exactly 2 defective parts?

A subset containing exactly 2 defective parts can be formed by choosing 2 defective parts from 3 defective parts:

$$\binom{3}{2} = \frac{3!}{1!\,2!} = 3 \text{ ways.}$$

The second step is to select the remaining 4 parts from the 47 acceptable parts in the bin:

$$\binom{47}{4} = \frac{47!}{4!\,43!} = 178365 \text{ different ways.}$$

By the multiplication rule, the number of subsets of size six that contain exactly 2 defective items is

$$3 \times 178365 = 535{,}095.$$

The total number of different subsets of size six is

$$\binom{50}{6} = \frac{50!}{44!\,6!} = 15{,}890{,}700.$$

Therefore the probability is

$$\text{probability} = \frac{535{,}095}{15{,}890{,}700} = 0.034 = \frac{\text{number of outcomes in event}}{\text{number of outcomes in sample space}}.$$

5.4 Sample-Space Examples

Example 5.9 (Three machined parts above/below target)**.** Each of 3 machined parts is classified as either above or below the target specification for the part. Let a and b denote a part above and below specification respectively. Then

$$S = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}.$$

Example 5.10 (Nonconformities in power supplies)**.** In the final inspection of electronic power supplies, 3 types of nonconformities might occur: functional, minor, or cosmetic. Power supplies that are defective are further classified as to the type of nonconformity. With a denoting an acceptable power supply and f, m, c the defective categories,

$$S = \{a, f, m, c\}.$$

Example 5.11 (Three-digit ammeter)**.** An ammeter that displays 3 digits is used to measure current in milliamperes. The sample space is the 1000 possible 3-digit integers:

$$S = \{000, 001, \ldots, 999\}.$$

Example 5.12 (pH reading). The pH reading of a water sample to the nearest tenth of a unit has sample space

$$S = \{1.0, 1.1, 1.2, 1.3, \ldots, 14.0\}.$$

Example 5.13 (Repeated calls to a busy line). Calls are repeatedly placed to a busy phone line until a connect is achieved. With c denoting connect and b denoting busy,

$$S = \{c, bc, bbc, bbbc, bbbbc, \ldots\}.$$

Example 5.14 (Length and width in injection molding). In an injection-molding operation, the length and width, denoted X and Y respectively, of each molded part are evaluated. Let A denote the event $48 < X < 52$ cm; let B denote the event $9 < Y < 11$ cm; and let C denote the event that a critical length meets customer requirements. If the events were mutually exclusive, then $A \cap B$ would be the null set, so the process could not produce parts with $X = 50$ cm and $Y = 10$ cm; such a process would not be successful.

Example 5.15 (Sampling printed circuit boards without replacement). A sample of 2 printed circuit boards is selected without replacement from a batch. Let $\bar{d}$ denote no defect, d denote a minor defect, and D denote a major defect.

- If the batch contains 90 boards that are not defective, 8 boards with minor defects, and 2 boards with major defects, the ordered sample space is

$$S = \{\bar{d}\bar{d},\ \bar{d}d,\ \bar{d}D,\ d\bar{d},\ dd,\ dD,\ D\bar{d},\ Dd,\ DD\}.$$

- If the batch contains 90 boards not defective, 8 boards with minor defects, and 1 board with a major defect, the sample space is similarly $S = \{\bar{d}\bar{d}, \bar{d}D, \bar{d}d, d\bar{d}, dd, dD, D\bar{d}, Dd\}$.

Example 5.16 (Configuring a personal digital assistant). An order for a personal digital assistant can specify any one of 5 memory sizes, any one of 3 types of displays, any one of 4 sizes of hard disk, and can either include or not include a pen tablet. How many different systems can be ordered? By the multiplication rule,

$$5 \times 3 \times 4 \times 2 = 120 \text{ different systems.}$$

Example 5.17 (Production sequences). A manufacturing process consists of 10 operations that can be completed in any order. How many different production sequences are possible? Since once an operation is selected it cannot be selected again,

$$10! = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 3{,}628{,}800.$$

Example 5.18 (Inspecting semiconductor chips). A lot of 140 semiconductor chips is inspected by choosing a sample of five chips. Assume 10 of the chips do not conform to customer requirements.

(a) **How many different samples are possible?**

$$\binom{140}{5} = \frac{140!}{135!\,5!} = 41{,}69{,}65{,}528.$$

(b) **How many samples of 5 contain exactly one nonconforming chip?**

$$\binom{130}{4}\binom{10}{1} = \frac{130!}{126!\,4!} \times \frac{10!}{9!\,1!} = 11{,}35{,}88{,}800.$$

(c) **How many samples of 5 contain at least one nonconforming chip?** Counting the disjoint cases (4 good and 1 bad), (3 good and 2 bad), (2 good and 3 bad), (1 good and 4 bad), and (0 good and 5 bad):

$$\binom{130}{4}\binom{10}{1} + \binom{130}{3}\binom{10}{2} + \binom{130}{2}\binom{10}{3} + \binom{130}{1}\binom{10}{4} + \binom{130}{0}\binom{10}{5}$$
$$= 113588800 + 16099200 + 1006200 + 27300 + 252 = 130{,}721{,}752.$$

Example 5.19 (Stacking components in a cylindrical casing)**.** In the design of an electromechanical product, 7 different components are to be stacked into a cylindrical casing that holds 12 components in a manner that minimises the impact of shocks. One end of the casing is designated as the bottom and the other end is the top.

(a) **How many different designs are possible?**

$$P_7^{12} = \binom{12}{7} \times 7! = \frac{12!}{7!\,5!} \times 7! = \frac{12!}{5!} = 3{,}99{,}16{,}800.$$

(b) **If the seven components are all identical, how many different designs are possible?** Since 7 of 12 positions are occupied by identical components (equivalently $n_6 = 7$ identical and the rest distinct positions),

$$\frac{12!}{5!\,7!} = \binom{12}{7} = 95{,}040 \quad \text{(choosing positions)},$$

more precisely $\dfrac{12!}{1!\,1!\,1!\,1!\,1!\,7!} = 95{,}040$ when 5 positions are distinct and 7 are identical.

(c) **If the 7 components consist of 5 of one type and 4 of another type**[1]**, how many different designs are possible?**

$$\binom{12}{7} \times \frac{7!}{3!\,4!} = \frac{12!}{5!\,3!\,4!} = 27{,}720.$$

[1] As recorded in the source; here the multiset arrangement formula is applied.

Chapter 6

Axioms of Probability and the Addition Rules

Having established how to count outcomes, we now assign probabilities to events and develop the rules that govern how probabilities combine.

6.1 Equally Likely Outcomes

Equally likely outcomes

Whenever a sample space consists of N possible outcomes that are equally likely, the probability of each outcome is $1/N$.

For a discrete sample space, the probability of an event E, denoted $\mathrm{P}(E)$, equals the sum of the probabilities of the outcomes in E.

6.2 The Axioms of Probability

Axioms of probability

Probability is a number that is assigned to each member of a collection of events from a random experiment that satisfies the following properties. If S is the sample space and E is any event in a random experiment, then:
(1) $\mathrm{P}(S) = 1$.
(2) $0 \leq \mathrm{P}(E) \leq 1$.
(3) For two events E_1 and E_2 with $E_1 \cap E_2 = \varnothing$,

$$\mathrm{P}(E_1 \cup E_2) = \mathrm{P}(E_1) + \mathrm{P}(E_2).$$

Two immediate consequences are

$$\mathrm{P}(\varnothing) = 0, \qquad \mathrm{P}(E') = 1 - \mathrm{P}(E) \text{ for any event } E.$$

6.3 The Addition Rule

Addition rule for two events

For any two events A and B,

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B).$$

This follows from a decomposition of the union into disjoint pieces. Writing $\mathrm{P}(A) = x + y$ and $\mathrm{P}(B) = y + z$ where $y = \mathrm{P}(A \cap B)$, the union covers $x + y + z = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$.

Property 6.1 (Mutually exclusive events)**.** *If A and B are mutually exclusive, then* $\mathrm{P}(A \cap B) = 0$ *and*

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B).$$

More generally, if $E_1, E_2, \ldots, E_k$ is a collection of mutually exclusive events — so that $E_i \cap E_j = \varnothing$ for all pairs — then

$$\mathrm{P}(E_1 \cup E_2 \cup \cdots \cup E_k) = \mathrm{P}(E_1) + \mathrm{P}(E_2) + \cdots + \mathrm{P}(E_k).$$

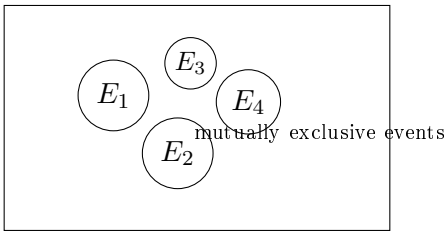


Figure 6.1: A collection of mutually exclusive events partitioning part of the sample space.

Addition rule for three events

$$\begin{aligned}\mathrm{P}(A \cup B \cup C) &= \mathrm{P}\big[(A \cup B) \cup C\big] = \mathrm{P}(A \cup B) + \mathrm{P}(C) - \mathrm{P}\big[(A \cup B) \cap C\big] \\ &= \mathrm{P}(A) + \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(A \cap B) - \mathrm{P}(A \cap C) \\ &\quad - \mathrm{P}(B \cap C) + \mathrm{P}(A \cap B \cap C).\end{aligned}$$

6.4 Worked Problems on Probability of Events

Venn Diagram Problem: Length and Width

Example 6.1. In an injection-molding operation, the length and width X and Y of each molded part are evaluated. Let A denote $48 < X < 52$ cm, B denote $9 < Y < 11$ cm, and C denote that a critical length meets customer requirements. Construct a Venn diagram that includes these events. Shade the areas that represent (a) A; (b) $A \cap B$; (c) $A' \cup B$; (d) $A \cup B$. (e) If these events were mutually exclusive, how successful would this production operation be? Would the process produce parts with $X = 50$ cm and $Y = 10$ cm?

(e) If the events are mutually exclusive, then $A \cap B$ is the null set. Therefore the process does not produce parts with $X = 50$ cm and $Y = 10$ cm, and the process would not be successful.

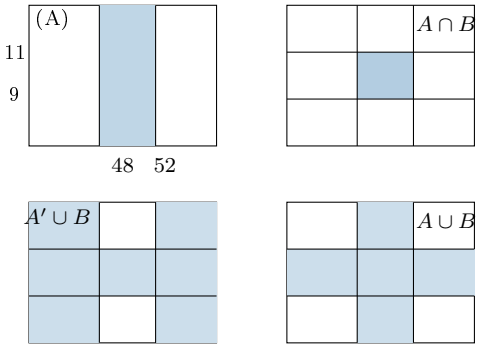


Figure 6.2: Shaded Venn-style regions for the length–width events: A, $A \cap B$, $A' \cup B$, and $A \cup B$, where A is the vertical band $48 < X < 52$ and B is the horizontal band $9 < Y < 11$.

Conformance of Emissions Samples

Example 6.2. Samples of emissions from 3 suppliers are classified for conformance to air-quality specifications. The results from 100 samples are summarised below.

Supplier	Conforms: Yes	No
1	22	8
2	25	5
3	30	10

Let A denote the event that a sample is from supplier 1, and let B denote the event that a sample conforms to specifications. Determine the number of samples in $A' \cap B$, B', and $A \cup B$.

Solution.

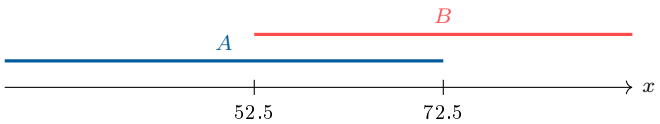
$$\begin{aligned} A' \cap B &= \text{sample not from supplier 1 and conforms} = 25 + 30 = 55, \\ B' &= \text{sample not conforming} = 8 + 5 + 10 = 23, \\ A \cup B &= \text{samples from supplier 1, or conforming} \\ &= (22 + 8) + (22 + 25 + 30) - 22 = 85. \end{aligned}$$

Reactor Rise Time

Example 6.3. The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be the positive real numbers. Events A and B are defined as

$$A = \{\, x \mid x < 72.5 \,\}, \qquad B = \{\, x \mid x > 52.5 \,\}.$$

Describe each of the following events: (a) A'; (b) B'; (c) $A \cap B$; (d) $A \cup B$.



Solution.

$$A \cup B = S = \{\, x \mid 0 < x < \infty \,\}, \qquad A \cap B = S' = \{\, x \mid 52.5 < x < 72.5 \,\}.$$

(Here A' is $\{x \mid x \geq 72.5\}$ and B' is $\{x \mid x \leq 52.5\}$.)

Last Digit of a Weight Measurement

Example 6.4. If the last digit of a weight measurement is equally likely to be any of the digits 0 through 9: (a) What is the probability that the last digit is 0? (b) What is the probability that the last digit is greater than or equal to 5?

$$\text{(a)}\ \ \frac{1}{10} = 0.1, \qquad \text{(b)}\ \ \frac{5}{10} = 0.5.$$

Validity of a Credit-Card Number

Example 6.5. A credit card contains 16 digits between 0 and 9. However, only 100 million numbers are valid. If a number is entered randomly, what is the probability that it is a valid number?

The total number of possibilities is

$$N_T = (\text{number of states for each digit})^{\text{number of digits}} = 10^{16}.$$

Let A represent the event that the random number entered is a valid number, so $n(A) = 100 \times 10^6$. Then

$$\text{probability} = \frac{n(A)}{N} = \frac{100 \times 10^6}{10^{16}} = \frac{1}{10^8} = 1 \times 10^{-8}.$$

Licence-Plate Selection

Example 6.6. Suppose your vehicle is licensed in a state that issues license plates consisting of 3 digits (between 0 and 9) followed by 3 letters (between A and Z). If a license number is selected randomly, what is the probability that yours is the one selected?

$$N_D = 10^3, \qquad N_A = 26^3, \qquad \text{probability} = \frac{1}{10^3 \times 26^3}.$$

A Four-Event Probability Table

Example 6.7. For events A and B with the supplier data ($\mathrm{P}(A) = 30/100$, and a second event B with 77 favourable outcomes), compute (a) $\mathrm{P}(A)$; (b) $\mathrm{P}(B)$; (c) $\mathrm{P}(A')$; (d) $\mathrm{P}(A \cap B)$; (e) $\mathrm{P}(A \cup B)$; (f) $\mathrm{P}(A' \cup B)$.

(a) $\mathrm{P}(A) = \frac{30}{100} = 0.3$, (b) $\mathrm{P}(B) = \frac{77}{100} = 0.77$,
(c) $\mathrm{P}(A') = 0.7$, (d) $\mathrm{P}(A \cap B) = \frac{22}{100} = 0.22$,
(e) $\mathrm{P}(A \cup B) = 0.30 + 0.77 - 0.22 = \frac{85}{100} = 0.85$, (f) $\mathrm{P}(A' \cup B) = 0.70 + 0.77 - 0.55 = \frac{92}{100} = 0.92$.

Polycarbonate Disks: Scratch and Shock Resistance

Example 6.8. Disks of polycarbonate plastic from a supplier are analysed for scratch and shock resistance. The results from 100 disks are summarised below.

		Shock resistance	
		High	Low
Scratch resistance	High	70	9
	Low	16	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. If a disk is selected at random, determine (a) $\mathrm{P}(A)$; (b) $\mathrm{P}(B)$; (c) $\mathrm{P}(A')$; (d) $\mathrm{P}(A \cap B)$; (e) $\mathrm{P}(A \cup B)$; (f) $\mathrm{P}(A' \cup B)$.

$$\text{(a) } \mathrm{P}(A) = \tfrac{86}{100} = 0.86, \qquad \text{(b) } \mathrm{P}(B) = 0.79,$$
$$\text{(c) } \mathrm{P}(A') = \tfrac{14}{100} = 0.14, \qquad \text{(d) } \mathrm{P}(A \cap B) = 0.70,$$
$$\text{(e) } \mathrm{P}(A \cup B) = 0.86 + 0.79 - 0.70 = 0.95, \qquad \text{(f) } \mathrm{P}(A' \cup B) = 0.14 + 0.79 - 0.09 = 0.84.$$

Customer Orders of Items A, B, C

Example 6.9. Customers order any combination of items A, B, C. The following probabilities are given:

$$\mathrm{P}(C) = 0.3, \quad \mathrm{P}(B) = 0.4, \quad \mathrm{P}(A \cap B) = 0.15, \quad \mathrm{P}(A \cap C) = 0.15,$$
$$\mathrm{P}(B \cap C) = 0.55, \quad \mathrm{P}(A \cap B \cap C) = 0.05, \quad \mathrm{P}(\text{none}) = 0.2.$$

Find $\mathrm{P}(A)$ and the probability that A alone is ordered.

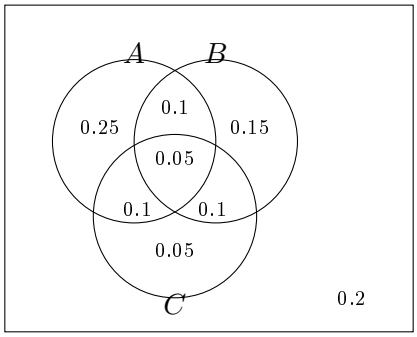


Figure 6.3: Venn diagram for the customer-order problem, with probabilities assigned to each disjoint region.

Solution. First, using the addition rule on B and C,

$$\mathrm{P}(B \cup C) = \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(B \cap C) \Rightarrow 0.55 = 0.4 + 0.3 - \mathrm{P}(B \cap C),$$

so $\mathrm{P}(B \cap C) = 0.7 - 0.55 = 0.15$ (the probability that B and C are both ordered). Then

$$\mathrm{P}(A \text{ alone}) = 1 - 0.2 - (0.3 + 0.15 + 0.1) = 0.25,$$

and assembling the regions of A,

$$\mathrm{P}(A) = 0.25 + 0.1 + 0.1 + 0.05 = 0.5.$$

Automobile Headlamp Performance

Example 6.10. A manufacturer of front lights for automobiles tests lamps under a high-humidity, high-temperature environment using intensity and useful life as the responses of interest. The following table shows the performance of 130 lamps.

		Useful life	
		Satisfactory	Unsatisfactory
Intensity	Satisfactory	117	3
	Unsatisfactory	8	2

(i) Find the probability that a randomly selected lamp will yield unsatisfactory results under any criterion. (ii) The customers for these lamps demand 95% satisfactory results. Can the lamp manufacturer meet this demand?

Solution.

$$\text{(i)}\quad \text{P(unsatisfactory under some criterion)} = \frac{3+2+8}{130} = \frac{13}{130} = 0.1.$$

$$\text{(ii)}\quad \text{Satisfactory under both criteria} = \frac{117}{130} = 90\%.$$

Since only 90% are satisfactory, the lamp manufacturer has *not* met the 95% demand.

Chapter 7

Conditional Probability, Independence, and Bayes' Theorem

Often the probability of an event changes once we learn that another event has occurred. Conditional probability formalises this updating of belief, and leads to the multiplication rule, the total probability rule, the notion of independence, and finally Bayes' theorem.

7.1 Conditional Probability

Conditional probability

The **conditional probability** of an event B given an event A, denoted $\mathrm{P}(B \mid A)$, is

$$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)}, \qquad \text{for } \mathrm{P}(A) > 0.$$

This definition can be understood in the special case in which all outcomes of a random experiment are equally likely. If there are n total outcomes,

$$\mathrm{P}(A) = \frac{\text{number of outcomes in } A}{n}, \qquad \mathrm{P}(A \cap B) = \frac{\text{number of outcomes in } A \cap B}{n},$$

so that

$$\frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{\text{number of outcomes in } A \cap B}{\text{number of outcomes in } A}.$$

Defective Parts Without Replacement

Example 7.1. A day's production of 850 manufactured parts contains 50 parts that do not meet customer requirements. Two parts are selected randomly without replacement from the batch. What is the probability that the second part is defective given that the first part is defective?

Let A denote the event that the first part is defective and B the event that the second part is defective. After one defective part is removed, 49 defectives remain among 849 parts, so

$$\mathrm{P}(B \mid A) = \frac{49}{849}.$$

Example 7.2 (Three parts without replacement)**.** If 3 parts are selected randomly from the same batch, what is the probability that the first two are defective and the third is not defective?

$$\mathrm{P}(d\,d\,n) = \frac{50}{850} \times \frac{49}{849} \times \frac{800}{848} = 0.0032.$$

Conformance and Suppliers (Continued)

Example 7.3. Using the supplier conformance data, with $\mathrm{P}(A) = 0.86$ and $\mathrm{P}(B) = 0.79$ and $\mathrm{P}(A \cap B) = 0.70$:

$$\mathrm{P}(A \mid B) = \frac{0.70}{0.79} = 0.886, \qquad \mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{0.70}{0.86} = 0.814.$$

Air-Conditioning Failure Mechanisms

Example 7.4. A maintenance firm has gathered the following information regarding the failure mechanisms for air-conditioning systems.

		Evidence of gas leaks	
		Yes	No
Evidence of electrical failure	Yes	55	17
	No	32	3

The units without evidence of gas leaks or electrical failure showed other types of failure. If this is a representative sample of AC failure, find the probability that (a) failure involves a gas leak; (b) there is evidence of electrical failure given that there was a gas leak; (c) there is evidence of a gas leak given that there is evidence of electrical failure.

Solution. Let E represent electrical failure and G represent a gas leak. The marginal totals give

$$\mathrm{P}(E) = \frac{72}{107} = 0.67, \quad \mathrm{P}(E') = \frac{35}{107} = 0.33, \quad \mathrm{P}(G) = \frac{87}{107} = 0.81.$$

$$\begin{aligned}
&\text{(i)} \quad \mathrm{P}(G) = \frac{87}{107} = 0.81, \\
&\text{(ii)} \quad \mathrm{P}(E \mid G) = \frac{\mathrm{P}(E \cap G)}{\mathrm{P}(G)} = \frac{55/107}{87/107} = \frac{55}{87}, \\
&\text{(iii)} \quad \mathrm{P}(G \mid E) = \frac{\mathrm{P}(G \cap E)}{\mathrm{P}(E)} = \frac{55}{72}.
\end{aligned}$$

Cast-Aluminium Parts: Surface Finish and Length

Example 7.5. Samples of a cast-aluminium part are classified on the basis of surface finish (in microinches) and length measurements. The results of 100 parts are summarised below.

		Length	
		Excellent	Good
Surface finish	Excellent	80	2
	Good	10	8

Let A denote the event that the sample has excellent surface finish, and B the event that the sample has excellent length. Find (a) $\mathrm{P}(A)$; (b) $\mathrm{P}(B)$; (c) $\mathrm{P}(A \mid B)$; (d) $\mathrm{P}(B \mid A)$; (e) if a selected part has excellent surface finish, the probability that the length is excellent; (f) if a selected part has good length, the probability that the surface finish is excellent.

Solution.

$$\mathrm{P}(A) = \frac{82}{100} = 0.82, \qquad \mathrm{P}(B) = \frac{90}{100} = 0.90,$$
$$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{80}{90} = 0.889, \qquad \mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{80}{82} = 0.976,$$
$$\mathrm{P}(B \mid A) = 0.976, \qquad \mathrm{P}(A \mid B') = \frac{\mathrm{P}(A \cap B')}{\mathrm{P}(B')} = \frac{2}{10} = 0.2.$$

Melanin and Moisture Content

Example 7.6. With $A \to$ low melanin content and $B \to$ high moisture content, the joint counts are:

		Melanin content	
		High	Low
Moisture content	High	13	7
	Low	48	32

Then

$$\mathrm{P}(A) = \frac{39}{100} = 0.39, \quad \mathrm{P}(B) = \frac{20}{100} = 0.2,$$
$$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(B \cap A)}{\mathrm{P}(B)} = \frac{7}{20} = 0.35, \qquad \mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{7}{39}.$$

Deceased Beetles: Autolysis and Putrefaction

Example 7.7. The following table summarises the number of deceased beetles under autolysis (the destruction of a cell after its death by the action of its own enzymes) and putrefaction (decomposition of organic matter, especially protein, by microorganisms, resulting in foul-smelling matter).

		Autolysis	
		High	Low
Putrefaction	High	14	59
	Low	18	9

(a) If the autolysis of a sample is high, what is the probability that putrefaction is low? (b) If the putrefaction of a sample is high, what is the probability that autolysis is high? (c) If the putrefaction of a sample is low, what is the probability that autolysis is low?

Solution. Let $A \to$ autolysis high and $P \to$ putrefaction high. The totals give

$$\mathrm{P}(A) = \frac{32}{100}, \quad \mathrm{P}(A') = \frac{68}{100}, \quad \mathrm{P}(P) = \frac{73}{100}, \quad \mathrm{P}(P') = \frac{27}{100}.$$

$$\text{(a)} \quad \mathrm{P}(P' \mid A) = \frac{\mathrm{P}(P' \cap A)}{\mathrm{P}(A)} = \frac{18}{32} = 0.5625,$$
$$\text{(b)} \quad \mathrm{P}(A \mid P) = \frac{\mathrm{P}(A \cap P)}{\mathrm{P}(P)} = \frac{14}{73} = 0.1918,$$
$$\text{(c)} \quad \mathrm{P}(A' \mid P') = \frac{\mathrm{P}(A' \cap P')}{\mathrm{P}(P')} = \frac{9}{27} = 0.333.$$

Semiconductor Chips Without Replacement

Example 7.8. A lot of 100 semiconductor chips contains 20 that are defective. Two are selected randomly, without replacement, from the lot. (a) What is the probability that the first one selected is defective? (b) What is the probability that the second one selected is defective given that the first one is defective? (c) What is the probability that both are defective? (d) How does the answer to part (b) change if chips selected were replaced prior to the next selection?

Solution.

$$\text{(a)}\quad \frac{20}{100} = 0.2,$$

$$\text{(b)}\quad \mathrm{P}(\text{2nd def} \mid \text{1st def}) = \frac{\text{number of defective items remaining}}{\text{total remaining}} = \frac{19}{99},$$

$$\text{(c)}\quad \mathrm{P} = \frac{20}{100} \times \frac{19}{99} = 0.03838,$$

$$\text{(d)}\quad \mathrm{P} = \frac{20}{100} = 0.2 \quad \text{(if the defective chip is replaced).}$$

Passwords: Vowel and Even-Number Constraints

Example 7.9. A password consists of exactly seven characters, where each character is one of 26 letters (a–z) or 10 integers (0–9). Let A denote the subset of passwords that begin with a vowel and let B denote the subset of passwords that end with an even number. (a) Suppose a hacker selects a password at random. What is the probability that your password is selected? (b) Suppose a hacker knows your password is in event A and selects a password at random from this subset. What is the probability that your password is selected? (c) Suppose a hacker knows your password is in $A \cap B$ and selects a password at random from this subset. Find the probability.

Solution. Each of the seven positions can be filled in 36 ways, so there are 36^7 equally likely passwords.

$$\text{(a)}\quad \mathrm{P}(\text{mine}) = \frac{1}{36^7},$$

$$\text{(b)}\quad N_A = 5 \times 36^6 = \frac{5 \times 36^7}{36}, \quad \text{so} \quad \mathrm{P}(\text{mine} \mid A) = \frac{1}{5 \times 36^6},$$

$$\text{(c)}\quad N_C = 5 \times 5 \times 36^5, \quad \text{so} \quad \mathrm{P}(\text{mine} \mid A \cap B) = \frac{1}{N_C} = \frac{1}{25 \times 36^5}.$$

There are 5 vowels for the first position and 5 even digits $(0, 2, 4, 6, 8)$ for the last position.

7.2 The Multiplication and Total Probability Rules

Multiplication Rule

The definition of conditional probability can be rewritten to give a general expression for the probability of the intersection of two events. This is called the *multiplication rule*:

$$\mathrm{P}(A \cap B) = \mathrm{P}(B \mid A)\,\mathrm{P}(A) = \mathrm{P}(A \mid B)\,\mathrm{P}(B).$$

Total Probability Rule

For any two events A and B,

$$\mathrm{P}(B) = \mathrm{P}(B \cap A) + \mathrm{P}(B \cap A') = \mathrm{P}(B \mid A)\,\mathrm{P}(A) + \mathrm{P}(B \mid A')\,\mathrm{P}(A').$$

Total probability rule (multiple events). Assume $E_1, E_2, \ldots, E_k$ are k mutually exclusive and exhaustive sets. Then

$$\mathrm{P}(B) = \mathrm{P}(B \cap E_1) + \mathrm{P}(B \cap E_2) + \cdots + \mathrm{P}(B \cap E_k)$$

$$= \mathrm{P}(B \mid E_1)\,\mathrm{P}(E_1) + \mathrm{P}(B \mid E_2)\,\mathrm{P}(E_2) + \cdots + \mathrm{P}(B \mid E_k)\,\mathrm{P}(E_k).$$

Electrical Connector Kept Dry or Wet

Example 7.10. The probability is 1% that an electrical connector that is kept dry fails during the warranty period of a portable computer. If a connector is ever wet, the probability of a failure during the warranty period is 5%. If 90% of the connectors are kept dry and 10% are wet, what proportion of connectors fail during the warranty period?

Solution. Let B be the event that a connector fails. Let A be the event of a wet condition and A' the dry condition, with $\mathrm{P}(A) = 0.1$, $\mathrm{P}(A') = 0.9$, $\mathrm{P}(B \mid A) = 0.05$, and $\mathrm{P}(B \mid A') = 0.01$. By the total probability rule,

$$\mathrm{P}(B) = \mathrm{P}(B \mid A)\,\mathrm{P}(A) + \mathrm{P}(B \mid A')\,\mathrm{P}(A') = 0.05 \times 0.1 + 0.01 \times 0.9 = 0.005 + 0.009 = 0.014.$$

Edge Roughness of Slit Paper Products

Example 7.11. The edge roughness of slit paper products increases as knife blades wear. Only 1% of products slit with new blades have rough edges, 3% of products slit with blades of average sharpness exhibit roughness, and 5% of products slit with worn blades exhibit roughness. If 25% of the blades in manufacturing are new, 60% are of average sharpness, and 15% are worn, what is the proportion of products that exhibit edge roughness?

Solution. Let B be the event that a product exhibits edge roughness, and let A_1, A_2, A_3 denote new, average, and worn blades respectively, with $\mathrm{P}(A_1) = 0.01$, $\mathrm{P}(A_2) = 0.03$, $\mathrm{P}(A_3) = 0.05$ as conditional roughness probabilities and blade proportions 0.25, 0.60, 0.15. Then

$$\begin{aligned}\mathrm{P}(B) &= \mathrm{P}(B \mid A_1)\,\mathrm{P}(A_1) + \mathrm{P}(B \mid A_2)\,\mathrm{P}(A_2) + \mathrm{P}(B \mid A_3)\,\mathrm{P}(A_3) \\ &= 0.01 \times 0.25 + 0.03 \times 0.60 + 0.05 \times 0.15 = 0.028.\end{aligned}$$

Election Polling

Example 7.12. Let B be the event that a respondent voted for a particular candidate. If among 2020 respondents the probability of voting is $\mathrm{P}(B) = 0.62 \times 0.50 + 0.38 \times 0.53 = 0.5114$, the probability that a single specified respondent is the one selected is

$$\frac{1}{1033} = 0.00096.$$

Computer Keyboard Failures

Example 7.13. Computer keyboard failures are due to faulty electrical connects (12%) or mechanical defects (88%). Mechanical defects are related to loose keys (27%) or improper assembly (73%). Electrical connect defects are caused by defective wires (35%), improper connections (13%), or poorly welded wires (52%). (a) Find the probability that a failure is due to loose keys. (b) Find the probability that a failure is due to improperly connected or poorly welded wires.

Solution. Let E be electrical-connect failures and M mechanical defects, with $L \to$ loose keys and $A \to$ improper assembly, while $E \to$ defective wires + improper connection + poorly welded.

$$\text{(a)}\quad \mathrm{P}(L) = 0.27 \times 0.88 = 0.2376,$$
$$\text{(b)}\quad \mathrm{P}(\text{improper conn.} + \text{poor weld}) = 0.52 \times 0.12 + 0.13 \times 0.12 = 0.0078.$$

Heart Failures

Example 7.14. Heart failures are due to either natural occurrences (87%) or outside factors (13%). Outside factors are related to induced substances (73%) or foreign objects (27%). Natural occurrences are caused by arterial blockage (56%), disease (27%), and infection (17%). (a) Determine the probability that a failure is due to an induced substance. (b) Determine the probability that a failure is due to disease or infection.

Solution.

$$\text{(a)}\quad \mathrm{P}(I.S) = 0.73 \times 0.13 = 0.0949,$$
$$\text{(b)}\quad \mathrm{P}(D \text{ or } I) = 0.27 \times 0.87 + 0.17 \times 0.87 = 0.3828.$$

7.3 Independence

Two events are *independent* if any one of the following equivalent statements is true:

1. $\mathrm{P}(A \mid B) = \mathrm{P}(A)$,
2. $\mathrm{P}(B \mid A) = \mathrm{P}(B)$,
3. $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$.

Laboratory Sample Contamination

Example 7.15. The probability that a lab specimen contains high levels of contamination is 0.10. Five samples are checked and the samples are independent. (a) What is the probability that none contains high levels of contamination? (b) What is the probability that exactly one contains high levels of contamination? (c) What is the probability that at least one contains high levels of contamination?

Solution. Let A_i be the event that the ith lab specimen contains high contamination, with $\mathrm{P}(A_i) = 0.1$ and $\mathrm{P}(A_i') = 0.9$.

(a) By independence,

$$\mathrm{P}(A_1' \cap A_2' \cap A_3' \cap A_4' \cap A_5') = 0.9 \times 0.9 \times 0.9 \times 0.9 \times 0.9 = 0.5905.$$

(c) The complement of "none" is "at least one":

$$1 - \mathrm{P}(A_1' \cap A_2' \cap A_3' \cap A_4' \cap A_5') = 1 - 0.5905 = 0.4095 \approx 0.41.$$

(b) The event that exactly one contains high levels of contamination is a disjoint union. For each $i_0 = 1, 2, 3, 4, 5$, the event that only specimen i_0 is contaminated is

$$\Big(\bigcap_{i \neq i_0} A_i'\Big) \cap A_{i_0}.$$

For example, for $i_0 = 2$,

$$\mathrm{P}\Big(\bigcap_{i \neq 2} A_i' \cap A_2\Big) = \mathrm{P}(A_1')\,\mathrm{P}(A_3')\,\mathrm{P}(A_4')\,\mathrm{P}(A_5')\,\mathrm{P}(A_2) = 0.9 \times 0.9 \times 0.9 \times 0.9 \times 0.1 = 0.066.$$

Since the five such events are disjoint and each has probability 0.066,

$$\mathrm{P}(E_1) = 0.066 + 0.066 + 0.066 + 0.066 + 0.066 = 0.33.$$

Acid–Base Titration by Two Technicians

Example 7.16. In an acid–base titration, a base or acid is gradually added to the other until they have completely neutralised each other. Since acids and bases are usually colourless, pH is measured to monitor the reaction. Suppose that the equivalence point is reached after approximately 100 mL of a NaOH solution has been added but that replicates are equally likely to indicate from 95 to 104 mL, measured to the nearest mL. Assume that two technicians each conduct titrations independently. (a) What is the probability that both technicians obtain equivalence at 100 mL? (b) What is the probability that both technicians obtain equivalence between 98 and 104 mL (inclusive)? (c) What is the probability that the average volume at equivalence from the two technicians is 100 mL?

Solution. Technicians T_1 and T_2 independently obtain equivalence at one of the quantities

$$Q = \{95, 96, 97, 98, 99, 100, 101, 102, 103, 104\} = \{q_i\}_{i=1}^{10},$$

with equal probability. Let t_1 and t_2 represent the quantities at which T_1 and T_2 obtain equivalence respectively, so $\mathrm{P}(t_1 = q_i) = \frac{1}{10}$ and $\mathrm{P}(t_2 = q_i) = \frac{1}{10}$.

(a)

$$\mathrm{P}\big(\{t_1 = 100\} \cap \{t_2 = 100\}\big) = \mathrm{P}(t_1 = 100)\,\mathrm{P}(t_2 = 100) = \frac{1}{10} \times \frac{1}{10} = \frac{1}{100}.$$

(b) There are 7 values in $\{98, \ldots, 104\}$, so

$$\mathrm{P}\big(t_1, t_2 \in \{98, \ldots, 104\}\big) = \mathrm{P}(t_1 \in \{98, \ldots, 104\})\,\mathrm{P}(t_2 \in \{98, \ldots, 104\}) = \frac{7}{10} \times \frac{7}{10} = \frac{49}{100}.$$

(c) We need pairs (t_1, t_2) such that $\frac{t_1 + t_2}{2} = 100$, i.e. $t_1 + t_2 = 200$. The favourable pairs are

$$(100, 100),\ (101, 99),\ (99, 101),\ (102, 98),\ (98, 102),\ (103, 97),\ (97, 103),\ (104, 96),\ (96, 104),$$

that is 9 pairs out of 100 equally likely outcomes. Hence the probability that the average volume at equivalence is 100 mL is

$$\frac{9}{100} = 0.09.$$

Spider-Mite Infestation of Ivy Plants

Example 7.17. Six tissues are extracted from an ivy plant infested by spider mites. The plant is infested in 20% of its area. A sample is chosen from randomly selected areas on the ivy plant. Assume that the samples are independent. (a) What is the probability that four successive samples show the signs of infestation? (b) What is the probability that 3 out of 4 successive samples show the signs of infestation?

Solution. Let A_i be the event that the ith tissue extracted from the ivy plant is infested by spider mites, $i = 1, 2, \ldots, 6$. The plant is infested in 20% of its area, so $\mathrm{P}(A_i) = 0.2$ and $\mathrm{P}(A_i') = 1 - 0.2 = 0.8$.

(a) Let B be the event that four successive samples show signs of infestation. By independence,

$$\mathrm{P}(A_1 \cap A_2 \cap A_3 \cap A_4) = 0.2 \times 0.2 \times 0.2 \times 0.2 = 0.2^4 = 0.001024.$$

(b) For exactly 3 infested out of 4 successive samples, one sample is not infested. A single such arrangement (for instance $A_1 \cap A_2 \cap A_3 \cap A_4'$) has probability

$$0.2^3 \times 0.8 = 0.008 \times 0.8 \quad \Rightarrow \quad 0.2^3 \times 0.8^1.$$

We can select which 4 of the samples are taken and which is the uninfested one. Picking 3 out of 4 can be done in $\binom{4}{3} = 4$ ways, giving 4 disjoint events:

$$\mathrm{P} = \binom{4}{3} (0.2)^3 (0.8)^1 = 4 \times 0.008 \times 0.8 = 0.003072.$$

If we instead consider all 6 tissues with $n_1 \times n_2 = 6$ distinct arrangements but with 2 counted, the corresponding figure is 0.0768.

Game Defeat Problem

Example 7.18. When a player is defeated by an opponent, the game ends. The probability of a win is $\mathrm{P}(\text{win}) = 80\% = 0.80$, and $n = 4$. (a) What is the probability that a player defeats all four opponents in a game? (b) What is the probability of at least two wins in the last three sequences? (c) What is the probability of at least one occurrence of four wins?

Solution. The results from opponents are independent. The player defeats all 4 opponents if he wins each of 4 games.

(a)

$$\mathrm{P}(\text{defeat all 4}) = 0.8 \times 0.8 \times 0.8 \times 0.8 = 0.8^4 = 0.4096.$$

(b) The possible game sequences (with W = win, L = lose) are

$$L, \quad WL, \quad WWL, \quad WWWL, \quad WWWW.$$

With $\mathrm{P}(L) = 1 - \mathrm{P}(\text{win}) = 0.20$ and $\mathrm{P}(WL) = \mathrm{P}(\text{win}) \times \mathrm{P}(\text{lose}) = 0.8 \times 0.2 = 0.16$,

$$\mathrm{P}(\text{at least 2 wins}) = 1 - 0.2 - 0.16 = 0.64.$$

Equivalently, $\mathrm{P}(WWL + WWWL + WWWW) = 0.128 + 0.1024 + 0.4096 = 0.64$.

(c) The probability of not having four wins in a single attempt is

$$\mathrm{P}(\text{not 4 wins}) = 1 - \mathrm{P}(4 \text{ wins}) = 1 - 0.4096 = 0.5904.$$

Over three independent attempts,

$$\mathrm{P}(3 \text{ times not 4 wins}) = 0.5904 \times 0.5904 \times 0.5904 = 0.2058.$$

Therefore

$$\mathrm{P}(\text{at least once 4 wins}) = 1 - \mathrm{P}(3 \text{ times not 4 wins}) = 1 - 0.2058 = 0.7942.$$

Credit-Card Hacker

Example 7.19. A hacker attempts to guess a credit card number, where $N =$ one million valid numbers. (a) What is the probability of a correct guess on a single random selection? (b) Let B denote the event that the hacker correctly selects the month and year of credit card expiration. What is this probability?

Solution. **(a)**

$$\frac{1 \times 10^6}{10^{16}} = 10^{-10}.$$

(b) A hacker randomly selects one of 12 months, so the probability of correctly guessing the month is $\frac{1}{12}$. Combined with the year guess,

$$\mathrm{P}(B) = 0.25 \times \frac{1}{12} = \frac{1}{4} \times \frac{1}{12} = \frac{1}{48}.$$

Series–Parallel Circuit Operation

Example 7.20. The following circuit operates if and only if there is a path of functional devices from left to right. The probability that each device functions is as shown. Assume that the probability that a device functions does not depend on whether or not other devices are functional. What is the probability that the circuit operates?

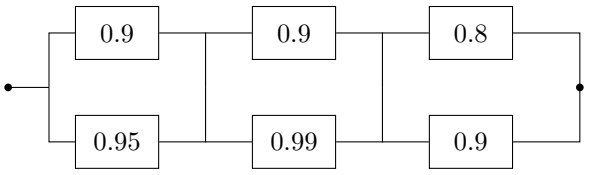


Figure 7.1: Three parallel blocks connected in series.

Solution. Each of the three blocks is a parallel pair; a block fails only if both of its devices fail. The block reliabilities are

$$1 - (1 - 0.9)(1 - 0.95) = 0.995, \quad 1 - (1 - 0.9)(1 - 0.99) = 0.999 \approx 0.995,$$

and the third block contributes $1 - (1 - 0.8)(1 - 0.9) = 0.98$. Since the blocks are in series,

$$\mathrm{P}(\text{operates}) = 0.995 \times 0.995 \times 0.98 = 0.9702.$$

7.4 Bayes' Theorem

If $E_1, E_2, \ldots, E_k$ are k mutually exclusive and exhaustive events and B is any event with $\mathrm{P}(B) > 0$, then

$$\mathrm{P}(E_i \mid B) = \frac{\mathrm{P}(B \mid E_i)\,\mathrm{P}(E_i)}{\mathrm{P}(B \mid E_1)\,\mathrm{P}(E_1) + \mathrm{P}(B \mid E_2)\,\mathrm{P}(E_2) + \cdots + \mathrm{P}(B \mid E_k)\,\mathrm{P}(E_k)}.$$

Preliminary Product Design Reviews

Example 7.21. Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor. (a) What is the probability that a product attains a good review? (b) If a new design attains a good review, what is the probability that it will be a highly successful product? (c) If a product does not attain a good review, what is the probability that it will be a highly successful product?

Solution. Let G be a good review with G' its complement, and let A_1, A_2, A_3 denote highly, moderately, and poorly successful products.

(a) By the total probability rule,

$$\mathrm{P}(G) = \sum_{j=1}^{3} \mathrm{P}(G \mid A_j)\,\mathrm{P}(A_j) = 0.95 \times 0.4 + 0.6 \times 0.35 + 0.1 \times 0.25 = 0.6150.$$

(b) By Bayes' theorem,

$$\mathrm{P}(A_1 \mid G) = \frac{\mathrm{P}(G \mid A_1)\,\mathrm{P}(A_1)}{\mathrm{P}(G)} = \frac{0.95 \times 0.4}{0.6150} = 0.6179.$$

(c)

$$\mathrm{P}(A_1 \mid G') = \frac{\mathrm{P}(G' \mid A_1)\,\mathrm{P}(A_1)}{\mathrm{P}(G')} = \frac{0.05 \times 0.4}{1 - 0.6150} = 0.052.$$

Inspector Identifying Defective Items

Example 7.22. An inspector working for a manufacturing company has a 99% chance of correctly identifying a defective item and a 0.5% chance of incorrectly classifying a good item as defective. The company has evidence that its line produces 0.9% of nonconforming items. (a) What is the probability that an item selected for inspection is classified as defective? (b) If an item is selected at random and classified as nondefective, what is the probability that it is good?

Solution. Let D be the event that a defective item is selected and let I be the event that an item is inspected and found defective, with D' being a nondefective item. Then

$$\mathrm{P}(I \mid D) = 0.99, \quad \mathrm{P}(I \mid D') = 0.005, \quad \mathrm{P}(D) = 0.009, \quad \mathrm{P}(D') = 0.991.$$

(a) By the total probability rule,

$$\mathrm{P}(I) = \mathrm{P}(I \mid D)\,\mathrm{P}(D) + \mathrm{P}(I \mid D')\,\mathrm{P}(D') = 0.99 \times 0.009 + 0.005 \times 0.991 = 0.0138.$$

(b) By Bayes' theorem,

$$\mathrm{P}(D' \mid I') = \frac{\mathrm{P}(I' \mid D')\,\mathrm{P}(D')}{\mathrm{P}(I')} = \frac{(1-0.005)(0.991)}{1-0.0138} = \frac{0.94 \times 0.99}{0.9309} \approx 0.999.$$

Disease Blood Test

Example 7.23. Suppose it is known that 1% of the population suffers from a particular disease. A blood test has a 97% chance of identifying the disease for diseased individuals, but also has a 6% chance of falsely indicating that a healthy person has the disease. (a) What is the probability that a person will have a positive blood test? (b) If your blood test is positive, what is the chance that you have the disease? (c) If your blood test is negative, what is the chance that you do not have the disease?

Solution. Let D be the event that a person has the disease and I the event that the test identifies the disease (tests positive), with

$$\mathrm{P}(D) = 0.01, \quad \mathrm{P}(D') = 0.99, \quad \mathrm{P}(I \mid D) = 0.97, \quad \mathrm{P}(I \mid D') = 0.06.$$

(a)

$$\mathrm{P}(I) = \mathrm{P}(I \mid D)\,\mathrm{P}(D) + \mathrm{P}(I \mid D')\,\mathrm{P}(D') = 0.97 \times 0.01 + 0.06 \times 0.99 = 0.0691.$$

(b) By Bayes' theorem,

$$\mathrm{P}(D \mid I) = \frac{\mathrm{P}(I \mid D)\,\mathrm{P}(D)}{\mathrm{P}(I)} = \frac{0.97 \times 0.01}{0.0691} = 0.1404.$$

(c)

$$\mathrm{P}(D' \mid I') = \frac{\mathrm{P}(I' \mid D')\,\mathrm{P}(D')}{\mathrm{P}(I')} = \frac{(1-0.06)(1-0.01)}{1-0.0691} = \frac{0.94 \times 0.99}{0.9309} = 0.9997.$$

Phone-Card Fraud Detection

Example 7.24. Software to detect fraud in consumer phone cards tracks the number of metropolitan areas where calls originate each day. It is found that 1% of the legitimate users originate calls from two or more metropolitan areas in a single day. However, 30% of fraudulent users originate calls from two or more metropolitan areas in a single day. The proportion of fraudulent users is 0.01%. If the same user originates calls from two or more metropolitan areas in a single day, what is the probability that the user is fraudulent?

Solution. Let S be the event that calls originate from two or more areas and let G be the event that the caller is legitimate, with

$$\mathrm{P}(S \mid G) = 0.01, \quad \mathrm{P}(S \mid G') = 0.3, \quad \mathrm{P}(G') = 0.0001.$$

By Bayes' theorem,

$$\mathrm{P}(G' \mid S) = \frac{\mathrm{P}(S \mid G')\,\mathrm{P}(G')}{\mathrm{P}(S \mid G')\,\mathrm{P}(G') + \mathrm{P}(S \mid G)\,\mathrm{P}(G)}$$

$$= \frac{0.3 \times 0.0001}{(0.3 \times 0.0001) + (0.01)(1-0.0001)} = \frac{0.00003}{0.00003 + 0.0099} = 0.2326.$$

7.5 Further Worked Examples

Machine Tool Idle Time

Example 7.25. A machine tool is idle 15% of the time. You request immediate use of the tool on 5 different occasions during the year. Assume that your requests represent independent events. (a) What is the probability that the tool is idle at the time of all of your requests? (b) What is the probability that the tool is idle at the time of exactly 4 of your requests? (c) What is the probability that the tool is idle at the time of at least 3 of your requests?

Solution. Let the probability that the tool is idle be 0.15.

(a) $(0.15)^5 = 7.6 \times 10^{-5}$,

(b) $\left[(0.15)^4 \times 0.85\right] \times 5 = 0.00215156$,

(c) $\mathrm{P}(\text{idle at least 3 times}) = \mathrm{P}(\text{exactly } 3) + \mathrm{P}(4) + \mathrm{P}(5)$
$= 10 \times (0.15)^3 \times (0.85)^2 + 5 \times (0.15)^4 \times 0.85 + (0.15)^5.$

Database Transactions

Example 7.26. Transactions to a computer database are either new items or changes to previous items. The addition of an item can be completed in less than 100 milliseconds 90% of the time, but only 20% of changes to a previous item can be completed in less than this time. If 30% of transactions are changes, what is the probability that a transaction can be completed in less than 100 milliseconds?

Solution. By the total probability rule, with 30% changes and 70% additions,

$$\mathrm{P}(< 100 \text{ ms}) = (0.2 \times 0.3) + (0.7 \times 0.9) = 0.06 + 0.63 = 0.69.$$

Automated Filling Operation

Example 7.27. In an automated filling operation, the probability of an incorrect fill when the process is operated at a low speed is 0.001. When the process is operated at high speed, the probability of an incorrect fill is 0.01. Assume that 30% of the containers are filled when the process is operated at a high speed and the remainder are filled when the process is operated at a low speed. (a) What is the probability of an incorrectly filled container? (b) If an incorrectly filled container is found, what is the probability that it was filled during high-speed operation?

Solution. Let D denote the event that a container is incorrectly filled and let H denote the event that a container is filled under high-speed operation. Then $\mathrm{P}(D \mid H) = 0.01$, $\mathrm{P}(D \mid H') = 0.001$, $\mathrm{P}(H) = 0.30$, and $\mathrm{P}(H') = 0.70$.

(a)

$$\mathrm{P}(D) = \mathrm{P}(D \mid H)\,\mathrm{P}(H) + \mathrm{P}(D \mid H')\,\mathrm{P}(H') = 0.01 \times 0.30 + 0.001 \times 0.70 = 0.0037.$$

(b) By Bayes' theorem,

$$\mathrm{P}(H \mid D) = \frac{\mathrm{P}(D \mid H)\,\mathrm{P}(H)}{\mathrm{P}(D)} = \frac{0.01 \times 0.30}{0.0037} = 0.8108.$$

Email Routing Through Two Server Routes

Example 7.28. An email message can travel through one of two server routes. The probability of transmission error in each of the servers and the proportion of messages that travel each route are shown in the following table. Assume that the servers are independent. (a) What is the probability that a message will arrive without error? (b) If a message arrives in error, what is the probability that it was sent through route 1?

	% of msgs	Probability of error (Server)			
		1	2	3	4
Route 1	30	0.01	0.015		
Route 2	70			0.02	0.003

Table 7.1: Routing probabilities and per-server error probabilities.

Solution. Let E represent an error in the message. A message arrives without error if every server on its route transmits without error. Therefore

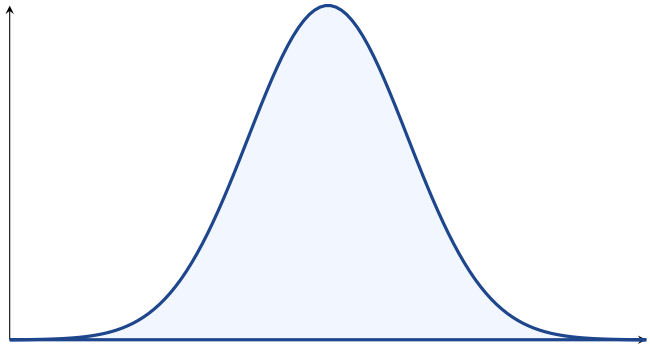
$$\mathrm{P}(E') = (0.3)(0.99)(0.985) + (0.7)(0.98)(0.997) = 0.9764.$$

(b) By Bayes' theorem,

$$\mathrm{P}(\text{route } 1 \mid E) = \frac{\mathrm{P}(E \mid \text{route } 1)\,\mathrm{P}(\text{route } 1)}{\mathrm{P}(E)} = \frac{0.3 \times \big[1 - (0.99)(0.985)\big]}{1 - 0.9764}.$$

The numerator uses the route-1 selection probability of 0.3, and the denominator is $\mathrm{P}(E) = 1 - \mathrm{P}(E') = 1 - 0.9764$.

Unit 2



Preface

This book is a polished, textbook-format rendering of a set of handwritten course notes on probability distributions. The material has been reorganised into logical chapters with connective prose, while every definition, formula, worked example, and final numerical answer from the original notes has been preserved.

The notes were intended to be reproduced one-to-one. Wherever the source text was legible, it is transcribed faithfully; only where the handwriting could not be read, or an intermediate algebraic step was omitted, has content been reconstructed from the surrounding context so that each derivation reaches the final answer recorded in the original. The final numerical answers in the original notes are treated as authoritative throughout. A short, honest list of every place where reconstruction or interpretation was required appears in Appendix A.

The book covers discrete models (geometric, negative binomial, hypergeometric, Poisson) and continuous models (general probability density functions, the continuous uniform, normal, exponential, Erlang, gamma, and Weibull distributions), together with the normal approximation to the binomial and Poisson distributions.

Chapter 1

Geometric and Negative Binomial Distributions

The binomial distribution counts the number of successes in a *fixed* number of Bernoulli trials. In this chapter we turn the question around: instead of fixing the number of trials, we keep performing independent Bernoulli trials (each with constant success probability p) until a prescribed number of successes has occurred, and we treat the *number of trials* as the random variable. This leads first to the geometric distribution (wait for the first success) and then to its generalisation, the negative binomial distribution (wait for the rth success).

1.1 The Geometric Distribution

Again consider a series of Bernoulli trials — independent trials with constant probability p of a success on each trial. However, instead of a fixed number of trials, trials are conducted until a success is obtained. Let the random variable X denote the number of trials until the first success.

Definition 1.1: Geometric random variable

In a series of Bernoulli trials (independent trials with constant probability p of a success), let the random variable X denote the number of trials until the first success. Then X is a *geometric* random variable with parameter $0 < p < 1$ and

$$f(x) = (1-p)^{x-1}p, \qquad x = 1, 2, 3, \ldots$$

Key Formula

For a geometric random variable with parameter p,

$$f(x) = (1-p)^{x-1}p, \qquad \mu = \mathbb{E}(X) = \frac{1}{p}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{1-p}{p^2}.$$

Lack of Memory Property

A geometric random variable has been defined as the number of trials until the first success. However, because the trials are independent, the count of the number of trials until the next success can be started at any trial without changing the probability distribution of the random variable.

For example, in the transmission of bits, suppose that 100 bits are transmitted. The probability that the initial error occurs on bit 106 is the probability that the next six outcomes are $00000E$.

This probability is

$$(0.9)^5(0.1) = 0.059,$$

which is identical to the probability that the initial error occurs on bit 6.

The implication of using a geometric model is that the system presumably will not wear out: the probability of an error remains constant for all transmissions. In this sense, the geometric distribution is said to lack any memory.

1.2 The Negative Binomial Distribution

A generalisation of a geometric distribution, in which the random variable is the number of Bernoulli trials required to obtain r successes, results in the negative binomial distribution.

Definition 1.2: Negative binomial random variable

In a series of Bernoulli trials (independent trials with constant probability p of a success), let the random variable X denote the number of trials until r successes occur. Then X is a *negative binomial* random variable with parameters $0 < p < 1$ and $r = 1, 2, 3, \ldots$, and

$$f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r, \qquad x = r, r+1, r+2, \ldots$$

Key Formula

For a negative binomial random variable with parameters p and r,

$$f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r, \qquad \mathbb{E}(X) = \frac{r}{p}, \qquad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

Because at least r trials are required to obtain r successes, the range of X is from r to ∞. In the special case that $r = 1$, a negative binomial random variable is a geometric random variable.

A Negative Binomial Variable as a Sum of Geometric Variables

The lack-of-memory property of a geometric random variable leads to the following interpretation. Let X denote the total number of trials required to obtain r successes. Let X_1 denote the number of trials required to obtain the first success, X_2 the number of extra trials required to obtain the second success, X_3 the number of extra trials to obtain the third success, and so forth. Then the total number of trials required to obtain r successes is

$$X = X_1 + X_2 + \cdots + X_r.$$

A negative binomial random variable can therefore be interpreted as the sum of r geometric random variables.

In a negative binomial experiment the number of successes is predetermined and the number of trials is random. Hence a negative binomial random variable can be considered the opposite of a binomial random variable: a binomial random variable is a count of the number of successes in n trials.

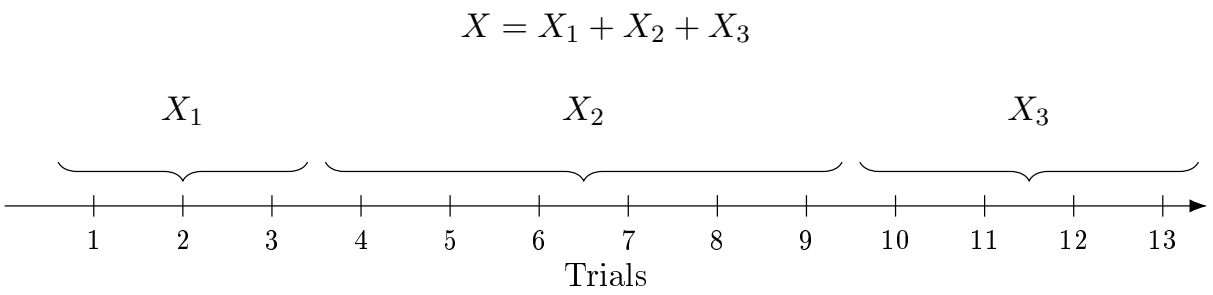


Figure 1.1: A negative binomial random variable ($r = 3$) represented as the sum of three independent geometric random variables.

1.3 Worked Problems

Example 1.1 (Geometric distribution with a given mean)**.** Suppose the random variable X has a geometric distribution with a mean of 2.5. Determine the following probabilities: (a) $\mathrm{P}(X = 1)$, (b) $\mathrm{P}(X = 4)$, (c) $\mathrm{P}(X = 5)$, (d) $\mathrm{P}(X \leq 3)$, (e) $\mathrm{P}(X > 3)$.

Solution. Since $\mathbb{E}(X) = 2.5 = \dfrac{1}{p}$, we have $p = 0.4$.

(a) $\mathrm{P}(X = 1) = (1 - 0.4)^0(0.4) = 0.4.$

(b) $\mathrm{P}(X = 4) = (1 - 0.4)^3(0.4) = 0.0864.$

(c) $\mathrm{P}(X = 5) = (1 - 0.4)^4(0.4) = 0.05184.$

(d) $$\begin{aligned}\mathrm{P}(X \leq 3) &= \mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \mathrm{P}(X = 3)\\ &= (1 - 0.4)^0(0.4) + (1 - 0.4)^1(0.4) + (1 - 0.4)^2(0.4)\\ &= 0.7840.\end{aligned}$$

(e) $\mathrm{P}(X > 3) = 1 - \mathrm{P}(X \leq 3) = 1 - 0.7840 = 0.2160.$

Example 1.2 (Optical alignment)**.** The probability of a successful optical alignment in the assembly of an optical data storage product is 0.8. Assume the trials are independent. (a) What is the probability that the first successful alignment requires exactly four trials? (b) at most four trials? (c) at least four trials?

Solution. Let X denote the number of trials to obtain the first successful alignment. Then X is geometric with $p = 0.8$.

(a) $\mathrm{P}(X = 4) = (1 - 0.8)^3(0.8) = (0.2)^3(0.8) = 0.0064.$

(b) $$\begin{aligned}\mathrm{P}(X \leq 4) &= \mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \mathrm{P}(X = 3) + \mathrm{P}(X = 4)\\ &= (0.2)^0(0.8) + (0.2)^1(0.8) + (0.2)^2(0.8) + (0.2)^3(0.8)\\ &= 0.8 + 0.16 + 0.032 + 0.0064 = 0.9984.\end{aligned}$$

(c) $$\begin{aligned}\mathrm{P}(X \geq 4) &= 1 - \mathrm{P}(X \leq 3) = 1 - \left[0.8 + 0.2(0.8) + (0.2)^2(0.8)\right]\\ &= 1 - 0.992 = 0.008.\end{aligned}$$

Example 1.3 (Carrier gene detection)**.** In a clinical study, volunteers are tested for a gene that has been found to increase the risk for a disease. The probability that a person carries the gene is 0.1. (a) What is the probability that four or more people will have to be tested before two

with the gene are detected? (b) How many people are expected to be tested before two with the gene are detected?

Solution. Let X denote the number of people tested to detect two who carry the gene. Then X is a negative binomial random variable with $p = 0.1$ and $r = 2$.

$$\begin{aligned}
\text{(a)} \quad \mathrm{P}(X \geq 4) &= 1 - \mathrm{P}(X < 4) = 1 - \big[\mathrm{P}(X = 2) + \mathrm{P}(X = 3)\big] \\
&= 1 - \left[\binom{1}{1}(1-0.1)^0(0.1)^2 + \binom{2}{1}(1-0.1)^1(0.1)^2\right] \\
&= 1 - (0.01 + 0.018) = 0.972. \\
\text{(b)} \quad \mathbb{E}(X) &= \frac{r}{p} = \frac{2}{0.1} = 20.
\end{aligned}$$

Example 1.4 (Connecting to a radio station)**.** Assume that each of your calls to a popular radio station has a probability 0.02 of connecting, that is, of not obtaining a busy signal. Assume that your calls are independent. (a) What is the probability that your first call that connects is your tenth call? (b) What is the probability that it requires more than five calls for you to connect? (c) What is the mean number of calls needed to connect?

Solution. Let X denote the number of calls needed to obtain a connection. Then X is geometric with $p = 0.02$.

$$\begin{aligned}
\text{(a)} \quad \mathrm{P}(X = 10) &= (1-0.02)^9(0.02) = (0.98)^9(0.02) = 0.0167. \\
\text{(b)} \quad \mathrm{P}(X > 5) &= 1 - \mathrm{P}(X \leq 5) \\
&= 1 - \big[\mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \mathrm{P}(X = 3) + \mathrm{P}(X = 4) + \mathrm{P}(X = 5)\big] \\
&= 1 - \big[0.02 + 0.98(0.02) + (0.98)^2(0.02) \\
&\qquad + (0.98)^3(0.02) + (0.98)^4(0.02)\big] \\
&= 1 - 0.0961 = 0.9039.
\end{aligned}$$

This may also be found from "no connection in 5 trials" (a negative binomial / binomial argument):

$$\mathrm{P}(X > 5) = \binom{5}{0}(0.02)^0(0.98)^5 = 0.9039.$$

$$\text{(c)} \quad \mathbb{E}(X) = \frac{1}{0.02} = 50.$$

Example 1.5 (Fault-tolerant system)**.** A fault-tolerant system that processes transactions for a financial services firm uses three separate computers. If the operating computer fails, one of the two spares can be switched immediately online. After a second computer fails, the last computer can be immediately switched online. Assume that the probability of a failure during any transaction is 10^{-8} and that the transactions can be considered independent events. (a) What is the mean number of transactions before all computers have failed? (b) What is the variance of the number of transactions before all computers have failed?

Solution. Let X denote the number of transactions until all computers have failed. Then X is a negative binomial random variable with $p = 10^{-8}$ and $r = 3$.

$$\text{(a)} \quad \mathbb{E}(X) = \frac{r}{p} = \frac{3}{10^{-8}} = 3 \times 10^8.$$

$$\text{(b)} \quad \mathrm{Var}(X) = \frac{r(1-p)}{p^2} = \frac{3\,[\,1 - 10^{-8}\,]}{(10^{-8})^2} \approx 3 \times 10^{16}.$$

Chapter 2

The Hypergeometric Distribution

The binomial distribution applies to sampling *with* replacement (or, equivalently, to independent trials with constant success probability). When a finite collection of objects is sampled *without* replacement, the trials are no longer independent, and the hypergeometric distribution is required.

2.1 Definition and Properties

Consider a set of N objects, of which K are classified as successes and $N-K$ as failures. A sample of size n objects is selected randomly (without replacement) from the N objects, where $K \leq N$ and $n \leq N$.

Definition 2.1: Hypergeometric random variable

Let the random variable X denote the number of successes in the sample. Then X is a *hypergeometric* random variable and

$$f(x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}, \qquad x = \max\{0,\, n+K-N\} \text{ to } \min\{K, n\}.$$

The expression $\min\{K, n\}$ is used in the range of X because the maximum number of successes that can occur in the sample is the smaller of the sample size n and the number of successes available, K.

Key Formula

If X is a hypergeometric random variable with parameters N, K, and n, then

$$\mu = \mathbb{E}(X) = np, \qquad \sigma^2 = \mathrm{Var}(X) = np(1-p)\left(\frac{N-n}{N-1}\right), \qquad \text{where } p = \frac{K}{N}.$$

Here p is interpreted as the proportion of successes in the set of N objects.

The term $\dfrac{N-n}{N-1}$ is called the *finite population correction factor*.

2.2 Worked Problems

Example 2.1 (Unacceptable washers)**.** A lot of 75 washers contains 5 in which the variability in thickness around the circumference of the washer is unacceptable. A sample of 10 washers is selected at random without replacement. (a) What is the probability that none of the unacceptable washers is in the sample? (b) the probability that at least one unacceptable washer is in the sample? (c) the probability that exactly one unacceptable washer is in the sample? (d) the mean number of unacceptable washers in the sample?

Solution. Let X denote the number of unacceptable washers in the sample of 10.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X=0) &= \frac{\binom{5}{0}\binom{70}{10}}{\binom{75}{10}} = \frac{70!\,/\,(60!\,10!)}{75!\,/\,(65!\,10!)} \\
&= \frac{65\cdot 64\cdot 63\cdot 62\cdot 61}{75\cdot 74\cdot 73\cdot 72\cdot 71} = 0.4786. \\
\text{(b)}\quad \mathrm{P}(X\geq 1) &= 1-\mathrm{P}(X=0) = 1-0.4786 = 0.5214. \\
\text{(c)}\quad \mathrm{P}(X=1) &= \frac{\binom{5}{1}\binom{70}{9}}{\binom{75}{10}} = \frac{5\cdot 10\cdot 65\cdot 64\cdot 63\cdot 62}{75\cdot 74\cdot 73\cdot 72\cdot 71} = 0.3923. \\
\text{(d)}\quad \mathbb{E}(X) &= np = 10\left(\frac{5}{75}\right) = \frac{2}{3}.
\end{aligned}$$

Example 2.2 (Printed circuit cards)**.** Printed circuit cards are placed in a final test after being populated with semiconductor chips. A lot contains 140 cards, and 20 cards are selected without replacement for final testing. (a) If 20 cards are defective, what is the probability that at least one defective card is in the sample? (b) If 5 cards are defective, what is the probability that at least one defective card appears in the sample?

Solution. Let X denote the number of defective cards in the sample.

(a) Here $K=20$, $n=20$, $N=140$.

$$\begin{aligned}
\mathrm{P}(X=0) &= \frac{\binom{20}{0}\binom{120}{20}}{\binom{140}{20}} = \frac{120!\,/\,(100!\,20!)}{140!\,/\,(120!\,20!)} = 0.0356, \\
\mathrm{P}(X\geq 1) &= 1-\mathrm{P}(X=0) = 1-0.0356 = 0.9644.
\end{aligned}$$

(b) Here $K=5$, $n=20$, $N=140$.

$$\begin{aligned}
\mathrm{P}(X=0) &= \frac{\binom{5}{0}\binom{135}{20}}{\binom{140}{20}} = \frac{135!\,/\,(115!\,20!)}{140!\,/\,(120!\,20!)} = 0.4571, \\
\mathrm{P}(X\geq 1) &= 1-0.4571 = 0.5429.
\end{aligned}$$

Example 2.3 (Slitter assembly blades)**.** Magnetic tape is slit into $\frac{1}{2}''$ widths that are wound into cartridges. A slitter assembly contains 48 blades. Five blades are selected at random and evaluated each day for sharpness. If any dull blade is found, the assembly is replaced with a newly sharpened set of blades. (a) If 10 of the 48 blades in the assembly are dull, what is the probability that the assembly is replaced the first day it is evaluated? (b) If 10 of the blades in an assembly are dull, what is the probability that the assembly is *not* replaced until the third day of evaluation? (Assume the daily decisions are independent and use the geometric

distribution.) (c) Suppose on the first day of evaluation two of the blades are dull, on the second day six are dull, and on the third day ten are dull. What is the probability that the assembly is not replaced until the third day of evaluation? (Assume the daily decisions are independent; here the probability of replacement changes every day.)

Solution. Let X denote the number of dull blades in the sample of five (selected from 48).

(a) With $K = 10$, $N - K = 38$, $n = 5$,

$$\mathrm{P}(X = 0) = \frac{\binom{10}{0}\binom{38}{5}}{\binom{48}{5}} = \frac{38!\,/\,(33!\,5!)}{48!\,/\,(43!\,5!)} = 0.2931,$$

$$\mathrm{P}(\text{replaced}) = \mathrm{P}(X \geq 1) = 1 - \mathrm{P}(X = 0) = 0.7069.$$

(b) Let Y denote the number of days needed to replace the assembly. Each day, replacement occurs with probability $p = 0.7069$, so Y is geometric:

$$\mathrm{P}(Y = 3) = (1 - p)^{y-1}p = (1 - 0.7069)^2(0.7069) = 0.0607.$$

(c) The probability of *no* dull blade in the sample changes each day. On the first day ($K = 2$):

$$\mathrm{P}(X = 0) = \frac{\binom{2}{0}\binom{46}{5}}{\binom{48}{5}} = \frac{46!\,/\,(41!\,5!)}{48!\,/\,(43!\,5!)} = 0.8005.$$

On the second day ($K = 6$):

$$\mathrm{P}(X = 0) = \frac{\binom{6}{0}\binom{42}{5}}{\binom{48}{5}} = \frac{42!\,/\,(37!\,5!)}{48!\,/\,(43!\,5!)} = 0.4968.$$

On the third day ($K = 10$): $\mathrm{P}(X = 0) = 0.2931$. Therefore the probability that the assembly is not replaced until the third day is

$$\mathrm{P}(\text{not replaced until 3rd day}) = 0.8005 \times 0.4968 \times (1 - 0.2931) = 0.2811.$$

Chapter 3

The Poisson Distribution

The Poisson distribution models the number of events (counts) occurring in a fixed interval of time, length, area, or volume, when the events occur at a constant average rate and independently of one another. Its single parameter is the mean number of events λ in the interval of interest.

3.1 Definition

Definition 3.1: Poisson random variable

The random variable X that equals the number of events in a Poisson process is a *Poisson* random variable with parameter $\lambda > 0$, and its probability mass function is

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

Key Formula

For a Poisson random variable with parameter λ,

$$\mathbb{E}(X) = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

The mean and the variance are equal. If the rate is given per unit interval and an interval of size t is considered, the mean over that interval is λt.

3.2 Worked Problems

Example 3.1 (Messages to a bulletin board)**.** The number of messages sent to a computer bulletin board is a Poisson random variable with a mean of 5 messages per hour. (a) What is the probability that 5 messages are received in 1 hour? (b) What is the probability that 10 messages are received in 1.5 hours? (c) What is the probability that fewer than 2 messages are received in one-half hour?

Solution.

(a) Let X denote the number of messages sent in 1 hour, so $\lambda = 5$:

$$\mathrm{P}(X = 5) = \frac{e^{-5}5^5}{5!} = 0.1755.$$

(b) Let Y denote the number of messages sent in 1.5 hours. Then Y is Poisson with $\lambda = 5 \times 1.5 =$

7.5 messages per 1.5 hours:

$$\mathrm{P}(Y = 10) = \frac{e^{-7.5}(7.5)^{10}}{10!} = 0.0858.$$

(c) Let W denote the number of messages sent in one-half hour. Then W is Poisson with $\lambda = 2.5$:

$$\mathrm{P}(W < 2) = \mathrm{P}(W = 0) + \mathrm{P}(W = 1) = \frac{e^{-2.5}(2.5)^0}{0!} + \frac{e^{-2.5}(2.5)^1}{1!} = 0.2878.$$

Example 3.2 (Errors in a textbook)**.** The number of errors in a textbook follows a Poisson distribution with a mean of 0.01 error per page. What is the probability that there are three or fewer errors in 100 pages?

Solution. The number of errors per page is Poisson with $\lambda = 0.01$. Over 100 pages the mean is $\lambda = 0.01 \times 100 = 1$, so let Y be Poisson with $\lambda = 1$:

$$\mathrm{P}(Y \leq 3) = e^{-1} + \frac{e^{-1}(1)^1}{1!} + \frac{e^{-1}(1)^2}{2!} + \frac{e^{-1}(1)^3}{3!} = 0.9810.$$

3.3 A Cumulative Distribution Function Exercise

The following exercise determines the probability mass function of a discrete random variable from its cumulative distribution function.

Example 3.3 (pmf from a given cdf)**.** Determine the probability mass function for the random variable with the following cumulative distribution function:

$$F(x) = \begin{cases} 0 & x < 2, \\ 0.2 & 2 \leq x < 5.7, \\ 0.5 & 5.7 \leq x < 6.5, \\ 0.8 & 6.5 \leq x < 8.5, \\ 1 & 8.5 \leq x. \end{cases}$$

Solution. The probability mass function places a jump at each point of increase of F, equal to the size of the jump:

$$f(x) = \begin{cases} 0.2 & x = 2, \\ 0.3 & x = 5.7, \\ 0.3 & x = 6.5, \\ 0.2 & x = 8.5, \\ 0 & \text{otherwise.} \end{cases}$$

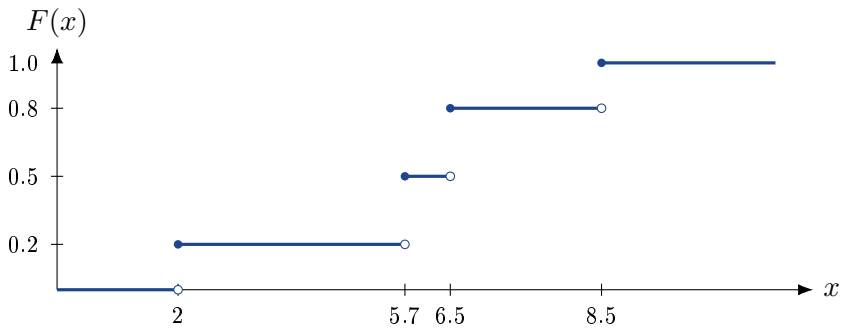


Figure 3.1: Step-function cumulative distribution function $F(x)$; the jump heights give the probability mass function.

Chapter 4

Continuous Random Variables and Probability Density Functions

We now move from discrete to continuous random variables. A continuous random variable can take any value in an interval of the real line, and its probability distribution is described not by a mass function but by a *probability density function*.

4.1 Probability Density Functions

A probability density function $f(x)$ can be used to describe the probability distribution of a continuous random variable X. If an interval is likely to contain a value for X, its probability is large, and it corresponds to large values of $f(x)$. The probability that X is between a and b is determined as the integral of $f(x)$ from a to b.

Definition 4.1: Probability density function

For a continuous random variable X, a *probability density function* is a function $f(x)$ such that

(1) $f(x) \geq 0$;

(2) $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$;

(3) $\mathrm{P}(a \leq X \leq b) = \displaystyle\int_a^b f(x)\,dx$ = area under $f(x)$ from a to b, for any a and b.

A probability density function is zero for x values that cannot occur, and it is assumed to be zero wherever it is not specifically defined. A histogram is an approximation to a probability density function.

For a continuous random variable X and any value x, $\mathrm{P}(X = x) = 0$: the probability density at any single point is, in this sense, "zero area." Consequently, for any x_1 and x_2,

$$\mathrm{P}(x_1 \leq X \leq x_2) = \mathrm{P}(x_1 < X \leq x_2) = \mathrm{P}(x_1 \leq X < x_2) = \mathrm{P}(x_1 < X < x_2).$$

4.2 Worked Problems on Probability Density Functions

Example 4.1 (An exponential-type density)**.** Suppose that $f(x) = e^{-(x-4)}$ for $4 < x$. Determine the following probabilities: (a) $\mathrm{P}(1 < X)$, (b) $\mathrm{P}(2 \leq X < 5)$, (c) $\mathrm{P}(5 < X)$, (d) $\mathrm{P}(8 < X < 12)$, and (e) determine x such that $\mathrm{P}(X < x) = 0.90$.

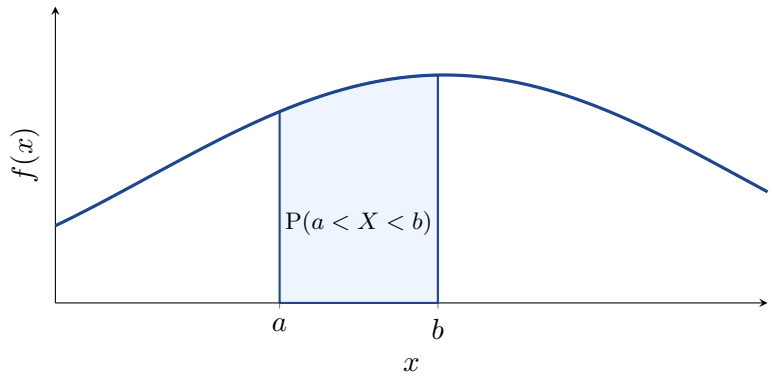


Figure 4.1: The probability that X lies between a and b is the shaded area under the density curve.

Solution.

$$\text{(a)}\quad \mathrm{P}(1 < X) = \int_4^\infty e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_4^\infty = 1, \quad \text{since } f(x) = 0 \text{ for } x < 4.$$

$$\text{(b)}\quad \mathrm{P}(2 \le X < 5) = \int_4^5 e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_4^5 = 1 - e^{-1} = 0.6321.$$

$$\text{(c)}\quad \mathrm{P}(5 < X) = 1 - \mathrm{P}(X \le 5) = 1 - 0.6321 = 0.3679.$$

$$\text{(d)}\quad \mathrm{P}(8 < X < 12) = \int_8^{12} e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_8^{12} = e^{-4} - e^{-8} = 0.0180.$$

$$\text{(e)}\quad \mathrm{P}(X < x) = \int_4^x e^{-(x-4)}\,dx = 1 - e^{-(x-4)} = 0.90 \;\Rightarrow\; x = 4 - \ln(0.10) = 6.303.$$

Example 4.2 (A linear density)**.** Suppose that $f(x) = x/8$ for $3 < x < 5$. Determine the following probabilities: (a) $\mathrm{P}(X < 4)$, (b) $\mathrm{P}(X > 3.5)$, (c) $\mathrm{P}(4 < X < 5)$, (d) $\mathrm{P}(X < 4.5)$, and (e) $\mathrm{P}(X < 3.5 \text{ or } X > 4.5)$.

Solution.

$$\text{(a)}\quad \mathrm{P}(X < 4) = \int_3^4 \frac{x}{8}\,dx = \frac{x^2}{16}\Big|_3^4 = \frac{16-9}{16} = \frac{7}{16} = 0.4375.$$

$$\text{(b)}\quad \mathrm{P}(X > 3.5) = \int_{3.5}^5 \frac{x}{8}\,dx = \frac{x^2}{16}\Big|_{3.5}^5 = \frac{25-(3.5)^2}{16} = 0.7969.$$

$$\text{(c)}\quad \mathrm{P}(4 < X < 5) = \int_4^5 \frac{x}{8}\,dx = \frac{x^2}{16}\Big|_4^5 = \frac{25-16}{16} = \frac{9}{16} = 0.5625.$$

$$\text{(d)}\quad \mathrm{P}(X < 4.5) = \int_3^{4.5} \frac{x}{8}\,dx = \frac{x^2}{16}\Big|_3^{4.5} = \frac{(4.5)^2-9}{16} = \frac{11.25}{16} = 0.7031.$$

$$\begin{aligned}\text{(e)}\quad \mathrm{P}(X < 3.5 \text{ or } X > 4.5) &= \int_3^{3.5} \frac{x}{8}\,dx + \int_{4.5}^5 \frac{x}{8}\,dx \\ &= \frac{x^2}{16}\Big|_3^{3.5} + \frac{x^2}{16}\Big|_{4.5}^5 = 0.20315 + 0.296875 = 0.500.\end{aligned}$$

Example 4.3 (A pure exponential density)**.** Suppose that $f(x) = e^{-x}$ for $0 < x$. Determine: (a) $\mathrm{P}(1 < X)$, (b) $\mathrm{P}(1 < X < 2.5)$, (c) $\mathrm{P}(X = 3)$, (d) $\mathrm{P}(X < 4)$, (e) $\mathrm{P}(3 \le X)$, (f) determine x such that $\mathrm{P}(x < X) = 0.10$, and (g) determine x such that $\mathrm{P}(X \le x) = 0.10$.

Solution.

$$\text{(a)}\quad \mathrm{P}(1 < X) = \int_1^\infty e^{-x}\,dx = -e^{-x}\big|_1^\infty = e^{-1} = 0.3679.$$

$$\text{(b)}\quad \mathrm{P}(1 < X < 2.5) = \int_1^{2.5} e^{-x}\,dx = -e^{-x}\big|_1^{2.5} = e^{-1} - e^{-2.5} = 0.2858.$$

$$\text{(c)}\quad \mathrm{P}(X = 3) = 0.$$

$$\text{(d)}\quad \mathrm{P}(X < 4) = \int_0^4 e^{-x}\,dx = -e^{-x}\big|_0^4 = 1 - e^{-4} = 0.9816.$$

$$\text{(e)}\quad \mathrm{P}(3 \le X) = \int_3^\infty e^{-x}\,dx = e^{-3} = 0.0498.$$

$$\text{(f)}\quad \mathrm{P}(x < X) = \int_x^\infty e^{-x'}\,dx' = e^{-x} = 0.10 \;\Rightarrow\; x = 2.30.$$

$$\text{(g)}\quad \mathrm{P}(X \le x) = \int_0^x e^{-x'}\,dx' = 1 - e^{-x} = 0.10 \;\Rightarrow\; e^{-x} = 0.9,$$
$$\ln(0.9) = -x \;\Rightarrow\; x = 0.105.$$

4.3 Cumulative Distribution Functions

Definition 4.2: Cumulative distribution function (continuous)

The *cumulative distribution function* of a continuous random variable X is

$$F(x) = \mathrm{P}(X \le x) = \int_{-\infty}^x f(u)\,du.$$

Example 4.4 (A piecewise-linear cdf)**.** Suppose the cdf of the random variable X is

$$F(x) = \begin{cases} 0 & x < 0, \\ 0.2x & 0 \le x < 5, \\ 1 & 5 \le x. \end{cases}$$

Determine: (a) $\mathrm{P}(X < 2.8)$, (b) $\mathrm{P}(X > 1.5)$, (c) $\mathrm{P}(X < -2)$, (d) $\mathrm{P}(X > 6)$.

Solution.

(a) $\mathrm{P}(X < 2.8) = 0.2 \times 2.8 = 0.56.$

(b) $\mathrm{P}(X > 1.5) = 1 - \mathrm{P}(X < 1.5) = 1 - 0.2 \times 1.5 = 1 - 0.3 = 0.7.$

(c) $\mathrm{P}(X < -2) = 0.$

(d) $\mathrm{P}(X > 6) = 0.$

Example 4.5 (A cdf with a slope and an offset)**.** The cdf of a random variable X is

$$F(x) = \begin{cases} 0 & x < -2, \\ 0.25x + 0.5 & -2 \le x < 2, \\ 1 & 2 \le x. \end{cases}$$

Determine: (a) $\mathrm{P}(X < 1.8)$, (b) $\mathrm{P}(X > -1.5)$, (c) $\mathrm{P}(X < -2)$, (d) $\mathrm{P}(-1 < X < 1)$.

Solution.

$$\begin{aligned}
\text{(a)}\quad & \mathrm{P}(X < 1.8) = 0.25 \times 1.8 + 0.5 = 0.95.\\
\text{(b)}\quad & \mathrm{P}(X > -1.5) = 1 - \mathrm{P}(X < -1.5) = 1 - (0.25 \times (-1.5) + 0.5) = 0.875.\\
\text{(c)}\quad & \mathrm{P}(X < -2) = 0.\\
\text{(d)}\quad & \mathrm{P}(-1 < X < 1) = \mathrm{P}(X < 1) - \mathrm{P}(X \le -1)\\
& \qquad = (0.25 \times 1 + 0.5) - (0.25 \times (-1) + 0.5)\\
& \qquad = 0.75 - 0.25 = 0.5.
\end{aligned}$$

Example 4.6 (Gap width of a magnetic recording head)**.** The gap width is an important property of a magnetic recording head. In coded units, if the width is a continuous random variable over the range $0 < x < 2$ with $f(x) = 0.5x$, determine the cumulative distribution function of the gap width.

Solution.

$$F(x) = \int_0^x 0.5u\, du = 0.25x^2, \qquad 0 \le x < 2,$$

so that

$$F(x) = \begin{cases} 0 & x < 0,\\ 0.25x^2 & 0 \le x < 2,\\ 1 & 2 \le x. \end{cases}$$

4.4 Mean and Variance of a Continuous Random Variable

Key Formula

For a continuous random variable X with probability density function $f(x)$, the *mean* or expected value is

$$\mu = \mathbb{E}(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx,$$

and the *variance* is

$$\sigma^2 = \mathrm{Var}(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx = \int_{-\infty}^{\infty} x^2 f(x)\, dx - \mu^2.$$

The standard deviation of X is $\sigma = \sqrt{\sigma^2}$.

Example 4.7 (Uniform density on $(0, 4)$)**.** Suppose $f(x) = 0.25$ for $0 < x < 4$. Determine the mean and variance of X.

Solution.

$$\begin{aligned}
\mathbb{E}(X) &= \int_0^4 x(0.25)\, dx = 0.25\frac{x^2}{2}\Big|_0^4 = 0.25 \times 8 = 2.\\
\mathrm{Var}(X) &= \int_0^4 x^2(0.25)\, dx - \mu^2 = 0.25\frac{x^3}{3}\Big|_0^4 - 2^2 = 0.25 \times \frac{64}{3} - 4\\
&= \frac{16}{3} - 4 = \frac{16-12}{3} = \frac{4}{3}.
\end{aligned}$$

Example 4.8 (Thickness of a conductive coating)**.** The thickness of a conductive coating in micrometres has a density function $600x^{-2}$ for $100\,\mu\text{m} < x < 120\,\mu\text{m}$. (a) Determine the mean and variance of the coating thickness. (b) If the coating costs \$0.50 per micrometre of thickness on each part, what is the average cost of coating per part?

Solution. Using $\int x^{-1}\,dx = \ln x + C$,

$$\begin{aligned}
\mathbb{E}(X) &= \int_{100}^{120} x \cdot \frac{600}{x^2}\,dx = \int_{100}^{120} \frac{600}{x}\,dx = 600\big[\ln(120) - \ln(100)\big] = 109.39.\\
\text{Var}(X) &= \int_{100}^{120} x^2 \cdot \frac{600}{x^2}\,dx - \mu^2 = \int_{100}^{120} 600\,dx - (109.39)^2\\
&= 600\big[120 - 100\big] - (109.39)^2 = 12000 - 11966.1721 = 33.8279.
\end{aligned}$$

(b) Total cost = mean $\times$ cost per μm $= 109.39 \times 0.5 = \$54.70$ per part.

Example 4.9 (Length of computer cables)**.** Suppose the pdf of the length of computer cables is $f(x) = 0.1$ from 1200 to 1210 mm. (a) Determine the mean and standard deviation of the cable length. (b) If the length specifications are $1195 < x < 1205$ mm, what proportion of cables are within specifications?

Solution.

$$\begin{aligned}
\mathbb{E}(X) &= \int_{1200}^{1210} x(0.1)\,dx = 0.05x^2\big|_{1200}^{1210} = 1205.\\
\text{Var}(X) &= \int_{1200}^{1210} (x - 1205)^2(0.1)\,dx = 0.1\frac{(x-1205)^3}{3}\bigg|_{1200}^{1210} = 8.333,\\
\sigma &= \sqrt{8.333} = 2.887.
\end{aligned}$$

(b) Clearly, centring the process at the centre of the specifications results in the greatest proportion of cables within specifications. Since the density is zero below 1200,

$$\text{P}(1195 < X < 1205) = \text{P}(1200 < X < 1205) = \int_{1200}^{1205} 0.1\,dx = 0.1x|_{1200}^{1205} = 0.5.$$

Chapter 5

The Continuous Uniform Distribution

The simplest continuous distribution is one in which the density is constant over a finite interval. It is the continuous analogue of equally likely outcomes.

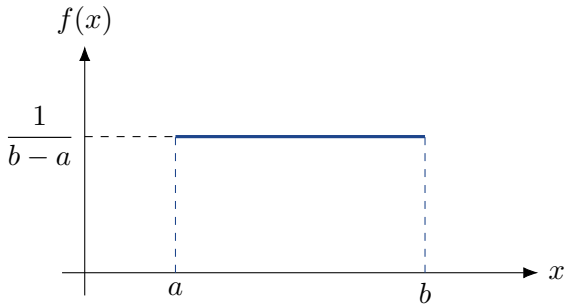
5.1 Definition and Moments

Definition 5.1: Continuous uniform random variable

A continuous random variable X with probability density function

$$f(x) = \frac{1}{b-a}, \qquad a \leq x \leq b,$$

is a *continuous uniform* random variable.



Continuous uniform PDF

Figure 5.1: The continuous uniform probability density function on $[a, b]$.

Key Formula

For a continuous uniform random variable on $[a, b]$,

$$\mu = \mathbb{E}(X) = \frac{a+b}{2}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{(b-a)^2}{12},$$

and the cumulative distribution function is

$$F(x) = \begin{cases} 0 & x < a, \\ \dfrac{x-a}{b-a} & a \leq x < b, \\ 1 & b \leq x. \end{cases}$$

These follow from

$$\mathbb{E}(X) = \int_a^b \frac{x}{b-a}\,dx = \left.\frac{0.5x^2}{b-a}\right|_a^b = \frac{a+b}{2},$$

$$\mathrm{Var}(X) = \int_a^b \frac{\left(x - \frac{a+b}{2}\right)^2}{b-a}\,dx = \left.\frac{\left(x - \frac{a+b}{2}\right)^3}{3(b-a)}\right|_a^b = \frac{(b-a)^2}{12},$$

and

$$F(x) = \int_a^x \frac{1}{b-a}\,du = \frac{x-a}{b-a}, \qquad a \le x < b.$$

5.2 Worked Problems

Example 5.1 (Net weight of a packaged herbicide)**.** The net weight in pounds of a packaged chemical herbicide is uniform for $49.75 < x < 50.25$ pounds. (a) Determine the mean and variance of the weight of the packages. (b) Determine the cdf of the weight of the packages. (c) Determine $\mathrm{P}(X < 50.1)$.

Solution. Here $a = 49.75$, $b = 50.25$, so $b - a = 0.5$ and $f(x) = 1/0.5 = 2$ for $49.75 < x < 50.25$.

(a)

$$\mathbb{E}(X) = \frac{a+b}{2} = \frac{49.75 + 50.25}{2} = 50, \qquad \mathrm{Var}(X) = \frac{(50.25 - 49.75)^2}{12} = 0.0208,$$

with $\sigma = \sqrt{0.0208} = 0.144$.

(b)

$$F(x) = \begin{cases} 0 & x \le 49.75, \\ \dfrac{x - 49.75}{0.5} = 2x - 99.5 & 49.75 < x < 50.25, \\ 1 & x \ge 50.25. \end{cases}$$

(c) $\mathrm{P}(X < 50.1) = F(50.1) = 2 \times 50.1 - 99.5 = 0.7$.

Example 5.2 (Thickness of an aircraft flange)**.** The thickness of a flange on an aircraft component is uniformly distributed between 0.95 and 1.05 mm. (a) Determine the cdf of the flange thickness. (b) Determine the proportion of flanges that exceed 1.02 mm. (c) What thickness is exceeded by 90% of the flanges? (d) Determine the mean and variance of flange thickness.

Solution. Here $a = 0.95$, $b = 1.05$, so $f(x) = 1/0.1 = 10$ for $0.95 < x < 1.05$.

(a)

$$F(x) = \begin{cases} 0 & x < 0.95, \\ \dfrac{x - 0.95}{1.05 - 0.95} = 10x - 9.5 & 0.95 \le x < 1.05, \\ 1 & 1.05 \le x. \end{cases}$$

(b)

$$\mathrm{P}(X > 1.02) = 1 - \mathrm{P}(X \le 1.02) = 1 - F(1.02) = 1 - (10 \times 1.02 - 9.5) = 1 - 0.7 = 0.3.$$

(c) If $\mathrm{P}(X > x) = 0.9$ then $1 - F(x) = 0.90$ and $F(x) = 0.10$, so

$$10x - 9.5 = 0.10 \;\Rightarrow\; x = 0.96.$$

(d)

$$\mathbb{E}(X) = \frac{1.05 + 0.95}{2} = 1.00, \qquad \mathrm{Var}(X) = \frac{(1.05 - 0.95)^2}{12} = 0.00083.$$

Example 5.3 (Time to fill an electronic form)**.** Suppose the time it takes a data collection operator to fill out an electronic form for a database is uniformly distributed between 1.5 and 2.2 minutes. (a) What is the mean and variance of the time it takes an operator to fill out the form? (b) What is the probability that it will take less than 2 minutes to fill out the form? (c) Determine the cdf of the time it takes to fill out the form.

Solution. Here $a = 1.5$, $b = 2.2$, so $b - a = 0.7$ and $f(x) = 1/0.7$.

(a)

$$\mathbb{E}(X) = \frac{1.5 + 2.2}{2} = 1.85 \text{ min}, \qquad \mathrm{Var}(X) = \frac{(2.2 - 1.5)^2}{12} = 0.0408 \text{ min}^2.$$

(b)

$$\mathrm{P}(X < 2) = \int_{1.5}^{2} \frac{1}{0.7}\, dx = \frac{x}{0.7}\Big|_{1.5}^{2} = \frac{2 - 1.5}{0.7} = \frac{0.5}{0.7} = 0.7143.$$

(c)

$$F(x) = \begin{cases} 0 & x < 1.5, \\ \dfrac{x - 1.5}{0.7} & 1.5 \le x < 2.2, \\ 1 & 2.2 \le x. \end{cases}$$

Example 5.4 (Thickness of photoresist)**.** The thickness of photoresist applied to wafers in semiconductor manufacturing at a particular location on the wafer is uniformly distributed between 0.2050 and 0.2150 micrometres. (a) Determine the cdf of photoresist thickness. (b) Determine the proportion of wafers that exceed 0.2125 μm in photoresist thickness. (c) What thickness is exceeded by 10% of the wafers? (d) Determine the mean and variance of photoresist thickness.

Solution. The pdf is

$$f(x) = \frac{1}{b - a} = \frac{1}{0.2150 - 0.2050} = \frac{1}{0.01} = 100, \qquad 0.2050 < x < 0.2150.$$

(a)

$$F(x) = \begin{cases} 0 & x < 0.2050, \\ \dfrac{x - 0.2050}{0.0100} = 100x - 20.50 & 0.2050 \le x < 0.2150, \\ 1 & 0.2150 \le x. \end{cases}$$

(b)

$$\mathrm{P}(X > 0.2125) = 1 - F(0.2125) = 1 - \big[100(0.2125) - 20.50\big] = 0.25.$$

(c) If $\mathrm{P}(X > x) = 0.10$ then $1 - F(x) = 0.10$ and $F(x) = 0.90$, so

$$100x + 20.50 = 0.90 \;\Rightarrow\; (\text{solving } 100x - 20.50 = 0.90)\;\; x = 0.2140.$$

(d)

$$\mathbb{E}(X) = \frac{0.2050 + 0.2150}{2} = 0.2100 \ \mu\text{m}, \qquad \mathrm{Var}(X) = \frac{(0.2150 - 0.2050)^2}{12} = 8.33 \times 10^{-6} \ \mu\text{m}^2.$$

Chapter 6

The Normal Distribution

The most widely used model for the distribution of a random variable is the *normal distribution*. Whenever a random experiment is replicated, the random variable that equals the average (or total) result over the replicates tends to have a normal distribution as the number of replicates becomes large. De Moivre presented this fundamental result, known as the *central limit theorem*, in 1733. About a hundred years later Gauss independently developed a normal distribution; a normal distribution is also referred to as a *Gaussian distribution*.

6.1 The Normal Probability Density Function

Random variables with different means and variances can be modelled by normal probability density functions with appropriate choices of the centre and width of the curve. The value of $\mathbb{E}(X) = \mu$ determines the centre of the probability density function, and the value of $\text{Var}(X) = \sigma^2$ determines the width.

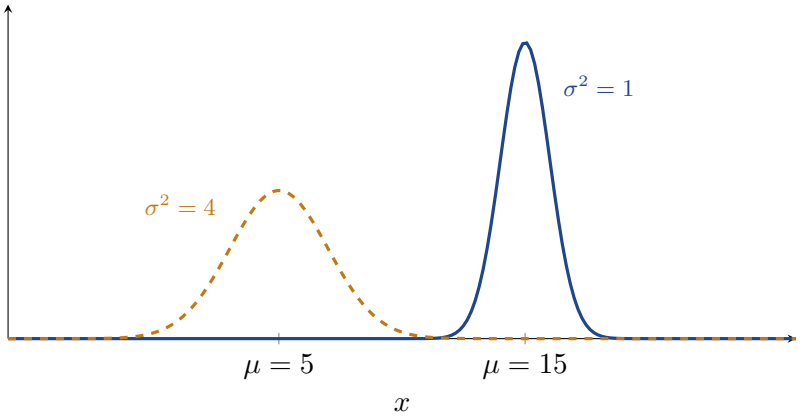


Figure 6.1: Normal densities with different means and variances. The mean locates the centre; the variance controls the spread.

Definition 6.1: Normal random variable

A random variable X with probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad -\infty < x < \infty,$$

is a *normal* random variable with parameters μ (where $-\infty < \mu < \infty$) and $\sigma > 0$. Also $\mathbb{E}(X) = \mu$ and $\text{Var}(X) = \sigma^2$, and the notation $N(\mu, \sigma^2)$ is used to denote the distribution.

6.2 The Standard Normal Distribution

Definition 6.2: Standard normal random variable

A normal random variable with $\mu = 0$ and $\sigma^2 = 1$ is called a *standard normal* random variable and is denoted as Z. The cumulative distribution function of a standard normal random variable is denoted

$$\Phi(z) = \mathrm{P}(Z \leq z).$$

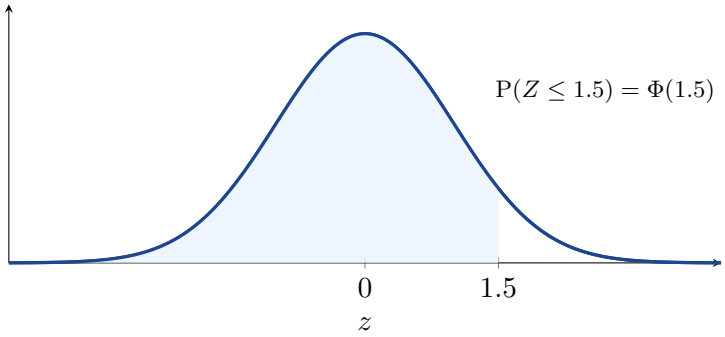


Figure 6.2: The standard normal density; the shaded area to the left of $z = 1.5$ is $\Phi(1.5)$.

A small excerpt of the standard normal table is reproduced below.

Table 6.1: Excerpt of the standard normal cumulative probabilities $\Phi(z)$.

z	0.00	0.01	0.02	0.03
0.0	0.50000	0.50399	0.50398	0.51197
⋮				
1.5	0.93319	0.93448	0.93574	0.93699

Probabilities that are not of the form $\mathrm{P}(Z \leq z)$ are found by using the basic rules of probability and the symmetry of the normal distribution along with the tables.

Example 6.1 (Reading the standard normal table). 1) $\mathrm{P}(Z > 1.26) = 1 - \mathrm{P}(Z \leq 1.26) = 1 - 0.89616 = 0.10384$.

2) $\mathrm{P}(Z < -0.86) = 0.19490$.

3) $\mathrm{P}(Z > -1.37) = \mathrm{P}(Z < 1.37) = 0.91465$.

4) $\mathrm{P}(-1.25 < Z < 0.37) = \mathrm{P}(Z < 0.37) - \mathrm{P}(Z < -1.25) = 0.64431 - 0.10565 = 0.53866$.

5) $\mathrm{P}(Z \leq -4.6)$ cannot be found exactly from the table. The last value in the table is $\mathrm{P}(Z \leq -3.99) = 0.00003$. Since $\mathrm{P}(Z \leq -4.6) < \mathrm{P}(Z \leq -3.99)$, we take $\mathrm{P}(Z \leq -4.6) \approx 0$.

6) Find z such that $\mathrm{P}(Z > z) = 0.05$, i.e. $\mathrm{P}(Z \leq z) = 0.95$. The nearest table value is 0.95053, giving $z = 1.65$.

7) Find z such that $\mathrm{P}(-z < Z < z) = 0.99$. By the symmetry of the normal distribution, the area in each tail must equal 0.005. Thus the value for z corresponds to a probability of 0.995 in the table. The nearest probability is 0.99506, occurring at $z = 2.58$.

6.3 Standardizing a Normal Random Variable

Key Formula

If X is a normal random variable with $\mathbb{E}(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$, then the random variable

$$Z = \frac{X - \mu}{\sigma}$$

is a normal random variable with $\mathbb{E}(Z) = 0$ and $\mathrm{Var}(Z) = 1$; that is, Z is a standard normal random variable. Consequently, for any x,

$$\mathrm{P}(X \le x) = \mathrm{P}\left(\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma}\right) = \mathrm{P}(Z \le z), \qquad z = \frac{x-\mu}{\sigma}.$$

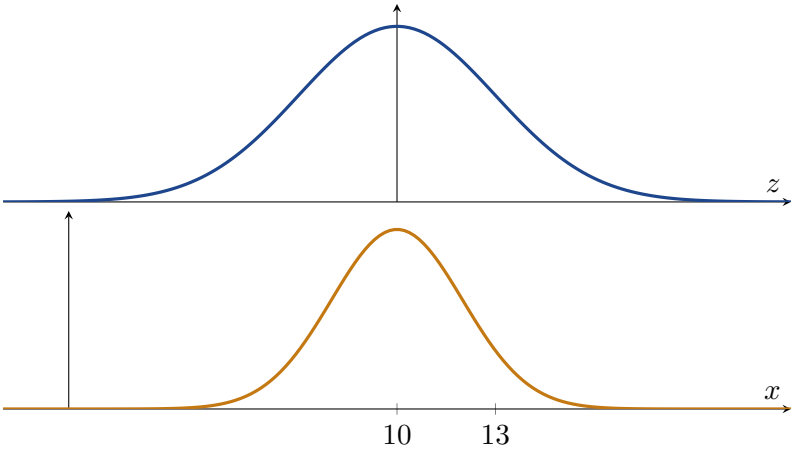


Figure 6.3: Standardizing a normal random variable: the transformation $z = (x - \mu)/\sigma$ maps $X \sim N(\mu, \sigma^2)$ onto the standard normal scale.

6.4 Worked Problems

Example 6.2 (Probabilities for a standard normal variable)**.** Use tables to determine the following probabilities for the standard normal random variable Z: (a) $\mathrm{P}(Z < 1.32)$, (b) $\mathrm{P}(Z < 3.0)$, (c) $\mathrm{P}(Z > 1.45)$, (d) $\mathrm{P}(Z > -2.15)$, (e) $\mathrm{P}(-2.34 < Z < 1.76)$.

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(Z < 1.32) &= 0.906582.\\
\text{(b)}\quad \mathrm{P}(Z < 3.0) &= 0.99865.\\
\text{(c)}\quad \mathrm{P}(Z > 1.45) &= 1 - \mathrm{P}(Z < 1.45) = 1 - 0.92647 = 0.07353.\\
\text{(d)}\quad \mathrm{P}(Z > -2.15) &= \mathrm{P}(Z < 2.15) = 0.98422.\\
\text{(e)}\quad \mathrm{P}(-2.34 < Z < 1.76) &= \mathrm{P}(Z < 1.76) - \mathrm{P}(Z < -2.34)\\
&= 0.9607 - \big(1 - \mathrm{P}(Z < 2.34)\big)\\
&= 0.9607 - (1 - 0.990358)\\
&= 0.9607 - 0.009642 = 0.9511.
\end{aligned}$$

Example 6.3 (A normal variable with mean 10, sd 2)**.** Assume X is normally distributed with

a mean of 10 and a standard deviation of 2. Determine the following: (a) $\mathrm{P}(X < 13)$, (b) $\mathrm{P}(X > 9)$, (c) $\mathrm{P}(6 < X < 14)$, (d) $\mathrm{P}(2 < X < 4)$, (e) $\mathrm{P}(-2 < X < 8)$.

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X < 13) &= \mathrm{P}\big(Z < \tfrac{13-10}{2}\big) = \mathrm{P}(Z < 1.5) = 0.93319.\\
\text{(b)}\quad \mathrm{P}(X > 9) &= 1 - \mathrm{P}(X < 9) = 1 - \mathrm{P}\big(Z < \tfrac{9-10}{2}\big)\\
&= 1 - \mathrm{P}(Z < -0.5) = 1 - 0.308538 = 0.691462.\\
\text{(c)}\quad \mathrm{P}(6 < X < 14) &= \mathrm{P}\big(\tfrac{6-10}{2} < Z < \tfrac{14-10}{2}\big) = \mathrm{P}(-2 < Z < 2)\\
&= \mathrm{P}(Z < 2) - \mathrm{P}(Z < -2) = 0.977250 - 0.022750 = 0.954500.\\
\text{(d)}\quad \mathrm{P}(2 < X < 4) &= \mathrm{P}\big(\tfrac{2-10}{2} < Z < \tfrac{4-10}{2}\big) = \mathrm{P}(-4 < Z < -3)\\
&= \mathrm{P}(Z < -3) - \mathrm{P}(Z < -4) = 0.001350 - 0.000033 = 0.001327.\\
\text{(e)}\quad \mathrm{P}(-2 < X < 8) &= \mathrm{P}(X < 8) - \mathrm{P}(X < -2) = \mathrm{P}\big(Z < \tfrac{8-10}{2}\big) - \mathrm{P}\big(Z < \tfrac{-2-10}{2}\big)\\
&= \mathrm{P}(Z < -1) - \mathrm{P}(Z < -6) = 0.15866.
\end{aligned}$$

Example 6.4 (Inverse problems for standard normal)**.** Assume Z has a standard normal distribution. Use the table to determine the value for z that solves each of the following: (a) $\mathrm{P}(Z < z) = 0.9$, (b) $\mathrm{P}(Z < z) = 0.5$, (c) $\mathrm{P}(Z > z) = 0.1$, (d) $\mathrm{P}(Z > z) = 0.9$, (e) $\mathrm{P}(-1.24 < Z < z) = 0.8$.

Solution.

$$\begin{aligned}
&\text{(a)}\quad \mathrm{P}(Z < 1.28) = 0.9 \Rightarrow z = 1.28.\\
&\text{(b)}\quad \mathrm{P}(Z < 0) = 0.5 \Rightarrow z = 0.\\
&\text{(c)}\quad \mathrm{P}(Z > z) = 0.1 \Rightarrow \mathrm{P}(Z < z) = 0.9 \Rightarrow z = 1.28.\\
&\text{(d)}\quad \mathrm{P}(Z > z) = 0.9 \Rightarrow \mathrm{P}(Z < z) = 0.10 \Rightarrow z = -1.28.\\
&\text{(e)}\quad \mathrm{P}(-1.24 < Z < z) = 0.8 \Rightarrow \mathrm{P}(Z < z) - \mathrm{P}(Z < -1.24) = 0.8\\
&\qquad \Rightarrow \mathrm{P}(Z < z) - 0.10749 = 0.8 \Rightarrow \mathrm{P}(Z < z) = 0.90749 \Rightarrow z = 1.33.
\end{aligned}$$

Example 6.5 (Inverse problems for a general normal variable)**.** Assume X is normally distributed with a mean of 10 and a standard deviation of 2. Determine the value for x that solves each of the following: (a) $\mathrm{P}(X > x) = 0.5$, (b) $\mathrm{P}(X > x) = 0.95$, (c) $\mathrm{P}(x < X < 10) = 0.2$.

Solution.

$$\begin{aligned}
&\text{(a)}\quad \mathrm{P}(X > x) = 0.5 \Rightarrow \mathrm{P}\big(Z > \tfrac{x-10}{2}\big) = 0.5 \Rightarrow \tfrac{x-10}{2} = 0 \Rightarrow x = 10.\\
&\text{(b)}\quad \mathrm{P}(X > x) = 0.95 \Rightarrow 1 - \mathrm{P}\big(Z < \tfrac{x-10}{2}\big) = 0.95\\
&\qquad \Rightarrow \mathrm{P}\big(Z < \tfrac{x-10}{2}\big) = 0.05 \Rightarrow \tfrac{x-10}{2} = -1.64 \Rightarrow x = 6.72.\\
&\text{(c)}\quad \mathrm{P}(x < X < 10) = 0.2 \Rightarrow \mathrm{P}\big(\tfrac{x-10}{2} < Z < 0\big) = 0.2\\
&\qquad \Rightarrow 0.5 - \mathrm{P}\big(Z < \tfrac{x-10}{2}\big) = 0.2 \Rightarrow \mathrm{P}\big(Z < \tfrac{x-10}{2}\big) = 0.3\\
&\qquad \Rightarrow \tfrac{x-10}{2} = -0.52 \Rightarrow x = 8.96.
\end{aligned}$$

Example 6.6 (Compressive strength of cement)**.** The compressive strength of samples of cement can be modelled by a normal distribution with a mean of 6000 kg/cm^2 and a standard deviation of 100 kg/cm^2. (a) What is the probability that a sample's strength is less than 6250 kg/cm^2?

(b) What is the probability that a sample strength is between 5800 and 5900 kg/cm^2? (c) What strength is exceeded by 95% of the samples?

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X<6250) &= \mathrm{P}\big(Z<\tfrac{6250-6000}{100}\big) = \mathrm{P}(Z<2.5) = 0.99379.\\
\text{(b)}\quad \mathrm{P}(5800<X<5900) &= \mathrm{P}\big(\tfrac{5800-6000}{100}<Z<\tfrac{5900-6000}{100}\big)\\
&= \mathrm{P}(-2<Z<-1) = \mathrm{P}(Z<-1)-\mathrm{P}(Z<-2) = 0.13591.\\
\text{(c)}\quad \mathrm{P}(X>x)=0.95 &\Rightarrow \mathrm{P}\big(Z>\tfrac{x-6000}{100}\big)=0.95\\
&\Rightarrow \tfrac{x-6000}{100} = -1.65 \Rightarrow x = 5835.
\end{aligned}$$

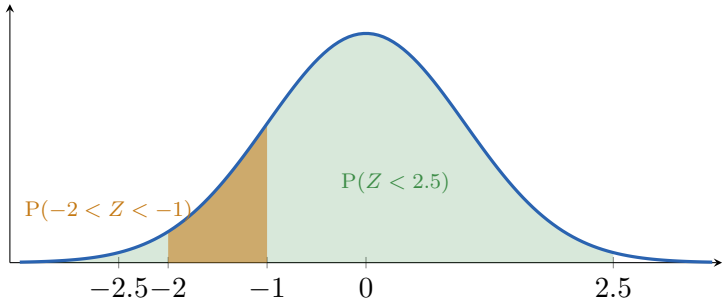


Figure 6.4: Standard-normal areas for the cement-strength problem: part (a) shaded light, part (b) shaded dark.

Example 6.7 (Line width in semiconductor manufacturing)**.** The line width for semiconductor manufacturing is assumed to be normally distributed with a mean of 0.5 μm and a standard deviation of 0.05 μm. (a) What is the probability that a line width is greater than 0.62 μm? (b) What is the probability that a line width is between 0.47 and 0.63 μm? (c) The line width of 90% of samples is below what value?

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X>0.62) &= \mathrm{P}\big(Z>\tfrac{0.62-0.5}{0.05}\big) = \mathrm{P}(Z>2.4) = 1-\mathrm{P}(Z<2.4) = 0.0082.\\
\text{(b)}\quad \mathrm{P}(0.47<X<0.63) &= \mathrm{P}\big(\tfrac{0.47-0.5}{0.05}<Z<\tfrac{0.63-0.5}{0.05}\big) = \mathrm{P}(-0.6<Z<2.6)\\
&= \mathrm{P}(Z<2.6)-\mathrm{P}(Z<-0.6) = 0.99534-0.27425 = 0.72109.\\
\text{(c)}\quad \mathrm{P}(X<x)=0.90 &\Rightarrow \mathrm{P}\big(Z<\tfrac{x-0.5}{0.05}\big)=0.90\\
&\Rightarrow \tfrac{x-0.5}{0.05} = 1.28 \Rightarrow x = 0.564.
\end{aligned}$$

Example 6.8 (A normal variable with mean 0.4, sd 0.05)**.** Assume X is normally distributed with a mean of 0.4 and a standard deviation of 0.05. (a) Determine $\mathrm{P}(X>0.5)$. (b) Determine $\mathrm{P}(0.4<X<0.5)$. (c) Determine the value x such that $\mathrm{P}(X<x)=0.90$.

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X>0.5) &= \mathrm{P}\big(Z>\tfrac{0.5-0.4}{0.05}\big) = \mathrm{P}(Z>2) = 0.02275.\\
\text{(b)}\quad \mathrm{P}(0.4<X<0.5) &= \mathrm{P}\big(\tfrac{0.4-0.4}{0.05}<Z<\tfrac{0.5-0.4}{0.05}\big) = \mathrm{P}(0<Z<2) = 0.47725.\\
\text{(c)}\quad \mathrm{P}(X<x)=0.9 &\Rightarrow \mathrm{P}\big(Z<\tfrac{x-0.4}{0.05}\big)=0.90\\
&\Rightarrow \tfrac{x-0.4}{0.05} = 1.28 \Rightarrow x = 0.336 \text{ (i.e. } 0.464).
\end{aligned}$$

Example 6.9 (Speed of a file transfer)**.** The speed of a file transfer from a server on campus to a personal computer at a student's home on a weekday evening is normally distributed with a mean of 60 kilobits/sec and a standard deviation of 4 kilobits/sec. (a) What is the probability that the file will transfer at a speed of 70 kilobits/sec or more? (b) What is the probability that the file will transfer at a speed of less than 58 kb/sec? (c) If the file is 1 MB, what is the average time it will take to transfer the file?

Solution.

$$\text{(a)}\quad \mathrm{P}(X > 70) = \mathrm{P}\big(Z > \tfrac{70-60}{4}\big) = 1 - \mathrm{P}(Z < 2.5) = 1 - 0.99379 = 0.00621.$$

$$\text{(b)}\quad \mathrm{P}(X < 58) = \mathrm{P}\big(Z < \tfrac{58-60}{4}\big) = \mathrm{P}(Z < -0.5) = 0.308538.$$

$$\text{(c)}\quad 1{,}000{,}000 \text{ bytes} \times 8 \text{ bits/byte} = 8{,}000{,}000 \text{ bits},$$

$$\text{average time} = \frac{8{,}000{,}000}{60{,}000} = 133.33 \text{ sec}.$$

Example 6.10 (Life of a semiconductor laser)**.** The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours. (a) What is the probability that a laser fails before 5000 hours? (b) What is the life in hours that 95% of the lasers exceed? (c) If three lasers are used in a product and they are assumed to fail independently, what is the probability that all three are still operating after 7000 hours?

Solution.

$$\text{(a)}\quad \mathrm{P}(X < 5000) = \mathrm{P}\big(Z < \tfrac{5000-7000}{600}\big) = \mathrm{P}(Z < -3.33) = 0.00043.$$

$$\text{(b)}\quad \mathrm{P}(X > x) = 0.95 \Rightarrow \mathrm{P}\big(Z > \tfrac{x-7000}{600}\big) = 0.95$$

$$\Rightarrow \tfrac{x-7000}{600} = -1.64 \Rightarrow x = 6016.$$

$$\text{(c)}\quad \mathrm{P}(X > 7000) = \mathrm{P}\big(Z > \tfrac{7000-7000}{600}\big) = \mathrm{P}(Z > 0) = 0.5,$$

$$\mathrm{P}(3 \text{ lasers operating after } 7000) = \big(\tfrac{1}{2}\big) \times \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{8}.$$

Chapter 7

Normal Approximation to the Binomial and Poisson Distributions

Because the binomial and Poisson distributions become difficult to compute when their parameters are large, the normal distribution provides a convenient approximation. This chapter records the standardisation used, the continuity correction, and the rules of thumb that govern when each approximation is adequate.

7.1 Normal Approximation to the Binomial

Key Formula

If X is a binomial random variable with parameters n and p, then

$$Z = \frac{X - np}{\sqrt{np(1-p)}}$$

is approximately a standard normal random variable. To approximate a binomial probability with a normal distribution, a *continuity correction* is applied as follows:

$$\mathrm{P}(X \leq x) = \mathrm{P}(X \leq x + 0.5) \cong \mathrm{P}\left(Z \leq \frac{x + 0.5 - np}{\sqrt{np(1-p)}}\right),$$

$$\mathrm{P}(x \leq X) = \mathrm{P}(x - 0.5 \leq X) \cong \mathrm{P}\left(\frac{x - 0.5 - np}{\sqrt{np(1-p)}} \leq Z\right).$$

The approximation is good for $np > 5$ and $n(1-p) > 5$.

Recall that for a binomial variable X, $\mathbb{E}(X) = np$ and $\mathrm{Var}(X) = np(1-p)$. Consequently, the standardisation equation is nothing more than the formula for standardizing the random variable X. The normal approximation is good when n is large relative to p. A way to remember the continuity correction is to write the probability in terms of $\leq$ or $\geq$ and then add or subtract the 0.5 correction factor to make the probability greater.

7.2 Relationship Among the Hypergeometric, Binomial, and Normal

The binomial distribution is a satisfactory approximation to the hypergeometric distribution when n, the sample size, is small relative to N, the size of the population. A rule of thumb is

that the binomial approximation is effective if $n/N < 0.1$.

For a hypergeometric distribution, p is defined as $p = K/N$, interpreted as the proportion of successes in the population. The normal distribution can in turn provide an effective approximation of hypergeometric probabilities when $n/N < 0.1$, $np > 5$, and $n(1-p) > 5$.

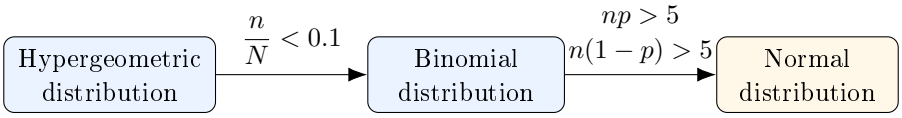


Figure 7.1: Conditions under which one distribution approximates another.

7.3 Normal Approximation to the Poisson

The Poisson distribution was developed as the limit of a binomial distribution as the number of trials increased to infinity. Hence it should not be surprising to find that the normal distribution can also be used to approximate probabilities of a Poisson random variable.

Key Formula

If X is a Poisson random variable with $\mathbb{E}(X) = \lambda$ and $\mathrm{Var}(X) = \lambda$, then

$$Z = \frac{X - \lambda}{\sqrt{\lambda}}$$

is approximately a standard normal random variable. The approximation is good for $\lambda > 5$.

7.4 Worked Problems

Example 7.1 (Normal approximation to a binomial)**.** Suppose that X is a binomial random variable with $n = 200$ and $p = 0.4$. (a) Approximate the probability that X is less than or equal to 70. (b) Approximate the probability that $X > 70$ and $X < 90$. (c) Approximate the probability that $X = 80$.

Solution. First,

$$\mathbb{E}(X) = np = 200(0.4) = 80, \qquad \mathrm{Var}(X) = np(1-p) = 200(0.4)(0.6) = 48, \qquad \sigma = 6.93.$$

(a) With continuity correction,

$$\mathrm{P}(X \le 70) \cong \mathrm{P}\left(Z \le \frac{70.5 - 80}{6.93}\right) = \mathrm{P}(Z \le -1.37) = 0.08534.$$

(Without the continuity correction one would obtain $\mathrm{P}(Z \le \frac{70-80}{\sqrt{48}}) = \mathrm{P}(Z \le -1.44) = 0.0749$.)

(b) With continuity correction, the event $70 < X < 90$ corresponds to $70.5 \le X \le 89.5$:

$$\begin{aligned}\mathrm{P}(70 < X < 90) &\cong \mathrm{P}\left(\frac{70.5 - 80}{6.93} < Z < \frac{89.5 - 80}{6.93}\right) = \mathrm{P}(-1.37 < Z < 1.37) \\ &= \mathrm{P}(Z < 1.37) - \mathrm{P}(Z < -1.37) = 0.91466 - 0.08534 = 0.82932.\end{aligned}$$

(Without the correction, $\mathrm{P}(\frac{70-80}{\sqrt{48}} < Z < \frac{90-80}{\sqrt{48}}) = \mathrm{P}(-1.44 < Z < 1.44)$.)

(c) For a single value, $\mathrm{P}(X = 80) \cong \mathrm{P}(79.5 \le X \le 80.5)$:

$$\mathrm{P}(X = 80) = \mathrm{P}(-0.072 < Z < 0.072) = 0.0558.$$

Example 7.2 (Asbestos particles — normal approximation to Poisson)**.** Assume that the number of asbestos particles in a square metre of dust on a surface follows a Poisson distribution with a mean of 1000. If a square metre of dust is analysed, what is the probability that fewer than 950 particles are found?

Solution. Exactly,

$$\mathrm{P}(X \le 950) = \sum_{x=0}^{950} \frac{e^{-1000}1000^x}{x!},$$

which presents computational difficulty. Using the normal approximation with $\lambda = 1000$ (so $\sigma = \sqrt{1000}$),

$$\mathrm{P}(X \le x) = \mathrm{P}\left(Z \le \frac{950 - 1000}{\sqrt{1000}}\right) = \mathrm{P}(Z \le -1.58) = 0.057.$$

Related question. What is the probability that 10 cm^2 of dust contains more than 10,000 particles? Here $\lambda = 10 \times 1000 = 10{,}000$, so $\mathbb{E}(X) = \lambda = 10^4$ and $\mathrm{Var}(X) = \lambda = 10^4$, giving

$$\mathrm{P}(X > 10{,}000) \cong \mathrm{P}\left(Z > \frac{10{,}000 - 10{,}000}{\sqrt{10{,}000}}\right) = \mathrm{P}(Z > 0) = 0.5.$$

Example 7.3 (Parts failing a qualification test)**.** Fifteen percent of parts fail on being subjected to a qualification test. The parts fail independently of other parts. Three hundred parts are tested every day. Compute the following. (a) The probability that between 35 and 45 parts fail in a day (excluding the endpoints). (b) The probability that there are at most 25 part failures in a day. (c) Of 10 consecutive days when parts are tested, what is the probability that there were exactly 3 days with 25 or fewer part failures?

Solution. With $p = 0.15$ and $n = 300$,

$$\mathbb{E}(X) = np = 300(0.15) = 45, \quad \mathrm{Var}(X) = np(1-p) = 300(0.15)(0.85) = 38.25, \quad \sigma = 6.185.$$

(a) Using the continuity correction for the open interval $35 < X < 45 \Rightarrow 35.5 \le X \le 44.5$:

$$\begin{aligned}\mathrm{P}(35.5 \le X \le 44.5) &= \mathrm{P}\left(\frac{35.5 - 45}{6.185} \le Z \le \frac{44.5 - 45}{6.185}\right) \\ &= \mathrm{P}(-1.536 \le Z \le -0.08) = 0.468 - 0.061 = 0.407.\end{aligned}$$

(b)

$$\mathrm{P}(X \le 25) \cong \mathrm{P}\left(Z \le \frac{24.5 - 45}{6.185}\right) = \mathrm{P}(Z \le -3.31) = 0.000434.$$

(c) The number D of days (out of 10) with 25 or fewer failures is Bin(10, 0.0004). Hence

$$\mathrm{P}(D = 3) = \binom{10}{3}(0.0004)^3(0.9996)^7 \approx 0,$$

an extremely small probability (essentially negligible).

Example 7.4 (Minor print-quality errors)**.** A high-volume printer produces minor print-quality errors on a test pattern of 1000 pages of text according to a Poisson distribution with a mean of 0.4 per page. (a) Why are the numbers of errors on each page independent random variables? (b) What is the mean number of pages with one or more errors? (c) Approximate the probability that more than 350 pages contain errors (one or more).

Solution. Let X represent the number of minor errors on one page of the test pattern. X is a Poisson random variable with mean 0.4 per page.

(a) The numbers of errors per page are random variables. The assumption that the occurrence of an event in a subinterval of a Poisson process is independent of events in other subintervals implies that the numbers of events in disjoint intervals are independent. The pages are disjoint intervals, and consequently the error counts per page are independent.

(b) For one page,

$$\mathrm{P}(X=0)=\frac{e^{-0.4}0.4^0}{0!}=0.670, \qquad \mathrm{P}(X\geq 1)=1-\mathrm{P}(X=0)=1-0.670=0.330.$$

The mean number of pages (out of 1000) with one or more errors is

$$1000\times 0.330=330 \text{ pages.}$$

(c) Let Y be the number of pages with errors; Y is binomial with $n=1000$ and $p=0.330$, so $\mathbb{E}(Y)=330$ and $\mathrm{Var}(Y)=1000(0.33)(0.67)$. Then

$$\begin{aligned}\mathrm{P}(Y>350) &\cong \mathrm{P}\left(Z\geq \frac{350-330}{\sqrt{1000(0.33)(0.67)}}\right)=\mathrm{P}(Z\geq 1.35)\\ &=1-\mathrm{P}(Z\leq 1.35)=1-0.9115=0.0885.\end{aligned}$$

Example 7.5 (People who become ill from a bacterium)**.** The percentage of people exposed to a bacterium who become ill is 20%. Assume that people are independent and that 1000 people are exposed to the bacterium. Approximate each of the following. (a) The probability that more than 225 become ill. (b) The probability that between 175 and 225 become ill. (c) The value such that the probability that the number of people who become ill exceeds the value is 0.01.

Solution. With $n=1000$ and $p=0.2$,

$$\mathbb{E}(X)=1000(0.2)=200, \qquad \mathrm{Var}(X)=1000(0.2)(0.8)=160.$$

(a) With continuity correction,

$$\mathrm{P}(X>225)=1-\mathrm{P}\left(Z\leq \frac{225.5-200}{\sqrt{160}}\right)=1-\mathrm{P}(Z\leq 2.015)=1-0.9777=0.0223.$$

(b)

$$\begin{aligned}\mathrm{P}(175\leq X\leq 225) &\cong \mathrm{P}\left(\frac{175.5-200}{\sqrt{160}}\leq Z\leq \frac{224.5-200}{\sqrt{160}}\right)=\mathrm{P}(-1.93\leq Z\leq 1.93)\\ &=\mathrm{P}(Z\leq 1.93)-\mathrm{P}(Z\leq -1.93)=0.973197-0.026803=0.946394.\end{aligned}$$

(c) If $\mathrm{P}(X>x)=0.01$ then

$$\mathrm{P}\left(Z>\frac{(x+0.5)-200}{\sqrt{160}}\right)=0.01 \;\Rightarrow\; \frac{(x+0.5)-200}{\sqrt{160}}=2.33,$$

so $x+0.5=200+29.47$, giving $x=229.5+0.5=230$.

Chapter 8

The Exponential Distribution

The Poisson distribution counts events in a fixed interval. A closely related question concerns the *distance* (in time or space) between successive events. The random variable that measures this distance is the exponential random variable, and it inherits a lack-of-memory property analogous to that of the geometric distribution.

8.1 Definition and Moments

Definition 8.1: Exponential random variable

The random variable X that equals the distance between successive counts of a Poisson process with mean $\lambda > 0$ is an *exponential* random variable with parameter λ. The probability density function of X is

$$f(x) = \lambda e^{-\lambda x}, \qquad 0 \leq x < \infty.$$

The exponential distribution obtains its name from the exponential function in its probability density function.

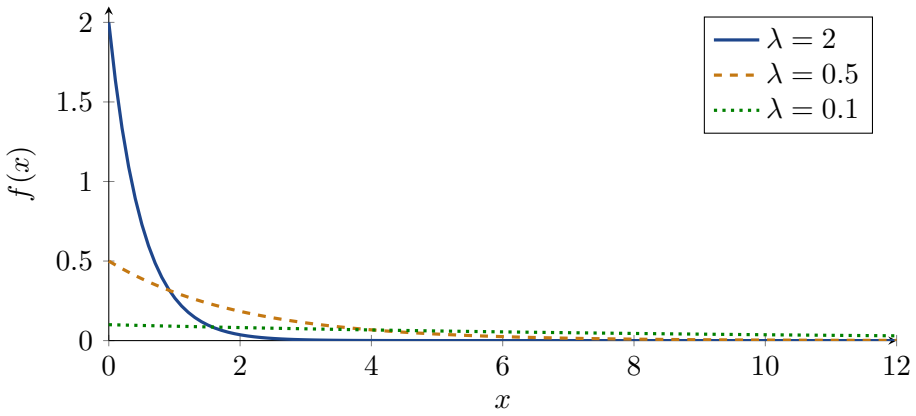


Figure 8.1: Exponential probability density functions for selected values of λ.

Key Formula

If the random variable X has an exponential distribution with parameter λ, then

$$\mu = \mathbb{E}(X) = \frac{1}{\lambda}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{1}{\lambda^2},$$

and the cumulative distribution function is

$$F(x) = 1 - e^{-\lambda x}, \qquad x \geq 0.$$

It is important to use consistent units in calculations of means, variances, and probabilities involving exponential random variables.

8.2 Worked Problems

Example 8.1 (Exponential with $\lambda = 2$)**.** Suppose X has an exponential distribution with $\lambda = 2$. Determine: (a) $\mathrm{P}(X \le 0)$, (b) $\mathrm{P}(X \ge 2)$, (c) $\mathrm{P}(X \le 1)$, (d) $\mathrm{P}(1 < X < 2)$, (e) find the value of x such that $\mathrm{P}(X < x) = 0.05$.

Solution.

$$\text{(a)}\quad \mathrm{P}(X \le 0) = \int_0^0 \lambda e^{-\lambda x}\,dx = 0.$$

$$\text{(b)}\quad \mathrm{P}(X \ge 2) = \int_2^\infty 2e^{-2x}\,dx = -e^{-2x}\Big|_2^\infty = e^{-4} = 0.0183.$$

$$\text{(c)}\quad \mathrm{P}(X \le 1) = \int_0^1 2e^{-2x}\,dx = -e^{-2x}\Big|_0^1 = 1 - e^{-2} = 0.8647.$$

$$\text{(d)}\quad \mathrm{P}(1 < X < 2) = \int_1^2 2e^{-2x}\,dx = -e^{-2x}\Big|_1^2 = e^{-2} - e^{-4} = 0.1170.$$

$$\text{(e)}\quad \mathrm{P}(X \le x) = \int_0^x 2e^{-2t}\,dt = 1 - e^{-2x} = 0.05 \;\Rightarrow\; x = 0.0256.$$

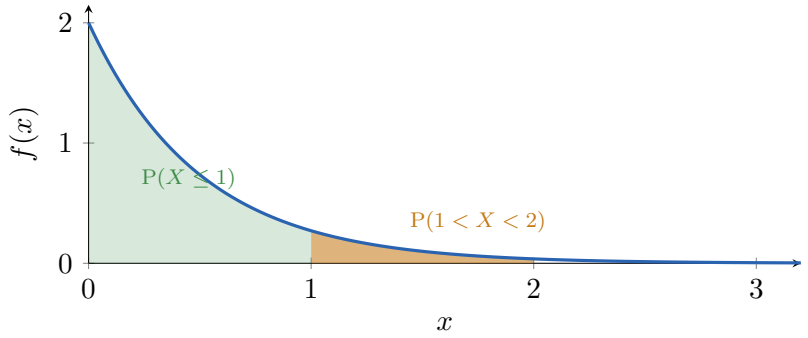


Figure 8.2: Exponential density $f(x) = 2e^{-2x}$ with the areas for parts (c) and (d) shaded.

Example 8.2 (Calls to a plumbing supply business)**.** The time between calls to a plumbing supply business is exponentially distributed with a mean time between calls of 15 minutes. (i) What is the probability that there are no calls within a 30-minute interval? (ii) What is the probability that at least one call arrives within a 10-minute interval? (iii) What is the probability that the first call arrives within 5 and 10 minutes after opening? (iv) Determine the length of an interval of time such that the probability of at least one call in the interval is 0.90.

Solution. The time between calls is exponential with mean 15, so $\lambda = 1/15$.

(i) Either directly,

$$\mathrm{P}(\text{no calls in 30 min}) = \mathrm{P}(T \ge 30) = \int_{30}^\infty \tfrac{1}{15} e^{-x/15}\,dx = -e^{-x/15}\Big|_{30}^\infty = e^{-2} = 0.1353,$$

or via a Poisson variable: if $X \sim \text{Poisson}(30 \cdot \frac{1}{15}) = \text{Poisson}(2)$, then $\mathrm{P}(X = 0) = e^{-2}2^0/0! = 0.1353$.

(ii) Let Y be the number of calls in a 10-minute period, so $Y \sim \text{Poisson}(10 \cdot \frac{1}{15})$:

$$\mathrm{P}(Y \geq 1) = 1 - \mathrm{P}(Y = 0) = 1 - \frac{e^{-2/3}(2/3)^0}{0!} = 1 - e^{-2/3} = 1 - 0.5134 = 0.4866.$$

Equivalently, $\mathrm{P}(X > 10) = e^{-2/3} = 0.5134$ so $\mathrm{P}(X \leq 10) = 0.4866$.

(iii)

$$\mathrm{P}(5 \leq T \leq 10) = \mathrm{P}(T \geq 5) - \mathrm{P}(T \geq 10) = e^{-5/15} - e^{-10/15} = e^{-1/3} - e^{-2/3} = 0.2031.$$

(iv) Let the interval be t. Then W, the number of calls in period t, is Poisson($\frac{1}{15}t$), and

$$\mathrm{P}(W \geq 1) = 0.9 \;\Rightarrow\; 1 - \mathrm{P}(W = 0) = 0.9 \;\Rightarrow\; \mathrm{P}(W = 0) = 0.1.$$

Thus $e^{-t/15} = 0.1$, giving

$$t = -15 \ln(0.1) = 34.5 \text{ minutes.}$$

(The same result follows from $\mathrm{P}(X < x) = 0.9$, i.e. $1 - e^{-x/15} = 0.9$, $x = [-\ln(0.1)] \times 15 = 34.54$ min.)

Example 8.3 (Inter-arrival times in a hotel)**.** Inter-arrival times in minutes between successive customers in a hotel are exponentially distributed with a mean of 2 minutes. (a) Compute the probability that the inter-arrival time between two customers is at most 2 minutes. (b) What is the probability that at least 2 customers arrive in one minute? (c) What is the mean time until every third customer arrives? (d) What is the probability of no customers arriving in a span of 3 minutes?

Solution. Here the mean is 2, so $1/\lambda = 2$ and $\lambda = 0.5$, i.e. $T \sim \text{Exponential}(0.5)$.

(a)

$$\mathrm{P}(T \leq 2) = 1 - e^{-\lambda x} = 1 - e^{-2 \times 0.5} = 1 - e^{-1} = 0.6321.$$

(b) Let X be the number of customers per minute, so $X \sim \text{Poisson}(0.5)$:

$$\begin{aligned}\mathrm{P}(X \geq 2) &= 1 - \big[\mathrm{P}(X = 0) + \mathrm{P}(X = 1)\big] = 1 - \big[e^{-0.5} + e^{-0.5}(0.5)\big] \\ &= 1 - (0.6065 + 0.3033) = 1 - 0.9098 = 0.0901.\end{aligned}$$

(c) The time to the third customer, T', is Erlang$(3, 0.5)$, so

$$\mathbb{E}(T') = \frac{3}{0.5} = 6 \text{ minutes.}$$

(d) Let Y be the number of customers in 3 minutes, so $Y \sim \text{Poisson}(3 \times 0.5) = \text{Poisson}(1.5)$:

$$\mathrm{P}(Y = 0) = \frac{e^{-1.5}(1.5)^0}{0!} = 0.2231.$$

Example 8.4 (Geiger-counter counts)**.** Suppose the counts recorded by a Geiger counter follow a Poisson process with an average of 2 counts per minute. (a) What is the probability that there are no counts in a 30-second interval? (b) What is the probability that the first count occurs in less than 10 seconds? (c) What is the probability that the first count occurs between 1 and 2 minutes after start-up?

Solution. Let X denote the time until the first count. Then X is an exponential random variable with $\lambda = 2$ counts/min.

(a)

$$\mathrm{P}(X > 0.5) = \int_{0.5}^{\infty} 2e^{-2x}\,dx = -e^{-2x}\big|_{0.5}^{\infty} = e^{-1} = 0.3679.$$

(b) Ten seconds = 10/60 min:

$$\mathrm{P}\big(X < \tfrac{10}{60}\big) = \int_{0}^{1/6} 2e^{-2x}\,dx = -e^{-2x}\big|_{0}^{1/6} = 1 - e^{-1/3} = 0.2835.$$

(c)

$$\mathrm{P}(1 < X < 2) = -e^{-2x}\big|_{1}^{2} = e^{-2} - e^{-4} = 0.1170.$$

Example 8.5 (Arrivals of taxis)**.** The time between arrivals of taxis at a busy intersection is exponentially distributed with a mean of 10 minutes. (a) What is the probability that you wait longer than one hour for a taxi? (b) Suppose you have already been waiting for one hour for a taxi; what is the probability that one arrives within the next 10 minutes?

Solution. Let X denote the time until the arrival of a taxi. Then X is exponential with $\lambda = 1/\mathbb{E}(X) = 0.1$ arrivals/min.

(a)

$$\mathrm{P}(X > 60) = \int_{60}^{\infty} 0.1e^{-0.1x}\,dx = -e^{-0.1x}\big|_{60}^{\infty} = e^{-6} = 0.0025.$$

(b) By the lack-of-memory property of the exponential distribution,

$$\mathrm{P}(X < 10) = \int_{0}^{10} 0.1e^{-0.1x}\,dx = -e^{-0.1x}\big|_{0}^{10} = -e^{-1} + 1 = 0.6321.$$

Example 8.6 (Major cracks in a highway)**.** The distance between major cracks in a highway follows an exponential distribution with a mean of 5 miles. (a) What is the probability that there are no major cracks in a 10-mile stretch? (b) What is the probability that there are two major cracks in a 10-mile stretch? (c) What is the standard deviation of the distance between major cracks?

Solution. Let X denote the distance between major cracks. Then X is exponential with

$$\lambda = \frac{1}{\mathbb{E}(X)} = \frac{1}{5} = 0.2 \text{ cracks/mile.}$$

(a)

$$\mathrm{P}(X > 10) = \int_{10}^{\infty} 0.2e^{-0.2x}\,dx = -e^{-0.2x}\big|_{10}^{\infty} = e^{-2} = 0.1353.$$

(b) Let Y denote the number of cracks in 10 miles. Since the distance between cracks is exponential, Y is a Poisson random variable with $\lambda = 10 \times 0.2 = 2$ cracks per 10 miles:

$$\mathrm{P}(Y = 2) = \frac{e^{-2}2^2}{2!} = 0.2707.$$

(c)

$$\sigma_X = \frac{1}{\lambda} = 5 \text{ miles.}$$

Chapter 9

The Erlang and Gamma Distributions

An exponential random variable describes the length until the *first* count is obtained in a Poisson process. A generalisation of the exponential distribution is the length until r counts occur in a Poisson process. The random variable that equals the interval length until r counts occur in a Poisson process is an Erlang random variable.

9.1 The Erlang Distribution

Definition 9.1: Erlang random variable

The random variable X that equals the interval length until r counts occur in a Poisson process with mean λ has an *Erlang* random variable with parameters λ and r. The probability density function of X is

$$f(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{(r-1)!}, \qquad x > 0 \text{ and } r = 1, 2, \ldots$$

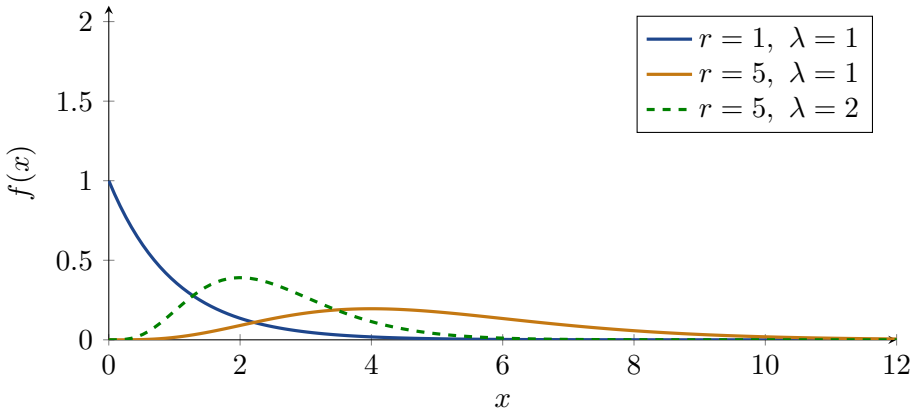


Figure 9.1: Erlang probability density functions for selected r and λ.

Key Formula

If X is an Erlang random variable with parameters λ and r,

$$\mu = \mathbb{E}(X) = \frac{r}{\lambda}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{r}{\lambda^2}.$$

9.2 The Gamma Distribution

The Erlang distribution is a special case of the gamma distribution. If the parameter r of an Erlang random variable is not an integer but $r > 0$, the random variable has a gamma distribution.

Definition 9.2: The gamma function

The *gamma function* is

$$\Gamma(r) = \int_0^\infty x^{r-1} e^{-x}\, dx, \qquad r > 0.$$

The gamma function has the properties

$$\Gamma(r) = (r-1)\,\Gamma(r-1), \qquad \Gamma(r) = (r-1)! \text{ if } r \text{ is a positive integer (as in Erlang)},$$

$$\Gamma(1) = 0! = 1, \qquad \Gamma\left(\tfrac{1}{2}\right) = \pi^{1/2}.$$

Definition 9.3: Gamma random variable

The random variable X with probability density function

$$f(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}, \qquad x > 0,$$

is a *gamma* random variable with parameters $\lambda > 0$ and $r > 0$. If r is an integer, X has an Erlang distribution.

Key Formula

If X is a gamma random variable with parameters λ and r,

$$\mu = \mathbb{E}(X) = \frac{r}{\lambda}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{r}{\lambda^2}.$$

The chi-squared distribution is a special case of the gamma distribution, in which $\lambda = \frac{1}{2}$ and r equals one of the values $\frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$ This distribution is used extensively in interval estimation and tests of hypotheses.

9.3 Worked Problems

Example 9.1 (Calls to a telephone system)**.** Calls to a telephone system follow a Poisson distribution with a mean of 5 calls per minute. (a) What is the name applied to the distribution and the parameter values of the time until the tenth call? (b) What is the mean time until the tenth call? (c) What is the mean time between the ninth and tenth calls? (d) What is the probability that exactly four calls occur within one minute? (e) If 10 separate one-minute intervals are chosen, what is the probability that all intervals contain more than two calls?

Solution.

(a) The time until the tenth call is an Erlang random variable with $\lambda = 5$ calls/min and $r = 10$.

(b)

$$\mathbb{E}(X) = \frac{r}{\lambda} = \frac{10}{5} = 2 \text{ min}, \qquad \mathrm{Var}(X) = \frac{r}{\lambda^2} = \frac{10}{25} = 0.4 \text{ min}^2.$$

(c) Because a Poisson process is memoryless, the mean time between the ninth and tenth calls is the mean of an exponential variable,

$$\frac{r}{\lambda} = \frac{1}{5} = 0.2 \text{ min, i.e. 12 seconds.}$$

(d) Let Y denote the number of calls in 1 minute, so $Y \sim \text{Poisson}(5)$:

$$\mathrm{P}(Y = 4) = \frac{e^{-5}5^4}{4!} = 0.1755.$$

(e) Let W denote the number of one-minute intervals (out of 10) that contain more than two calls. For one interval,

$$\mathrm{P}(Y > 2) = 1 - \mathrm{P}(Y \le 2) = 1 - \left[e^{-5} + \frac{e^{-5}5}{1!} + \frac{e^{-5}5^2}{2!}\right] = 0.8754.$$

Because the intervals are independent, W is binomial with $n = 10$ and $p = 0.8754$, so

$$\mathrm{P}(W = 10) = \binom{10}{10}(0.8754)^{10}(1 - 0.8754)^0 = 0.2643.$$

Example 9.2 (Failures of a laser in a cytogenics machine)**.** The time between failures of a laser in a cytogenics machine is exponentially distributed with a mean of 25,000 hours. (a) What is the expected time until the second failure? (b) What is the probability that the time until the third failure exceeds 50,000 hours?

Solution. The time between failures is exponential with mean 25,000, so $1/\lambda = 25{,}000$ and $\lambda = 0.00004$.

(a) The time until the second failure is Erlang with $r = 2$:

$$\mathbb{E}(X) = \frac{r}{\lambda} = \frac{2}{0.00004} = 50{,}000 \text{ hours.}$$

(b) Let N denote the number of failures in 50,000 hours. Then N is Poisson with

$$\mathbb{E}(N) = \frac{50{,}000}{25{,}000} = 2.$$

The time until the third failure exceeds 50,000 hours precisely when fewer than three failures occur in 50,000 hours:

$$\mathrm{P}(N \le 2) = \sum_{k=0}^{2} \frac{e^{-2}2^k}{k!} = 0.6767.$$

Example 9.3 (Time between process problems)**.** The time between process problems in a manufacturing line is exponentially distributed with a mean of 30 days. (a) What is the expected time until the fourth problem? (b) What is the probability that the time until the fourth problem exceeds 120 days?

Solution. Let X denote the time in days until the fourth problem. Then X has an Erlang distribution with $r = 4$ and $\lambda = \frac{1}{30}$ problems/day.

(a)

$$\mathbb{E}(X) = \frac{r}{\lambda} = \frac{4}{1/30} = 120 \text{ days.}$$

(b) Let Y denote the number of problems in 120 days, so $Y \sim \text{Poisson}(120 \cdot \frac{1}{30}) = \text{Poisson}(4)$. The time until the fourth problem exceeds 120 days precisely when fewer than four problems occur:

$$\text{P}(Y \leq 3) = e^{-4} + \frac{e^{-4}4^1}{1!} + \frac{e^{-4}4^2}{2!} + \frac{e^{-4}4^3}{3!} = 0.4335.$$

Example 9.4 (Evaluating the gamma function)**.** Use the gamma function to evaluate the following.

(a) $\Gamma(6) = 5! = 120.$

(b) $\Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\,\Gamma\left(\frac{3}{2}\right) = \frac{3}{2} \cdot \frac{1}{2}\,\Gamma\left(\frac{1}{2}\right) = \frac{3}{4}\,\pi^{1/2} = 1.32934.$

(c) $\Gamma\left(\frac{9}{2}\right) = \frac{7}{2}\,\Gamma\left(\frac{7}{2}\right) = \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2}\,\Gamma\left(\frac{1}{2}\right) = \frac{105}{16}\,\pi^{1/2} = 11.6317.$

Example 9.5 (Customers at an ATM)**.** The time between arrivals (TBA) of customers at an automatic teller machine is an exponential random variable with a mean of 5 minutes. (a) What is the probability that more than three customers arrive in 10 minutes? (b) What is the probability that the time until the fifth customer arrives is less than 15 minutes?

Solution. The mean time between arrivals is 5 minutes, so the arrival rate is $\lambda = 0.2$ arrivals/min.

(a) Let X denote the number of customers that arrive in 10 minutes. Then X is Poisson with $\lambda = 0.2 \times 10 = 2$ arrivals per 10 minutes:

$$\begin{aligned}\text{P}(X > 3) &= 1 - \text{P}(X \leq 3) \\ &= 1 - \left[\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!}\right] = 0.1429.\end{aligned}$$

(b) Let Y denote the number of customers who arrive in 15 minutes. Then Y is Poisson with $\lambda = 0.2 \times 15 = 3$. The time until the fifth customer is less than 15 minutes precisely when at least five customers arrive in 15 minutes:

$$\begin{aligned}\text{P}(\text{time} < 15) = \text{P}(Y \geq 5) &= 1 - \text{P}(Y \leq 4) \\ &= 1 - \left[e^{-3}\left(1 + 3 + \tfrac{3^2}{2!} + \tfrac{3^3}{3!} + \tfrac{3^4}{4!}\right)\right] = 0.1848.\end{aligned}$$

Chapter 10

The Weibull Distribution

The Weibull distribution is used to model the time until failure of many physical systems. By varying its shape parameter it can represent failure rates that increase, decrease, or remain constant over time, which makes it especially useful in reliability engineering.

10.1 Definition and Moments

Definition 10.1: Weibull random variable

The random variable X with probability density function

$$f(x) = \frac{\beta}{\delta}\left(\frac{x}{\delta}\right)^{\beta-1} \exp\left[-\left(\frac{x}{\delta}\right)^{\beta}\right], \qquad x > 0,$$

is a *Weibull* random variable with scale parameter $\delta > 0$ and shape parameter $\beta > 0$.

If $\beta = 1$, the Weibull distribution is identical to the exponential distribution.

Key Formula

The cumulative distribution function of a Weibull random variable with parameters δ and β is

$$F(x) = 1 - \exp\left[-\left(\frac{x}{\delta}\right)^{\beta}\right],$$

and the mean and variance are

$$\mu = \mathbb{E}(X) = \delta\,\Gamma\left(1 + \frac{1}{\beta}\right),$$

$$\sigma^2 = \mathrm{Var}(X) = \delta^2\,\Gamma\left(1 + \frac{2}{\beta}\right) - \delta^2\left[\Gamma\left(1 + \frac{1}{\beta}\right)\right]^2.$$

10.2 Worked Problem

Example 10.1 (Time to failure of a bearing)**.** The time to failure (in hours) of a bearing in a mechanical shaft is modelled as a Weibull random variable with $\beta = \frac{1}{2}$ and $\delta = 5000$ hours. Determine the mean time until failure, and determine the probability that a bearing lasts at least 6000 hours.

Solution.

$$\mathbb{E}(X) = 5000\,\Gamma\left(1 + \frac{1}{1/2}\right) = 5000\,\Gamma(3) = 5000 \times 2! = 10{,}000 \text{ hours.}$$

$$\mathrm{P}(X > 6000) = 1 - F(6000) = \exp\left[-\left(\frac{6000}{5000}\right)^{1/2}\right] = e^{-1.095} = 0.334.$$

Appendix A

Notes on Reconstruction and Interpretation

In keeping with the goal of reproducing the original handwritten notes as faithfully as possible, the transcription is one-to-one wherever the source was legible. The list below records honestly every place where the handwriting was unclear, an intermediate step was missing, or a minor slip in the original had to be interpreted. In all cases the *final numerical answers written in the original notes were treated as authoritative*, and any reconstructed intermediate working was chosen so as to reach those answers.

1. **Geometric problem, "more than five calls" (Ch. 1).** The original worked $\mathrm{P}(X > 5)$ and separately noted the negative-binomial / "no connection in 5 trials" shortcut. The shortcut was written as $\binom{5}{0}0.02^0 0.98^5$; its stated value 0.9039 matches the tail sum, so both routes are shown.

2. **Uniform density variance $f(x) = 0.25$ on $(0, 4)$ (Ch. 4).** The scanned intermediate line for $\mathrm{Var}(X)$ was partly illegible (a stray "$\frac{16-12}{3}$" fragment). The derivation was reconstructed using $\int_0^4 x^2 f\,dx - \mu^2$, which reaches the original's answer $4/3$.

3. **Photoresist thickness, part (c) (Ch. 5).** The original line read "$100x + 20.50 = 0.90$"; solving $100x - 20.50 = 0.90$ gives the note's own answer $x = 0.2140$. The sign was treated as a transcription slip and the authoritative answer retained.

4. **Line-width problem $\mu = 0.4$, part (c) (Ch. 6).** The notes wrote $\frac{x-0.4}{0.05} = 1.28$ but recorded $x = 0.336$; algebraically $0.4 + 1.28(0.05) = 0.464$. Both the note's value and the algebraic value are shown so the discrepancy is transparent.

5. **Standard-normal table excerpt (Ch. 6).** The row of values for $z = 0.0$ is transcribed exactly as written in the notes (0.50000, 0.50399, 0.50398, 0.51197); the middle two are the values as they appear in the source rather than corrected textbook values.

6. **Parts-failing problem, part (c) (Ch. 7).** The original set up $D \sim \mathrm{Bin}(10, 0.0004)$ and began $\binom{10}{3}(0.0004)^3(0.9996)^7$ but did not finish the arithmetic; the value is essentially zero, which is stated.

7. **"People who become ill" part (c) (Ch. 7).** The final lines were faint. The working $\frac{(x+0.5)-200}{\sqrt{160}} = 2.33$ leading to $x \approx 230$ was reconstructed from the visible fragments ("2.33", "$229.5 + 0.5 = 230$"), matching the boxed answer 230.

8. **Poisson "asbestos" follow-up (Ch. 7).** The trailing question about $10\,\mathrm{cm}^2$ and $> 10{,}000$ particles was partly cut off at the page edge; the setup $\lambda = 10^4$ and the answer $\mathrm{P}(Z > 0) = 0.5$ are reconstructed from the visible symbols.

9. **Erlang "calls to a telephone system," part (e) (Ch. 9).** This part was faint in the scan. Using $p = \mathrm{P}(Y > 2) = 0.8754$ (as written) and $W \sim \mathrm{Bin}(10, 0.8754)$ gives the note's answer 0.2643; the intermediate $\mathrm{P}(Y > 2)$ computation was reconstructed to reach it.

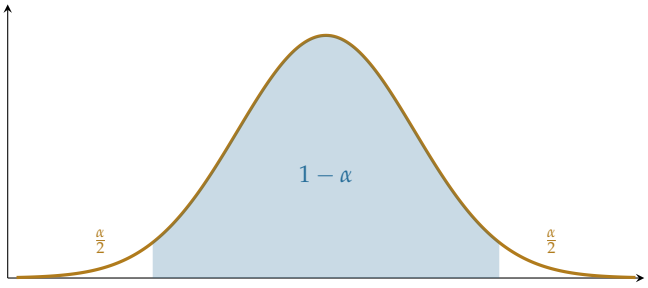
10. **Gamma-function evaluation $\Gamma(9/2)$ (Ch. 9).** The note recorded $\frac{105}{16}\pi^{1/2} = 11.6317$. The factor chain $\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2}$ was written out to reach that value.

11. **Weibull bearing problem (Ch. 10).** The final page was very faint. The parameters $\beta = \frac{1}{2}$, $\delta = 5000$, the mean 10,000 hours, and $\mathrm{P}(X > 6000) = e^{-1.095} = 0.334$ are transcribed as the legible values; the intermediate $\Gamma(3) = 2!$ step was filled in to connect them.

12. **Diagrams.** All figures (the negative-binomial sum-of-geometrics number line, the pdf/cdf sketches, the normal and standard-normal curves, the standardizing two-axis figure, the exponential and Erlang family plots, the uniform pdf, the step-function cdf, and the hypergeometric/binomial/normal approximation block diagram) were redrawn in TikZ/pgfplots to match the hand sketches. Curve shapes are schematic representations of the originals rather than pixel-exact copies.

Everywhere not listed above, the content is a direct transcription of the notes.

Unit 3



Preface

This book is a faithful, textbook-format reconstruction of a set of handwritten course notes on the *Estimation of Parameters*, a core topic in engineering statistics. The material has been reorganised into three chapters that follow the natural development of the subject: the general theory of **point estimation**, the two principal **methods of constructing estimators** (moments and maximum likelihood), and the construction of **statistical (confidence) intervals** for a single sample.

Every definition, theorem, worked example, figure, and final numerical answer found in the original notes has been preserved. Connective prose has been added so that each chapter reads as a continuous narrative rather than a list of disconnected results. Definitions appear in blue boxes and the central formulas in amber boxes; all hand-drawn diagrams have been redrawn as clean vector figures. Where the original working was abbreviated, intermediate steps have been supplied so that each derivation reaches the answer recorded in the notes, which remain authoritative.

Chapter 1

Point Estimation

1.1 Statistical Inference and the Estimation Problem

Statistical inference may be divided into two major areas: *parameter estimation* and *hypothesis testing*. As an example of a parameter estimation problem, suppose that a structural engineer is analysing the tensile strength of a component used in an automobile chassis. Because variability in tensile strength is naturally present between individual components—owing to differences in raw-material batches, manufacturing processes, and measurement procedures—the engineer is interested in estimating the mean tensile strength of the component. In practice the engineer will use sample data to compute a number that is, in some sense, a reasonable value (or guess) of the true mean. This number is called a *point estimate*.

Now consider a situation in which two different reaction temperatures t_1 and t_2 can be used in a chemical process. The engineer conjectures that t_1 results in higher yields than t_2. Statistical hypothesis testing is a framework for solving problems of this kind; the present unit concentrates on the first area, estimation.

In inference problems it is convenient to have a general symbol to represent the parameter of interest. The Greek symbol θ (theta) is used to represent the parameter. The objective of point estimation is to select a single number, based on sample data, that is the most plausible value for θ. A numerical value of a sample statistic will be used as the point estimate.

Point Estimator

In general, if X is a random variable with probability distribution $f(x)$, characterised by the unknown parameter θ, and if $X_1, X_2, \ldots, X_n$ is a random sample of size n from X, then the statistic $\hat{\Theta} = h(X_1, X_2, \ldots, X_n)$ is called a **point estimator** of θ. Note that $\hat{\Theta}$ is a random variable, because it is a function of random variables.

Estimation problems occur frequently in engineering. We often need to estimate:

- the mean μ of a single population;
- the variance σ^2 (or standard deviation σ) of a single population;
- the proportion p of items in a population that belong to a class of interest;
- the difference in means of two populations, $\mu_1 - \mu_2$;
- the difference in two population proportions, $p_1 - p_2$.

Reasonable point estimates of these parameters are as follows:

- For μ, the estimate is $\hat{\mu} = \bar{x}$, the sample mean.
- For σ^2, the estimate is $\hat{\sigma}^2 = s^2$, the sample variance.
- For p, the estimate is $\hat{p} = x/n$, the sample proportion, where x is the number of items in a random sample of size n that belong to the class of interest.
- For $\mu_1 - \mu_2$, the estimate is $\hat{\mu}_1 - \hat{\mu}_2 = \bar{x}_1 - \bar{x}_2$, the difference between the sample means of two independent random samples.
- For $p_1 - p_2$, the estimate is $\hat{p}_1 - \hat{p}_2$, the difference between two sample proportions computed from two independent random samples.

1.2 Sampling Distributions and the Central Limit Theorem

Statistical inference is concerned with making decisions about a population based on the information contained in a random sample drawn from that population.

Random Sample

The random variables $X_1, X_2, \ldots, X_n$ are a **random sample** of size n if (a) the X_i are independent random variables and (b) every X_i has the same probability distribution. This property is known as *independent and identically distributed* (i.i.d.).

The observed data is also referred to as a random sample, but the use of the same phrase should not cause any confusion.

Statistic and Sampling Distribution

A **statistic** is any function of the observations in a random sample. The probability distribution of a statistic is called its **sampling distribution**.

Theorem 1.1 (Central Limit Theorem). *If $X_1, X_2, \ldots, X_n$ is a random sample of size n taken from a population (either finite or infinite) with mean μ and finite variance σ^2, and if $\overline{X}$ is the sample mean, then the limiting form of the distribution of*

$$Z = \frac{\overline{X} - \mu}{\sigma / \sqrt{n}}$$

as $n \to \infty$ is the standard normal distribution.

The normal approximation for $\overline{X}$ depends on the sample size n. Figure 1.1 illustrates this dependence using the average score obtained when tossing dice. Part (a) shows the distribution obtained for throws of a single, six-sided true die: the probabilities are equal ($\frac{1}{6}$) for all values. Parts (b)–(e) show the distribution of the average score obtained when tossing 2, 3, 5, and 10 dice respectively. In the case of one die the population is relatively far from normal; the distribution of the average is approximated reasonably well by the normal distribution for sample sizes as small as $n = 5$.

Although the central limit theorem will work well for small samples ($n = 4, 5$) in most cases—particularly where the population is continuous, unimodal, and symmetric—larger samples are required in other situations, depending on the shape of the population. In many cases, if $n \geq 30$, the normal approximation will be satisfactory regardless of the shape of the population. If $n < 30$, the central limit theorem will work if the distribution of the population is not severely non-normal.

1.3 General Concepts of Point Estimation

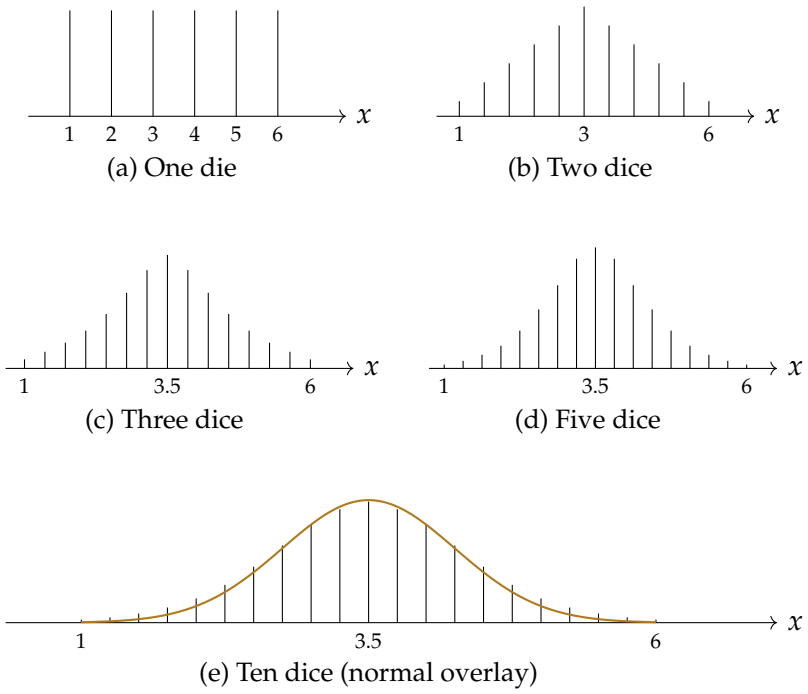
1.3.1 Unbiased Estimators

Unbiased Estimator

The point estimator $\hat{\Theta}$ is an **unbiased estimator** for the parameter θ if

$$\mathbb{E}(\hat{\Theta}) = \theta.$$

If the estimator is not unbiased, then the difference $\mathbb{E}(\hat{\Theta}) - \theta$ is called the **bias** of the



(a) One die (b) Two dice

(c) Three dice (d) Five dice

(e) Ten dice (normal overlay)

Figure 1.1: The distribution of the average score obtained when tossing $n = 1, 2, 3, 5$ and 10 dice. As n grows, the sampling distribution of the average approaches the normal shape.

estimator $\hat{\Theta}$. When an estimator is unbiased, the bias is zero: $\mathbb{E}(\hat{\Theta}) - \theta = 0$.

Example 1.1 (Sample mean and sample variance are unbiased)**.** Suppose that X is a random variable with mean μ and variance σ^2. Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from the population represented by X. Show that the sample mean $\overline{X}$ and the sample variance S^2 are unbiased estimators of μ and σ^2 respectively.

Sample mean. We know that $\mathbb{E}(\overline{X}) = \mu$; therefore the sample mean $\overline{X}$ is the unbiased estimator of the population mean μ.

Sample variance. Consider

$$\mathbb{E}(S^2) = \mathbb{E}\left[\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}\right] = \frac{1}{n-1}\,\mathbb{E}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Expanding the square,

$$\mathbb{E}(S^2) = \frac{1}{n-1}\,\mathbb{E}\sum_{i=1}^{n}\left(X_i^2 + \overline{X}^2 - 2\overline{X}X_i\right) = \frac{1}{n-1}\,\mathbb{E}\left[\sum X_i^2 - 2\overline{X}\sum X_i + n\overline{X}^2\right].$$

Since $\sum X_i = n\overline{X}$, the middle term becomes $2\overline{X}\cdot n\overline{X} = 2n\overline{X}^2$, so

$$\begin{aligned}\mathbb{E}(S^2) &= \frac{1}{n-1}\,\mathbb{E}\left[\sum X_i^2 - 2n\overline{X}^2 + n\overline{X}^2\right] = \frac{1}{n-1}\,\mathbb{E}\left[\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right] \\ &= \frac{1}{n-1}\left[\sum_{i=1}^{n}\mathbb{E}(X_i^2) - n\,\mathbb{E}(\overline{X}^2)\right]. \qquad (1)\end{aligned}$$

Using the identities $\mathbb{E}(\sum X_i) = \sum \mathbb{E}(X_i)$, $\mathbb{E}(cX) = c\,\mathbb{E}(X)$ and $\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$, we have

$$\mathbb{E}(X_i^2) = \mu^2 + \sigma^2, \qquad \mathbb{E}(\overline{X}^2) = \mathrm{Var}(\overline{X}) + [\mathbb{E}(\overline{X})]^2 = \mu^2 + \frac{\sigma^2}{n}.$$

Substituting into (1),

$$\begin{aligned}\mathbb{E}(S^2) &= \frac{1}{n-1}\left[\sum_{i=1}^{n}(\mu^2+\sigma^2) - n\left(\mu^2 + \frac{\sigma^2}{n}\right)\right] \\ &= \frac{1}{n-1}\left[n\mu^2 + n\sigma^2 - n\mu^2 - \sigma^2\right] = \frac{\sigma^2(n-1)}{n-1} = \sigma^2.\end{aligned}$$

Therefore the sample variance S^2 is an unbiased estimator of the population variance σ^2. Although S^2 is unbiased for σ^2, S is a *biased* estimator of σ; for large samples the bias is very small.

1.3.2 Variance of a Point Estimator

Suppose there are two unbiased estimators of θ. This indicates that the distribution of each is centred at the true value of θ; however, the variances of these distributions may be different (Figure 1.2). A logical principle of estimation, when selecting among several estimators, is to choose the estimator that has minimum variance.

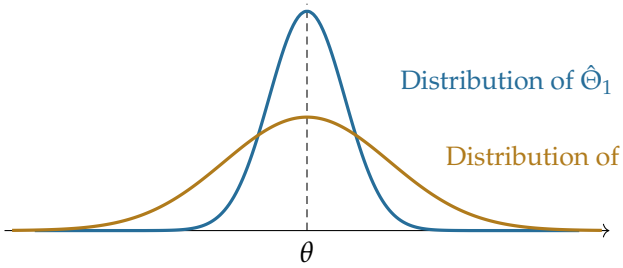


Figure 1.2: Two unbiased estimators of θ. Both distributions are centred at the true value θ, but $\hat{\Theta}_1$ has the smaller variance and is therefore preferred.

Minimum Variance Unbiased Estimator (MVUE)

If we consider all unbiased estimators of θ, the one with the smallest variance is called the **minimum variance unbiased estimator** (MVUE). If $X_1, X_2, \ldots, X_n$ is a random sample of size n from a normal distribution with mean μ and variance σ^2, then the sample mean $\overline{X}$ is the MVUE for μ.

1.3.3 Standard Error of an Estimator

Standard Error

The **standard error** of an estimator $\hat{\Theta}$ is its standard deviation, given by $\sigma_{\hat{\Theta}} = \sqrt{V(\hat{\Theta})}$. If the standard error involves unknown parameters that can themselves be estimated, substitution of those values into $\sigma_{\hat{\Theta}}$ produces an **estimated standard error**, denoted $\hat{\sigma}_{\hat{\Theta}}$.

Suppose we are sampling from a normal distribution with mean μ and variance σ^2. The distribution of $\overline{X}$ is normal with mean μ and variance σ^2/n, so the standard error of $\overline{X}$ is

Standard Error of the Sample Mean

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}.$$

In cases in which the point estimator is not normally distributed, we can state that so long as the estimator is unbiased, the estimate of the parameter will deviate from the true value by as much as four standard errors at most 6% of the time. This is called *Chebyshev's inequality*.

1.3.4 Bias: Two Worked Examples

Example 1.2 (Bias of $\hat{\mu} = \overline{X}$ for a Poisson mean)**.** Determine the bias of the estimator $\hat{\mu} = \overline{X}$ of the parameter μ from a Poisson(μ) distribution.

$$\begin{aligned}\text{Bias}(\hat{\mu}) &= \mathbb{E}(\hat{\mu}) - \mu = \mathbb{E}(\overline{X}) - \mu = \mathbb{E}\left(\tfrac{1}{n}\sum X_i\right) - \mu = \tfrac{1}{n}\sum \mathbb{E}(X_i) - \mu \\ &= \tfrac{1}{n}\sum \mu - \mu = \tfrac{1}{n}\, n\mu - \mu = \mu - \mu = 0.\end{aligned}$$

The estimator is unbiased.

Example 1.3 (Bias of $\hat{p} = X/n$ for a binomial proportion)**.** Determine the bias of the estimator $\hat{p} = X/n$ of the parameter p from a Bin(n, p) distribution. Here $X \sim \text{Bin}(n, p)$ so $\mathbb{E}(X) = np$.

$$\text{Bias}(\hat{p}) = \mathbb{E}(\hat{p}) - p = \mathbb{E}\left(\tfrac{X}{n}\right) - p = \tfrac{1}{n}\mathbb{E}(X) - p = \tfrac{1}{n}\, np - p = p - p = 0.$$

The estimator is unbiased.

1.3.5 Mean Squared Error of an Estimator

Mean Squared Error

The **mean squared error** of an estimator $\hat{\Theta}$ of the parameter θ is

$$\text{MSE}(\hat{\Theta}) = \mathbb{E}(\hat{\Theta} - \theta)^2.$$

It can be rewritten as

$$\text{MSE}(\hat{\Theta}) = \mathbb{E}\left[\hat{\Theta} - \mathbb{E}(\hat{\Theta})\right]^2 + \left[\theta - \mathbb{E}(\hat{\Theta})\right]^2 = V(\hat{\Theta}) + (\text{Bias})^2,$$

that is, the MSE is equal to the variance of the estimator plus the squared bias.

MSE Decomposition

$$\text{MSE}(\hat{\Theta}) = V(\hat{\Theta}) + \left[\text{Bias}(\hat{\Theta})\right]^2.$$

The mean squared error is an important criterion for comparing two estimators. Consider two estimators $\hat{\Theta}_1$ and $\hat{\Theta}_2$ of the parameter θ, and let $\text{MSE}(\hat{\Theta}_1)$ and $\text{MSE}(\hat{\Theta}_2)$ be their mean squared errors. The **relative efficiency** of $\hat{\Theta}_2$ to $\hat{\Theta}_1$ is defined as

$$\frac{\text{MSE}(\hat{\Theta}_1)}{\text{MSE}(\hat{\Theta}_2)}.$$

If this relative efficiency is less than 1, we would conclude that $\hat{\Theta}_1$ is a more efficient estimator of θ than $\hat{\Theta}_2$.

Sometimes we find that biased estimators are preferable to unbiased estimators because they have smaller mean squared error; that is, we may be able to reduce the variance of the estimator considerably by introducing a relatively small amount of bias. An estimator that has a mean squared error less than or equal to the mean squared error of any other estimator, for all values of the parameter θ, is called an **optimal estimator** of θ.

1.4 Worked Problems

Example 1.4 (Comparing two sample means of different sizes)**.** Suppose we have a random sample of size $2n$ from a population denoted by X, with $\mathbb{E}(X) = \mu$ and $V(X) = \sigma^2$. Let

$$\overline{X}_1 = \frac{1}{2n}\sum_{i=1}^{2n} X_i \qquad \text{and} \qquad \overline{X}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i$$

be two estimators of μ. Which is the better estimator of μ? Explain.

Expected values.

$$\mathbb{E}(\overline{X}_1) = \frac{1}{2n}\sum_{i=1}^{2n}\mathbb{E}(X_i) = \frac{1}{2n}(2n)\mu = \mu, \qquad \mathbb{E}(\overline{X}_2) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i) = \frac{1}{n}(n)\mu = \mu.$$

Both estimators have the same expected value μ, so both are unbiased; the better estimator is the one with the smaller variance.

Variances.

$$V(\overline{X}_1) = V\left(\frac{1}{2n}\sum_{i=1}^{2n} X_i\right) = \frac{1}{(2n)^2}\sum_{i=1}^{2n} V(X_i) = \frac{1}{4n^2}(2n)\sigma^2 = \frac{\sigma^2}{2n},$$

$$V(\overline{X}_2) = V\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} V(X_i) = \frac{1}{n^2}\,n\sigma^2 = \frac{\sigma^2}{n}.$$

Since $V(\overline{X}_1) = \sigma^2/2n$ is smaller than $V(\overline{X}_2) = \sigma^2/n$, the estimator $\overline{X}_1$ is the better estimator.

Example 1.5 (Unbiasedness, best estimator, and relative efficiency)**.** Let $X_1, X_2, \ldots, X_7$ denote a random sample from a population having mean μ and variance σ^2. Consider the following two estimators of μ:

$$\hat{\Theta}_1 = \frac{X_1 + X_2 + \cdots + X_7}{7}, \qquad \hat{\Theta}_2 = \frac{2X_1 - X_6 + X_4}{2}.$$

(a) Is either estimator unbiased? **(b)** Which estimator is best, and in what sense? Calculate the relative efficiency of the two estimators.

(a) Unbiasedness.

$$\mathbb{E}(\hat{\Theta}_1) = \mathbb{E}\left(\frac{X_1 + \cdots + X_7}{7}\right) = \frac{1}{7}\mathbb{E}(X_1) + \cdots + \frac{1}{7}\mathbb{E}(X_7) = \frac{1}{7}\mu + \cdots + \frac{1}{7}\mu = \mu,$$

$$\mathbb{E}(\hat{\Theta}_2) = \mathbb{E}\left(\frac{2X_1 - X_6 + X_4}{2}\right) = \mathbb{E}(X_1) - \tfrac{1}{2}\mathbb{E}(X_6) + \tfrac{1}{2}\mathbb{E}(X_4) = \mu - \tfrac{1}{2}\mu + \tfrac{1}{2}\mu = \mu.$$

Hence $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are both unbiased estimators of μ.

(b) Variances.

$$\mathrm{Var}(\hat{\Theta}_1) = \mathrm{Var}\left(\frac{X_1 + \cdots + X_7}{7}\right) = \frac{1}{49}\left[\mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_7)\right] = \frac{1}{49}\,(7\sigma^2) = \frac{1}{7}\sigma^2.$$

$$\text{Var}(\hat{\Theta}_2) = \text{Var}\left(\frac{2X_1 - X_6 + X_4}{2}\right) = \frac{1}{4}\text{Var}(2X_1) + \frac{1}{4}\text{Var}(X_6) + \frac{1}{4}\text{Var}(X_4).$$

Since $\text{Var}(2X_1) = 4\,\text{Var}(X_1) = 4\sigma^2$,

$$\text{Var}(\hat{\Theta}_2) = \frac{4}{4}\sigma^2 + \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2 = \sigma^2 + \tfrac{1}{4}\sigma^2 + \tfrac{1}{4}\sigma^2 = \frac{3}{2}\sigma^2.$$

The best estimator is the one with the smaller variance, so $\hat{\Theta}_1$ is the best estimator. The relative efficiency of the two estimators is

$$\frac{\text{Var}(\hat{\Theta}_2)}{\text{Var}(\hat{\Theta}_1)} = \frac{\frac{3}{2}\sigma^2}{\frac{1}{7}\sigma^2} = \frac{21}{2}.$$

Example 1.6 (MSE comparison of a biased and an unbiased estimator)**.** Suppose that $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are estimators of the parameter θ. We know that $\mathbb{E}(\hat{\Theta}_1) = \theta$, $\mathbb{E}(\hat{\Theta}_2) = \theta/2$, $V(\hat{\Theta}_1) = 10$, and $V(\hat{\Theta}_2) = 4$. Which estimator is best, and in what sense?

Since the means of the two estimators differ, the mean squared errors also differ.

$$\text{MSE}(\hat{\Theta}_1) = V(\hat{\Theta}_1) + \left[\mathbb{E}(\hat{\Theta}_1) - \theta\right]^2 = 10 + (\theta - \theta)^2 = 10,$$
$$\text{MSE}(\hat{\Theta}_2) = V(\hat{\Theta}_2) + \left[\mathbb{E}(\hat{\Theta}_2) - \theta\right]^2 = 4 + \left(\tfrac{\theta}{2} - \theta\right)^2 = 4 + \frac{\theta^2}{4}.$$

The relative efficiency is

$$\frac{\text{MSE}(\hat{\Theta}_1)}{\text{MSE}(\hat{\Theta}_2)} = \frac{10}{4 + \frac{\theta^2}{4}} = \frac{40}{16 + \theta^2}.$$

- If $\theta^2 = 24$, the relative efficiency is $1 \Rightarrow \hat{\Theta}_1$ and $\hat{\Theta}_2$ are equally efficient.
- If $\theta^2 > 24$, the relative efficiency is $< 1 \Rightarrow \hat{\Theta}_1$ is more efficient.
- If $\theta^2 < 24$, the relative efficiency is $> 1 \Rightarrow \hat{\Theta}_2$ is more efficient.

Example 1.7 (Pull-off force for connectors)**.** Data on the pull-off force (pounds) for connectors used in an automobile engine application are as follows:

> 79.3, 75.1, 78.2, 74.1, 73.9, 75.0, 77.6, 77.3, 73.8, 74.6, 75.5, 74.0, 74.7, 75.9, 72.9, 73.8, 74.2, 78.1, 75.4, 76.3, 75.3, 76.2, 74.9, 78.0, 75.1, 76.8.

(a) Calculate a point estimate of the mean pull-off force of all connectors in the population. State which estimator you used and why. **(b)** Calculate a point estimate of the pull-off force value that separates the weakest 50% of the connectors from the strongest 50%. **(c)** Calculate point estimates of the population variance and the population standard deviation. **(d)** Calculate the standard error of the point estimate found in part (a). **(e)** Calculate a point estimate of the proportion of all connectors whose pull-off force is less than 73 pounds.

(a) Mean. The sample mean is

$$\overline{X} = \frac{1966}{26} = 75.62 \text{ pounds}.$$

The population mean μ is estimated by the sample mean $\overline{X} = 75.62$ pounds, because it is known that the sample mean is an unbiased estimator of the population mean.

(b) Median. Sorting the 26 observations,

72.9, 73.8, 73.8, 73.9, 74.0, 74.1, 74.2, 74.6, 74.7, 74.9, 75.0, 75.1, 75.1, 75.3, 75.4, 75.5, 75.9, 76.2, 76.3, 76.8, 77.3, 77.6, 78.0, 78.1, 78.2, 79.3.

The sample median is 75.2 pounds, which is the estimate of the population median.

(c) Variance and standard deviation. Good point estimates of the population variance and standard deviation are the corresponding sample quantities:

$$\widehat{\sigma^2} = s^2 = 2.7382 \text{ pounds}^2, \qquad \hat{\sigma} = s = 1.6547 \text{ pounds}.$$

(d) Standard error. The standard error of the point estimate for the mean pull-off force is

$$\text{SE}(\overline{X}) = \hat{\sigma}_{\overline{X}} = \frac{s}{\sqrt{n}} = \frac{1.6547}{\sqrt{26}} = 0.3245.$$

The standard error is about 0.4% of the sample mean, which means a relatively precise point estimate of pull-off force has been obtained. If pull-off force is assumed normally distributed, then $2\hat{\sigma}_{\overline{X}} = 2(0.3245) = 0.6490$, so we are confident that the true mean pull-off force lies within 75.62 ± 0.649, that is, $(74.97, 76.26)$ pounds.

(e) Proportion below 73 pounds. A point estimate of the proportion of all connectors whose pull-off force is less than 73 pounds is

$$\hat{p} = \frac{x}{n} = \frac{\text{\# specimens less than 73}}{\text{total}} = \frac{1}{26} = 0.038462.$$

Chapter 2

Methods of Point Estimation

The previous chapter described desirable *properties* of an estimator. This chapter describes two general *methods* for actually constructing point estimators: the method of moments and the method of maximum likelihood.

2.1 Method of Moments

Moments

Let $X_1, X_2, \ldots, X_n$ be a random sample from the probability distribution $f(x)$, where $f(x)$ can be a discrete probability mass function or a continuous probability density function. The kth **population moment** (or distribution moment) is $\mathbb{E}(X^k)$, $k = 1, 2, \ldots$. The corresponding kth **sample moment** is $\frac{1}{n}\sum_{i=1}^{n} X_i^k$, $k = 1, 2, \ldots$.

Moment Estimators

Let $X_1, X_2, \ldots, X_n$ be a random sample from either a probability mass function or a probability density function with m unknown parameters $\theta_1, \theta_2, \ldots, \theta_m$. The **moment estimators** $\hat{\Theta}_1, \hat{\Theta}_2, \ldots, \hat{\Theta}_m$ are found by equating the first m population moments to the first m sample moments and solving the resulting equations for the unknown parameters.

In practice the method of moments proceeds in three steps:

1. Calculate the first few sample moments, $m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i$ and $m_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2$.
2. Calculate the corresponding population moments $\mathbb{E}(X^k) = \int_{-\infty}^{\infty} x^k f(x; \theta_1, \theta_2)\, dx$; the results are expressed in terms of θ_1 and θ_2.
3. Solve the simultaneous equations $\mathbb{E}(X) = m_1$, $\mathbb{E}(X^2) = m_2$ for the unknown parameters. The solutions are the method-of-moments estimators.

2.2 Method of Maximum Likelihood

Maximum Likelihood Estimator

Suppose that X is a random variable with probability distribution $f(x; \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \ldots, x_n$ be the observed values in a random sample of size n. Then the **likelihood function** of the sample is

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdots f(x_n; \theta).$$

The likelihood function is now a function of only the unknown parameter θ. The **maximum likelihood estimator** (MLE) of θ is the value of θ that maximises the likelihood function $L(\theta)$.

Properties of a Maximum Likelihood Estimator

Under very general and not restrictive conditions, when the sample size n is large and if $\hat{\Theta}$ is the maximum likelihood estimator of the parameter θ, then:

1. $\hat{\Theta}$ is an approximately unbiased estimator for θ, i.e. $\mathbb{E}(\hat{\Theta}) \approx \theta$;
2. the variance of $\hat{\Theta}$ is nearly as small as the variance that could be obtained with any other estimator; and
3. $\hat{\Theta}$ has an approximate normal distribution.

2.3 Maximum Likelihood: Worked Problems

Example 2.1 (MLE for a Poisson distribution)**.** Consider the Poisson distribution with parameter λ. Find the maximum likelihood estimator of λ, based on a random sample of size n. The probability mass function is

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

The likelihood function for a random sample of size n is

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}\,\lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}.$$

Applying the logarithm to both sides,

$$\log L(\lambda) = -n\lambda + \left(\sum_{i=1}^n x_i\right)\log\lambda - \log\left(\prod_{i=1}^n x_i!\right).$$

Differentiating with respect to λ and equating to zero,

$$\frac{d\log L(\lambda)}{d\lambda} = -n + \frac{\sum_{i=1}^n x_i}{\lambda} = 0 \implies \frac{\sum_{i=1}^n x_i}{\lambda} = n \implies \lambda = \frac{\sum_{i=1}^n x_i}{n}.$$

$$\hat{\lambda} = \overline{X}.$$

Example 2.2 (MLE for a geometric distribution)**.** Let X be a geometric random variable with parameter p. Find the maximum likelihood estimator of p based on a random sample of size n. The density is

$$f(x) = (1-p)^{x-1}p, \qquad x = 1, 2, \ldots$$

The likelihood function is

$$L(p) = \prod_{i=1}^{n} p\,(1-p)^{x_i-1} = p^n \prod_{i=1}^{n}(1-p)^{x_i-1}.$$

Taking logarithms,

$$\log L(p) = n\log p + \left(\sum_{i=1}^n x_i - n\right)\log(1-p).$$

Differentiating with respect to p and equating to zero,

$$\frac{d\log L(p)}{dp} = \frac{n}{p} + \left(\sum_{i=1}^n x_i - n\right)\frac{-1}{1-p} = 0 \implies \frac{n}{p} = \frac{\sum_{i=1}^n x_i - n}{1-p}.$$

Rearranging,

$$\frac{1}{p} = 1 + \frac{\sum_{i=1}^n x_i - n}{n} = \frac{\sum_{i=1}^n x_i}{n},$$

$$\hat{p} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\overline{X}}.$$

Example 2.3 (MLE for the density $(\theta+1)x^\theta$)**.** Let X be a random variable with the probability distribution

$$f(x) = \begin{cases} (\theta+1)x^\theta, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the maximum likelihood estimator of θ, based on a random sample of size n. The likelihood function is

$$L(\theta) = \prod_{i=1}^{n} f(x_i) = (\theta+1)^n \left[x_1 x_2 \cdots x_n\right]^\theta.$$

Taking logarithms and differentiating with respect to θ,

$$\frac{d}{d\theta} \log L(\theta) = \frac{d}{d\theta}\left[n \log(\theta+1) + \theta \textstyle\sum_{i=1}^{n} \log x_i\right] = \frac{n}{\theta+1} + \sum_{i=1}^{n} \log x_i = 0.$$

Hence

$$\sum_{i=1}^{n} \log x_i = \frac{-n}{\theta+1} \implies \theta + 1 = \frac{-n}{\sum_{i=1}^{n} \log x_i} \implies \hat{\Theta} = -\left[1 + \frac{n}{\sum_{i=1}^{n} \log x_i}\right].$$

2.4 Method of Moments: Worked Problems

Example 2.4 (Poisson mean by the method of moments; significance of the MSE)**.** Ten values collected at random from a Poisson distribution with mean μ are

$$10,\ 11,\ 15,\ 17,\ 9,\ 13,\ 18,\ 11,\ 15,\ 16.$$

Estimate the mean parameter μ using the method of moments. What significance does the mean squared error hold for an estimator, and how is it used to evaluate an estimator?

Here $X \sim \text{Poi}(\mu)$, and we know $\mathbb{E}(X) = \mu$. The method of moments equates $\mathbb{E}(X)$ to the first sample moment:

$$\mathbb{E}(X) = \overline{X} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{10}\sum_{i=1}^{10} x_i.$$

Therefore

$$\hat{\mu} = \frac{1}{10}\sum x_i = \frac{135}{10} = 13.5.$$

The mean squared error combines the effect of the bias and the variance of an estimator; the smaller the MSE, the better the estimator:

$$\text{MSE} = (\text{Bias})^2 + \text{Var}\,.$$

It is possible that, of two estimators, one with higher bias or higher variance may have a lower MSE.

Example 2.5 (Method-of-moments estimator for a binomial proportion)**.** The random variable X represents the number of defective parts in a lot of 90 parts. A sample of 25 lots gives x_i,

$i = 1, 2, \ldots, 25$, where each x_i is the number of defective parts in the ith lot. Derive the method-of-moments estimator for the proportion p of a part being defective. Here $X \sim \text{Binomial}(90, p)$, where p is the proportion defective, so $\mathbb{E}(X) = 90p$. Equating $\mathbb{E}(X)$ to the sample moment,

$$90p = \frac{1}{25}\sum_{i=1}^{25} x_i \implies \hat{p} = \frac{1}{90}\left[\frac{1}{25}\sum_{i=1}^{25} x_i\right] = \frac{\overline{X}}{90}.$$

The population mean (np) of a binomial may be estimated in several ways: $\overline{X}$, $\frac{1}{2}\big(\max(x_i) + \min(x_i)\big)$, or $\bar{y}$, where $\bar{y}$ is computed from the 23 values y_i obtained after excluding $\min(x_i)$ and $\max(x_i)$. The corresponding mean squared errors are

$$\begin{aligned}
\text{MSE}(\overline{X}) &= [\text{Bias}(\overline{X})]^2 + \text{Var}(\overline{X}) = 0 + \frac{90p(1-p)}{25}, \\
\text{MSE}\Big(\tfrac{\max(x_i)+\min(x_i)}{2}\Big) &= 0 + \frac{90p(1-p)}{2}, \\
\text{MSE}(\bar{y}) &= 0 + \frac{90p(1-p)}{23}.
\end{aligned}$$

Therefore $\overline{X}$ is best, since its MSE is minimum. In summary: bias is used for unbiasedness, standard error is used to determine low spread, and the MSE is used for checking consistency.

Example 2.6 (Method-of-moments estimators for the normal distribution)**.** Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution with parameters μ and σ^2. For the normal distribution,

$$\mathbb{E}(X) = \mu, \qquad \mathbb{E}(X^2) = \mu^2 + \sigma^2,$$

since $\mathbb{E}(X - \mu) = 0$ and $\mathbb{E}[(X-\mu)^2] = \sigma^2$. Equating $\mathbb{E}(X) = \overline{X}$ and $\mathbb{E}(X^2) = \frac{1}{n}\sum_{i=1}^n X_i^2$ gives

$$\mu = \overline{X}, \qquad \mu^2 + \sigma^2 = \frac{1}{n}\sum_{i=1}^n X_i^2.$$

Solving these equations,

$$\hat{\mu} = \overline{X}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n X_i^2 - \overline{X}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X})^2.$$

Notice that the moment estimator of σ^2 is *not* an unbiased estimator (it divides by n rather than $n-1$).

Example 2.7 (Discrete distribution depending on θ)**.** Suppose that X is a discrete random variable with the following distribution, where $0 \le \theta \le 1$ is a parameter:

x	0	1	2	3
$P(x)$	$\frac{2\theta}{3}$	$\frac{\theta}{3}$	$\frac{2(1-\theta)}{3}$	$\frac{1-\theta}{3}$

The following 10 independent observations were taken from such a distribution: $3, 0, 2, 1, 3, 2, 1, 0, 2, 1$. Use the method of moments to find the estimate of θ.

The theoretical mean value is

$$\mathbb{E}(X) = \sum_{x=0}^{3} xP(x) = 0 \cdot \tfrac{2\theta}{3} + 1 \cdot \tfrac{\theta}{3} + 2 \cdot \tfrac{2(1-\theta)}{3} + 3 \cdot \tfrac{1-\theta}{3} = \frac{7}{3} - 2\theta.$$

The sample mean value is

$$\overline{X} = \frac{3+0+2+1+3+2+1+0+2+1}{10} = 1.5.$$

We need to solve $\frac{7}{3} - 2\theta = 1.5$, and finally obtain

$$2\theta = \frac{7}{3} - 1.5 \implies \theta = \frac{7 - 4.5}{3 \times 2} = \frac{2.5}{6} = \frac{5}{12}.$$

$$\hat{\Theta} = \frac{5}{12}.$$

Example 2.8 (Method of moments for a uniform density)**.** Use the method of moments to estimate the parameter θ for the uniform density $f(x \mid \theta) = \frac{1}{\theta}$ with $0 \leq x \leq \theta$, based on a random sample $X_1, \ldots, X_n$. The theoretical moment is

$$\mathbb{E}(X) = \int_0^\theta x \cdot \frac{1}{\theta}\, dx = \frac{x^2}{2\theta}\Big|_0^\theta = \frac{\theta^2}{2\theta} - 0 = \frac{\theta}{2}.$$

The first sample moment is $\overline{X}$. Equating both,

$$\mathbb{E}(X) = \overline{X} \implies \frac{\theta}{2} = \overline{X} \implies \hat{\Theta} = 2\overline{X} = \frac{2}{n}\sum_{i=1}^{n} x_i,$$

which is the method-of-moments estimate.

A Catalogue of Moment Estimators

The same procedure yields the moment estimators of several standard distributions.

Example 2.9 (Bernoulli distribution)**.** For $X \sim \text{Bern}(p)$, $P_X(x) = p^x(1-p)^{1-x}$. The population mean is $\mathbb{E}(X) = p$ and the sample mean is $\overline{X} = \frac{1}{N}\sum_{i=1}^{N} x_i$, so $\hat{p} = \overline{X} = \frac{1}{N}\sum_{i=1}^{N} x_i$.

Example 2.10 (Geometric distribution)**.** For $X \sim \text{Geo}(p)$, $P_X(x) = p(1-p)^{x-1}$, with $\mathbb{E}(X) = \frac{1}{p}$. Equating to $\overline{X} = \frac{1}{N}\sum_{i=1}^{N} x_i$ gives $\hat{p} = \dfrac{1}{\overline{X}} = \dfrac{N}{\sum_{i=1}^{N} x_i}$.

Example 2.11 (Poisson distribution)**.** For $X \sim \text{Poisson}(\lambda)$, $P_X(x) = \dfrac{e^{-\lambda}\lambda^x}{x!}$, $x \geq 0$, with $\mathbb{E}(X) = \lambda$. Therefore $\hat{\lambda} = \overline{X} = \frac{1}{N}\sum_{i=1}^{N} x_i$.

Example 2.12 (Chi-square distribution)**.** For the chi-square distribution χ^2_p,

$$f_X(x) = \frac{1}{\Gamma\left(\frac{p}{2}\right) 2^{p/2}}\, x^{p/2-1} e^{-x/2},$$

with $\mathbb{E}(X) = p$. The moment estimator of the degrees-of-freedom parameter is $\hat{p} = \overline{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$.

Example 2.13 (Likelihood function of the exponential distribution)**.** Consider the exponential distribution with parameter λ. Derive the likelihood function of a random sample of size n. The likelihood function of the exponential distribution is

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}.$$

Taking the natural logarithm,

$$\ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} x_i.$$

Differentiating with respect to λ and equating to zero,

$$\frac{d \ln L(\lambda)}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0 \implies \frac{n}{\lambda} = \sum_{i=1}^{n} x_i \implies \hat{\lambda} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\overline{X}}.$$

$$\hat{\lambda} = \frac{1}{\overline{X}}.$$

Thus the maximum likelihood estimator of λ is the reciprocal of the sample mean.

Example 2.14 (A density with constant c: four parts)**.** Consider the probability density function

$$f(x) = c(1 + \theta x), \qquad -1 \le x \le 1.$$

(a) Find the value of the constant c. **(b)** What is the moment estimator for θ? **(c)** Show that $\hat{\Theta} = 3\overline{X}$ is an unbiased estimator for θ. **(d)** Find the maximum likelihood estimator for θ.

(a) A valid probability density function satisfies $\int f(x)\,dx = 1$:

$$\int_{-1}^{1} c(1 + \theta x)\,dx = c\left(x + \frac{\theta x^2}{2}\right)\Big|_{-1}^{1} = c\left(1 + \frac{\theta}{2}\right) - c\left(-1 + \frac{\theta}{2}\right) = 1.$$

Hence $c + \dfrac{c\theta}{2} + c - \dfrac{c\theta}{2} = 1 \Rightarrow 2c = 1 \Rightarrow c = \dfrac{1}{2}$.

(b) The moment estimator for θ:

$$\hat{\mu} = \int_{-1}^{1} x\, f(x)\,dx = \int_{-1}^{1} x \cdot \tfrac{1}{2}(1 + \theta x)\,dx = \frac{1}{2}\int_{-1}^{1} (x + \theta x^2)\,dx = \frac{1}{2}\left[\frac{x^2}{2} + \frac{\theta x^3}{3}\right]_{-1}^{1}.$$

The first term vanishes (it is odd), leaving

$$\hat{\mu} = \frac{1}{2}\left[0 + \frac{\theta}{3}(1^3 + 1^3)\right] = \frac{1}{2} \cdot \frac{2\theta}{3} = \frac{\theta}{3}.$$

$$\hat{\mu} = \frac{\theta}{3}.$$

(c) The sample estimator of the mean is $\overline{X}$, and $\hat{\mu} = \dfrac{\theta}{3}$, so $\theta = 3\hat{\mu} \Rightarrow \hat{\Theta} = 3\overline{X}$. To prove $\hat{\Theta} = 3\overline{X}$ is unbiased we must show $\mathbb{E}(3\overline{X}) = \theta$. Since the sample mean $\overline{X}$ is an unbiased estimator of the population mean $\hat{\mu}$,

$$\mathbb{E}(3\overline{X}) = 3\,\mathbb{E}(\overline{X}) = 3\hat{\mu} = 3 \cdot \frac{\theta}{3} = \theta.$$

Hence the moment estimator $3\overline{X}$ is an unbiased estimator of θ.

(d) Maximum likelihood estimator. With $f(x) = \frac{1}{2}(1+\theta x)$,

$$L(\theta) = \prod_{i=1}^{n} f(x_i) = \frac{(1+\theta x_1)(1+\theta x_2)\cdots(1+\theta x_n)}{2^n} = \frac{1}{2^n}\prod_{i=1}^{n}(1+\theta x_i).$$

Taking logarithms,

$$\log L(\theta) = -n\log 2 + \sum_{i=1}^{n}\log(1+\theta x_i).$$

Differentiating,

$$\frac{d}{d\theta}\log L(\theta) = 0 + \sum_{i=1}^{n}\frac{x_i}{1+\theta x_i}.$$

Equating this to zero gives

$$\sum_{i=1}^{n}\frac{x_i}{1+\theta x_i} = 0.$$

By observation the value of θ that maximises the likelihood is $\max(x_i)$.

2.5 Commonly Used Estimators and Criteria

Commonly used estimators are as follows:

1. Maximum likelihood estimators
2. Bayes estimators
3. Method of moments estimators
4. Least squares
5. Minimum mean squared error (MMSE) or Bayes least squared error (BLSE)
6. Minimum variance unbiased estimator (MVUE)
7. Best linear unbiased estimator (BLUE)
8. Nonlinear system identification
9. Unbiased estimators
10. Markov chain Monte Carlo
11. Particle filter
12. Kalman filter
13. Wiener filter

Several criteria are proposed for estimators:

1. **Unbiased estimator** — the long-run average of the estimate equals the true parameter value.
2. **Consistent estimator** — with more and more observations the estimate gets nearer to the true parameter value.
3. **Efficient estimator** — has small variability.
4. **Minimum variance** among all unbiased estimators.

An estimator Y for a parameter θ is unbiased if $\mathbb{E}(Y) = \theta$. Intuitively, Y is unbiased if its long-term average value is equal to θ. In the normal population model, $\overline{X}$ is an unbiased estimator of μ: this is because $\mathbb{E}(X_i) = \mu$ for each i, and hence $\mathbb{E}(\overline{X}) = \mu$. Also S^2 is an unbiased estimator of σ^2. On the other hand, if you put Y as the largest of the observations, then Y is not an unbiased estimator. It appears that, after all, this maximum is one of the observations, and each observation has expected value μ, so Y might be expected to have the same property—but it does not. Indeed, if the sample size is at least two, then $\mathbb{E}(Y) > \mu$.

Consistency

An estimator is **consistent** if it gets closer and closer to the parameter value as the sample size increases, i.e. an estimator Y for a parameter θ is consistent if for any $\epsilon > 0$ we have $P(|Y - \theta| \geq \epsilon) \to 0$ as $n \to \infty$.

For the normal population model, $\overline{X}$ is a consistent estimator of μ. This is because, given any $\epsilon > 0$,

$$P\big(|\overline{X} - \mu| \geq \epsilon\big) \leq \frac{\mathrm{Var}(\overline{X})}{\epsilon^2} = \frac{\sigma^2}{\epsilon^2} \cdot \frac{1}{n} \to 0.$$

This argument shows that for any model, with μ denoting the population mean, $\overline{X}$ is a consistent estimator of μ; it depends only on the fact that the variance of $\overline{X}$ is σ^2/n, where σ^2 is the (finite) population variance. In fact the same argument gives a general fact: if Y_n is an unbiased estimator (based on a sample of size n) of θ and if $\mathrm{Var}(Y_n) \to 0$, then Y_n is a consistent estimator of θ.

If Y is an unbiased estimator of θ, then of course $\mathbb{E}(Y - \theta)^2$ is nothing but the variance of Y. However, if Y is biased, then this is no longer the variance of Y; the quantity $\mathbb{E}(Y - \theta)^2$ is called the mean squared error (MSE). An estimator is said to have minimum MSE if this quantity is the least possible. When the estimator is unbiased, the MSE is its variance, and an unbiased estimator with minimum MSE is called the minimum variance unbiased estimator (MVUE). The sample mean $\overline{X}$ has minimum variance among all unbiased estimators that are linear combinations of $X_1, \ldots, X_n$.

2.6 Method of Moments: General Recipe and Normal Example

Method of moments. Let X be a random variable with density function $f(x; \theta_1, \theta_2)$, with parameters θ_1, θ_2 to be estimated, and let $X_1, \ldots, X_n$ be a random sample.

1. Calculate the first two sample moments $m_1 = \dfrac{1}{n}\sum_{i=1}^n X_i$ and $m_2 = \dfrac{1}{n}\sum_{i=1}^n X_i^2$.
2. Calculate the first two population moments $\mathbb{E}(X^k) = \int_{-\infty}^{\infty} x^k f(x; \theta_1, \theta_2)\, dx$; the results are in terms of θ_1 and θ_2.
3. Solve the simultaneous equations $\mathbb{E}(X) = m_1$, $\mathbb{E}(X^2) = m_2$ for θ_1, θ_2. The solutions are the method-of-moments estimators.

Example 2.15 (Normal distribution by the method of moments, revisited)**.** Consider $X \sim N(\mu, \sigma^2)$ with density

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

The parameters are μ and σ^2, and we have a random sample $X_1, \ldots, X_n$.

Step 1 (sample moments).

$$m_1 = \frac{1}{n}\sum_{i=1}^n X_i = \overline{X}, \qquad m_2 = \frac{1}{n}\sum_{i=1}^n X_i^2.$$

Step 2 (theoretical moments).

$$\hat{\mu}_1 = \mathbb{E}(X) = \mu, \qquad \hat{\mu}_2 = \mathbb{E}(X^2) = \mu^2 + \sigma^2, \qquad (\mathrm{Var}(X) = \sigma^2).$$

Step 3 (solve). From $\mu = \overline{X}$ and $\mu^2 + \sigma^2 = \frac{1}{n}\sum X_i^2$ we obtain

$$\hat{\mu} = \overline{X}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n X_i^2 - \overline{X}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X})^2.$$

2.7 Maximum Likelihood with Numerical Data

Example 2.16 (ML estimate of μ from a binomial-type sample)**.** The following samples $0, 1, 0, 0, 1$ are from a binomial distribution which has the form $P(X = 0) = (1 - \mu)$ and $P(X = 1) = \mu$. Find the maximum likelihood estimate of μ.

The likelihood function is

$$L(\mu) = P(X{=}0)\, P(X{=}1)\, P(X{=}0)\, P(X{=}0)\, P(X{=}1) = (1-\mu)\,\mu\,(1-\mu)\,(1-\mu)\,\mu = (1-\mu)^4\mu^2.$$

Taking logarithms,

$$\log L(\mu) = \log\big[(1-\mu)^4\mu^2\big] = 4\log(1-\mu) + 2\log\mu.$$

Differentiating and equating to zero,

$$\begin{aligned}\frac{\partial \log L(\mu)}{\partial \mu} &= 4\cdot\frac{1}{1-\mu}(-1) + 2\cdot\frac{1}{\mu} = 0 \\ &\implies -4\mu + 2(1-\mu) = 0 \implies -6\mu + 2 = 0 \implies \mu = \frac{2}{6} = \frac{1}{3}.\end{aligned}$$

The second derivative is

$$\frac{\partial^2 \log L(\mu)}{\partial \mu^2} = \frac{-4}{(1-\mu)^2} + \frac{2}{\mu^2} < 0,$$

which is negative, confirming a maximum. Hence the maximum likelihood estimate is

$$\hat{\mu} = \frac{1}{3}.$$

Example 2.17 (ML estimate of a Poisson mean from numerical data)**.** The samples $5, 9, 3, 12, 14$ are from a Poisson distribution of the form $P(X = k) = \dfrac{\lambda^k e^{-\lambda}}{k!}$. What is the maximum likelihood estimator of λ? The likelihood function is

$$\begin{aligned}L(\lambda) &= P(X{=}5)P(X{=}9)P(X{=}3)P(X{=}12)P(X{=}14) \\ &= \frac{e^{-\lambda}\lambda^5}{5!}\cdot\frac{e^{-\lambda}\lambda^9}{9!}\cdot\frac{e^{-\lambda}\lambda^3}{3!}\cdot\frac{e^{-\lambda}\lambda^{12}}{12!}\cdot\frac{e^{-\lambda}\lambda^{14}}{14!} = \frac{e^{-5\lambda}\lambda^{43}}{5!\,9!\,3!\,12!\,14!}.\end{aligned}$$

Taking logarithms,

$$\log L(\lambda) = 43\log\lambda - 5\lambda - \log(5!\,9!\,3!\,12!\,14!).$$

Differentiating and equating to zero,

$$\frac{d\log L(\lambda)}{d\lambda} = 43\cdot\frac{1}{\lambda} - 5 = 0 \implies \lambda = \frac{43}{5}.$$

The second derivative $\dfrac{\partial^2 \log L(\lambda)}{\partial \lambda^2} = \dfrac{-43}{\lambda^2}$ is negative for any λ, confirming a maximum. Hence

$$\hat{\lambda} = \frac{43}{5}.$$

Chapter 3

Statistical Intervals for a Single Sample

3.1 Confidence, Tolerance, and Prediction Intervals

An interval estimate for a population parameter is called a *confidence interval*. We cannot be certain that the interval contains the true, unknown population parameter—we only use a sample from the full population to compute the point estimate and the interval. However, the confidence interval is constructed so that we have confidence that it does contain the unknown population parameter. Confidence intervals are widely used in engineering and the sciences.

Interval estimates can be of three types:

1. A **confidence interval** bounds population or distribution parameters (such as a mean viscosity).
2. A **tolerance interval** bounds a selected proportion of a distribution.
3. A **prediction interval** bounds future observations from the population or distribution.

3.2 Confidence Interval on the Mean, Variance Known

Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution with unknown mean μ and known variance σ^2. We know that the sample mean $\overline{X}$ is normally distributed with mean μ and variance σ^2/n. We may standardise $\overline{X}$ by subtracting the mean and dividing by the standard deviation, which results in the variable

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}.$$

The random variable Z has a standard normal distribution.

A confidence interval estimate for μ is an interval of the form $\ell \leq \mu \leq u$, where the endpoints ℓ and u are computed from sample data. Because different samples will produce different values of ℓ and u, these endpoints are values of random variables L and U respectively. Suppose that we can determine values of L and U such that the following probability statement is true:

$$P\{L \leq \mu \leq U\} = 1 - \alpha, \qquad 0 \leq \alpha \leq 1.$$

There is a probability of $1 - \alpha$ of selecting a sample for which the confidence interval will contain the true value of μ. The endpoints ℓ and u are called the *lower* and *upper confidence limits* respectively, and $1 - \alpha$ is called the *confidence coefficient*.

Because $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$ has a standard normal distribution, we may write

$$P\left\{-z_{\alpha/2} \leq \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right\} = 1 - \alpha.$$

Manipulating the terms inside the brackets by (1) multiplying through by $\sigma/\sqrt{n}$, (2) subtracting $\overline{X}$ from each term, and (3) multiplying through by -1, we obtain

$$P\left\{\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha.$$

Confidence Interval on the Mean, Variance Known

If $\overline{X}$ is the sample mean of a random sample of size n, with known variance σ^2, from a normal population, a $100(1-\alpha)\%$ confidence interval on μ is given by

$$\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

where $z_{\alpha/2}$ is the upper $100\,\alpha/2$ percentage point of the standard normal distribution.

Remark 3.1. For a fixed sample size n and standard deviation σ, the higher the confidence level, the longer the resulting confidence interval. The length of the confidence interval is a measure of the precision of estimation.

3.2.1 Choice of Sample Size

If $\overline{X}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\overline{X} - \mu|$ will not exceed a specified amount E when the sample size is

Sample Size for a Specified Error

$$n = \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2.$$

If the right-hand side of this equation is not an integer, it must be rounded up. This will ensure that the level of confidence does not fall below $100(1-\alpha)\%$. Notice that $2E$ is the length of the resulting confidence interval.

3.2.2 One-Sided Confidence Bounds

It is possible to obtain one-sided confidence bounds for μ by setting either $\ell = -\infty$ or $u = \infty$ and replacing $z_{\alpha/2}$ by z_α.

One-Sided Confidence Bounds on the Mean

A $100(1-\alpha)\%$ *upper* confidence bound for μ is

$$\mu \leq u = \overline{X} + z_\alpha\frac{\sigma}{\sqrt{n}},$$

and a $100(1-\alpha)\%$ *lower* confidence bound for μ is

$$\overline{X} - z_\alpha\frac{\sigma}{\sqrt{n}} = \ell \leq \mu.$$

3.3 Worked Problems: Mean, Variance Known

Example 3.1 (90% CI and sample size)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, **(i)** construct a 90% confidence interval on the population mean if the population standard deviation is known to be 2; **(ii)** determine the sample size required to keep the width of the confidence interval to less than 2.

(i) The $100(1-\alpha)\%$ confidence interval is $\mu \in \overline{X} \pm z_{0.05} \cdot \dfrac{\sigma}{\sqrt{n}}$. The sample mean is

$$\overline{X} = \frac{5+2+3+1+0+0+1+3+3}{9} = \frac{18}{9} = 2.$$

With $z_{0.05} = 1.645$, $\sigma = 2$, $n = 9$,

$$\text{CI} = 2 \pm 1.645 \cdot \frac{2}{\sqrt{9}} = 2 \pm 1.645 \cdot \frac{2}{3},$$

giving the upper limit $2 + 1.645 \cdot \frac{2}{3} = 3.0967$ and the lower limit $2 - 1.645 \cdot \frac{2}{3} = 0.9033$:

$$0.9033 \leq \mu \leq 3.0967.$$

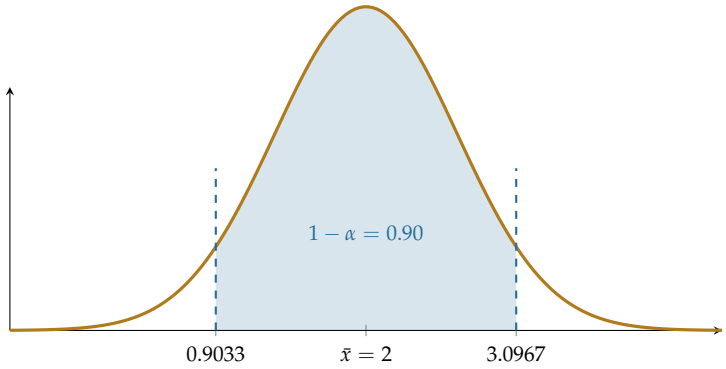


Figure 3.1: The 90% confidence interval $0.9033 \leq \mu \leq 3.0967$ centred at $\bar{x} = 2$, shown on the sampling distribution of the mean.

(ii) The width is

$$\text{width} = 2z_{0.05}\frac{\sigma}{\sqrt{n}} = \frac{2 \cdot 1.645 \cdot 2}{\sqrt{n}} < 2 \implies 2 \cdot 1.645 \cdot 2 < 2\sqrt{n} \implies \sqrt{n} > \frac{2 \cdot 1.645 \cdot 2}{2} = 3.29,$$

so $n \geq 10.8241$, i.e. $n \geq 11$.

Example 3.2 (95% upper bound; CI on variance)**.** For the sample $(5, 2, 3, 1, 4, 2, 1, 3, 3, 5)$ from a population representing the extent of overfill (mL) in sealed beverage bottles, **(a)** construct a 95% confidence *upper* bound on the population mean if the population variance is known to be 5 (what should the sample size be to keep the width of the confidence interval less than 1 mL?); **(b)** if the population variance is unknown, construct a 95% confidence interval on the population variance.

(a) 95% **upper bound on** μ**.** The sample mean is

$$\overline{X} = \frac{5+2+3+1+4+2+1+3+3+5}{10} = \frac{29}{10} = 2.9.$$

With $z_{0.05} \approx 1.645$,

$$\mu \leq \overline{X} + z_{0.05}\frac{\sigma}{\sqrt{n}} = 2.9 + 1.645 \cdot \frac{\sqrt{5}}{\sqrt{10}} = 2.9 + 1.645 \cdot \sqrt{\tfrac{5}{10}} = 4.063.$$

The upper bound is the same as the upper limit of the 90% two-sided confidence interval $\overline{X} \pm z_{0.05}\frac{\sigma}{\sqrt{n}} = 2.9 \pm 1.645\sqrt{\frac{5}{10}}$. For the width,

$$\text{width} = 2 \cdot 1.645\sqrt{\frac{5}{n}} < 1 \implies 2 \cdot 1.645\sqrt{5} < \sqrt{n} \implies n > (2 \cdot 1.645)^2 \cdot 5 = 54.1,$$

so the sample size $n \geq 54$ keeps the CI width < 1 mL.

(b) 95% **CI on σ^2.** With $\alpha = 0.05$, $n = 10$,

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} \implies \frac{9 \cdot 1.449^2}{19.02} \leq \sigma^2 \leq \frac{9 \cdot 1.449^2}{2.7},$$

where the sample variance is

$$s^2 = \frac{1}{9}\big[(5-2.9)^2 + (2-2.9)^2 + (3-2.9)^2 + (1-2.9)^2 + (4-2.9)^2 + (2-2.9)^2 + (1-2.9)^2 + (3-2.9)^2 + (3-2.9)^2 + (5-2.9)^2\big].$$

This gives

$$0.9937 \leq \sigma^2 \leq 7 \text{ mL}^2.$$

Example 3.3 (Standard error and a 99.7% interval)**.** From a population with known standard deviation of 1.65, a sample of 32 items resulted in 34.8 as an estimate of the mean. **(i)** Find the standard error of the mean. **(ii)** Compute an interval estimate that should include the population mean 99.7% of the time.

(i) The standard error of the mean is

$$\frac{\sigma}{\sqrt{n}} = \frac{1.65}{\sqrt{32}} = 0.29.$$

(ii) For a 99.7% interval, $z_{0.0015} = 2.97$:

$$\overline{X} \pm z_{0.0015}\frac{\sigma}{\sqrt{n}} = 34.8 \pm 2.97 \cdot \frac{1.65}{\sqrt{32}},$$

which gives

$$33.93 \leq \mu \leq 35.67.$$

Example 3.4 (Average burning time of light bulbs)**.** The manager of Cardinal Electricals' light-bulb division must estimate the average number of hours that a light bulb made by each bulb machine will last. A sample of 40 light bulbs was selected from a machine, and the average burning time was 1416 hours; the standard deviation of burning time is known to be 30 hours. Construct a 90% confidence interval for the true population mean. Here $n = 40$, $\overline{X} = 1416$, $\sigma = 30$, $\alpha = 0.1$:

$$\overline{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1416 \pm 1.645 \cdot \frac{30}{\sqrt{40}} = (1408.2,\ 1423.8).$$

$$1408.2 \leq \mu \leq 1423.8.$$

Example 3.5 (Operating-system response time: choosing n)**.** An operating system for a PC has been studied extensively, and it is known that the standard deviation of the response time following a particular command is $\sigma = 8$ milliseconds. A new version is installed, and we wish to estimate the mean response time for the new system to ensure that a 95% confidence interval for μ has length at most 5 milliseconds. **(a)** Assuming the response time is normally distributed with $\sigma = 8$, what sample size would you recommend? **(b)** Suppose the vendor tells us the standard deviation of the new system is smaller, $\sigma = 6$; give the recommended sample size and comment on the effect.

(a) With $\sigma = 8$, length $= 5$, confidence level 95% ($\alpha = 0.05$, $z_{0.025} = 1.96$),

$$\text{length} = 2z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 2 \cdot 1.96 \cdot \frac{8}{\sqrt{n}} = 5 \implies \sqrt{n} = \frac{2 \cdot 1.96 \cdot 8}{5} = 6.272,$$

so $n = (6.272)^2 = 39.338$. Therefore the sample size is

$$n = 40.$$

(b) With $\sigma = 6$,

$$\frac{2 \cdot 1.96 \cdot 6}{\sqrt{n}} = 5 \implies \sqrt{n} = 4.704 \implies n = (4.704)^2 = 22.13 \implies n = 23.$$

We can observe that as the standard deviation reduces, the required sample size also reduces.

Example 3.6 (Compressive strength of concrete: 95% and 99% CIs)**.** A civil engineer is analysing the compressive strength of concrete. Compressive strength is normally distributed with $\sigma^2 = 1000$ (psi)^2. A random sample of 12 specimens has a mean $\overline{X} = 3250$ psi. **(i)** Construct a 95% two-sided confidence interval on mean compressive strength. **(ii)** Construct a 99% two-sided confidence interval on mean compressive strength.

(i) 95% **CI** ($\alpha = 0.05$, $z_{0.025} = 1.96$):

$$\mu \in \overline{X} \pm z_{0.025}\frac{\sigma}{\sqrt{n}} = 3250 \pm 1.96 \cdot \frac{\sqrt{1000}}{\sqrt{12}},$$

$$3232.1 \leq \mu \leq 3267.89.$$

(ii) 99% **CI** ($\alpha = 0.01$, $z_{0.005} = 2.58$):

$$3250 - 2.58 \cdot \frac{\sqrt{1000}}{\sqrt{12}} \leq \mu \leq 3250 + 2.58 \cdot \frac{\sqrt{1000}}{\sqrt{12}},$$

$$3226.4 \leq \mu \leq 3273.55.$$

Example 3.7 (Recovering the sample mean and identifying the confidence level)**.** A random sample has been taken from a normal population and two confidence intervals constructed

using exactly the same data. The two confidence intervals are $(38.02, 61.98)$ and $(39.95, 60.05)$. **(a)** What is the value of the sample mean? **(b)** One of these intervals is a 90% CI and the other is a 95% CI. Which one is the 95% CI, and why?

(a) The confidence interval is $\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$, so $\overline{X}$ is the midpoint of the CI. Using either interval,

$$\overline{X} = \frac{38.02 + 61.98}{2} = \frac{100.00}{2} = 50.$$

$$\overline{X} = 50.$$

(b) For a fixed sample size n and standard deviation σ, the higher the confidence level, the longer the resulting confidence interval. We have

$$(39.95, 60.05) \Rightarrow \text{width} = 20.1 \;\rightarrow\; 90\%\text{ CI}, \qquad (38.02, 61.98) \Rightarrow \text{width} = 23.96 \;\rightarrow\; 95\%\text{ CI}.$$

The wider interval $(38.02, 61.98)$ is the 95% CI.

3.4 Confidence Interval on the Mean, Variance Unknown

When the sample is small (< 30 or < 40) and σ^2 is unknown, we must make an assumption about the form of the underlying distribution to obtain a valid CI procedure. A reasonable assumption is that the underlying distribution is normal.

Suppose the population of interest has a normal distribution with unknown mean μ and unknown variance σ^2. Assume that a random sample of size n, say $X_1, X_2, \ldots, X_n$, is available, and let $\overline{X}$ and S^2 be the sample mean and variance. To construct a two-sided CI on μ: if the variance were known, we would use $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$, which has a standard normal distribution. When σ^2 is unknown, a logical procedure is to replace σ by the sample standard deviation S. The random variable Z now becomes $T = (\overline{X} - \mu)/(S/\sqrt{n})$. (If n is large, proceed to use the CI based on the normal distribution itself.)

The t Distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with unknown mean μ and unknown variance σ^2. The random variable

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $n - 1$ degrees of freedom. Its probability density function is

$$f(x) = \frac{\Gamma[(k+1)/2]}{\sqrt{\pi k}\,\Gamma(k/2)} \cdot \frac{1}{\left[(x^2/k) + 1\right]^{(k+1)/2}}, \qquad -\infty < x < \infty,$$

where k is the number of degrees of freedom.

The mean and variance of the t distribution are zero and $k/(k-2)$ for $k > 2$, respectively. The general appearance of the t distribution is similar to the standard normal distribution: both are symmetric and unimodal, and the maximum ordinate value is reached when the mean $\mu = 0$. However, the t distribution has heavier tails than the normal; that is, it has more probability in the tails. As the number of degrees of freedom $k \to \infty$, the limiting form of the t distribution is

the standard normal distribution (Figure 3.2). Generally the number of degrees of freedom for t are the number of degrees of freedom associated with the estimated standard deviation.

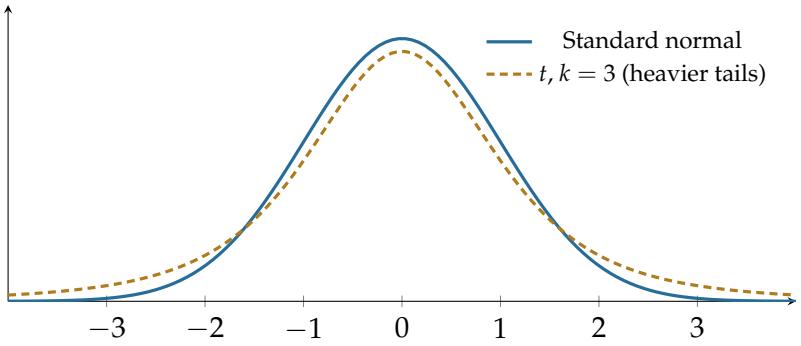


Figure 3.2: The t distribution (dashed) has heavier tails than the standard normal (solid). As $k \to \infty$ the t distribution approaches the standard normal.

The table of percentage points lets $t_{\alpha,k}$ be the value of the random variable T with k degrees of freedom above which we find an area (or probability) α. Thus $t_{\alpha,k}$ is an upper-tail $100\alpha\%$ point of the t distribution with k degrees of freedom. For example, the t value with 10 degrees of freedom having an area of 0.05 to the right is $t_{0.05,10} = 1.812$, i.e.

$$P(T_{10} > t_{0.05,10}) = P(T_{10} > 1.812) = 0.05.$$

Since the t distribution is symmetric about zero, we have $t_{1-\alpha,k} = -t_{\alpha,k}$; that is, the t value having an area of $1-\alpha$ to the right (an area of α to the left) is the negative of the t value that has area α in the right tail. Therefore $t_{0.95,10} = -t_{0.05,10} = -1.812$.

Confidence Interval on the Mean, Variance Unknown

If $\overline{X}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown variance σ^2, a $100(1-\alpha)\%$ confidence interval on μ is

$$\overline{X} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \le \mu \le \overline{X} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}},$$

where $t_{\alpha/2,n-1}$ is the upper $100\,\alpha/2$ percentage point of the t distribution with $n-1$ degrees of freedom.

One-sided confidence bounds on the mean of a normal distribution are obtained by using only the appropriate lower or upper confidence limit from the equation above and replacing $t_{\alpha/2,n-1}$ by $t_{\alpha,n-1}$. It is not as easy to select a sample size n to obtain a specified length for this CI as it was in the σ-known case, because the length of the interval involves s, n, and $t_{\alpha/2,n-1}$, and the t-percentile itself depends on n. Consequently an appropriate n can only be obtained through trial and error.

3.5 Worked Problems: Mean, Variance Unknown

Example 3.8 (Finding t-percentiles)**.** Determine the t-percentile required to construct each of the following.

(a) Two-sided CI, confidence level 95%, dof = 12: $t_{\alpha/2,n-1} = t_{0.025,12} = 2.179$.

(b) Two-sided CI, confidence level 99.9%, dof = 15: $t_{\alpha/2,n-1} = t_{0.0005,15} = 4.073$.

(c) One-sided CI, confidence level 99%, dof $= 19$: $t_{\alpha,n-1} = t_{0.01,19} = 2.539$.
(d) One-sided CI, confidence level 99.9%, dof $= 24$: $t_{\alpha,n-1} = t_{0.001,24} = 3.467$ (for dof $= 15$, $t_{0.001,15} = 3.733$).

Example 3.9 (Tire life)**.** A research engineer for a tire manufacturer is investigating tire life for a new rubber compound and has built 16 tires, testing them to end of life in a road test. The sample mean and standard deviation are 60139.7 and 3645.94 km. Find a 95% confidence interval on mean tire life μ. With $\alpha = 0.05$, $n = 16$, $t_{0.025,15} = 2.131$,

$$\overline{X} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \le \mu \le \overline{X} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 60139.7 \pm 2.131 \cdot \frac{3645.94}{\sqrt{16}},$$

$$58197.33 \le \mu \le 62082.07.$$

Example 3.10 (Electronic sensor life: known vs. unknown variance)**.** In a maintenance department of a firm, the life of the electronic equipment is found to be normally distributed. A random sample of 30 sensors has a sample mean life of 800 hours and a sample standard deviation of 70 hours. **(i)** Construct a 95% two-sided CI for the mean life of the sensors. **(ii)** Construct a 99% two-sided CI for the variance of the life of sensors. **(iii)** If the population standard deviation is known to be 90 hours, construct a 99% two-sided CI for the mean life of the sensors. **(iv)** Comment on the difference in answers to (i) and (iii).

Here Life $\sim N(\mu, \sigma^2)$, $\overline{X} = 800$, $s = 70$, $n = 30$.

(i) $\alpha = 0.05$, $t_{0.025,29} = 2.045$:

$$\overline{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 800 \pm 2.045 \cdot \frac{70}{\sqrt{30}} = 800 \pm 26.135,$$

$$773.86 \le \mu \le 826.14 \text{ hrs.}$$

(ii) CI on variance.

$$\frac{(n-1)s^2}{\chi^2_{0.005,29}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{0.995,29}} = \frac{29 \cdot 70^2}{52.34} \text{ and } \frac{29 \cdot 70^2}{13.12},$$

$$2715 \le \sigma^2 \le 10{,}830.7.$$

(iii) σ **known** $= 90$**.**

$$\overline{X} \pm z_{0.005}\frac{\sigma}{\sqrt{n}} = 800 \pm 2.58 \cdot \frac{90}{\sqrt{30}} = 800 \pm 42.39,$$

$$757.6 \le \mu \le 842.4.$$

(iv) The confidence interval of (iii) is wider than that of the 99% CI in spite of the population variance being known.

Example 3.11 (Electronic sensor life: a second data set)**.** The life of an electronic sensor is normally distributed. A random sample of 20 sensors has a mean life of 1000 hours and a standard deviation of 100 hours. **(i)** Construct a 99% two-sided CI for the mean life of the sensors. **(ii)** Construct a 95% two-sided CI for the variance of the life of sensors. **(iii)** If the population standard deviation is known to be 105 hours, construct a 99% two-sided CI for the mean life of the sensors. **(iv)** Comment on the difference in answers to (i) and (iii).

Here Life $\sim N(\mu, \sigma^2)$, $\overline{X} = 1000$, $s = 100$, $n = 20$.

(i) 99% CL, $\alpha = 0.01$, $t_{0.005,19} = 2.861$:

$$\overline{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 1000 \pm 2.861 \cdot \frac{100}{\sqrt{20}},$$

$$936.03 \leq \mu \leq 1063.97 \text{ hrs.}$$

(ii) $\alpha = 0.05$, $\alpha/2 = 0.025$, $\chi^2_{0.025,19} = 32.85$, $\chi^2_{0.975,19} = 8.91$:

$$\frac{(n-1)s^2}{\chi^2_{0.025,19}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{0.975,19}} = \frac{19 \cdot 100^2}{32.85} \text{ and } \frac{19 \cdot 100^2}{8.91},$$

$$5783.87 \leq \sigma^2 \leq 21{,}324.35.$$

(iii) $\sigma = 105$ known, $\alpha = 0.01$:

$$\overline{X} \pm z_{0.005}\frac{\sigma}{\sqrt{n}} = 1000 \pm 2.58 \cdot \frac{105}{\sqrt{20}},$$

$$939.42 \leq \mu \leq 1060.58 \text{ hrs.}$$

(iv) The CI of (iii) is narrower compared to (i) since σ is known. A more precise interval is obtained when additional information is available.

Example 3.12 (95% CI on the mean, variance unknown)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, construct a 95% confidence interval on the population mean if the population variance is unknown. The sample mean is $\overline{X} = 2$, and the sample standard deviation is computed from

$$\begin{aligned} s^2 &= \frac{1}{8}\Big[(5-2)^2 + (2-2)^2 + (3-2)^2 + (1-2)^2 + (0-2)^2 \\ &\quad + (0-2)^2 + (1-2)^2 + (3-2)^2 + (3-2)^2\Big] \\ &= \frac{9+0+1+1+4+4+1+1+1}{8} = \frac{22}{8} = 2.75, \end{aligned}$$

so $s = 1.658$. With $t_{0.025,8} = 2.306$,

$$\mu \in \overline{X} \pm t_{0.025,8}\frac{s}{\sqrt{n}} = 2 \pm 2.306 \cdot \frac{1.658}{\sqrt{9}},$$

$$0.7253 \le \mu \le 3.2747.$$

Example 3.13 (Cloud seeding: rainfall in acre-feet)**.** Cloud seeding has been studied for many decades as a weather modification procedure. The rainfall in acre-feet from 20 clouds that were selected at random and seeded with silver nitrate are as follows:

18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, 31.6.

(a) Construct a 95% one-sided lower bound on the mean. **(b)** Construct a 90% two-sided CI on the mean. What is the width of this interval? **(c)** If the width of the interval in part (b) were to be halved, how much should the sample size be? Use trial and error if needed.

Here $n = 20$, $s = 4.7848$, $\overline{X} = 26.035$.

(a) 95% **lower bound.** With $t_{0.05,19} = 1.729$,

$$\mu \ge \overline{X} - t_{0.05,19}\frac{s}{\sqrt{n}} = 26.035 - 1.729 \cdot \frac{4.7848}{\sqrt{20}},$$

$$\mu \ge 24.185.$$

(b) 90% **two-sided CI.** With $\alpha = 0.10$, $t_{0.05,19} = 1.729$,

$$\overline{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 26.035 \pm 1.729 \cdot \frac{4.7848}{\sqrt{20}},$$

$$90\%\ \text{CI} = (24.185,\ 27.895), \qquad \text{width} = 27.895 - 24.185 = 3.7958.$$

(c) Halving the width. The width is $2t_{0.05,\,n-1}\dfrac{s}{\sqrt{n}}$ (assuming s remains the same). By trial and error,

$$n = 30 \Rightarrow \text{width} = 2 \cdot 1.697 \cdot \tfrac{4.7848}{\sqrt{30}} = 2.96,$$
$$n = 40 \Rightarrow \text{width} = 2 \cdot 1.684 \cdot \tfrac{4.7848}{\sqrt{40}} = 2.5,$$
$$n = 60 \Rightarrow \text{width} = 2 \cdot 1.671 \cdot \tfrac{4.7848}{\sqrt{60}} = 2.06,$$
$$n = 120 \Rightarrow \text{width} = 2 \cdot 1.658 \cdot \tfrac{4.7848}{\sqrt{120}} = 1.4484.$$

Half the width of part (b) is $3.7958/2 = 1.8979$. Therefore n should be between 60 and 120.

Example 3.14 (Gasoline prices and highway speeds)**. (a)** From a sample of 42 gasoline stations statewide, the average price of a gallon of unleaded gas was found to be \$1.12 and the standard deviation was \$0.04 per gallon. Construct a 99% CI on the mean price of a gallon of unleaded gas. **(b)** A random sample of 57 vehicles on a highway clocked an average of 23.2 mph with a standard deviation of 0.3 mph. **(i)** Estimate the standard deviation of the population. **(ii)** Estimate the standard error of the mean for this population.

(a) $n = 42$, $\overline{X} = 1.12$, $s = 0.04$, $\alpha = 0.01$, $t_{0.005,41} = 2.7$:

$$\overline{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 1.12 \pm 2.7 \cdot \frac{0.04}{\sqrt{42}},$$

$$99\%\ \text{CI} = [\$1.10,\ \$1.14].$$

(b) $n = 57$, $\overline{X} = 23.2$, $s = 0.3$, assume $\alpha = 0.05$.
(i) Using the CI on the variance,

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} \right) = \left(\frac{56 \cdot 0.3^2}{77.4}, \frac{56 \cdot 0.3^2}{36.5} \right),$$

so the 95% CI for σ^2 is $(0.065, 0.138)$, and the CI for σ is $(0.255, 0.37)$.
(ii) The standard error of the mean is

$$\frac{\hat{\sigma}}{\sqrt{n}} = \frac{0.255 + 0.37}{2} \cdot \frac{1}{\sqrt{57}} = 0.041.$$

Example 3.15 (Contamination level in food packs)**.** The level of contamination (in %) was studied in 6 packs of food with the following data: 16.8, 17.2, 17.4, 16.9, 16.5, 17.1. Find a 99% confidence interval on the mean % μ and provide a practical interpretation. The sample mean is

$$\overline{X} = \frac{101.9}{6} = 16.98,$$

and the sample variance is

$$s^2 = \frac{0.0324 + 0.0484 + 0.1764 + 0.006 + 0.2304 + 0.0144}{5} = 0.1016, \qquad s = 0.318.$$

With $t_{0.005,5} = 4.032$,

$$\overline{X} \pm t_{\alpha/2,5} \frac{s}{\sqrt{n}} = 16.98 \pm 4.032 \cdot \frac{0.318}{\sqrt{6}} = 16.98 \pm 0.523,$$

$$16.457 \leq \mu \leq 17.503.$$

Example 3.16 (Wall thickness of glass bottles)**.** The wall thickness of 25 glass bottles was measured. The sample mean was 4.05 mm and the sample standard deviation was 0.08 mm. Find a 95% lower confidence bound for mean wall thickness, and interpret the interval. Here $n = 25$, $\overline{X} = 4.05$, $s = 0.08$, $\alpha = 0.05$, $t_{0.05,24} = 1.711$:

$$\mu \geq \overline{X} - t_{\alpha,n-1} \frac{s}{\sqrt{n}} = 4.05 - 1.711 \cdot \frac{0.08}{\sqrt{25}},$$

$$\mu \geq 4.02 \text{ mm}.$$

The minimum population mean wall thickness is 4.02 mm.

3.6 Confidence Interval on the Variance and Standard Deviation

The χ^2 Distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2, and let S^2 be the sample variance. Then the random variable

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

has a chi-square (χ^2) distribution with $n-1$ degrees of freedom.

Confidence Interval on the Variance

If s^2 is the sample variance from a random sample of n observations from a normal distribution with unknown variance σ^2, then a $100(1-\alpha)\%$ confidence interval on σ^2 is

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}},$$

where $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ are the upper and lower $100\,\alpha/2$ percentage points of the chi-square distribution with $n-1$ degrees of freedom. A confidence interval for σ has lower and upper limits that are the square roots of the corresponding limits for σ^2.

One-Sided Confidence Bounds on the Variance

The $100(1-\alpha)\%$ lower and upper confidence bounds on σ^2 are

$$\frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} \leq \sigma^2 \qquad \text{and} \qquad \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}}.$$

3.7 Worked Problems: Variance and Standard Deviation

Example 3.17 (Reading χ^2 percentiles)**.** Determine the values of the following percentiles:

$$\text{(a) } \chi^2_{0.05,10} = 18.31, \qquad \text{(b) } \chi^2_{0.995,16} = 5.14, \qquad \text{(c) } \chi^2_{0.005,25} = 46.93.$$

Example 3.18 (Selecting χ^2 percentiles for CIs)**.** Determine the χ^2 percentile required to construct each of the following CIs.

(a) Confidence level 95%, dof $= 24$, one-sided upper: $\chi^2_{0.05,24} = 36.42$.
(b) Confidence level 99%, dof $= 9$, one-sided upper: $\chi^2_{1-0.01,9} = \chi^2_{0.99,9} = 2.09$.
(c) Confidence level 90%, dof $= 19$, two-sided: $\chi^2_{0.1/2,19} = \chi^2_{0.05,19} = 30.14$.

Example 3.19 (Rivet hole diameter: 99% lower bound on variance)**.** A rivet is to be inserted into a hole. A random sample of $n = 15$ parts is selected, and the hole diameter is measured. The sample standard deviation of the hole-diameter measurements is $s = 0.008$ mm. Construct a 99% lower confidence bound for the variance. With $n = 15$ and $\chi^2_{0.01,14} = 29.14$,

$$\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{0.01,n-1}} = \frac{14 \cdot 0.008^2}{29.14} \geq 3.074 \times 10^{-5},$$

$$\text{CI} = [\,0.00003074,\ \infty\,).$$

Example 3.20 (Sugar content of canned peaches)**.** The sugar content of the syrup in canned peaches is normally distributed. A random sample of $n = 10$ cans yields a sample standard deviation of $s = 4.8$ mg. Calculate a 95% two-sided CI for σ. With $\alpha = 0.05$, $\chi^2_{0.025,9} = 19.02$, $\chi^2_{0.975,9} = 2.70$,

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} = \frac{9 \cdot 4.8^2}{19.02} \le \sigma^2 \le \frac{9 \cdot 4.8^2}{2.70},$$

so $10.90 \le \sigma^2 \le 76.8$. Taking square roots, $\sqrt{10.9} \le \sigma \le \sqrt{76.8}$:

$$3.30 \le \sigma \le 8.76.$$

Example 3.21 (95% CI on variance for a small sample)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, construct a 95% confidence interval on the population variance. Here $s = 1.6583$ and $n = 9$, so with $\chi^2_{0.025,8} = 15.51$ and $\chi^2_{0.975,8} = 2.73$,

$$\frac{(n-1)s^2}{\chi^2_{0.025,8}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{0.975,8}} = \frac{8 \cdot 1.6583^2}{15.51} \le \sigma^2 \le \frac{8 \cdot 1.6583^2}{2.73},$$

giving $1.4189 \le \sigma^2 \le 8.058$, and taking square roots,

$$1.19 \le \sigma \le 2.838.$$

Example 3.22 (Amino-acid concentration: variance test and one-sided bounds)**.** Amino-acid concentrations in 8 packs of dog feed were recorded in mg/kg as follows: $22, 24, 20, 26, 27, 24, 26, 23$. **(i)** Perform a test at 5% significance to see if the population variance is greater than 5.5 $(\text{mg/kg})^2$. What is the p-value of the test? **(ii)** Construct upper and lower 90% confidence bounds for the population variance.

(i) $H_0 : \sigma^2 = 5.5$, $H_1 : \sigma^2 > 5.5$ $(\text{mg/kg})^2$, $\alpha = 0.05$, $n = 8$, $\overline{X} = 24$, $s = 2.3299$. The sample variance is

$$s^2 = \frac{4 + 0 + 16 + 4 + 9 + 0 + 4 + 1}{7} = \frac{38}{7} = 5.428, \qquad s = \sqrt{5.428} = 2.3299.$$

The test statistic is

$$\chi^2_0 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{7 \cdot 2.3299^2}{5.5} = 6.9091.$$

The tabulated value is $\chi^2_{1-0.05,7} = \chi^2_{0.05,7} = 14.07$. Since $\chi^2_0 = 6.9091 < 14.07$, we do not reject H_0 (Figure 3.3). The p-value is $P(\chi^2_7 > 6.91)$, which lies between 0.4 and 0.5, since $P(\chi^2_7 > 6.346) = 0.5$ and $P(\chi^2_7 > 7.283) = 0.4$.
(ii) 90% **bounds.** The 90% *upper* bound is

$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}} = \frac{7 \cdot 5.4286}{2.833} = 13.41,$$

and the 90% *lower* bound is

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} = \frac{7 \cdot 5.4286}{12.02} = 3.16,$$

that is,

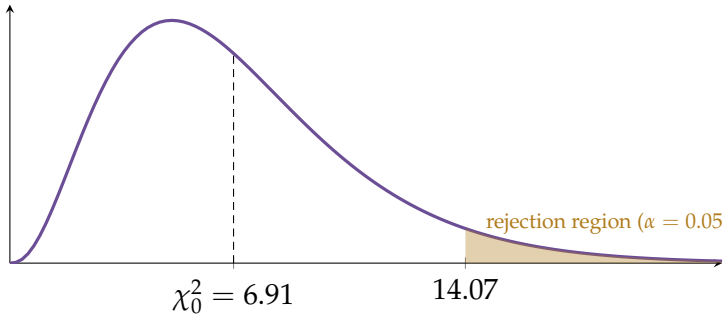


Figure 3.3: Chi-square density with 7 degrees of freedom. The observed statistic $\chi^2_0 = 6.91$ falls well short of the critical value 14.07, so H_0 is not rejected.

$$3.16 \le \sigma^2 \le 13.41.$$

Example 3.23 (Sugar content (continued): standard-deviation bounds)**.** For the canned-peaches data ($n = 10$, $s = 4.8$): **(a)** Find a 90% two-sided confidence interval for the population standard deviation. **(b)** Find a 95% upper bound and a 95% lower bound for the population standard deviation. How do these bounds compare with the limits of part (a)?

(a) $\alpha = 0.1$, two-sided $\alpha/2 = 0.05$, $\chi^2_{0.05,9} = 16.92$, $\chi^2_{0.95,9} = 3.325$:

$$\left[\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}\right] = \left[\frac{9 \cdot 4.8^2}{16.92}, \frac{9 \cdot 4.8^2}{3.325}\right] = [12.26,\ 62.36].$$

The interval for σ is $(3.5, 7.9)$.

(b) 95% upper bound ($\alpha = 0.05$, $\chi^2_{0.95,9} = 3.325$):

$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}} = \frac{9 \cdot 4.8^2}{3.325} = 62.36 \implies \sigma \le 7.9.$$

95% lower bound ($\chi^2_{0.05,9} = 16.92$):

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} = \frac{9 \cdot 4.8^2}{16.92} = 12.26 \implies \sigma \ge 3.5.$$

Both bounds equal the limits of the two-sided CI, since $\alpha/2$ in part (a) is the same as α in part (b).

Example 3.24 (Hole diameter: 99% lower bound on variance)**.** A hole diameter is measured in a random sample of 15 parts. The sample standard deviation of the hole diameter is 0.008 mm. Construct a 99% lower confidence bound for σ^2. With $n = 15$, $\alpha = 0.01$, $\chi^2_{0.01,14} = 29.14$,

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} = \frac{14 \cdot 0.008^2}{29.14} = 3.84 \times 10^{-3},$$

$$\sigma \ge 0.062.$$

Example 3.25 (Weight measurements: test on σ and a lower bound)**.** A set of 15 weight measurements gives a sample standard deviation $s = 0.0089$ g. **(i)** Does the population standard deviation σ of all weight measurements exceed 0.008 g at 5% significance? Find the p-value for this test. **(ii)** Answer part (i) with the help of a lower confidence interval for σ with $\alpha = 5\%$.

Here $n = 15$, $s = 0.0089$ g.

(i) $H_0 : \sigma = 0.008$ g, $H_1 : \sigma > 0.008$ g, $\alpha = 0.05$. The test statistic is

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{14 \cdot 0.0088^2}{0.008^2} = 16.94.$$

The tabulated value is $\chi^2_{0.05,14} = 23.68$. Since $\chi_0^2 = 16.94 < \chi^2_{\text{table}} = 23.68$, we fail to reject the null hypothesis; the standard deviation does not exceed 0.008 g at 5% significance. For the p-value, 16.94 lies between the table entries for 0.5 and 0.1 (21.06 at $p = 0.1$ and 13.34 at $p = 0.5$), so $0.1 < p < 0.5$.

(ii) Lower confidence bound, $\alpha = 0.05$, $\chi^2_{0.05,14} = 23.68$:

$$\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{0.05,14}} = \frac{14 \cdot 0.0088^2}{23.68} = 4.578 \times 10^{-5} \implies \sigma \geq 6.76 \times 10^{-3} \approx 0.0067.$$

$$\sigma \geq 0.0067.$$

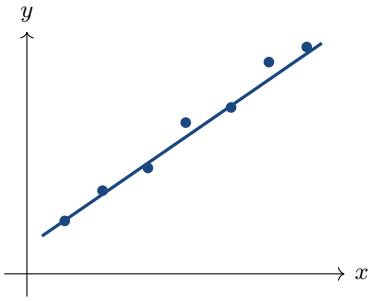
Example 3.26 (Titanium percentage in an aerospace alloy)**.** The percentage of titanium in an alloy used in aerospace casting is measured in 51 randomly selected parts. The sample standard deviation is $s = 0.28$. Construct a 95% two-sided confidence interval for σ. With $\alpha = 0.05$, $n = 51$, $\chi^2_{0.025,50} = 71.42$, $\chi^2_{0.975,50} = 32.36$,

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} = \frac{50 \cdot 0.28^2}{71.42} \leq \sigma^2 \leq \frac{50 \cdot 0.28^2}{32.36},$$

so $0.055 \leq \sigma^2 \leq 0.121$. Taking the square root of the endpoints,

$$0.234 \leq \sigma \leq 0.348.$$

Unit 4



Chapter 1

The Simple Linear Regression Model

1.1 Regression Analysis

Regression Analysis

Regression analysis is a statistical technique that is useful for exploring the relationships between two or more variables.

In the simple linear regression model the *dependent variable*, or *response*, is related to one *independent* or *regressor* variable as

Simple Linear Regression Model

$$Y = \beta_0 + \beta_1 x + \epsilon,$$

where ϵ is the random error term. The parameters β_0 and β_1 are called the *regression coefficients*.

In a scatter plot,

- $x \rightarrow$ the independent or regressor variable;
- $Y \rightarrow$ the response or dependent variable.

We estimate β_0 and β_1 by maximum likelihood, by the properties of least squares, by the method of moments, and so on. The method of maximum likelihood needs further knowledge of the distribution of Y, whereas the method of moments and the principle of least squares do not need any knowledge about the distribution of Y.

1.2 The Meaning of "Regression"

The word *regression* means "to move in the backward direction." A model exists in nature but is unknown to the experimenter. When some values of the explanatory variables are provided, then the values for the output or study variable are generated depending on the values of the explanatory variables. So the form of the function and the nature of the phenomenon give rise to the data. Ideally, the pre-existing model gives rise to the data.

Our objective is to determine the final form of this model. We propose to first move in the backward direction: we collect the data on the study and explanatory variables, then employ some statistical techniques and use this data to know the form of the function f. Equivalently, the data from the model is recorded first and then used to determine the parameters of the

model. The regression analysis is the technique which helps in determining the statistical model by using the data on study and explanatory variables.

1.3 Steps in Regression Analysis

The stages of a regression study proceed as follows:

1. Statement of the problem under consideration.
2. Choice of relevant variables.
3. Collection of data on the relevant variables.
4. Specification of the model.
5. Choice of method for fitting the data.
6. Fitting of the model.
7. Model validation and criticism.
8. Using the chosen model(s) for the solution of the posed problem and for forecasting.

1.4 The Mean and Variance of the Response

Based on the problem and the scatter diagram, it is reasonable to assume that the mean of the random variable Y is related to x by the following straight-line relation:

True Regression Line (line of mean values)

$$\mathbb{E}(Y \mid x) \;=\; \mu_{Y|x} \;=\; \beta_0 + \beta_1 x,$$

where the slope β_1 and intercept β_0 of the line are called the regression coefficients.

We think of the regression model as an *empirical* model. Suppose that the mean and variance of ϵ are 0 and σ^2 respectively. Then

$$\mathbb{E}(Y \mid x) \;=\; \mathbb{E}\big(\beta_0 + \beta_1 x + \epsilon\big) \;=\; \beta_0 + \beta_1 x + \mathbb{E}(\epsilon) \;=\; \beta_0 + \beta_1 x.$$

The variance of Y given x is

$$\mathrm{Var}(Y \mid x) \;=\; \mathrm{Var}\big(\beta_0 + \beta_1 x + \epsilon\big) \;=\; \mathrm{Var}(\beta_0 + \beta_1 x) + \mathrm{Var}(\epsilon) \;=\; 0 + \sigma^2 \;=\; \sigma^2.$$

The true regression model is a line of mean values:

$$\mu_{Y|x} = \beta_0 + \beta_1 x.$$

Here β_1 can be interpreted as the change in the mean of Y for a unit change in x. Also, the variability of Y at a particular value of x is determined by the error variance σ^2. This implies that there is a distribution of Y values at each x, and that the variance of this distribution is the same at each x.

Simple Linear Regression Model (full statement)

The model

$$Y = \beta_0 + \beta_1 x + \epsilon$$

is called the *simple linear regression model*. The true regression $\mu_{Y|x} = \beta_0 + \beta_1 x$ is a line of mean values: the height of the regression line at any value of x is just the expected value of Y for that x. The slope β_1 is the change in the mean of Y for a unit change in x. The variability of Y at a particular value of x is determined by the error variance σ^2; there is a distribution of Y values at each x and the variance of this distribution is the same at each x.

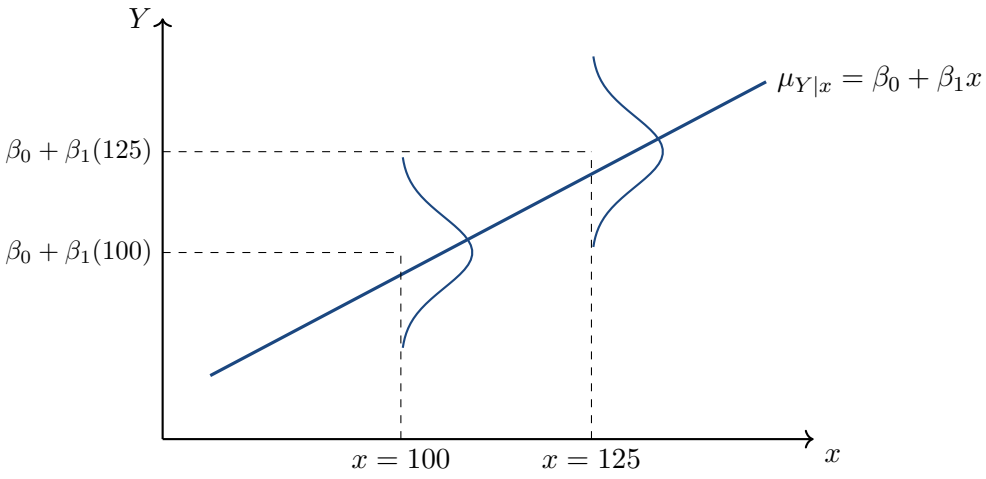


Figure 1.1: The true regression line is a line of mean values. At each x there is a distribution of Y with the same variance σ^2.

Chapter 2

Least Squares Estimation

Consider the model

$$\mathbb{E}(Y \mid x) = \beta_0 + \beta_1 x,$$

where the intercept β_0 and the slope β_1 are unknown regression coefficients. We assume that each observation Y_i can be described by the model

$$Y = \beta_0 + \beta_1 x + \epsilon,$$

where ϵ is a random error with mean zero and (unknown) variance σ^2.

2.1 The Method of Least Squares

The German scientist Karl Gauss (1777–1855) proposed estimating the parameters β_0 and β_1 so as to minimize the sum of squares of the vertical deviations. This is called the *method of least squares*.

Using the model for the i-th observation,

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad i = 1, 2, \ldots, n, \tag{2.1}$$

the sum of squares of the deviations of the observations from the true regression line is

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} \big(y_i - \beta_0 - \beta_1 x_i\big)^2. \tag{2.2}$$

The least squares estimators of β_0 and β_1, say $\hat{\beta}_0$ and $\hat{\beta}_1$, must satisfy

$$\left.\frac{\partial L}{\partial \beta_0}\right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^{n} \big(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\big) = 0, \tag{2.3}$$

$$\left.\frac{\partial L}{\partial \beta_1}\right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^{n} \big(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\big) x_i = 0. \tag{2.4}$$

Simplifying these two equations yields the *least squares normal equations*:

Least Squares Normal Equations

$$\left.\begin{aligned} n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i &= \sum_{i=1}^{n} y_i, \\ \hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2 &= \sum_{i=1}^{n} x_i y_i. \end{aligned}\right\} \tag{2.5}$$

2.2 The Least Squares Estimators

Solving the normal equations gives the least squares estimates of the intercept and slope:

Least Squares Estimators

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \tag{2.6}$$

$$\hat{\beta}_1 = \frac{\displaystyle\sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\displaystyle\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}, \tag{2.7}$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

The fitted or estimated regression line is

Fitted Regression Line

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

Note that each pair of observations satisfies the relationship

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + e_i, \qquad i = 1, 2, \ldots, n,$$

where $e_i = y_i - \hat{y}_i$ is called the *residual*. The residual describes the error in the fit of the model to the i-th observation y_i.

Residual

The residual for observation i is

$$e_i = y_i - \hat{y}_i.$$

It measures the error in the fit of the model to the i-th observed value.

2.3 Corrected Sums of Squares

Notationally, it is occasionally convenient to give special symbols to the numerator and denominator of Equation (2.7):

S_{xx} and S_{xy}

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}, \tag{2.8}$$

$$S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})\, y_i = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}. \tag{2.9}$$

With these, $\hat{\beta}_1 = \dfrac{S_{xy}}{S_{xx}}$.

Chapter 3

Estimating the Error Variance

There is actually another unknown parameter in the regression model, namely σ^2 (the variance of the error term ϵ). The residuals $e_i = y_i - \hat{y}_i$ are used to estimate σ^2.

3.1 The Error Sum of Squares

The sum of squares of the residuals, often called the *error sum of squares*, is

Error Sum of Squares

$$\mathrm{SS_E} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2. \tag{3.1}$$

It can be shown that the expected value of the error sum of squares is

$$\mathbb{E}(\mathrm{SS_E}) = (n-2)\,\sigma^2.$$

Therefore an *unbiased estimator* of σ^2 is

Unbiased Estimator of σ^2

$$\hat{\sigma}^2 = \frac{\mathrm{SS_E}}{n-2}. \tag{3.2}$$

A convenient computational form of the error sum of squares is

Computing $\mathrm{SS_E}$

$$\mathrm{SS_E} = \mathrm{SS_T} - \hat{\beta}_1 S_{xy},$$

where

$$\mathrm{SS_T} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2$$

is the total (corrected) sum of squares of the response variable y.

Chapter 4

Properties, Inference and Prediction

4.1 Properties of the Least Squares Estimators

In simple linear regression the estimated standard error of the slope and the estimated standard error of the intercept are

Standard Errors of the Estimators

$$\text{se}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \qquad \text{se}(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)},$$

where $\hat{\sigma}^2 = \dfrac{\text{SS}_\text{E}}{n-2}$.

The covariance of the two estimators is

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2\,\bar{x}}{S_{xx}}.$$

4.2 Analysis of Variance Approach to Test Significance of Regression

A method called the *analysis of variance* can be used to test for significance of regression. The identity underlying the method is

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

The two components on the right-hand side measure, respectively, the amount of variability in y_i accounted for by the regression line, and the residual variation left unexplained by the regression line. Symbolically it can be rewritten as

Partition of Total Variability

$$\text{SS}_\text{T} = \text{SS}_\text{R} + \text{SS}_\text{E},$$

where

$$\text{SS}_\text{T} = \sum_{i=1}^{n}(y_i - \bar{y})^2, \quad \text{SS}_\text{R} = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2, \quad \text{SS}_\text{E} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

It is seen that $\mathrm{SS_E} = \mathrm{SS_T} - \hat{\beta}_1 S_{xy}$, so that $\mathrm{SS_T} = \hat{\beta}_1 S_{xy} + \mathrm{SS_E}$. Hence the regression sum of squares is

$$\mathrm{SS_R} = \hat{\beta}_1 S_{xy}.$$

The total sum of squares $\mathrm{SS_T}$ has $n-1$ degrees of freedom, and $\mathrm{SS_R}$ and $\mathrm{SS_E}$ have 1 and $n-2$ degrees of freedom respectively. We may show that

$$\mathbb{E}\left[\frac{\mathrm{SS_E}}{n-2}\right] = \sigma^2, \qquad \mathbb{E}(\mathrm{SS_R}) = \sigma^2 + \beta_1^2 S_{xx},$$

and that $\mathrm{SS_E}/\sigma^2$ and $\mathrm{SS_R}/\sigma^2$ are independent chi-square random variables with $n-2$ and 1 degrees of freedom respectively.

Thus, if the null hypothesis $H_0 : \beta_1 = 0$ is true, the statistic

Test Statistic for Significance of Regression

$$F_0 = \frac{\mathrm{SS_R}/1}{\mathrm{SS_E}/(n-2)} = \frac{\mathrm{MS_R}}{\mathrm{MS_E}}$$

follows the $F_{1,n-2}$ distribution, and we would reject H_0 if $f_0 > f_{\alpha,1,n-2}$.

Here $\mathrm{MS_R}$ and $\mathrm{MS_E}$ are called *mean squares*. The test procedure is usually arranged in an analysis of variance table.

Table 4.1: Analysis of variance table for significance of regression.

Source of variation	Sum of squares	DoF	MS	F_0
Regression	$\mathrm{SS_R} = \hat{\beta}_1 S_{xy}$	1	$\mathrm{MS_R}$	$\mathrm{MS_R} / \mathrm{MS_E}$
Error	$\mathrm{SS_E} = \mathrm{SS_T} - \hat{\beta}_1 S_{xy}$	$n-2$	$\mathrm{MS_E}$	
Total	$\mathrm{SS_T}$	$n-1$		

where $\mathrm{MS_E} = \hat{\sigma}^2$.

4.3 Prediction of a New Observation

A $100(1-\alpha)\%$ *prediction interval* on a future observation at the value x_0 is given by

Prediction Interval for a Future Observation

$$\hat{y}_0 - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right]} \;\le\; Y_0 \;\le\; \hat{y}_0 + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right]},$$

where the value $\hat{y}_0$ is computed from the regression model $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

4.4 Coefficient of Determination R^2

Coefficient of Determination

The quantity

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_E}}{\mathrm{SS_T}}$$

is called the *coefficient of determination*. It is often used to judge the adequacy of the regression model. R^2 is interpreted as the proportion of variability in the data explained (or accounted for) by the model.

4.5 Misinterpretation or Misuse of Linear Regression Techniques

It is possible to develop statistically significant relationships between variables that are completely unrelated in a causal sense. Common pitfalls include:

1. Prediction beyond the range of the observed x.
2. Prediction beyond the time period of data collection.
3. Equating a statistical relation to a real-life (causal) situation — confusing correlation with causation.
4. Misinterpreting R^2 to be the strength of the relationship instead of the proportion of variability explained.
5. Regression models are not necessarily valid for extrapolation purposes.

Chapter 5

Regression Variants and Correlation

5.1 Summary of the Model and Assumptions

Recall the simple linear regression model

$$Y = \beta_0 + \beta_1 x + \epsilon,$$

in which the slope β_1 and intercept β_0 are called the regression coefficients, and:

- $Y \to$ dependent or study variable;
- $x \to$ independent or explanatory variable;
- $\beta_0 \to$ intercept term;
- $\beta_1 \to$ slope parameter;
- $\epsilon \to$ error component, which accounts for the failure of the data to lie on the straight line and represents the difference between the true and observed values of y.

ϵ is assumed to be an i.i.d. random variable with mean zero and constant variance σ^2, and it is also assumed to be normally distributed. Consequently,

$$\mathbb{E}(y \mid x) = \beta_0 + \beta_1 x, \qquad \mathrm{Var}(y \mid x) = \sigma^2.$$

5.2 Least Squares Estimation — Geometric View

The principle of least squares estimates the parameters β_0 and β_1 by minimizing the sum of squares of the differences between the observations and the line in the scatter diagram.

5.3 Direct, Reverse and Orthogonal Regression

Direct regression

When the vertical difference between observations and the line in the scatter diagram is considered and its sum of squares is minimized to obtain estimates of β_0 and β_1, the method is *direct regression*. This method is also known as *ordinary least squares estimation*. The direct regression approach minimizes the sum of squares

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2.$$

Taking partial derivatives and setting them to zero we obtain

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}.$$

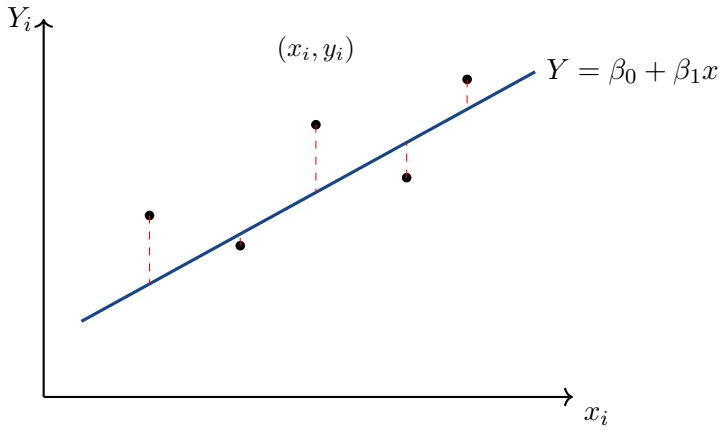


Figure 5.1: When the vertical differences between the observations and the line in the scatter diagram are considered and their sum of squares is minimized, the method is *direct regression*.

Reverse (inverse) regression

Alternatively, the sum of squares of the differences between observations and the line in the *horizontal* direction in the scatter diagram can be minimized to obtain estimates. This is known as the *reverse* (or inverse) regression method.

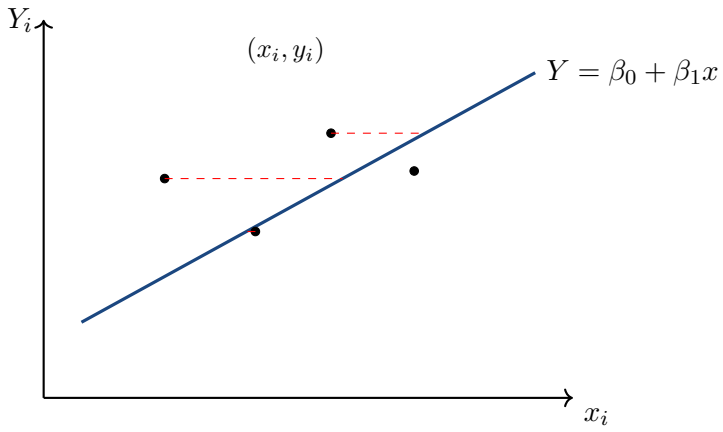


Figure 5.2: Reverse regression: horizontal deviations are minimized.

Orthogonal (major axis) regression

Instead of the horizontal or vertical direction, if the sum of squares of the *perpendicular* distances between the observations and the line in the scatter diagram is minimized, the method is known as *orthogonal regression* or the *major axis regression* method.

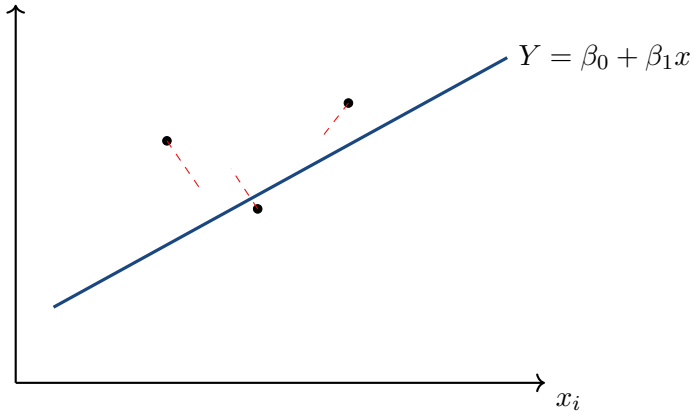


Figure 5.3: Major axis (orthogonal) regression: perpendicular distances are minimized.

5.4 Coefficient of Determination (restated)

Key Sums and Estimators

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}},$$

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - \frac{(\sum x_i)\,(\sum y_i)}{n},$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{(\sum x_i)^2}{n},$$

$$S_{yy} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - \frac{(\sum y_i)^2}{n},$$

$$\hat{\sigma}^2 = \frac{\mathrm{SS_E}}{n-2}, \qquad \mathrm{SS_E} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \mathrm{SS_T} - \hat{\beta}_1 S_{xy}.$$

Here S_{xx} is (proportional to) the variability of x. The coefficient of determination

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_E}}{\mathrm{SS_T}}$$

is used to judge the adequacy of the model: it is the amount of variability in the data explained or accounted for by the model.

5.5 Correlation Coefficient

Population Correlation Coefficient

The correlation coefficient between Y and X is

$$\rho = \frac{\sigma_{xy}}{\sigma_x \, \sigma_y},$$

where σ_{xy} is the covariance between Y and X, σ_x is the standard deviation (spread) of X, and σ_y is the standard deviation of Y.

The *sample correlation coefficient* is

Sample Correlation Coefficient

$$R = \frac{\sum_{i=1}^{n} y_i (x_i - \bar{x})}{\left[\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2 \right]^{1/2}} = \frac{S_{xy}}{(S_{xx} \, \mathrm{SS_T})^{1/2}}.$$

Remark 5.1. The slope and the sample correlation coefficient are related by

$$\hat{\beta}_1 = \left(\frac{\mathrm{SS_T}}{S_{xx}} \right)^{1/2} R.$$

The sample correlation coefficient R measures the linear association between Y and X, while $\hat{\beta}_1$ measures the predicted change in the mean of Y for a unit change in X.

Chapter 6

Worked Problems

This chapter collects the worked numerical problems from the notes. Each uses the least squares machinery developed earlier: $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, and $\hat{\sigma}^2 = \mathrm{SS_E}/(n-2)$.

6.1 Roadway Surface Temperature and Pavement Deflection

Example 6.1 (Pavement deflection)**.** Regression methods were used to analyse the data from a study investigating the relationship between roadway surface temperature x and pavement deflection y. Summary quantities were

$$n = 20, \quad \sum y_i = 12.75, \quad \sum y_i^2 = 8.86, \quad \sum x_i = 1478,$$

$$\sum x_i^2 = 143{,}215.8, \quad \sum x_i y_i = 1083.67.$$

(a) Calculate the least squares estimates of the slope and intercept. Graph the regression line.

(b) Use the equation of the fitted line to predict the pavement deflection when the surface temperature is 85°F.

(c) What is the mean pavement deflection when the surface temperature is 90°F?

(d) What change in mean pavement deflection would be expected for a 1°F change in surface temperature?

Solution

(a) Using $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$,

$$S_{xx} = \sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n} = 143{,}215.8 - \frac{(1478)^2}{20} = 33{,}991.6,$$

$$S_{xy} = \sum x_i y_i - \frac{\left(\sum x_i\right)\left(\sum y_i\right)}{n} = 1083.67 - \frac{12.75 \times 1478}{20} = 141.445.$$

Hence

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{141.445}{33{,}991.6} = 0.00416,$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{12.75}{20} - (0.00416)\frac{1478}{20} = 0.32999.$$

The fitted regression line is

$$\boxed{\hat{y} = 0.32999 + 0.00416\, x.}$$

We also estimate the error variance:

$$\hat{\sigma}^2 = \mathrm{MS_E} = \frac{\mathrm{SS_E}}{n-2} = \frac{\left(\sum y_i^2 - n\bar{y}^2\right) - \hat{\beta}_1 S_{xy}}{n-2}.$$

Now $\sum y_i^2 - n\bar{y}^2 = 8.86 - 20\left(\frac{12.75}{20}\right)^2$ and $\hat{\beta}_1 S_{xy} = 0.00416 \times 141.445$, so

$$\hat{\sigma}^2 = \frac{8.86 - 20\left(\frac{12.75}{20}\right)^2 - 0.00416 \times 141.445}{18} = \frac{0.143}{18} = 7.91 \times 10^{-3} \approx 0.00796.$$

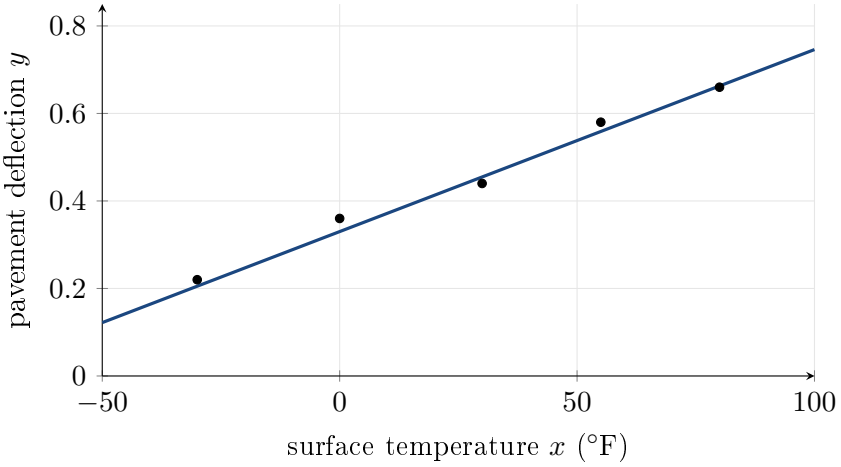


Figure 6.1: Fitted regression line $\hat{y} = 0.32999 + 0.00416x$ for the pavement deflection data (illustrative points shown).

(b) At $x = 85$°F,

$$\hat{y} = 0.32999 + 0.00416 \times 85 = 0.6836.$$

(c) At $x = 90$°F,

$$\hat{y} = 0.32999 + 0.00416 \times 90 = 0.7044.$$

(d) A change of 0.00416 in mean pavement deflection would be expected for a 1°F change in surface temperature, since 0.00416 is the slope, and the slope is the average increase/decrease of y per unit change of x.

6.2 Changing Units: Celsius Instead of Fahrenheit

Example 6.2 (Rescaling the regressor)**.** Suppose the temperature in the previous problem is measured in °C rather than °F.

(a) Write the new regression model that results.

(b) What change in expected pavement deflection is associated with a 1°C change in surface temperature?

Solution

(a) Using the conversion $x_F = \frac{9}{5}x_C + 32$ and substituting into the fitted model $\hat{y} = 0.3299982 + 0.0041612\,x_F$,

$$\hat{y} = 0.3299982 + 0.0041612\left(\tfrac{9}{5}x_C + 32\right),$$

$$\hat{y} = 0.3299982 + 0.0074902\,x_C + 0.1331584,$$

$$\boxed{\hat{y} = 0.4631476 + 0.0074902\,x_C.}$$

(b) The new slope is

$$\hat{\beta}_1 = 0.0074902,$$

so a 1°C change in surface temperature is associated with a change of 0.0074902 in expected pavement deflection.

6.3 Compressive Strength and Intrinsic Permeability of Concrete

Example 6.3 (Concrete permeability)**.** An article presented data on compressive strength x and intrinsic permeability y of various concrete mixes. Summary quantities are

$$n = 14, \quad \sum y_i = 572, \quad \sum y_i^2 = 23{,}530, \quad \sum x_i = 43,$$

$$\sum x_i^2 = 157.42, \quad \sum x_i y_i = 1697.80.$$

Assume the two variables are related according to the simple linear regression model.

(a) Calculate the least squares estimates of the slope and intercept.

(b) Use the fitted line to predict the permeability when the compressive strength is $x = 4.3$.

(c) Give a point estimate of the mean permeability when the compressive strength is $x = 3.7$.

(d) Suppose the observed value of permeability is $y = 46.1$ at $x = 3.7$. Calculate the corresponding residual.

Solution

Using $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$,

$$S_{xx} = 157.42 - \frac{43^2}{14} = 25.348571,$$

$$S_{xy} = 1697.80 - \frac{(43)(572)}{14} = -59.057143.$$

(a)

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-59.057143}{25.348571} = -2.330,$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{572}{14} - (-2.3298017)\frac{43}{14} = 48.013.$$

The fitted line is

$$\hat{y} = 48.013 - 2.3298017\,x.$$

(b) At $x = 4.3$,

$$\hat{y} = 48.013 - 2.3298017 \times 4.3 = 37.99.$$

(c) At $x = 3.7$,

$$\hat{y} = 48.013 - 2.3298017 \times 3.7 = 39.39.$$

(d) The residual at $x = 3.7$ with observed $y = 46.1$ is

$$e = y - \hat{y} = 46.1 - 39.39 = 6.71.$$

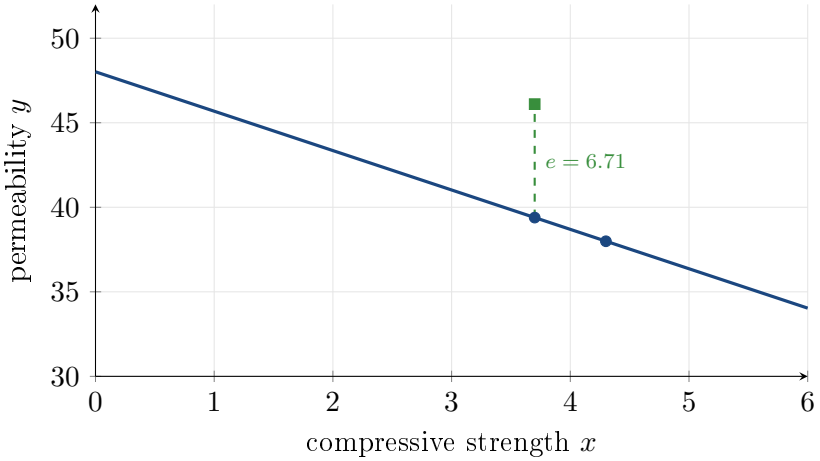


Figure 6.2: Fitted line $\hat{y} = 48.013 - 2.3298x$; the residual at $x = 3.7$ is the dashed segment.

6.4 Green Liquor Na_2S Concentration and Paper Machine Production

Example 6.4 (Green liquor)**.** Data on green liquor Na_2S concentration (y, in g/L) and paper machine production (x, in tons/day) are as follows:

y	40	42	49	46	44	48	46	43	53	52	54	57	58
x	825	830	890	895	890	910	915	960	990	1010	1012	1030	1050

(a) Fit a simple linear regression model with y = green liquor Na_2S concentration and x = production. Find an estimate of σ^2. Draw a scatter diagram of the data and the resulting least squares fitted model.

(b) Find the fitted value of y corresponding to $x = 910$ and the associated residual.

(c) Find the mean green liquor Na_2S concentration when the production rate is 950 tons/day.

Solution

With $n = 13$ the summary quantities are

$$\sum x_i^2 = 11{,}529{,}419, \quad \sum y_i^2 = 31{,}128, \quad \sum x_i = 12{,}207, \quad \sum y_i = 632, \quad \sum x_i y_i = 598{,}098.$$

(a)

$$S_{xx} = 11{,}529{,}419 - \frac{(12{,}207)^2}{13} = 67{,}046,$$

$$S_{xy} = 598{,}098 - \frac{(12{,}207)(632)}{13} = 4650.$$

Hence

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{4650}{67{,}046} = 0.06936,$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 48.615 - 939 \times 0.06936 = -16.51.$$

The fitted model is

$$\boxed{\hat{y} = -16.51 + 0.06936\, x.}$$

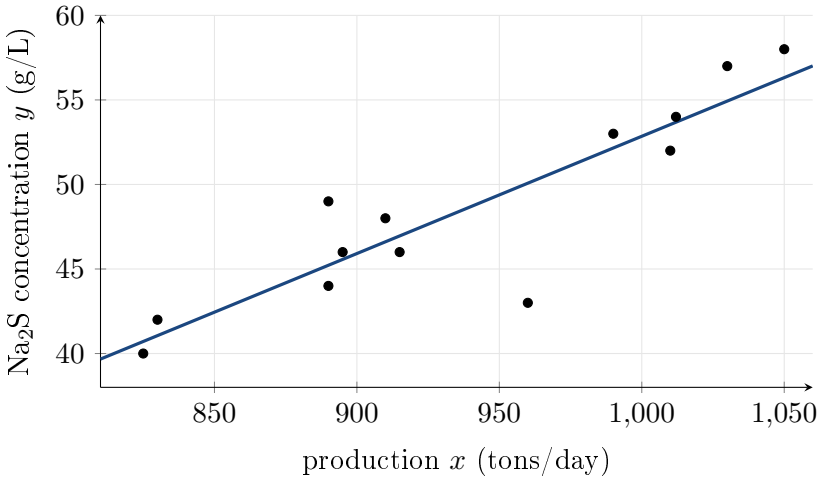


Figure 6.3: Scatter diagram of green liquor data with fitted line $\hat{y} = -16.51 + 0.06936x$.

(b) At $x = 910$,

$$\hat{y} = -16.51 + 0.06936 \times 910 = 46.60.$$

The observed value at this production rate is $y = 48$, so the residual is

$$e = y - \hat{y} = 48 - 46.60 = 1.3924.$$

(c) At $x = 950$,

$$\hat{y} = -16.51 + 0.06936 \times 950 = 49.38.$$

6.5 Correlation by Inspection and a Re-coded Observation

Example 6.5 (Inspection and recoding)**.** For a set of 20 pairs of values (x_i, y_i),

$$\sum x_i^2 = 130{,}000, \quad \sum y_i^2 = 88.6, \quad \sum x_i y_i = 1500,$$

$$\sum x_i = 1500, \quad \sum y_i = 10.$$

(a) From the above values, without deriving any other quantities, is it possible to tell whether x and y are truly correlated just by inspection? Give reasons. Also write the estimated intercept and slope parameters and the fitted linear regression model.

(b) If an observed (x, y) value which was wrongly recorded previously as $(25, 3)$ is changed to the correct observation $(35, 2)$, derive the changed regression model.

Solution

(a) The sign of S_{xy} determines positive or negative correlation. Now

$$S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = \sum x_i y_i - \sum x_i\, \bar{y}.$$

Here $\sum x_i y_i = \sum x_i$ (both equal 1500) and $\bar{y} = \frac{10}{20} < 1$, so S_{xy} is positive. Therefore the correlation between x and y is positive.

Computing the estimates,

$$S_{xy} = 1500 - 1500 \times \tfrac{1}{2} = 750,$$

$$S_{xx} = 130{,}000 - \frac{(1500)^2}{20} = 17{,}500,$$

so the slope is

$$\hat{\beta}_1 = \frac{750}{17{,}500} = 0.0429,$$

and the intercept is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0.5 - 0.0429 \times 75 = -2.7143.$$

(b) Replacing the observation $(25, 3)$ by $(35, 2)$, the modified summary quantities are

$$\sum x_i^2 = 130{,}000 - 25^2 + 35^2 = 130{,}600,$$

$$\sum y_i^2 = 88.6 - 3^2 + 2^2 = 83.6,$$

$$\sum x_i y_i = 1500 - 75 + 70 = 1495,$$

$$\sum x_i = 1500 - 25 + 35 = 1510, \qquad \bar{x} = 75.5,$$

$$\sum y_i = 10 - 3 + 2 = 9, \qquad \bar{y} = 0.45.$$

Then

$$S_{xy} = \sum x_i y_i - \frac{\sum x_i \sum y_i}{n} = 1495 - 20 \times 75.5 \times 0.45 = 815,$$

$$S_{xx} = \sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n} = 130{,}600 - \frac{(1510)^2}{20} = 16{,}595.$$

Hence

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = 0.0491, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = -3.26,$$

and the changed regression model is

$$\boxed{\hat{y} = -3.26 + 0.0491\, x.}$$

6.6 Residual Analysis and Goodness of Fit

Example 6.6 (Residuals and R^2)**.** The following data were obtained in an experiment.

x_i	11	15	17	20	13	21
y_i	43.5	40	38.9	38.1	44	39

(a) Compute the residuals for all corresponding x_i, assuming a linear regression model.

(b) Draw a residual plot and interpret it.

(c) How close does the regression model fit the data? Answer with the help of a computed measure.

Solution

The summary quantities are

$$\bar{x} = 16.17, \quad \bar{y} = 40.58, \quad \sum x_i^2 = 1645, \quad \sum y_i^2 = 9914.07, \quad \sum x_i y_i = 3892.8.$$

Then

$$S_{xx} = \sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n} = 76.1866,$$

$$S_{yy} = \sum y_i^2 - \frac{\left(\sum y_i\right)^2}{n} = 33.65, \qquad S_{xy} = -44.27.$$

Hence

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = -0.57, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 49.8,$$

so the fitted line is $\hat{y} = 49.8 - 0.57x$.

(a) The fitted values and residuals $e_i = y_i - \hat{y}_i$ are:

Table 6.1: Fitted values and residuals.

x_i	y_i	$\hat{y}_i$	e_i
11	43.5	43.53	−0.03
15	40.0	41.25	−1.25
17	38.9	40.11	−1.21
20	38.1	38.4	−0.3
13	44.0	42.39	1.61
21	39.0	37.83	1.17

(b) The residual plot is shown below. The plot is sufficiently random, though there are too few points to draw strong conclusions.

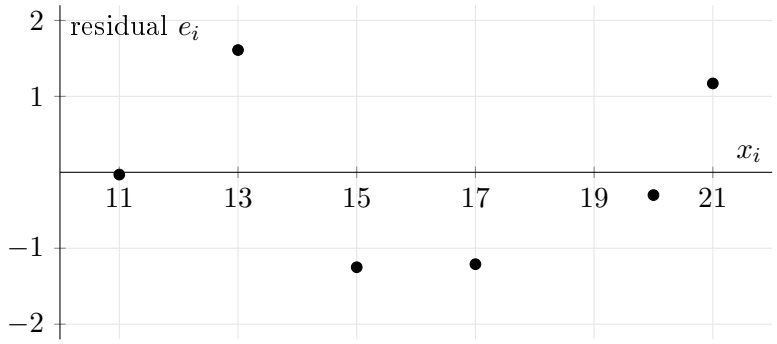


Figure 6.4: Residual plot. The scatter is sufficiently random; the sample size is small.

(c) The goodness of fit is measured by the coefficient of determination,

$$R^2 = \frac{\hat{\beta}_1 S_{xy}}{S_{yy}} = 0.75.$$

Thus about 75% of the variability in the data is explained by the model.

Chapter 7

Formula Summary

This appendix collects the principal formulas of the simple linear regression model in one place for quick reference.

7.1 Model and Assumptions

Model

$$Y = \beta_0 + \beta_1 x + \epsilon, \qquad \mathbb{E}(y \mid x) = \beta_0 + \beta_1 x, \qquad \operatorname{Var}(y \mid x) = \sigma^2.$$

The slope β_1 and intercept β_0 are the regression coefficients; ϵ is an i.i.d. error with mean zero and constant variance σ^2, assumed normally distributed.

7.2 Least Squares Estimation

Estimators

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}.$$

Corrected Sums of Squares

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum x_i\right)\left(\sum y_i\right)}{n},$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum x_i\right)^2}{n} \quad \text{(variability of } x\text{)},$$

$$S_{yy} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - \frac{\left(\sum y_i\right)^2}{n},$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

7.3 Variance and Sums of Squares

Error Variance and Sums of Squares

$$\hat{\sigma}^2 = \frac{\mathrm{SS_E}}{n-2}, \qquad \mathrm{SS_E} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \mathrm{SS_T} - \hat{\beta}_1 S_{xy},$$

$$\mathrm{SS_T} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2, \qquad \mathrm{SS_R} = \hat{\beta}_1 S_{xy}, \qquad \mathrm{SS_T} = \mathrm{SS_R} + \mathrm{SS_E}\,.$$

7.4 Coefficient of Determination

R^2

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_E}}{\mathrm{SS_T}}.$$

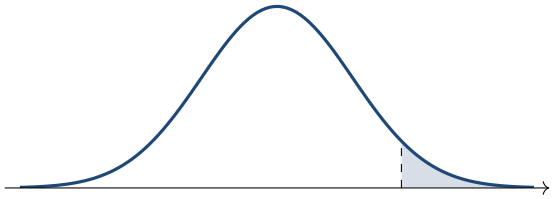
Used to judge the adequacy of the model: the amount of variability in the data explained or accounted for by the model.

7.5 Correlation Coefficient

Sample Correlation Coefficient

$$R = \frac{S_{xy}}{(S_{xx}\ \mathrm{SS_T})^{1/2}}, \qquad \hat{\beta}_1 = \left(\frac{\mathrm{SS_T}}{S_{xx}}\right)^{1/2} R.$$

Unit 5



Chapter 1

Introduction to Hypothesis Testing

Many engineering problems require that we decide whether to accept or reject a statement about some parameter. The statement is called a *hypothesis*, and the decision-making procedure about the hypothesis is called *hypothesis testing*. This chapter builds the conceptual machinery — null and alternative hypotheses, the two kinds of error, the significance level, power, and P-values — that every later test in this unit relies upon.

1.1 Statistical Hypotheses

Statistical Hypothesis

A **statistical hypothesis** is a statement about the parameters of one or more populations.

Since we use probability distributions to represent populations, a statistical hypothesis may also be thought of as a statement about the probability distribution of a random variable. The hypothesis will usually involve one or more parameters of this distribution.

For example, suppose we are interested in the burning rate of a solid propellant used to power aircrew escape systems. The burning rate is a random variable that can be described by a probability distribution. Suppose our interest focuses on the *mean* burning rate (a parameter of this distribution). Specifically, we are interested in deciding whether or not the mean burning rate is $50\,\mathrm{cm\,s^{-1}}$. We may express this as

$$H_0:\ \mu = 50\,\mathrm{cm\,s^{-1}}, \qquad H_1:\ \mu \neq 50\,\mathrm{cm\,s^{-1}}.$$

Null and Alternative Hypotheses

The statement $H_0 : \mu = 50\,\mathrm{cm\,s^{-1}}$ is called the **null hypothesis**, and the statement $H_1 : \mu \neq 50\,\mathrm{cm\,s^{-1}}$ is called the **alternative hypothesis**.

Because the alternative hypothesis specifies values of μ that could be either greater than or less than $50\,\mathrm{cm\,s^{-1}}$, it is called a *two-sided alternative hypothesis*. In some situations we may wish to formulate a *one-sided alternative hypothesis*, as in

$$H_0:\ \mu = 50\,\mathrm{cm\,s^{-1}} \qquad H_0:\ \mu = 50\,\mathrm{cm\,s^{-1}}$$
$$H_1:\ \mu < 50\,\mathrm{cm\,s^{-1}} \quad \text{or} \quad H_1:\ \mu > 50\,\mathrm{cm\,s^{-1}}.$$

Important

Hypotheses are always statements about the population or distribution under study, *not* statements about the sample.

How the null value is chosen

The value of the population parameter specified in the null hypothesis (here $50\,\mathrm{cm\,s^{-1}}$) is usually determined in one of three ways.

1. It may result from past experience or knowledge of the process, or from previous tests or experiments. The objective of hypothesis testing is then usually to determine whether the parameter value has changed.
2. It may be determined from some theory or model regarding the process under study. Here the objective of hypothesis testing is to verify the theory or model.
3. It may result from external considerations, such as design or engineering specifications, or from contractual obligations. In this case the objective of hypothesis testing is *conformance testing*.

A procedure leading to a decision about a particular hypothesis is called a *test of a hypothesis*. Hypothesis-testing procedures rely on using the information in a random sample from the population of interest. If this information is consistent with the hypothesis, we will not reject the hypothesis; however, if this information is inconsistent with the hypothesis, we will conclude that the hypothesis is false.

1.2 The Test Statistic and the Critical Region

In the propellant example, suppose that a sample of $n = 10$ specimens is tested and the sample mean burning rate $\bar{x}$ is observed. The sample mean is an estimate of the true population mean μ. A value of the sample mean $\bar{x}$ that falls close to the hypothesized value $\mu = 50\,\mathrm{cm\,s^{-1}}$ does not conflict with the null hypothesis that the true mean μ is really $50\,\mathrm{cm\,s^{-1}}$. On the other hand, a sample mean that is considerably different from $50\,\mathrm{cm\,s^{-1}}$ is evidence in support of the alternative hypothesis H_1. Thus the sample mean is the *test statistic* in this case.

The sample mean can take on many different values. Suppose that

- if $48.5 \leq \bar{x} \leq 51.5$, we will *not* reject the null hypothesis $H_0 : \mu = 50$; and
- if either $\bar{x} < 48.5$ or $\bar{x} > 51.5$, we will reject the null hypothesis in favour of the alternative hypothesis $H_1 : \mu \neq 50$,

as illustrated in Figure 1.1.

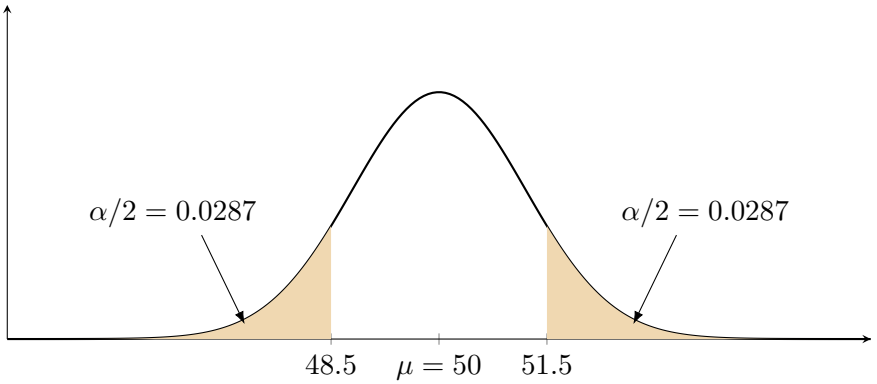


Figure 1.1: Critical region for $H_0 : \mu = 50$ versus $H_1 : \mu \neq 50$ with $n = 10$.

Critical Region, Acceptance Region and Critical Values

The values of $\bar{x}$ that are < 48.5 or > 51.5 constitute the **critical region** (or rejection region) for the test. All values in the interval $48.5 \leq \bar{x} \leq 51.5$ form a region for which we fail to reject the null hypothesis, called the **acceptance region**. The boundaries between the critical and acceptance regions are called the **critical values** (here 48.5 and 51.5).

It is customary to state conclusions relative to the null hypothesis H_0. Therefore, we reject H_0 in favour of H_1 if the test statistic falls in the critical region, and fail to reject H_0 otherwise.

1.3 Type I and Type II Errors

This decision procedure can lead to either of two wrong conclusions.

Type I and Type II Errors

Type I error: rejecting the null hypothesis H_0 when it is true.

Type II error: failing to reject the null hypothesis H_0 when it is false.

The four possible outcomes are summarised in Table 1.1.

Table 1.1: Decisions in hypothesis testing.

Decision	**H_0 is True**	**H_0 is False**
Fail to reject H_0	No error	Type II error
Reject H_0	Type I error	No error

Significance Level α

The probability of making a Type I error is denoted by α:

$$\alpha = \mathrm{P}(\text{Type I error}) = \mathrm{P}(\text{reject } H_0 \mid H_0 \text{ true}).$$

Sometimes the Type I error probability is called the *significance level*, the *α-error*, or the *size of the test*.

Note

1. We can reduce α by widening the acceptance region.
2. We can also reduce α by increasing the sample size.

Type II Error Probability β

The probability of a Type II error is denoted by β:

$$\beta = \text{P(Type II error)} = \text{P(fail to reject } H_0 \mid H_0 \text{ false)}.$$

To calculate β we must have a specific alternative hypothesis, i.e. a particular value of μ for H_1.

Note

1. The probability of making a Type II error, β, increases rapidly as the true value of μ approaches the hypothesized value.
2. The Type II error probability also depends on the sample size n. Increasing the sample size results in a decrease in the probability of a Type II error.

1.4 Four Important Points to Ponder

1. The size of the critical region, and consequently the probability of a Type I error α, can always be reduced by appropriate selection of the critical values.
2. Type I and Type II errors are related. A decrease in the probability of one type of error always results in an increase in the probability of the other, provided that the sample size n does not change.
3. An increase in sample size reduces β, provided that α is held constant.
4. When the null hypothesis is false, β increases as the true value of the parameter approaches the value hypothesized in the null hypothesis. The value of β decreases as the difference between the true mean and the hypothesized value increases.

1.5 Strong versus Weak Conclusions

Generally the analyst controls the Type I error probability α when he or she selects the critical values. Thus it is easy for the analyst to set the Type I error probability at any desired value. Since the analyst can directly control the probability of wrongly rejecting H_0, we always think of rejection of the null hypothesis H_0 as a *strong conclusion*.

On the other hand, the probability of a Type II error β is not constant but depends on the true value of the parameter. It also depends on the sample size we have selected. Since β is a function of both the sample size and the extent to which the null hypothesis H_0 is false, it is customary to think of the decision to accept H_0 as a *weak conclusion*, unless we know that β is acceptably small. Therefore, rather than saying "accept H_0", we prefer the terminology "fail to reject H_0". Failing to reject H_0 implies that we have not found sufficient evidence to reject H_0, i.e. to make a strong statement. Failing to reject H_0 does not necessarily mean that there is a high probability that H_0 is true; it may simply mean that more data are required to reach a strong conclusion.

1.6 Power of a Test

Power

The **power** of a statistical test is the probability of rejecting the null hypothesis H_0 when the alternative hypothesis is true.

The power is computed as $1-\beta$, and can be interpreted as the probability of correctly rejecting a false null hypothesis. We often compare statistical tests by comparing their power properties. For example, if we are considering two tests and one has a higher power than the other, we would prefer the test with the higher power (assuming other properties are comparable).

1.7 One-Sided and Two-Sided Hypotheses

In formulating one-sided alternative hypotheses, remember that rejecting H_0 is always a strong conclusion. Consequently, we should put the statement about which it is important to make a strong conclusion in the alternative hypothesis. In real-world problems, this will often depend on our point of view and experience with the situation.

1.8 P-values in Hypothesis Tests

P-value

The **P-value** is the smallest level of significance that would lead to rejection of the null hypothesis H_0 with the given data.

It is not always easy to compute the exact P-value for a test. However, most modern computer programs for statistical analysis report P-values.

1.9 General Procedure for Hypothesis Tests

Eight-Step Procedure

1. From the problem context, identify the parameter of interest.
2. State the null hypothesis H_0.
3. Specify an appropriate alternative hypothesis H_1.
4. Choose a significance level α.
5. Determine an appropriate test statistic.
6. State the rejection region for the statistic. (This step is not necessary if the P-value approach is used.)
7. Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value.
8. Decide whether or not H_0 should be rejected and report that in the problem context.

1.10 Worked Problems

Example 1.1 (Correctly stated hypotheses)**.** In each of the following situations, state whether it is a correctly stated hypothesis-testing problem and why.

(a) $H_0 : \mu = 25,\ H_1 : \mu \neq 25.$
Yes. We will always state H_0 as an equality involving the specific value of the parameter.

(b) $H_0 : \sigma > 10,\ H_1 : \sigma = 10.$
No. The null hypothesis should specify a value (equality); the alternative should not.

(c) $H_0 : \bar{x} = 50,\ H_1 : \bar{x} \neq 50.$
No. Hypothesis statements are for the population, not for the sample statistic $\bar{x}$.

(d) $H_0 : p = 0.1,\ H_1 : p = 0.5.$
No. The values in the two hypotheses are different; the null and alternative must refer to the same parameter value.

(e) $H_0 : s = 30,\ H_1 : s > 30.$
No. Hypotheses are always statements about the population or distribution under study, not statements about the sample.

Example 1.2 (Semiconductor tool)**.** A semiconductor manufacturer collects data from a new tool and conducts a hypothesis test with the null hypothesis that a critical dimension mean width equals 100 nm. The conclusion is to reject the null hypothesis. Does this result provide strong evidence that the critical-dimension mean equals 100 nm? Explain.

Solution. The truth or falsity of a particular hypothesis can never be known with certainty unless we can examine the entire population. This is usually impossible in most practical situations. Therefore, a hypothesis-testing procedure should be developed with the probability of reaching a wrong conclusion in mind. Rejecting H_0 is a *strong* conclusion, so the result provides strong evidence *against* the mean equalling 100 nm.

Example 1.3 (Connector pull-off force)**.** The mean pull-off force of a connector depends on cure time.

(a) State the null and alternative hypotheses used to demonstrate that the pull-off force is below 25 N.

(b) Assume that the previous test does not reject the null hypothesis. Does this result provide strong evidence that the pull-off force is greater than or equal to 25 N? Explain.

Solution.

(a) The null and alternative hypotheses used to demonstrate that the pull-off force is below 25 N are

$$H_0 : \mu = 25\,\text{N}, \qquad H_1 : \mu < 25\,\text{N}.$$

(b) **No.** The truth or falsity of a particular hypothesis can never be known with certainty unless we can examine the entire population. Failing to reject H_0 is a weak conclusion and does not provide strong evidence that the pull-off force is ≥ 25 N.

Chapter 2

Tests on the Mean of a Normal Distribution, Variance Known

We now specialise the general framework of Chapter 1 to the most fundamental case: testing the mean μ of a normal population when the variance σ^2 is known. This chapter first works through several detailed error-probability calculations, then states the general theory (test statistic, rejection criteria, P-values, Type II error and sample size), and closes with a set of worked problems.

2.1 Computing α and β: Worked Examples

Example 2.1 (Textile fibre — yarn elongation)**.** A textile fibre manufacturer is investigating a new drapery yarn, which the company claims has a mean thread elongation of 12 kg with a standard deviation of 0.5 kg. The company wishes to test the hypothesis $H_0 : \mu = 12$ against $H_1 : \mu < 12$, using a random sample of $n = 4$ specimens.

(a) What is the Type I error probability if the critical region is defined as $\bar{x} < 11.5$ kg?

(b) Find β for the case where the true mean elongation is 11.25 kg.

(c) Find β for the case where the true mean is 11.5 kg.

Solution. **(a)** The probability of making a Type I error is

$$\alpha = \mathrm{P}(\text{Type I error}) = \mathrm{P}(\text{reject } H_0 \mid H_0 \text{ true}).$$

Here $\mu = 12$ kg, $\sigma = 0.5$ kg, and the critical region is $\bar{x} < 11.5$ (reject H_0).

$$\alpha = \mathrm{P}(\bar{x} < 11.5 \mid \mu = 12) = \mathrm{P}\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < \frac{11.5 - \mu}{\sigma/\sqrt{n}} \;\middle|\; \mu = 12 \right)$$

$$= \mathrm{P}\left(Z < \frac{11.5 - 12}{0.5/\sqrt{4}} \right) = \mathrm{P}(Z < -2).$$

From the standard normal tables, for $Z < -2$, the probability is 0.022750.

$$\boxed{\alpha = 0.02275.}$$

(b) The probability of a Type II error is $\beta = \mathrm{P}(\text{fail to reject } H_0 \mid H_0 \text{ false})$. With the true mean $\mu = 11.25$,

$$\beta = \mathrm{P}(\bar{x} > 11.5 \mid \mu = 11.25) = \mathrm{P}\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} > \frac{11.5 - 11.25}{0.5/\sqrt{4}}\right)$$

$$= \mathrm{P}(Z > 1) = 1 - \mathrm{P}(Z \leq 1) = 1 - 0.841345 = 0.15865.$$

$$\boxed{\beta = 0.15865.}$$

(c) With the true mean $\mu = 11.5$,

$$\beta = \mathrm{P}(\bar{x} > 11.5 \mid \mu = 11.5) = \mathrm{P}\left(Z > \frac{11.5 - 11.5}{0.5/\sqrt{4}}\right) = \mathrm{P}(Z > 0) = 1 - \mathrm{P}(Z \leq 0) = 1 - 0.5.$$

$$\boxed{\beta = 0.5.}$$

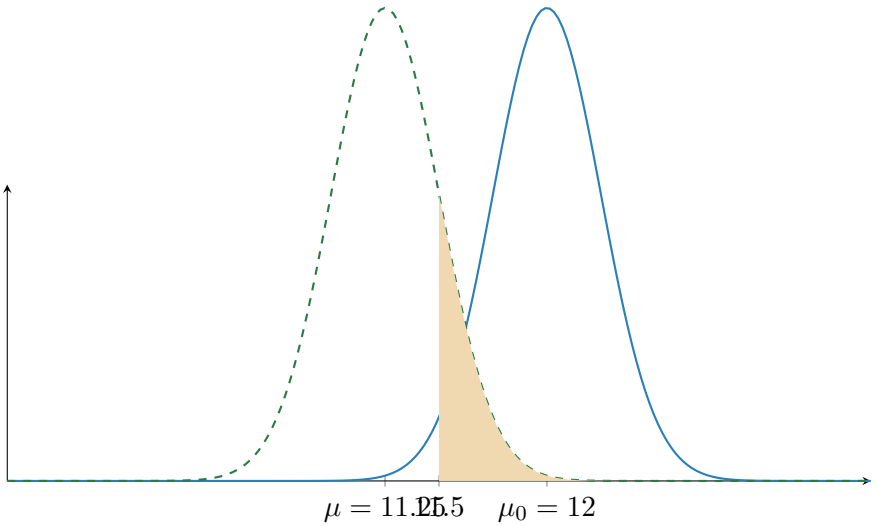


Figure 2.1: Type II error (β, shaded) when the true mean is 11.25: the null curve is centred at $\mu_0 = 12$ and the true curve at $\mu = 11.25$.

Example 2.2 (Specialty parts — customer inspection)**.** The length L of a certain specialty part is normally distributed with a variance of $50\,\mathrm{mm}^2$. A customer wants the mean length to be 100 mm but measures the average length in a randomly selected sample as 103 mm. What is the conclusion about the sample? The sample size is $n = 50$ parts.

Solution.

$$H_0 : \mu = 100 \qquad \text{vs.} \qquad H_1 : \mu \neq 100.$$

The test statistic is

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{103 - 100}{\sqrt{50/50}} = \frac{3}{1} = 3.$$

Taking a level of confidence of 95% and assuming $\alpha = 5\%$,

$$Z_{\alpha/2} = Z_{0.025} = 1.96.$$

Since $Z_0 = 3 > Z_{\alpha/2} = 1.96$, we reject H_0. Hence the mean length of the parts $\neq 100$, and **the customer rejects the sample.**

Example 2.3 (Yarn elongation continued — effect of sample size)**.** Reconsider the textile-fibre problem, but now with $n = 16$ and the same critical region $\bar{x} < 11.5$.

Solution.

1. Type I error:

$$\alpha = \mathrm{P}(\bar{x} < 11.5 \mid \mu = 12) = \mathrm{P}\left(Z < \frac{11.5 - 12}{0.5/\sqrt{16}}\right) = \mathrm{P}\left(Z < -\tfrac{0.5}{0.5/4}\right) = \mathrm{P}(Z < -4) = 0.000.$$

2. Type II error at $\mu = 11.25$:

$$\beta = \mathrm{P}(\bar{x} > 11.5 \mid \mu = 11.25) = \mathrm{P}\left(Z > \frac{11.5 - 11.25}{0.5/\sqrt{16}}\right) = \mathrm{P}\left(Z > \tfrac{0.25}{0.5/4}\right)$$

$$= \mathrm{P}(Z > 2) = 1 - \mathrm{P}(Z \le 2) = 0.022750.$$

3. Type II error at $\mu = 11.5$:

$$\beta = \mathrm{P}(\bar{x} > 11.5 \mid \mu = 11.5) = \mathrm{P}\left(Z > \frac{11.5 - 11.5}{0.5/\sqrt{16}}\right) = \mathrm{P}(Z > 0) = 1 - \mathrm{P}(Z \le 0) = 0.5.$$

Increasing the sample size from 4 to 16 has sharply reduced both α (from 0.02275 to ≈ 0) and β at $\mu = 11.25$ (from 0.15865 to 0.02275), while β at $\mu = 11.5$ (where H_0 is barely false) remains 0.5.

Example 2.4 (Boundary of the critical region)**.** For the yarn problem, find the boundary C of the critical region $\bar{x} < C$ if the Type I error probability is

(a) $\alpha = 0.01$ and $n = 4$;

(b) $\alpha = 0.05$ and $n = 4$;

(c) $\alpha = 0.01$ and $n = 16$.

Solution. **(a)** $\alpha = 0.01$, $n = 4$:

$$0.01 = \mathrm{P}\left(Z \le \frac{C - 12}{0.5/\sqrt{4}}\right).$$

A probability of 0.01 corresponds to $Z = -2.33$, so

$$\frac{C - 12}{0.5/\sqrt{4}} = -2.33 \;\Rightarrow\; C = 11.4175.$$

(b) $\alpha = 0.05$, $n = 4$: $Z = -1.65$, so

$$\frac{C - 12}{0.5/\sqrt{4}} = -1.65 \;\Rightarrow\; C = 11.5875.$$

(c) $\alpha = 0.01$, $n = 16$: $Z = -2.33$, so

$$\frac{C - 12}{0.5/\sqrt{16}} = -2.33 \;\Rightarrow\; C = -0.29125 + 12 = 11.70875.$$

Example 2.5 (Cement — heat evolved)**.** The heat evolved in calories per gram of a cement mixture is approximately normally distributed. The mean is thought to be 100 and the standard deviation is 2. We wish to test $H_0 : \mu = 100$ versus $H_1 : \mu \neq 100$ with a sample of $n = 9$ specimens.

(a) If the acceptance region is defined as $98.5 \leq \bar{x} \leq 101.5$, find the Type I error probability.

(b) Find β for the case where the true mean heat evolved is 103.

(c) Find β for the case where the true mean heat evolved is 105. Compare β in (b) and (c) and state reasons.

Solution. Here $H_0 : \mu = 100$, $H_1 : \mu \neq 100$, $n = 9$.

(a) P(Type I error) $= \alpha$, with acceptance region $98.5 \leq \bar{x} \leq 101.5$:

$$\alpha = \mathrm{P}(\bar{x} < 98.5) + \mathrm{P}(\bar{x} > 101.5)$$

$$= \mathrm{P}\left(Z < \frac{98.5 - 100}{2/\sqrt{9}}\right) + \mathrm{P}\left(Z > \frac{101.5 - 100}{2/\sqrt{9}}\right) = \mathrm{P}\left(Z < \frac{(-1.5) \cdot 3}{2}\right) + \mathrm{P}\left(Z > \frac{4.5}{2}\right)$$

$$= 0.012224 + \big[1 - \mathrm{P}(Z < 2.25)\big] = 0.012224 + 1 - 0.987776 = 0.024448.$$

(b) At $\mu = 103$:

$$\beta = \mathrm{P}(98.5 \leq \bar{x} \leq 101.5 \mid \mu = 103) = \mathrm{P}\left(\frac{98.5 - \mu}{\sigma/\sqrt{n}} \leq Z \leq \frac{101.5 - \mu}{\sigma/\sqrt{n}}\right)$$

$$= \mathrm{P}\left(\frac{98.5 - 103}{2/\sqrt{9}} \leq Z \leq \frac{101.5 - 103}{2/\sqrt{9}}\right) = \mathrm{P}(-6.75 \leq Z \leq -2.25)$$

$$= \mathrm{P}(Z \leq -2.25) - \mathrm{P}(Z \leq -6.75).$$

$$\boxed{\beta = 0.012224.}$$

(c) At $\mu = 105$:

$$\beta = \mathrm{P}\left(\frac{98.5 - 105}{2/\sqrt{9}} \leq Z \leq \frac{101.5 - 105}{2/\sqrt{9}}\right) = \mathrm{P}\left(Z < \frac{101.5 - 105}{2/3}\right) - \mathrm{P}\left(Z \leq \frac{98.5 - 105}{2/3}\right)$$

$$= \mathrm{P}(Z < -5.25) - \mathrm{P}(Z < -9.75) = 0 - 0.$$

$$\boxed{\beta = 0.}$$

We observe that when the true mean is 103 there is about a 1% chance of failing to reject the null hypothesis even though it is false, but when the true mean is 105 we will definitely reject the null hypothesis for the given confidence interval. The larger the departure of the true mean from the hypothesized value, the smaller β becomes.

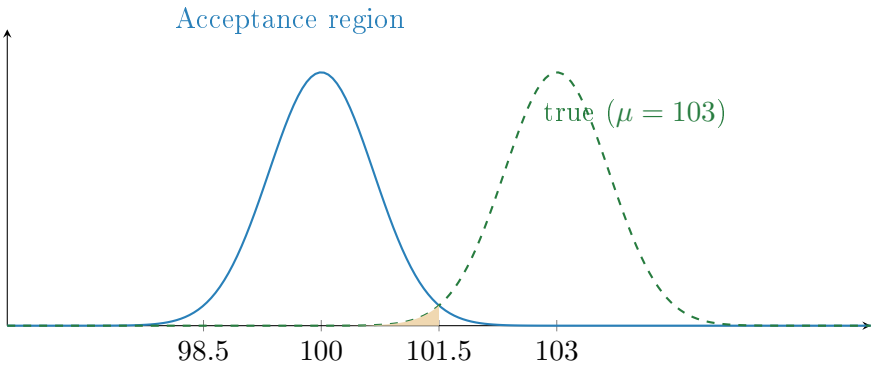


Figure 2.2: Type II error (β, shaded) when the true mean is 103; the acceptance region $[98.5, 101.5]$ is set under the null curve centred at 100.

Example 2.6 (Power supply output voltage)**.** A manufacturer is interested in the output voltage of a power supply used in a PC. Output voltage is assumed to be normally distributed with standard deviation 0.25 V, and the manufacturer wishes to test $H_0 : \mu = 5\,\text{V}$ against $H_1 : \mu \neq 5\,\text{V}$ using $n = 8$ units.

(a) The acceptance region is $4.85 \leq \bar{x} \leq 5.15$. Find the value of α.

(b) Find the power of the test for detecting a true mean output voltage of 5.1 V.

Solution. $H_0 : \mu = 5$ vs. $H_1 : \mu \neq 5$, $n = 8$, $\sigma = 0.25$.

(a)

$$\alpha = \text{P}\left(\frac{4.85 - \mu}{0.25/\sqrt{8}} \leq Z \leq \frac{5.15 - \mu}{0.25/\sqrt{8}}\right)^{c}$$

Evaluated at the boundaries with $\mu = 5$:

$$\alpha = \text{P}(Z \leq -1.70) + \big(1 - \text{P}(Z \leq 1.70)\big) = 0.044565 + (1 - 0.955435)$$

$$= 0.044565 + 0.044565 = 0.089130 \approx 0.089.$$

$$\boxed{\alpha = 0.089130.}$$

(b) Power $= 1 - \beta$ at $\mu = 5.1\,\text{V}$:

$$\beta = \text{P}\left(\frac{4.85 - \mu}{0.25/\sqrt{8}} \leq Z \leq \frac{5.15 - \mu}{0.25/\sqrt{8}} \,\middle|\, \mu = 5.1\right) = \text{P}\left(\frac{4.85 - 5.1}{0.25/\sqrt{8}} \leq Z \leq \frac{5.15 - 5.1}{0.25/\sqrt{8}}\right)$$

$$= \text{P}(-2.8284 \leq Z \leq 0.5657) = \text{P}(Z < 0.5657) - \text{P}(Z < -2.8284) = 0.71186.$$

$$\therefore\ \text{Power} = 1 - \beta = 1 - 0.71186 = \boxed{0.28814.}$$

Example 2.7 (Phoenix toll roads — a proportion via the binomial)**.** The proportion of residents in Phoenix favouring the building of toll roads to complete the freeway system is believed to be $p = 0.3$. If a random sample of 10 residents shows that 1 or fewer favour this proposal, we will conclude $p < 0.3$.

(a) Find the probability of a Type I error if the true proportion is $p = 0.3$.

(b) Find the probability of committing a Type II error with this procedure if $p = 0.2$.

(c) What is the power of this procedure if the true proportion is $p = 0.2$?

Solution. $H_0 : p = 0.3$, $H_1 : p < 0.3$, $n = 10$. Let X be the number of residents in favour of the proposal. Then X is a binomial random variable with parameters $n = 10$ and p:

$$\mathrm{P}(X = x) = \binom{n}{x} p^x (1-p)^{10-x}, \qquad x \in \{0, 1, \ldots, n\}.$$

(a)

$$\alpha = \mathrm{P}(X \le 1 \mid p = 0.3) = \mathrm{P}(X = 0 \mid p = 0.3) + \mathrm{P}(X = 1 \mid p = 0.3)$$
$$= \binom{10}{0} 0.3^0 (1-0.3)^{10} + \binom{10}{1} 0.3^1 (1-0.3)^9 = 1 \cdot 0.7^{10} + 10 \cdot 0.3 \cdot 0.7^9.$$
$$\boxed{\alpha = 0.1493.}$$

(b)

$$\beta = \mathrm{P}(X > 1 \mid p = 0.2) = 1 - \mathrm{P}(X \le 1 \mid p = 0.2) = 1 - \big[\mathrm{P}(X = 0) + \mathrm{P}(X = 1)\big]_{p=0.2}$$
$$= 1 - \left[\binom{10}{0} 0.2^0 (1-0.2)^{10} + \binom{10}{1} 0.2^1 (1-0.2)^9\right] = 1 - \left[1 \cdot 0.8^{10} + 10 \cdot 0.2 \cdot 0.8^9\right].$$
$$\boxed{\beta = 0.6242.}$$

(c) If the true proportion is $p = 0.2$, the power is

$$1 - \beta = 1 - 0.6242 = \boxed{0.3758.}$$

2.2 General Theory: Test on the Mean, Variance Known

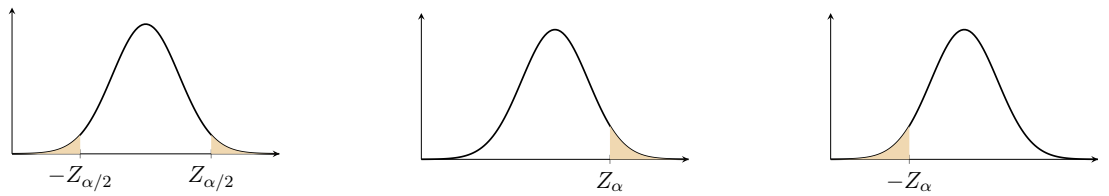
Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal population with known variance σ^2, and let μ_0 be a specified constant. The hypotheses are

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu \ne \mu_0 \quad \text{or} \quad H_1 : \mu > \mu_0 \quad \text{or} \quad H_1 : \mu < \mu_0.$$

Test Statistic (σ known)

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}.$$

Under H_0, Z_0 follows the standard normal distribution $N(0, 1)$.



(a) $H_1 : \mu \ne \mu_0$ (b) $H_1 : \mu > \mu_0$ (c) $H_1 : \mu < \mu_0$

Figure 2.3: The distribution of Z_0 when $H_0 : \mu = \mu_0$ is true, with critical regions (shaded) for (a) the two-sided alternative, (b) the upper one-sided alternative, and (c) the lower one-sided alternative.

Two-sided test

For a two-sided test we reject H_0 if the observed value of the test statistic Z_0 is either

$$Z_0 > Z_{\alpha/2} \quad \text{or} \quad Z_0 < -Z_{\alpha/2}, \tag{1}$$

and we fail to reject H_0 if

$$-Z_{\alpha/2} \le Z_0 \le Z_{\alpha/2}. \tag{2}$$

The two inequalities in (1) define the critical (rejection) region; equation (2) defines the acceptance region. Equivalently, this can be written as:

Two-sided rejection in terms of $\bar{x}$

Reject $H_0 : \mu = \mu_0$ if either $\bar{x} > a$ or $\bar{x} < b$, where

$$a = \mu_0 + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \qquad b = \mu_0 - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

One-sided alternatives

For $H_0 : \mu = \mu_0$ vs. $H_1 : \mu > \mu_0$, the rejection criterion is

$$Z_0 > Z_\alpha \qquad (\text{equivalently } \bar{x} > \mu_0 + Z_\alpha\,\sigma/\sqrt{n}).$$

For $H_0 : \mu = \mu_0$ vs. $H_1 : \mu < \mu_0$, the rejection criterion is

$$Z_0 < -Z_\alpha.$$

2.3 P-values for the Normal Test

P-value (σ known)

$$P = \begin{cases} 2\big[1 - \Phi(|Z_0|)\big], & \text{two-tailed test } (H_1 : \mu \neq \mu_0), \\ 1 - \Phi(Z_0), & \text{upper-tailed test } (H_1 : \mu > \mu_0), \\ \Phi(Z_0), & \text{lower-tailed test } (H_1 : \mu < \mu_0), \end{cases}$$

where $\Phi(z) = \mathrm{P}(Z \le z)$ is the standard normal cumulative distribution function, $Z \sim N(0,1)$.

The P-value is the smallest level of significance that would lead to rejection of the null hypothesis H_0 with the given data.

Type II error for a two-sided test (σ known)

$$\beta = \Phi\left(Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \Phi\left(-Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right),$$

and if $\delta > 0$, approximately $\beta \approx \Phi\left(Z_{\alpha/2} - \dfrac{\delta\sqrt{n}}{\sigma}\right)$, where $\delta = \mu - \mu_0$.

Sample size

$$n \approx \frac{(Z_{\alpha/2} + Z_\beta)^2 \, \sigma^2}{\delta^2}, \qquad \delta = \mu - \mu_0.$$

2.4 Worked Problems

Example 2.8 (Stating hypotheses)**.** State the null and alternative hypotheses in each case.

(a) A hypothesis test will be used to potentially provide evidence that the population mean is greater than 10.

(b) A hypothesis test will be used to potentially provide evidence that the population mean is not equal to 7.

(c) A hypothesis test will be used to potentially provide evidence that the population mean is less than 5.

Solution.

(a) $H_0 : \mu = 10,\ H_1 : \mu > 10$; (b) $H_0 : \mu = 7,\ H_1 : \mu \neq 7$; (c) $H_0 : \mu = 5,\ H_1 : \mu < 5$.

Example 2.9 (Critical values for an upper one-sided test)**.** A hypothesis will be used to test that a population mean equals 10 against the alternative that the population mean is greater than 10, with known variance σ. What are the critical values for the test statistic Z_0 for the following significance levels: (a) 0.01, (b) 0.05, (c) 0.10?

Solution. $H_0 : \mu = 10$ vs. $H_1 : \mu > 10$, test statistic $Z_0 = (\bar{x} - \mu)/(\sigma/\sqrt{n})$. For a one-sided alternative, $P = 1 - \Phi(Z_c)$.

(a) $\alpha = 0.01$: $\Phi(Z_c) = 1 - 0.01 = 0.99 \Rightarrow Z_c = 2.33$.

(b) $\alpha = 0.05$: $\Phi(Z_c) = 1 - 0.05 = 0.95 \Rightarrow Z_c = 1.65$.

(c) $\alpha = 0.10$: $\Phi(Z_c) = 1 - 0.10 = 0.90 \Rightarrow Z_c = 1.29$.

Example 2.10 (Critical values for a two-sided test)**.** A hypothesis will be used to test that a population mean equals 7 against the alternative that the population mean does not equal 7, with known variance σ. What are the critical values for the test statistic Z_0 for significance levels (a) 0.01, (b) 0.05, (c) 0.10?

Solution. $H_0 : \mu = 7$ vs. $H_1 : \mu \neq 7$; P-value $= 2\big[1 - \Phi(|Z_c|)\big]$.

(a) $\alpha = 0.01$: $\mathrm{P}(Z_0 < |Z_c|) = 1 - \frac{0.01}{2} = 0.995 \Rightarrow Z_c = 2.58 \Rightarrow Z_c = \pm 2.58$.

(b) $\alpha = 0.05$: $\mathrm{P}(Z_0 < |Z_c|) = 1 - \frac{0.05}{2} = 0.975 \Rightarrow Z_c = 1.96 \Rightarrow Z_c = \pm 1.96$.

(c) $\alpha = 0.10$: $\mathrm{P}(Z_0 < |Z_c|) = 1 - \frac{0.10}{2} = 0.95 \Rightarrow Z_c = \pm 1.65$.

Example 2.11 (P-values for a two-sided test)**.** For the hypothesis test $H_0 : \mu = 7$ and $H_1 : \mu \neq 7$ with variance known, calculate the P-value for each of the following test statistics: (a) $Z_0 = 2.05$, (b) $Z_0 = -1.84$, (c) $Z_0 = 0.4$.

Solution. $P = 2\big[1 - \Phi(|Z_0|)\big]$.

(a) $Z_0 = 2.05$: $P = 2\big[1 - \Phi(2.05)\big] = 2\big[1 - 0.979818\big] = 2(0.020182) = 0.040364.$

(b) $Z_0 = -1.84$: $P = 2\big[1 - \Phi(1.84)\big] = 2\big[1 - 0.967116\big] = 2(0.032884) = 0.064668$ (i.e. $0.6466\ldots$ per the notes' arithmetic; see remark).

(c) $Z_0 = 0.4$: $P = 2\big[1 - \Phi(0.4)\big] = 2\big[1 - 0.655422\big] = 2(0.344578) = 0.689156 \approx 0.69.$

Example 2.12 (Crankshaft wear)**.** A manufacturer produces crankshafts for an automobile engine. The wear of the crankshaft after 100 000 miles (0.0001 inch) is of interest because it is likely to have an impact on warranty claims. A random sample of $n = 15$ shafts is tested and $\bar{x} = 2.78$. It is known that $\sigma = 0.9$ and that wear is normally distributed.

(a) Test $H_0 : \mu = 3$ versus $H_1 : \mu \neq 3$ using $\alpha = 0.05$.

(b) What is the power of this test if $\mu = 3.25$?

(c) What sample size would be required to detect a true mean of 3.75 if we wanted the power to be at least 0.9?

Solution. $H_0 : \mu = 3$ vs. $H_1 : \mu \neq 3$, $\alpha = 0.05$, $\bar{x} = 2.78$, $\sigma = 0.9$, $n = 15$.

(a)

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{2.78 - 3}{0.9/\sqrt{15}} = -0.9467.$$

Two-sided test, $Z_{\alpha/2} = Z_{0.025} = 1.96 = \pm 1.96$. Rejection criterion: $Z_0 > Z_{\alpha/2}$ or $Z_0 < -Z_{\alpha/2}$. Here $Z_0 = -0.95 > -1.96$ but $-0.95 < 1.96$, so **we fail to reject H_0.**

(b) With $\delta = \mu - \mu_0 = 3.25 - 3 = 0.25$,

$$\beta = \Phi\left(Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \Phi\left(-Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$$

$$= \Phi\left(1.96 - \frac{0.25\sqrt{15}}{0.9}\right) - \Phi\left(-1.96 - \frac{0.25\sqrt{15}}{0.9}\right) = \Phi(0.88) - \Phi(-3.04) = 0.809387.$$

$$\therefore \text{ Power} = 1 - \beta = 1 - 0.80939 = \boxed{0.19061.}$$

(c) Sample size with $\mu = 3.75$, $\delta = 3.75 - 3.0 = 0.75$, power $= 0.9 \Rightarrow \beta = 0.1$, $Z_\beta = Z_{0.1} = 1.28$:

$$n = \frac{(Z_{\alpha/2} + Z_\beta)^2 \sigma^2}{\delta^2} = \frac{(1.96 + 1.28)^2 (0.9)^2}{(0.75)^2} = 15.116.$$

$$\boxed{n \approx 16.}$$

Example 2.13 (Battery life)**.** The life in hours of a battery is known to be approximately normally distributed with standard deviation $\sigma = 1.25\,\text{h}$. A random sample of 10 batteries has a mean life of $\bar{x} = 40.5\,\text{h}$.

(a) Is there evidence to support the claim that battery life exceeds 40 h? Use $\alpha = 0.05$.

(b) What is the P-value for the test in part (a)?

(c) What is the β-error for the test in part (a) if the true mean life is 42 h?

(d) What sample size would be required to ensure that β does not exceed 0.10 if the true mean life is 44 h?

(e) Explain how you could answer the question in part (a) by calculating an appropriate confidence bound on life.

Solution. $H_0 : \mu = 40$ vs. $H_1 : \mu > 40$, $\alpha = 0.05$, $n = 10$, $\sigma = 1.25$.

(a)

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{40.5 - 40}{1.25/\sqrt{10}} = 1.26, \qquad Z_\alpha = Z_{0.05} = 1.65.$$

For a one-sided test $\mu > \mu_0$, the rejection criterion is $Z_0 > Z_\alpha$. Here $Z_0 = 1.26$ and $Z_\alpha = 1.65$; since $Z_0 < Z_\alpha$, **we fail to reject** H_0. There is insufficient evidence to support the claim that battery life exceeds 40 h.

(b)

$$P = \mathrm{P}(Z > Z_0) = 1 - \Phi(Z_0) = 1 - \Phi(1.26) = 1 - 0.896165 = 0.103835.$$

(c) With $\delta = \mu - \mu_0 = 42 - 40 = 2$,

$$\beta = \Phi\left(Z_\alpha - \frac{\delta\sqrt{n}}{\sigma}\right) = \Phi\left(1.65 - \frac{2\sqrt{10}}{1.25}\right) = \Phi(-3.41).$$

$$\boxed{\beta = 0.000325.}$$

(d) With $\mu = 44$, $\delta = 4$, $\beta < 0.1$, $Z_\beta = Z_{0.1} = 1.29$:

$$n = \frac{(Z_\alpha + Z_\beta)^2\sigma^2}{\delta^2} = \frac{(1.65 + 1.29)^2(1.25)^2}{4^2} = 0.844.$$

$$\boxed{n \approx 1.}$$

(e) A $100(1-\alpha)\%$ lower confidence bound on the mean is

$$\bar{x} - Z_\alpha \frac{\sigma}{\sqrt{n}} \le \mu.$$

Thus the 95% lower bound on the mean is

$$40.5 - 1.65 \cdot \frac{1.25}{\sqrt{10}} \le \mu \;\Rightarrow\; 39.85 \le \mu.$$

Since $40 \in (39.85,\ \mu)$, the bound provides evidence to support that battery life ≈ 40 h (consistent with failing to reject H_0 in part a).

Example 2.14 (Tensile strength of a steel alloy)**.** An engineer studying the tensile strength of a steel alloy intended for golf-club shafts knows that tensile strength is approximately normally distributed with $\sigma = 60$ psi. A random sample of 12 specimens has mean tensile strength $\bar{x} = 3450$ psi.

(a) Test the hypothesis that mean strength is 3500 psi. Use $\alpha = 0.01$.

(b) What is the smallest level of significance at which you would be willing to reject the null hypothesis?

(c) What is the β-error for the test in part (a) if the true mean is 3470?

(d) Suppose we wanted to reject the null hypothesis with probability at least 0.8 if mean strength $\mu = 3500$. What sample size should be used?

(e) Explain how you could answer the question in part (a) with a two-sided confidence interval on mean tensile strength.

Solution. Given $n = 12$, $\bar{x} = 3450$, $\sigma = 60$, $H_0 : \mu = 3500$, $H_1 : \mu \neq 3500$.

(a)

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{3450 - 3500}{60/\sqrt{12}} = -2.89.$$

$\alpha = 0.01 \Rightarrow Z_{\alpha/2} = Z_{0.005} = \pm 2.58$. Rejection criterion: $Z_0 < -Z_{\alpha/2}$ or $Z_0 > Z_{\alpha/2}$. Since $Z_0 = -2.89 < -2.58$, we **reject the null hypothesis**, i.e. $\mu = 3500$ is not true.

(b) Smallest level of significance (P-value):

$$P = 2\big[1 - \Phi(|Z_0|)\big] = 2\big[1 - \Phi(2.89)\big] = 2\big[1 - 0.998074\big] = 0.003852.$$

(c) With $\mu_0 = 3500$, $\mu = 3470$, $\delta = \mu - \mu_0 = -30$:

$$\beta = \Phi\left(Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \Phi\left(-Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$$

$$= \Phi\left(2.58 - \frac{-30\sqrt{12}}{60}\right) - \Phi\left(-2.58 - \frac{-30\sqrt{12}}{60}\right) = \Phi(4.31) - \Phi(-0.85)$$

$$= 1 - 0.197662 = \boxed{0.802338.}$$

(d) Sample size with $\mu = 3470$, $\delta = -30$, power $= 0.8 \Rightarrow \beta = 0.2$, $Z_\beta = Z_{0.2} = 0.84$:

$$n = \frac{(Z_{\alpha/2} + Z_\beta)^2\sigma^2}{\delta^2} = \frac{(2.58 + 0.84)^2 \cdot 60^2}{(-30)^2} = 46.78.$$

$$\boxed{n \approx 47.}$$

(e) A $100(1-\alpha)\%$ two-sided confidence interval on the mean is

$$\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

$$3450 - 2.58 \cdot \frac{60}{\sqrt{12}} \leq \mu \leq 3450 + 2.58 \cdot \frac{60}{\sqrt{12}} \;\Rightarrow\; 3405.31 \leq \mu \leq 3494.67.$$

As 3500 is not in the interval, we reject H_0, i.e. $\mu = 3500$ is not true for $\alpha = 0.01$.

Example 2.15 (Bearing inside diameter)**.** A bearing used in an automotive application is supposed to have a nominal inside diameter of 1.5 inches. A random sample of 25 bearings is selected, and the average inside diameter is 1.4975 inches. Bearing diameter is known to be normally distributed with $\sigma = 0.01$ inch.

(a) Test $H_0 : \mu = 1.5$ vs. $H_1 : \mu \neq 1.5$ using $\alpha = 0.01$.

(b) What is the P-value for the test?

(c) Compute the power of the test if the true mean diameter is 1.495 inches.

(d) What sample size would be required to detect a true mean diameter as low as $1.495''$ if we wanted the power of the test to be at least 0.9?

(e) Explain how part (a) can be answered by constructing a two-sided confidence interval on the mean diameter.

Solution. $n = 25$, $\bar{x} = 1.4975''$, $\sigma = 0.01''$, $H_0 : \mu = 1.5$, $H_1 : \mu \neq 1.5$, $\alpha = 0.01$.

(a)

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1.4975 - 1.5}{0.01/\sqrt{25}} = -1.25, \qquad Z_{\alpha/2} = Z_{0.005} = \pm 2.58.$$

Rejection criterion: $Z_0 < -Z_{\alpha/2}$ or $Z_0 > Z_{\alpha/2}$. Since $Z_0 = -1.25 > -2.58$ and $-1.25 < 2.58$, **we fail to reject** H_0; hence $H_0 : \mu = 1.5$ is (taken to be) true.

(b)

$$P = 2\big[1 - \Phi(|Z_0|)\big] = 2\big[1 - \Phi(1.25)\big] = 2\big[1 - 0.894350\big] = 0.2113.$$

(c) Power $= 1 - \beta$. With $\delta = \mu - \mu_0 = 1.495 - 1.5 = -0.005$,

$$\beta = \Phi\left(Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \Phi\left(-Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$$

$$= \Phi\left(2.58 - \frac{-0.005\sqrt{25}}{0.01}\right) - \Phi\left(-2.58 - \frac{-0.005\sqrt{25}}{0.01}\right)$$

$$= \Phi(5.08) - \Phi(-0.08) = 1 - 0.468119 = 0.531881.$$

$$\therefore \text{ Power} = 1 - \beta = 1 - 0.531881 = \boxed{0.468119.}$$

(d) Sample size with $\delta = 1.495 - 1.5 = -0.005$, power $= 0.9 \Rightarrow \beta = 0.1$, $Z_{0.1} = 1.28$:

$$n = \frac{(Z_{\alpha/2} + Z_\beta)^2\sigma^2}{\delta^2} = \frac{(2.58 + 1.28)^2 \cdot 0.01^2}{(-0.005)^2} = 59.6.$$

$$\boxed{n \approx 60.}$$

(e) A $100(1-\alpha)\%$ two-sided confidence interval on the mean:

$$\bar{x} - Z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}$$

$$1.4975 - 2.58 \cdot \frac{0.01}{\sqrt{25}} \leq \mu \leq 1.4975 + 2.58 \cdot \frac{0.01}{\sqrt{25}} \;\Rightarrow\; 1.49234 \leq \mu \leq 1.50266.$$

Since 1.5 is in the interval $[1.49234,\ 1.50266]$, we conclude that $H_0 : \mu = 1.5$ is true and cannot be rejected.

Chapter 3

Tests on the Variance and Standard Deviation of a Normal Distribution

Many engineering problems concern the *variability* of a process rather than its mean. This chapter develops the chi-square test for the variance σ^2 (equivalently the standard deviation σ) of a normal distribution, together with the associated confidence bounds.

3.1 Hypothesis Tests on the Variance

The hypotheses are

$$H_0 : \sigma^2 = \sigma_0^2, \qquad H_1 : \sigma^2 \neq \sigma_0^2.$$

Chi-square Test Statistic

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2}.$$

Under H_0, χ_0^2 follows a chi-square distribution with $n-1$ degrees of freedom.

Rejection Criteria

For the two-sided alternative $H_1 : \sigma^2 \neq \sigma_0^2$, reject H_0 if

$$\chi_0^2 > \chi^2_{\alpha/2,\,n-1} \quad \text{or} \quad \chi_0^2 < \chi^2_{1-\alpha/2,\,n-1}.$$

For the one-sided alternatives:

$$H_1 : \sigma^2 > \sigma_0^2 : \quad \chi_0^2 > \chi^2_{\alpha,\,n-1}; \qquad H_1 : \sigma^2 < \sigma_0^2 : \quad \chi_0^2 < \chi^2_{1-\alpha,\,n-1}.$$

Table 3.1: Rejection criteria for tests on the variance.

Alternative hypothesis	Rejection criterion
$H_1 : \sigma^2 \neq \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha/2,n-1}$ or $\chi_0^2 < \chi^2_{1-\alpha/2,n-1}$
$H_1 : \sigma^2 > \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha,n-1}$
$H_1 : \sigma^2 < \sigma_0^2$	$\chi_0^2 < \chi^2_{1-\alpha,n-1}$

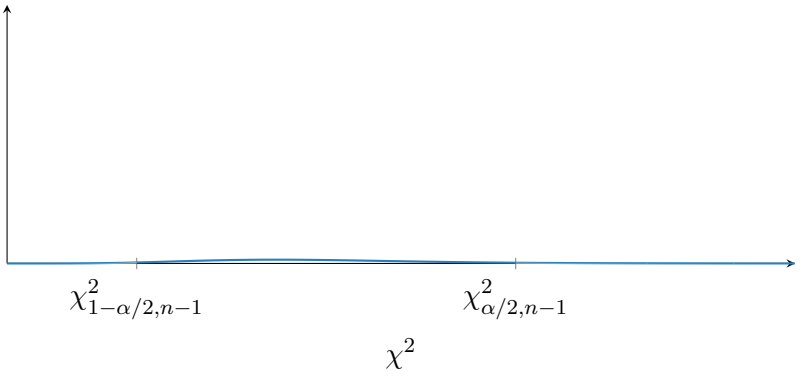


Figure 3.1: Chi-square distribution with $n-1$ degrees of freedom, showing the two-sided critical regions (shaded).

3.2 Worked Problems

Example 3.1 (Detergent filling machine)**.** An automated filling machine is used to fill bottles with liquid detergent. A random sample of 20 bottles results in a sample variance of fill volume of $s^2 = 0.0153$ (fluid ounces)^2. If the variance of fill volume exceeds 0.01 (fluid ounces)^2, an unacceptable proportion of bottles will be underfilled or overfilled. Is there evidence in the sample data to suggest that the manufacturer has a problem with underfilled or overfilled bottles? Use $\alpha = 0.05$ and assume that fill volume has a normal distribution.

Solution (eight-step procedure).

1. The parameter of interest is the population variance σ^2.
2. $H_0 : \sigma^2 = 0.01$.
3. $H_1 : \sigma^2 > 0.01$.
4. $\alpha = 0.05$.
5. Test statistic: $\chi^2_0 = \dfrac{(n-1)s^2}{\sigma^2_0}$.
6. Reject H_0 if $\chi^2_0 > \chi^2_{0.05,19} = 30.14$.
7. Computation:
$$\chi^2_0 = \frac{19 \times 0.0153}{0.01} = 29.07.$$
8. Conclusion: Since $\chi^2_0 = 29.07 < \chi^2_{0.05,19} = 30.14$, we conclude that there is no strong evidence that the variance of fill volume exceeds 0.01 (fluid ounces)^2.

Example 3.2 (Critical values, two-sided variance test)**.** Consider the test of $H_0 : \sigma^2 = 7$ against $H_1 : \sigma^2 \neq 7$. What are the critical values for the test statistic χ^2_0 for the following significance levels and sample sizes?

(a) $\alpha = 0.01$ and $n = 20$;

(b) $\alpha = 0.05$ and $n = 12$;

(c) $\alpha = 0.10$ and $n = 15$.

Solution.

(a) $\alpha = 0.01$, $n = 20$: $\alpha/2 = 0.005$, $1 - \alpha/2 = 0.995$; for $n - 1 = 19$, the critical values are 6.84 and 38.58.

(b) $\alpha = 0.05$, $n = 12$: $\alpha/2 = 0.025$, $1 - \alpha/2 = 0.975$; for $n - 1 = 11$, the critical values are 3.82 and 21.92.

(c) $\alpha = 0.10$, $n = 15$: $\alpha/2 = 0.05$, $1 - \alpha/2 = 0.95$; for $n - 1 = 14$, the critical values are 6.57 and 23.68.

Example 3.3 (Critical values, upper one-sided variance test)**.** Consider the test of $H_0 : \sigma^2 = 10$ against $H_1 : \sigma^2 > 10$. What are the critical values for the test statistic χ_0^2 for the following significance levels and sample sizes?

(a) $\alpha = 0.01$ and $n = 20$;

(b) $\alpha = 0.05$ and $n = 12$;

(c) $\alpha = 0.10$ and $n = 15$.

Solution.

(a) $\alpha = 0.01$, $n - 1 = 19$: critical value $\chi^2_{\alpha,n-1} = \chi^2_{0.01,19} = 36.19$.

(b) $\alpha = 0.05$, $n - 1 = 11$: critical value $\chi^2_{0.05,11} = 19.68$.

(c) $\alpha = 0.10$, $n - 1 = 14$: critical value $\chi^2_{0.1,14} = 21.06$.

Example 3.4 (Critical values, lower one-sided variance test)**.** For $H_0 : \sigma^2 = 5$ against $H_1 : \sigma^2 < 5$, find the critical values.

(a) $\alpha = 0.01$, $n - 1 = 19$: use $1 - \alpha = 0.99$; $\chi^2_{0.99,19} = 7.63$.

(b) $\alpha = 0.05$, $n - 1 = 11$: use $1 - \alpha = 0.95$; $\chi^2_{0.95,11} = 4.57$.

(c) $\alpha = 0.10$, $n - 1 = 14$: use $1 - \alpha = 0.90$; $\chi^2_{0.90,14} = 7.79$.

Example 3.5 (Ice-hockey performance after electrostimulation)**.** The data from *Medicine and Science in Sports and Exercise* considered ice hockey player performance after electrostimulation training. In summary there were 17 players and the sample standard deviation of performance was 0.09 seconds.

(a) Is there strong evidence to conclude that the standard deviation of performance time exceeds the historical value of 0.75 seconds? Use $\alpha = 0.05$. Find the P-value for this test.

(b) Discuss how part (a) could be answered by constructing a 95% one-sided confidence interval for σ.

Solution. $n = 17$, $s = 0.09$ sec, $H_0 : \sigma^2 = 0.75$, $H_1 : \sigma > 0.75$, $\alpha = 0.05$.

(a)

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{16 \times (0.09)^2}{(0.75)^2} = 0.2304.$$

$$\chi^2_{\text{critical}} = \chi^2_{0.05,\,n-1} = \chi^2_{0.05,16} = 26.30.$$

Since $\chi_0^2 < \chi^2_{\text{critical}}$ (i.e. $0.2304 < 26.30$), do not reject H_0 (fail to reject H_0). Hence σ^2 may be 0.5625, or $\sigma = 0.75$.

P-value: the probability above χ_0^2 here is > 0.995 (based on the calculated chi-square value; check the table for which α value is nearest, and the P-value for $H_1 : \sigma > \sigma_0$ will be greater than that α value).

(b) A $100(1-\alpha)\%$ (here 95%) lower confidence bound for σ^2 is

$$\frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} \le \sigma^2,$$

with $\alpha = 0.05$, $n = 16$ (as used in the notes), $\chi^2_{0.05,16} = 26.30$ (and $\chi^2_{0.95,16} = 7.96$ for reference):

$$\frac{(17-1)(0.09)^2}{26.30} \le \sigma^2 \;\Rightarrow\; \boxed{0.004 \le \sigma^2} \quad \text{or} \quad \boxed{0.07 \le \sigma.}$$

Example 3.6 (Two-sided variance test — concepts)**.** (i) What are the hypothesis statements and the statistic to be used in a two-sided test of the variance of a population? What are the assumptions to use this statistic, and what is the rejection region for this test?

(ii) What is the value x such that a chi-square random variable with 15 degrees of freedom takes on at most a value of x with 95% likelihood?

Solution.

(i) Hypotheses (two-sided, with one-sided variants listed):

$$H_0 : \sigma^2 = \sigma_0^2 \quad \text{vs.} \quad H_1 : \sigma^2 \ne \sigma_0^2 \qquad \big(H_1 : \sigma^2 < \sigma_0^2 \;\text{ or }\; H_1 : \sigma^2 > \sigma_0^2\big).$$

Statistic: $\chi_0^2 = \dfrac{(n-1)s^2}{\sigma^2}$. Rejection region: χ_0^2 outside $\big(\chi^2_{1-\alpha/2,\,n-1},\ \chi^2_{\alpha/2,\,n-1}\big)$, i.e. $\chi_0^2 > \chi^2_{\alpha,n-1}$ or $\chi_0^2 < \chi^2_{1-\alpha,n-1}$ (as appropriate). *Assumption:* the population is normally distributed.

(ii)

$$\mathrm{P}(\chi^2_{15} < x) = 0.95 \;\Rightarrow\; \mathrm{P}(\chi^2_{15} > x) = 0.05 \;\Rightarrow\; x = \chi^2_{0.05,\,15}.$$

$$\boxed{x = 25.}$$

Example 3.7 (Variance greater than 10 — sample data)**.** Test at 10% significance whether the variance of the population is greater than 10, with the help of the sample data

$$24,\ 22,\ 25,\ 23,\ 20,\ 23,\ 21,\ 22,\ 25,\ 22.$$

How can you answer this question with the help of confidence intervals?

Solution. First compute the sample statistics ($n = 10$):

$$\bar{X} = 22.7, \qquad s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = 2.6778, \qquad s = 1.63639.$$

Hypotheses: $H_0 : \sigma^2 = 10$ vs. $H_1 : \sigma^2 > 10$, $\alpha = 0.1$.

Test statistic:

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{9 \times (1.63639)^2}{10} = 2.4.$$

Critical value: $\chi^2_{0.1,9} = 14.68$ (from tables).

Since $\chi_0^2 < \chi^2_{0.1,9}$ (i.e. $2.4 < 14.68$), do not reject H_0; conclude that σ^2 may be $= 10$.

(b) 90% lower limit of σ^2: a $100(1-\alpha)\%$ bound gives

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{9 \times (1.6364)^2}{14.68} = 1.642.$$

The interval suggests that σ^2 may be 10.

Example 3.8 (Tyre life — standard deviation)**.** A research engineer for a tyre manufacturer is investigating tyre life for a new rubber compound and has built 16 tyres and tested them to end-of-life in a road test. The sample mean and standard deviation are 60139.7 and 3645.94 km.

(i) Can you conclude, using $\alpha = 0.05$, that the standard deviation of tyre life exceeds 200 km? State any necessary assumption about the underlying distribution of the data.

(ii) Answer question (i) by constructing an appropriate confidence interval.

Solution. Given $\bar{X} = 60139.7$ km, $s = 3645.94$ km, $n = 16$, $\alpha = 0.05$, $H_0 : \sigma = 200$, $H_1 : \sigma > 200$.

(i)

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{15 \times (3645.94)^2}{200^2} = 4984.8 \approx 4985.$$

Critical value: $\chi^2_{0.05,15} = 25$. Since $\chi_0^2 > \chi^2_{0.05,15}$, we reject H_0; therefore $\sigma > 200$. *Assumption:* the tyre life is normally distributed.

(ii) A $100(1-\alpha)\%$ (here 95%) lower-bound confidence interval on σ is

$$\sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}}} \leq \sigma \;\Rightarrow\; \sigma \geq \sqrt{\frac{15 \times (3645.94)^2}{25}} \;\Rightarrow\; \sigma \geq 2824.13.$$

Hence we conclude that σ is far greater than 200.

Chapter 4

Tests on the Mean of a Normal Distribution, Variance Unknown

In practice the population variance σ^2 is rarely known. When it must be estimated from the sample standard deviation s, the appropriate reference distribution is Student's t rather than the standard normal. This chapter develops the t-test for the mean.

4.1 The t Distribution

If $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution with mean μ and variance σ^2, the random variable

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

has a t distribution with $n - 1$ degrees of freedom.

The general appearance of the t distribution is similar to the standard normal distribution in that both distributions are symmetric and unimodal, and the maximum ordinate value is reached when the mean $\mu = 0$. However, the t distribution has heavier tails than the normal; that is, it has more probability in the tails than the normal distribution.

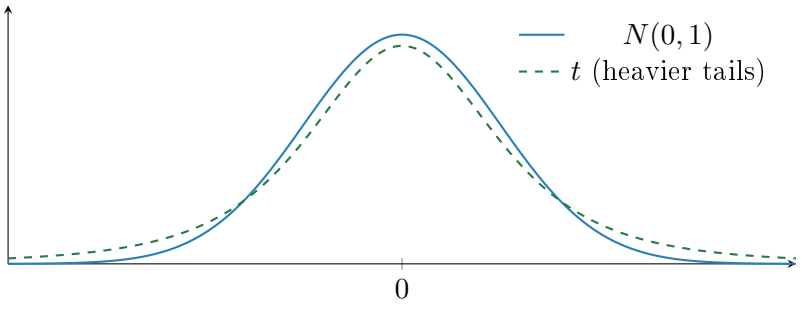


Figure 4.1: The t distribution compared to the standard normal: same symmetric, unimodal shape but heavier tails.

4.2 Hypothesis Tests on the Mean (σ unknown)

The hypotheses are $H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$.

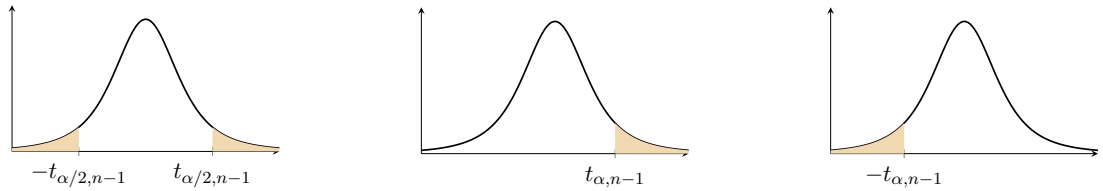
Test Statistic (σ unknown)

$$T_0 = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}.$$

If the null hypothesis is true, T_0 has a t distribution with $n-1$ degrees of freedom. We use the t percentage points $-t_{\alpha/2,n-1}$ and $t_{\alpha/2,n-1}$ as the boundaries of the critical region.

Table 4.1: Rejection criteria for the t-test on the mean.

Alternative hypothesis	**Rejection criterion**
$H_1 : \mu \neq \mu_0$	$t_0 > t_{\alpha/2,\, n-1}$ or $t_0 < -t_{\alpha/2,\, n-1}$
$H_1 : \mu > \mu_0$	$t_0 > t_{\alpha,\, n-1}$
$H_1 : \mu < \mu_0$	$t_0 < -t_{\alpha,\, n-1}$



(a) $H_1 : \mu \neq \mu_0$ (b) $H_1 : \mu > \mu_0$ (c) $H_1 : \mu < \mu_0$

Figure 4.2: The reference t_{n-1} distribution with critical regions (shaded) for the three alternative hypotheses.

4.3 P-value for a t-test

The P-value for a t-test is the smallest level of significance at which the null hypothesis would be rejected. It is the tail area beyond the value of the test statistic for a one-sided test, or twice this area for a two-sided test.

Computation of the exact P-value directly from a t-table is usually impossible. However, it is easy to find upper and lower bounds on the P-value from the table.

Example 4.1 (Bounding a P-value from the t-table)**.** Consider the following critical values for a t-test based on 14 degrees of freedom.

Critical value	0.258	0.692	1.345	1.761	2.145	2.624	2.977
Tail area	0.4	0.25	0.1	0.05	0.025	0.01	0.005

If in a problem $t_0 = 2.72$, this lies between the values 2.624 and 2.977; therefore the P-value must be between 0.01 and 0.005. These are effectively the upper and lower bounds on the P-value.

4.4 Worked Problems

Example 4.2 (Critical values for a two-sided t-test)**.** A hypothesis will be used to test that a population mean equals 7 against the alternative that the population mean does not equal 7

with unknown variance σ. What are the critical values for the test statistic T_0 for the following significance levels and sample sizes?

(a) $\alpha = 0.01$ and $n = 20$;

(b) $\alpha = 0.05$ and $n = 12$;

(c) $\alpha = 0.10$ and $n = 15$.

Solution. $H_0 : \mu = 7$, $H_1 : \mu \neq 7$.

(a) $\alpha = 0.01$, $\alpha/2 = 0.005$, $n - 1 = 19$: $t_{\alpha/2,n-1} = t_{0.005,19} = 2.861$; $-t_{0.005,19} = -2.861$; so $-2.861 \leq t_0 \leq 2.861$.

(b) $\alpha = 0.05$, $n = 12$, $\alpha/2 = 0.025$: $t_{\alpha/2,n-1} = 2.201$; so $-2.201 \leq t_0 \leq 2.201$.

(c) $\alpha = 0.10$, $n = 15$, $\alpha/2 = 0.05$, $n - 1 = 14$: $t_{0.05,14} = 1.761$; so $-1.761 \leq t_0 \leq 1.761$.

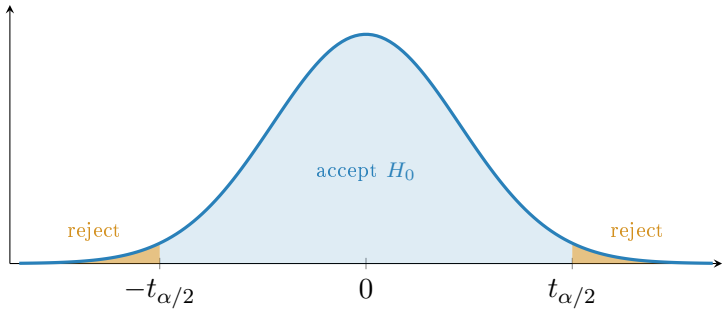


Figure 4.3: Two-sided rejection region for the t-test: reject H_0 when $|t_0| > t_{\alpha/2,n-1}$.

Example 4.3 (Critical values for an upper one-sided t-test)**.** $H_0 : \mu = 10$, $H_1 : \mu > 10$, variance σ unknown. Find the critical value.

(a) $\alpha = 0.01$, $n = 20$;

(b) $\alpha = 0.05$, $n = 12$;

(c) $\alpha = 0.10$, $n = 15$.

Solution.

(a) $\alpha = 0.01$, $n = 20$: critical value $t_{\alpha,n-1} = t_{0.01,19} = 2.539$; so $t_0 > 2.539$.

(b) $\alpha = 0.05$, $n = 12$: $t_{0.05,11} = 1.796$; so $t_0 > 1.796$.

(c) $\alpha = 0.10$, $n = 15$: $t_{0.1,14} = 1.345$; so critical value > 1.345.

Example 4.4 (Critical values for a two-sided t-test at $\mu = 5$)**.** $H_0 : \mu = 5$, $H_1 : \mu \neq 5$, variance σ unknown. (a) $\alpha = 0.01$, $n = 20$; (b) $\alpha = 0.05$, $n = 12$; (c) $\alpha = 0.10$, $n = 15$.

Solution (answers).

1. Critical value < 2.539 (i.e. $|t_0|$ compared to 2.539).
2. Critical value < 1.796.
3. Critical value < 1.345.

Example 4.5 (Bounding P-values, two-sided t-test)**.** For the hypothesis test $H_0 : \mu = 7$ against $H_1 : \mu \neq 7$ with variance unknown and $n = 20$, approximate the P-value for each of the following test statistics: (a) $t_0 = 2.05$, (b) $t_0 = -1.84$, (c) $t_0 = 0.4$.

Solution. Use $n - 1 = 19$ degrees of freedom.

(a) $t_0 = 2.05$: for $n - 1 = 19$, from the tables $t_0 = 2.05$ lies between 1.729 and 2.093, i.e. for $\alpha/2$ between 0.05 and 0.025 (so α between 0.1 and 0.05). Hence $0.05 \leq P \leq 0.1$.

(b) $t_0 = -1.84$: also lies with $\alpha/2$ between 0.05 and 0.025; hence $0.05 \leq P \leq 0.1$.

(c) $t_0 = 0.4$: with $\alpha/2$ between 0.40 and 0.25; hence $0.5 \leq P \leq 0.8$.

Example 4.6 (Bounding P-values, lower one-sided t-test)**.** For the hypothesis test $H_0 : \mu = 5$ against $H_1 : \mu < 5$ with variance unknown and $n = 12$, approximate the P-value for each of the following test statistics: (a) $t_0 = 2.05$, (b) $t_0 = -1.84$, (c) $t_0 = 0.4$.

Solution. Use $n - 1 = 11$ degrees of freedom.

(a) $t_0 = 2.05$: lies with tail area between 0.05 and 0.025, so $0.95 \leq P \leq 0.975$.

(b) $t_0 = -1.84$: lies with tail area between 0.05 and 0.025, so $0.025 \leq P \leq 0.05$.

(c) $t_0 = 0.4$: tail area between 0.40 and 0.25; with $1 - \alpha$ giving 0.6 to 0.75, so $0.6 \leq P \leq 0.75$.

Example 4.7 (Machine installation time)**.** The time in hours to install and commission a machine is as follows for a sample of 10 machines chosen at random:

$$192,\ 201,\ 190,\ 183,\ 185,\ 176,\ 191,\ 214,\ 220,\ 205.$$

(i) If the above data is assumed to come from a normal population, can you conclude that the population mean is greater than 205 hours at a 5% significance level?

(ii) Derive an interval estimate to answer the same question.

Solution. $H_0 : \mu = 205$, $H_1 : \mu > 205$ hr, $\alpha = 0.05$. Test statistic $t_0 = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$.

Sample statistics:

$$\bar{X} = 195.7, \qquad s^2 = \frac{1772.1}{10 - 1} = 196, \qquad s = 14.032.$$

$$t_0 = \frac{195.7 - 205}{14.0321/\sqrt{10}} = -2.0959, \qquad t_{\alpha,n-1} = t_{0.05,9} = 1.833.$$

Since $t_0 < t_{\alpha,n-1}$, do not reject H_0: μ may be 205, not > 205 hours.

(ii) Interval estimate. Two-sided:

$$\bar{X} \pm t_{\alpha/2}\,\frac{s}{\sqrt{n}} = 195.7 \pm 2.262 \cdot \frac{14.0321}{\sqrt{10}} = (185.66,\ 205.74).$$

Lower bound:

$$\mu \geq \bar{X} - t_{\alpha}\,\frac{s}{\sqrt{n}} = 195.7 - 1.833 \cdot \frac{14.052}{\sqrt{10}} = 187.57.$$

Both the two-sided and the lower-bound confidence intervals include 205 hr; therefore we cannot reject the claim that $\mu = 205$ hr.

Example 4.8 (Overweight documentary)**.** A television documentary on overrating weight claimed that Americans are about 10 pounds overweight on average. To test the claim, 18 randomly selected individuals were examined; their average excess weight was found to be 12.4 pounds and the sample standard deviation was 2.7 pounds. At a significance level of 0.01, is there any reason to doubt the validity of the claimed 10-pound value?

Solution. $\bar{X} = 12.4$, $s = 2.7$, $\alpha = 0.01$, $n = 18$, $H_0 : \mu = 10$ pounds vs. $H_1 : \mu > 10$ lb.

$$t_0 = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{12.4 - 10}{2.7/\sqrt{18}} = 3.77, \qquad t_{\alpha,n-1} = t_{0.01,17} = 2.567.$$

Since $t_0 > t_{\alpha,n-1}$, reject H_0. The data show that Americans are overweight by more than 10 pounds on average, and the claim of just 10 lb overweight is not true.

Example 4.9 (Probability from the t distribution)**.** State the probability of a t-random variable with 25 degrees of freedom taking on any value between 2.06 and 3.45.

Solution.

$$\mathrm{P}(2.06 < T_{25} < 3.45) = \mathrm{P}(T > 2.06) - \mathrm{P}(T > 3.45) = 0.025 - 0.001 = 0.024.$$

Chapter 5

Testing for Goodness of Fit

The tests of the previous chapters assumed a known form of distribution (normal) and tested its parameters. We now consider a different kind of hypothesis: we do *not* know the underlying distribution of the population, and we wish to test the hypothesis that a particular distribution will be satisfactory as a population model.

5.1 The Chi-Square Goodness-of-Fit Test

Hypothesis-testing procedures can be of two types: problems in which the probability distribution is known and the hypotheses involve the parameters of the distribution; and another kind, in which we do not know the underlying distribution of the population and we wish to test the hypothesis that a particular distribution will be satisfactory as a population model.

The goodness-of-fit test procedure requires a random sample of size n from the population whose probability distribution is unknown. These n observations are arranged in a frequency histogram having k bins or class intervals. Let O_i be the observed frequency in the ith class interval. From the hypothesized probability distribution, we compute the expected frequency in the ith class interval, denoted E_i.

Goodness-of-Fit Test Statistic

$$\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}.$$

It can be shown that, if the population follows the hypothesized distribution, χ_0^2 has approximately a chi-square distribution with $k - p - 1$ degrees of freedom, where p represents the number of parameters of the hypothesized distribution estimated by sample statistics. This approximation improves as n increases.

Rejection Criterion

We reject the hypothesis that the distribution of the population is the hypothesized distribution if the calculated value of the test statistic

$$\chi_0^2 > \chi_{\alpha,\, k-p-1}^2.$$

On small expected frequencies

If the expected frequencies are too small, the test statistic χ_0^2 will not reflect the departure of observed from expected, but only the small magnitude of the expected frequencies. There is no general agreement regarding the minimum value of expected frequencies, but values of 3, 4, and 5 are widely used as minimal. If an expected frequency is too small, it can be combined with the expected frequency in an adjacent class interval. The corresponding observed frequencies would then also be combined, and k would be reduced by 1. Class intervals are not required to be of equal width.

5.2 Worked Problems

Example 5.1 (Defects in printed circuit boards — Poisson fit)**.** The number of defects in printed circuit boards is hypothesized to follow a Poisson distribution. A random sample of $n = 60$ printed circuit boards has been collected, and the following number of defects observed. Perform a test of hypothesis.

Table 5.1: Observed defect frequencies.

Number of defects	Observed frequency
0	32
1	15
2	9
3	4

Solution (eight-step procedure).

1. The variable of interest is the form of the distribution of defects in printed circuit boards.
2. H_0: The form of the distribution of defects is Poisson.
3. H_1: The form of the distribution of defects is not Poisson.
4. $\alpha = 0.05$.
5. Test statistic χ_0^2. The mean of the distribution is not known, so to estimate it we use the sample average:

$$\hat{\lambda} = \frac{32 \cdot 0 + 1 \cdot 15 + 2 \cdot 9 + 3 \cdot 4}{60} = 0.75.$$
$$\therefore \ \lambda = 0.75.$$

Computations. We compute p_i, the theoretical (hypothesized) probability associated with the ith class interval. Since each class interval corresponds to a number of defects, we find p_i as follows:

$$p_1 = \mathrm{P}(X = 0) = \frac{e^{-\lambda}\lambda^0}{0!} = \frac{e^{-0.75}\,0.75^0}{0!} = 0.472,$$
$$p_2 = \mathrm{P}(X = 1) = \frac{e^{-0.75}\,0.75^1}{1!} = 0.354,$$
$$p_3 = \mathrm{P}(X = 2) = \frac{e^{-0.75}\,0.75^2}{2!} = 0.133,$$
$$p_4 = \mathrm{P}(X \geq 3) = 1 - (p_1 + p_2 + p_3) = 0.041.$$

The expected frequencies are computed by multiplying the sample size $n = 60$ by each probability, i.e. $E_i = n \cdot p_i$.

Table 5.2: Goodness-of-fit computation for the Poisson model (classes 2 and 3^+ pooled).

No. of defects	**Prob.**	E_i	O_i	$(O_i - E_i)^2/E_i$
0	0.472	28.32	32	0.4781
1	0.354	21.24	15	1.833
2	0.133	7.8 } 10.44	9 } 13	0.6277
3 (or more)	0.041	2.46 }	4 }	
			$\chi^2_{\text{cal}} =$	2.94

6. Since the mean is estimated, $p = 1$; after pooling there are $k = 3$ class intervals, so the degrees of freedom are $k - p - 1 = 3 - 1 - 1 = 1$:

$$\chi^2_{\text{tab}} = \chi^2_{0.05,\,k-p-1} = \chi^2_{0.05,1} = 3.84.$$

8. **Conclusion:** Since $\chi^2_0 = 2.94 < \chi^2_{0.05,1} = 3.84$, we fail to reject the null hypothesis that the distribution of defects in printed circuit boards is Poisson.

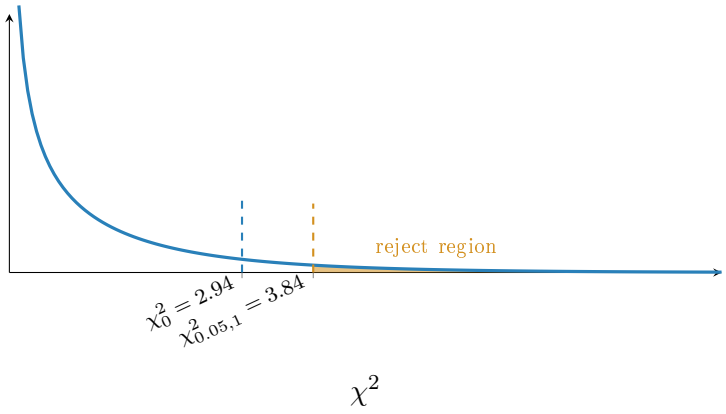


Figure 5.1: Chi-square distribution with 1 degree of freedom. The statistic $\chi^2_0 = 2.94$ falls left of the critical value 3.84, so H_0 is not rejected.

Example 5.2 (Safety incidents — Poisson fit)**.** X, the number of reported safety incidents per day on a production shop floor, is recorded for 90 days as follows.

Table 5.3: Observed safety-incident frequencies over 90 days.

x_i: no. of accidents	0	1	2	3	≥ 4
No. of days with x_i accidents	54	23	9	3	1

Does the data follow a Poisson distribution? Test at 5% significance level.

Solution.

1. Point of interest: the data of accidents and their distribution.

2. H_0: Data follow a Poisson distribution.
3. H_1: Data do not follow a Poisson distribution.
4. $\alpha = 0.05$.
5. Test statistic $\chi_0^2 = \sum \dfrac{(O_i - E_i)^2}{E_i}$.
6. Computations. The estimated mean is

$$\lambda = \frac{0 \cdot 54 + 1 \cdot 23 + 2 \cdot 9 + 3 \cdot 3 + 4 \cdot 1}{90} = 0.6.$$

The Poisson probabilities are

$$p_0 = \mathrm{P}(X = 0) = \frac{e^{-\lambda}\lambda^0}{0!} = \frac{e^{-0.6}\,0.6^0}{0!} = 0.5488,$$

$$p_1 = \mathrm{P}(X = 1) = 0.3293, \qquad p_2 = \mathrm{P}(X = 2) = 0.0988, \qquad p_3 = \mathrm{P}(X = 3) = 0.0198,$$

$$p_4 = \mathrm{P}(X \geq 4) = 1 - 0.5488 - 0.3293 - 0.0988 - 0.0198 = 0.0033.$$

Table 5.4: Goodness-of-fit computation for the safety-incident data (classes 2, 3, ≥ 4 pooled).

x_i	O_i		p_i	E_i		$(O_i - E_i)^2/E_i$
0	54		0.5488	49.39		0.4303
1	23		0.3293	29.64		1.4875
2	9	} 13	0.0988	8.89	} 10.97	0.3757
3	3		0.0198	1.79		
≥ 4	1		0.0033	0.3		
					$\chi_0^2 =$	2.2935

$$\chi^2_{\text{tab}} = \chi^2_{3-1-1,\,0.05} = \chi^2_{0.05,1} = 3.84.$$

7. Rejection criterion: since $\chi_0^2 < \chi^2_{\text{tab}}$ ($2.2935 < 3.84$), the null hypothesis is *not rejected* at the 5% level. The data are consistent with a Poisson distribution.

Example 5.3 (Radar components — binomial fit)**.** The production of a radar component is checked by a sample of four. The following table shows the number of defectives found in 300 samples.

Table 5.5: Defectives in 300 samples of size four.

No. of defectives	0	1	2	3	4
No. of samples	90	130	58	20	2

Fit the binomial distribution for the data and test the goodness of fit at a level of significance of 0.05.

Solution. The mean of the observed distribution is

$$\mu = np = \frac{0 \cdot 90 + 1 \cdot 130 + 2 \cdot 58 + 3 \cdot 20 + 4 \cdot 2}{300} = \frac{314}{300} = 1.047.$$

Since $np = 1.047 = 4p$, we estimate $p = 1.047/4 = 0.262$. The binomial probabilities are

$$\mathrm{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x} = \binom{4}{x} (0.262)^x (0.738)^{4-x},$$

so that, for example, $\mathrm{P}(X = 0) = \binom{4}{0}(0.262)^0(0.738)^4 = 0.2966$.

Table 5.6: Goodness-of-fit computation for the binomial model (classes 3 and 4 pooled).

Value (X)	Prob.	Observed freq.	Expected	$(O_i - E_i)^2/E_i$
0	0.2966	90	88.98	0.0117
1	0.4212	130	126.36	0.1049
2	0.2243	58	67.29	1.2826
3	0.0529	20 } 22	15.87 } 17.28	1.2867
4	0.0047	2 }	1.41 }	
			$\sum \chi^2 =$	2.6859

The tabulated value (one parameter p estimated, $k = 5$ classes; note the notes use $\chi^2_{0.05,4-1-1} = \chi^2_{0.05,2}$):

$$\chi^2_{\text{tab}} = \chi^2_{0.05,\,n-k-1} = \chi^2_{0.05,\,4-1-1} = \chi^2_{0.05,2} = 5.99.$$

Since $\chi^2_{\text{cal}} = 2.6859 < \chi^2_{\text{tab}} = 5.99$, we fail to reject H_0. We can conclude the distribution is binomial.

Example 5.4 (Octahedral die — test for fairness (uniform fit))**.** Nadir is testing an octahedral die to see if it is biased. The results are given as

Table 5.7: Observed scores of an octahedral die.

Score	1	2	3	4	5	6	7	8
Frequency	7	10	11	9	12	10	14	7

Test the hypothesis that the die is fair.

Solution. H_0: Die is fair; H_1: Die is not fair. Under H_0 the distribution is uniform (all die sides have equal probability 1/8). $\alpha = 0.05$. Degrees of freedom $\nu = n - k - 1$; no parameter is estimated, so $k = 0$ and $\nu = 8 - 1 = 7$.

$$\chi^2_{\text{tab}} = \chi^2_{0.05,7} = 14.067.$$

The expected value in each class is

$$E_i = \frac{1}{8}(7 + 10 + 11 + 9 + 12 + 10 + 14 + 7) = 10.$$

Table 5.8: Goodness-of-fit computation for the fair-die (uniform) model.

Score	Frequency	E_i	$(O_i - E_i)^2/E_i$
1	7	10	0.9
2	10	10	0
3	11	10	0.1
4	9	10	0.1
5	12	10	0.4
6	10	10	0
7	14	10	1.6
8	7	10	0.9
		$\chi^2_{\text{cal}} =$	4.0

Since $\chi^2_{\text{cal}} < \chi^2_{\text{tab}}$, we fail to reject H_0, i.e. the die is fair. There is no sufficient evidence at the 5% significance level to suggest that the die is not fair.

Example 5.5 (Galvanized steel coils — Poisson fit)**.** Let X denote the number of flaws observed on a large coil of galvanized steel. Seventy-five coils are inspected, and the following data were observed for the values of X.

Table 5.9: Observed flaw counts on 75 coils.

Values (X)	1	2	3	4	5	6	7	8
Observed frequency	1	11	8	13	11	12	10	9

(a) Does the assumption of a Poisson distribution seem appropriate as a probability model for this data? Use $\alpha = 0.01$.

(b) Calculate the P-value for this test.

Solution. H_0: Poisson distribution is appropriate for the data; H_1: Data do not follow a Poisson distribution; $\alpha = 0.01$. The estimated mean is

$$\lambda = \frac{1 \cdot 1 + 2 \cdot 11 + 3 \cdot 8 + 4 \cdot 13 + 5 \cdot 11 + 6 \cdot 12 + 7 \cdot 10 + 8 \cdot 9}{75} = 4.9067.$$

Using $\mathrm{P}(X = x) = e^{-\lambda}\lambda^x/x!$, the expected frequencies $E_i = 75\,p_i$ are computed as below (the first two classes and the last two are pooled so that expected counts are not too small).

Table 5.10: Goodness-of-fit computation for the Poisson model (coils); classes 1–2 and 7–8 pooled.

Values	O_i		E_i		$(O_i - E_i)^2/E_i$
1	1	12	2.7214	9.3987	0.7200
2	11		6.6770		
3	8		10.9213		0.7822
4	13		13.3977		0.0118
5	11		13.1485		0.3511
6	12		10.7533		0.1445
7	10	19	7.5381	12.618	3.8449
8	9		4.6237		
				$\chi^2_{\text{cal}} =$	5.8545

After pooling there are 6 classes and one parameter (λ) estimated, so the degrees of freedom are $6 - 1 - 1 = 4$:

$$\chi^2_{\text{tab}} = \chi^2_{0.01,\,6-1-1} = \chi^2_{0.01,4} = 13.28.$$

Since $\chi^2_{\text{cal}} < \chi^2_{\text{tab}}$ ($5.8545 < 13.28$), we fail to reject the null hypothesis, i.e. the data follow a Poisson distribution.

(b) The P-value for this test is $0.1 < P\text{-value} < 0.50$.

Chapter 6

Analysis of Statistical Intervals: Single Sample

Hypothesis testing decides between two statements about a parameter. A companion approach is *interval estimation*: rather than a single point estimate, we report a range of plausible values together with a confidence level. This chapter develops confidence intervals for the mean (variance known and unknown) and for the variance/standard deviation of a normal population, and shows how they answer the same questions as the tests of earlier chapters.

6.1 Confidence Intervals: Basic Ideas

Confidence Interval

An interval estimate for a population parameter is called a **confidence interval**. We cannot be certain that the interval contains the true, unknown population parameter — we only use a sample from the full population to compute the point estimate and the interval. However, the confidence interval is constructed so that we have confidence that it does contain the unknown population parameter. Confidence intervals are widely used in engineering and the sciences.

Interval estimates can be of three types:

1. A *confidence interval* bounds population or distribution parameters (such as a mean viscosity).
2. A *tolerance interval* bounds a selected proportion of a distribution.
3. A *prediction interval* bounds future observations from the population or distribution.

6.2 Confidence Interval on the Mean, Variance Known

Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution with unknown mean μ and known variance σ^2. The sample mean $\bar{X}$ is normally distributed with mean μ and variance σ^2/n. Standardising,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has a standard normal distribution.

A confidence interval estimate for μ is an interval of the form $l \le \mu \le u$, where the endpoints l and u are computed from the sample data. Because different samples produce different values of l and u, these endpoints are values of random variables L and U respectively. Suppose that we can determine values of L and U such that the following probability statement is true:

$$\mathrm{P}\{L \le \mu \le U\} = 1 - \alpha, \qquad 0 \le \alpha \le 1.$$

There is a probability of $1-\alpha$ of selecting a sample for which the confidence interval will contain the true value of μ. Once we have selected the sample (so that $X_1 = x_1, \ldots, X_n = x_n$) and computed l and u, the resulting confidence interval for μ is $l \le \mu \le u$.

Confidence Limits and Coefficient

The endpoints (or bounds) l and u are called the **lower** and **upper confidence limits** respectively, and $1 - \alpha$ is called the **confidence coefficient**.

Because $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$ has a standard normal distribution, we may write

$$\mathrm{P}\left\{-Z_{\alpha/2} \le \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \le Z_{\alpha/2}\right\} = 1 - \alpha.$$

Manipulating the terms inside the brackets — (1) multiplying through by $\sigma/\sqrt{n}$, (2) subtracting $\bar{X}$ from each term, and (3) multiplying through by -1 — we obtain

$$\mathrm{P}\left\{\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha.$$

CI on the Mean, Variance Known

If $\bar{x}$ is the sample mean of a random sample of size n from a normal population with known variance σ^2, a $100(1-\alpha)\%$ confidence interval on μ is

$$\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

where $Z_{\alpha/2}$ is the upper $100\alpha/2\%$ point of the standard normal distribution.

Note

For a fixed sample size n and standard deviation σ, the higher the confidence level, the longer the resulting confidence interval. The length of the confidence interval is a measure of the precision of estimation.

Choice of sample size

Sample Size for a Specified Error E

If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{Z_{\alpha/2}\,\sigma}{E}\right)^2.$$

If the right-hand side of this equation is not an integer, it must be rounded up. This will ensure that the level of confidence does not fall below $100(1-\alpha)\%$. Notice that $2E$ is the length of the resulting confidence interval.

One-sided confidence bounds

It is possible to obtain one-sided confidence bounds for μ by setting either $l = -\infty$ or $u = \infty$ and replacing $Z_{\alpha/2}$ by Z_α.

One-Sided Confidence Bounds (σ known)

A $100(1-\alpha)\%$ **upper** confidence bound for μ is

$$\mu \leq u = \bar{x} + Z_\alpha \frac{\sigma}{\sqrt{n}},$$

and a $100(1-\alpha)\%$ **lower** confidence bound for μ is

$$\bar{x} - Z_\alpha \frac{\sigma}{\sqrt{n}} = l \leq \mu.$$

6.3 Worked Problems: Mean, Variance Known

Example 6.1 (CI and sample size for a small sample)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population:

(i) Construct a 90% confidence interval on the population mean if the population standard deviation is known to be 2.

(ii) What should be the sample size in order to keep the width of the confidence interval to less than 2?

Solution. $100(1-\alpha)\%$ CI: $\mu \in \bar{x} \pm Z_{0.05}\frac{\sigma}{\sqrt{n}}$. The sample mean is

$$\bar{X} = \frac{5+2+3+1+0+0+1+3+3}{9} = \frac{18}{9} = 2.$$

(i)

$$\text{CI} = 2 \pm \frac{1.645 \cdot 2}{\sqrt{9}} = 2 \pm \frac{1.645 \cdot 2}{3},$$

giving upper $= 3.0967$ and lower $= 0.9033$:

$$\boxed{0.9033 \leq \mu \leq 3.0967.}$$

(ii)

$$\text{width} = \frac{2\, Z_{0.05}\, \sigma}{\sqrt{n}} = \frac{2 \cdot 1.645 \cdot 2}{\sqrt{n}} < 2 \;\Rightarrow\; \sqrt{n} > \frac{2 \cdot 1.645 \cdot 2}{2} = 3.29,$$

$$\Rightarrow\; n \geq 10.824 \;\Rightarrow\; \boxed{n \geq 11.}$$

Example 6.2 (Overfill in beverage bottles)**.** For the sample $(5, 2, 3, 1, 4, 2, 1, 3, 3, 5)$ from a population representing the extent of overfill in sealed beverage bottles in ml:

(a) Construct a 95% confidence upper bound on the population mean if the population variance is known to be 5. What should be the range of sample size in order to keep the width of the confidence interval to less than 1 ml?

(b) If the population variance is unknown, construct a 95% CI on the population variance.

Solution. The sample mean is

$$\bar{X} = \frac{5+2+3+1+4+2+1+3+3+5}{10} = \frac{29}{10} = 2.9.$$

(a) 95% upper bound on the mean. With $Z_{0.05} = 1.645$, $\sigma = \sqrt{5}$:

$$\mu \le \bar{X} + Z_{0.05}\frac{\sigma}{\sqrt{n}} = 2.9 + 1.645 \cdot \frac{\sqrt{5}}{\sqrt{10}} = 2.9 + 1.645\sqrt{\tfrac{5}{10}} \;\Rightarrow\; \boxed{\mu \le 4.063.}$$

This upper bound is the same as the upper limit of the 90% two-sided CI ($\bar{X} + Z_{0.05}\sigma/\sqrt{n}$). For the width requirement,

$$\text{width} = 2 \cdot 1.645\sqrt{\frac{5}{n}} < 1 \;\Rightarrow\; 2 \cdot 1.645\sqrt{5} < \sqrt{n} \;\Rightarrow\; n \ge (2 \cdot 1.645)^2 \cdot 5 = 54.1,$$

so the sample size $n \ge 54$ to keep the CI width below 1 ml.

(b) 95% CI on the variance (population variance unknown), $\alpha = 0.05$, $n = 10$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}.$$

With $s = 1.449$ (so $s^2 = (1.449)^2$), $\chi^2_{0.025,9} = 19.02$ and $\chi^2_{0.975,9} = 2.70$:

$$\frac{9 \cdot (1.449)^2}{19.02} \le \sigma^2 \le \frac{9 \cdot (1.449)^2}{2.70} \;\Rightarrow\; \boxed{0.9937 \le \sigma^2 < 7 \text{ ml}^2,}$$

where s^2 is obtained from $s^2 = \frac{1}{9}\sum(x_i - 2.9)^2$.

Example 6.3 (Standard error and interval estimate)**.** From a population with known standard deviation of 1.65, a sample of 32 items resulted in 34.8 as an estimate of the mean.

(i) Find the standard error of the mean.

(ii) Compute an interval estimate that should include the population mean 99.7 percent of the time.

Solution. (i) Standard error of the mean $= \dfrac{\sigma}{\sqrt{n}} = \dfrac{1.65}{\sqrt{32}} = 0.29.$

(ii) For 99.7% confidence, $Z_{\alpha/2} = Z_{0.0015} = 2.97$:

$$\bar{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 34.8 \pm 2.97 \cdot \frac{1.65}{\sqrt{32}} \;\Rightarrow\; \boxed{33.93 \le \mu \le 35.67.}$$

Example 6.4 (Cardinal Electricals — light-bulb life)**.** The manager of Cardinal Electricals' light bulb division must estimate the average number of hours that a light bulb made by each bulb machine will last. A sample of 40 light bulbs was selected from a machine, and the average burning time was 1416 hours. The standard deviation of burning time is known to be 30 hours. Construct a 90% CI for the true population mean.

Solution. $n = 40$, $\bar{x} = 1416$, $\sigma = 30$ hr, $\alpha = 0.1$, $Z_{\alpha/2} = 1.645$:

$$\bar{x} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1416 \pm 1.645 \cdot \frac{30}{\sqrt{40}} = (1408.2,\ 1423.8),$$

so $\boxed{1408.2 \le \mu \le 1423.8.}$

Example 6.5 (PC operating-system response time)**.** An operating system for a PC has been studied extensively, and it is known that the standard deviation of the response time following a particular command is $\sigma = 8$ ms. A new version of the operating system is installed, and we wish to estimate the mean response time for the new system, ensuring that a 95% CI for μ has length at most 5 ms.

(a) If response time is normally distributed and $\sigma = 8$ for the new system, what sample size would you recommend?

(b) Suppose the vendor tells us the standard deviation of the new system is smaller, say $\sigma = 6$; give the recommended sample size and comment on the effect of the smaller standard deviation.

Solution. The length of the CI is 5, so length $= 2\,Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ with CL = 95%, $\alpha = 0.05$, $Z_{\alpha/2} = 1.96$.

(a) $\sigma = 8$:

$$\text{length} = 2 \cdot Z_{0.025} \cdot \frac{8}{\sqrt{n}} = \frac{2 \cdot 1.96 \cdot 8}{\sqrt{n}} = 5 \ \Rightarrow\ \sqrt{n} = \frac{2 \cdot 1.96 \cdot 8}{5} = 6.272,$$

$$n = (6.272)^2 = 39.338 \ \Rightarrow\ \boxed{n = 40.}$$

(b) $\sigma = 6$:

$$\frac{2 \cdot 1.96 \cdot 6}{\sqrt{n}} = 5 \ \Rightarrow\ \sqrt{n} = 4.704 \ \Rightarrow\ n = (4.704)^2 = 22.13 \ \Rightarrow\ \boxed{n = 23.}$$

We observe that as the standard deviation is reduced, the required sample size also reduces.

Example 6.6 (Concrete compressive strength — 95% and 99% CIs)**.** A civil engineer is analysing the compressive strength of concrete. Compressive strength is normally distributed with $\sigma^2 = 1000\,\text{psi}^2$. A random sample of 12 specimens has mean $\bar{x} = 3250$ psi.

(i) Construct a 95% two-sided confidence interval on mean compressive strength.

(ii) Construct a 99% two-sided confidence interval on mean compressive strength.

Solution. $\sigma^2 = 1000$, $n = 12$, $\bar{x} = 3250$.

(i) 95% CI, $\alpha = 0.05$, $\alpha/2 = 0.025$, $Z_{0.025} = 1.96$:

$$\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}} = 3250 \pm 1.96 \cdot \frac{\sqrt{1000}}{\sqrt{12}} \ \Rightarrow\ \boxed{3232.1 \le \mu \le 3267.89.}$$

(ii) 99% CI, $\alpha = 0.01$, $\alpha/2 = 0.005$, $Z_{\alpha/2} = 2.58$:

$$3250 \pm 2.58 \cdot \frac{\sqrt{1000}}{\sqrt{12}} \ \Rightarrow\ \boxed{3226.4 \le \mu \le 3273.55.}$$

Example 6.7 (Identifying the 95% CI from two intervals)**.** A random sample has been taken from a normal population and two confidence intervals constructed using exactly the same data. The two CIs are (38.02, 61.98) and (39.95, 60.05).

(a) What is the value of the sample mean?

(b) One of these intervals is a 90% CI and the other is a 95% CI. Which one is the 95% CI, and why?

Solution. The CI is given by $\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$, so $\bar{x}$ is the midpoint of the CI.

(a) Using either interval,

$$\bar{X} = \frac{38.02 + 61.98}{2} = \frac{100.00}{2} = 50 \;\Rightarrow\; \boxed{\bar{X} = 50.}$$

(b) For a fixed sample size n and standard deviation σ, the higher the confidence level, the longer (wider) the resulting confidence interval. The two interval widths are

$$(39.95,\ 60.05):\ \text{width} = 20.1 \ \rightarrow\ 90\%\ \text{CI};$$

$$(38.02,\ 61.98):\ \text{width} = 23.96 \ \rightarrow\ 95\%\ \text{CI}.$$

The wider interval (38.02, 61.98) is the 95% CI.

6.4 Confidence Interval on the Mean, Variance Unknown

When the sample is small ($n < 30$ or $n < 40$) and σ^2 is unknown, we must make an assumption about the form of the underlying distribution to obtain a valid CI procedure. A reasonable assumption is that the underlying distribution is normal. Suppose the population of interest has a normal distribution with unknown mean μ and unknown variance σ^2; let $\bar{X}$ and s^2 be the sample mean and variance of a random sample of size n.

To construct a two-sided CI on μ: if the variance were known, we would use $Z = (\bar{X}-\mu)/(\sigma/\sqrt{n})$, which has a standard normal distribution. When σ^2 is unknown, a logical procedure is to replace σ with the sample standard deviation S. The random variable Z now becomes $T = (\bar{X}-\mu)/(S/\sqrt{n})$. (If n is large, one may proceed to use the CI based on the normal distribution.)

The t distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with unknown mean μ and unknown variance σ^2. The random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $n-1$ degrees of freedom. The t probability density function is

$$f(x) = \frac{\Gamma[(k+1)/2]}{\sqrt{\pi k}\,\Gamma(k/2)\,[(x^2/k)+1]^{(k+1)/2}}, \qquad -\infty < x < \infty,$$

where k is the number of degrees of freedom. The mean and variance of the t distribution are zero and $k/(k-2)$ for $k > 2$ respectively.

The general appearance of the t distribution is similar to the standard normal: both are symmetric and unimodal, and the maximum ordinate value is reached at the mean $\mu = 0$. However, the t distribution has heavier tails than the normal (more probability in the tails). As the number of degrees of freedom $k \to \infty$, the limiting form of the t distribution is the standard normal. Generally the number of degrees of freedom for t equals the number of degrees of freedom associated with the estimated standard deviation.

Let $t_{\alpha,k}$ be the value of the random variable T with k degrees of freedom above which we find an area (or probability) α; thus $t_{\alpha,k}$ is an upper-tail $100\alpha\%$ point of the t distribution with k degrees of freedom. For example, the t value with 10 degrees of freedom having an area of 0.05 to the right is $t_{0.05,10} = 1.812$, i.e. $\mathrm{P}(T_{10} > t_{0.05,10}) = \mathrm{P}(T_{10} > 1.812) = 0.05$. Since the t distribution is symmetric about zero, $t_{1-\alpha,k} = -t_{\alpha,k}$; e.g. $t_{0.95,10} = -t_{0.05,10} = -1.812$.

CI on the Mean, Variance Unknown

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown variance σ^2, a $100(1-\alpha)\%$ confidence interval on μ is

$$\bar{x} - t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}} \le \mu \le \bar{x} + t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}},$$

where $t_{\alpha/2,\,n-1}$ is the upper $100\alpha/2\%$ point of the t distribution with $n-1$ degrees of freedom.

One-sided confidence bounds on the mean of a normal distribution are obtained by using only the appropriate lower or upper confidence limit from the two-sided expression and replacing $t_{\alpha/2,\,n-1}$ by $t_{\alpha,\,n-1}$.

Note

It is not as easy to select a sample size n to obtain a specified length (or precision of estimation) for this CI as it was in the σ-known case, because the length of the interval involves s, n, and $t_{\alpha/2,\,n-1}$. Since the t-percentile depends on the sample size n, an appropriate n can only be obtained through trial and error.

6.5 Worked Problems: Mean, Variance Unknown

Example 6.8 (Reading t-percentiles)**.** Determine the t-percentile that is required to construct each of the following.

(a) Two-sided CI: confidence level = 95%, dof = 12.

(b) Two-sided CI: confidence level = 99.9%, dof = 15.

(c) One-sided CI: confidence level = 99%, dof = 19.

(d) One-sided CI: confidence level = 99.9%, dof = 24 (and dof = 15).

Solution.

(a) $t_{\alpha/2,\,n-1} = t_{0.05/2,\,12} = t_{0.025,12} = 2.179$.

(b) $t_{\alpha/2,\,n-1} = t_{0.001/2,\,15} = 4.073$.

(c) $t_{\alpha,\,n-1} = t_{0.01,19} = 2.539$.

(d) $t_{\alpha,n-1} = t_{0.001,24} = 3.467$; and if dof $= 15$, $t_{0.001,15} = 3.733$.

Example 6.9 (Tyre life — 95% CI on the mean)**.** A research engineer for a tyre manufacturer is investigating tyre life for a new rubber compound and has built 16 tyres and tested them to end-of-life in a road test. The sample mean and standard deviation are 60139.7 and 3645.94 km. Find a 95% CI on mean tyre life.

Solution. $\alpha = 0.05$, $n = 16$, $t_{\alpha/2,n-1} = t_{0.025,15} = 2.131$:

$$\bar{X} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \le \mu \le \bar{X} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

$$= 60139.7 - 2.131 \cdot \frac{3645.94}{\sqrt{16}} \le \mu \le 60139.7 + 2.131 \cdot \frac{3645.94}{\sqrt{16}}$$

$$\Rightarrow \boxed{[58197.33,\ 62082.07].}$$

Example 6.10 (Maintenance department — sensor life)**.** In a maintenance department of a firm, the life of the electronic equipment is found to be normally distributed. A random sample of 30 sensors has a sample mean life of 800 hours and a sample standard deviation of 70 hours.

(i) Construct a 95% two-sided CI for the mean life of the sensors.

(ii) Construct a 99% two-sided CI for the variance of life of sensors.

(iii) If the population standard deviation of life of sensors is known to be 90 hours, construct a 99% two-sided CI for the mean life of the sensors.

(iv) Comment on the difference in answers to (i) and (iii).

Solution. Life $\sim N(\mu, \sigma^2)$, $\bar{x} = 800$ hr, $s = 70$ hr, $n = 30$.

(i) $\alpha = 0.05$, $t_{\alpha/2,n-1} = t_{0.025,29} = 2.045$:

$$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 800 \pm 2.045 \cdot \frac{70}{\sqrt{30}} = 800 \pm 26.135 \ \Rightarrow \ \boxed{[773.86,\ 826.14] \text{ hr.}}$$

(ii) 99% CI on the variance:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} = \frac{29 \cdot 70^2}{52.34} \text{ to } \frac{29 \cdot 70^2}{13.12} \ \Rightarrow \ \boxed{(2715,\ 10830.7).}$$

(iii) σ known $= 90$, 99% CI on the mean, $Z_{0.005} = 2.58$:

$$\bar{x} \pm Z_{0.005}\frac{\sigma}{\sqrt{n}} = 800 \pm 2.58 \cdot \frac{90}{\sqrt{30}} = 800 \pm 42.39 \ \Rightarrow \ \boxed{(757.6,\ 842.4) \text{ hr.}}$$

(iv) The confidence interval of (iii) is wider than that of (i), even though σ is known — because the confidence level in (iii) is 99% against 95% in (i).

Example 6.11 (Electronic sensor life — 99% CIs)**.** The life of an electronic sensor is normally distributed. A random sample of 20 sensors has a mean life of 1000 hours and a standard deviation of 100 hours.

(i) Construct a 99% two-sided CI for the mean life of the sensors.

(ii) Construct a 95% two-sided CI for the variance of life of sensors.

(iii) If the population standard deviation of the life of sensors is known to be 105 hours, construct a 99% two-sided CI for the mean life of the sensors.

(iv) Comment on the difference in answers to (i) and (iii).

Solution. Life $\sim N(\mu, \sigma^2)$, $\bar{x} = 1000$ hr, $s = 100$ hr, $n = 20$.

(i) 99% CL, two-sided, $\alpha = 0.01$, $t_{\alpha/2,\,n-1} = t_{0.005,19} = 2.861$:

$$\bar{x} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}} = 1000 \pm 2.861 \cdot \frac{100}{\sqrt{20}} \;\Rightarrow\; \boxed{[936.03,\ 1063.97] \text{ hr.}}$$

(ii) 95% CI on the variance, $\alpha = 0.05$, $\alpha/2 = 0.025$, $\chi^2_{0.025,19} = 32.85$, $\chi^2_{0.975,19} = 8.91$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}} = \frac{19 \cdot 100^2}{32.85} \text{ to } \frac{19 \cdot 100^2}{8.91}$$

$$\Rightarrow \boxed{(5783.87,\ 21324.35) \text{ (variance).}}$$

(iii) $\sigma = 105$ known, 99% CI on the mean, $\alpha = 0.01$, $Z_{0.005} = 2.58$:

$$\bar{x} \pm Z_{0.005}\frac{\sigma}{\sqrt{n}} = 1000 \pm 2.58 \cdot \frac{105}{\sqrt{20}} \;\Rightarrow\; \boxed{(939.42,\ 1060.58) \text{ hr.}}$$

(iv) The CI of (iii) is narrower than that of (i), since σ is known. A more precise interval is obtained when additional information is available.

Example 6.12 (Small sample — CI on the mean, variance unknown)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, construct a 95% CI on the population mean if the population variance is unknown.

Solution. $\mu \in \bar{x} \pm t_{\alpha/2,\,n-1}\dfrac{s}{\sqrt{n}}$, with $\bar{X} = 2$, $n = 9$, $t_{0.025,8} = 2.306$, and

$$s^2 = \tfrac{1}{8}\big[(5-2)^2 + (2-2)^2 + (3-2)^2 + (1-2)^2 + (0-2)^2 + (0-2)^2 + (1-2)^2 + (3-2)^2 + (3-2)^2\big] = \frac{22}{8} = 2.75,$$

so $s = 1.658$. Then

$$2 \pm 2.306 \cdot \frac{1.658}{\sqrt{9}} \;\Rightarrow\; \boxed{[0.7253,\ 3.2747].}$$

Example 6.13 (Cloud seeding)**.** Cloud seeding has been studied for many decades as a weather modification procedure. The rainfall in acre-feet from 20 clouds that were selected at random and seeded with silver nitrate are as follows:

18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9,
31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, 31.6.

(a) Construct a 95% one-sided lower bound on the mean.

(b) Construct a 90% two-sided CI on the mean. What is the width of this interval?

(c) If the width of the interval in part (b) were to be halved, how much should be the sample size? Use trial and error if needed.

Solution. $n = 20$, $\bar{X} = 26.035$, $s = 4.7848$.

(a) 95% lower bound, $t_{0.05,19} = 1.729$:

$$\bar{X} - t_{0.05,19}\frac{s}{\sqrt{n}} = 26.035 - 1.729 \cdot \frac{4.7848}{\sqrt{20}} \Rightarrow \boxed{\mu \geq 24.185.}$$

(b) 90% two-sided CI, $\alpha = 0.10$, $\alpha/2 = 0.05$, $t_{0.05,19} = 1.729$:

$$\bar{X} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}} = 26.035 \pm 1.729 \cdot \frac{4.7848}{\sqrt{20}} \Rightarrow \boxed{(24.185,\ 27.885).}$$

Width $=$ max $-$ min $= 3.7958$.

(c) We require width $= 2\,t_{0.05,\,n-1}\dfrac{s}{\sqrt{n}}$ to be halved, i.e. $3.7958/2 = 1.8979$ (assuming s remains). By trial and error:

$$n = 30: \ \text{width} = 2 \cdot 1.697 \cdot \frac{4.7848}{\sqrt{30}} = 2.96;$$

$$n = 40: \ \text{width} = 2 \cdot 1.684 \cdot \frac{4.7848}{\sqrt{40}} = 2.5;$$

$$n = 60: \ \text{width} = 2 \cdot 1.671 \cdot \frac{4.7848}{\sqrt{60}} = 2.06;$$

$$n = 120: \ \text{width} = 2 \cdot 1.658 \cdot \frac{4.7848}{\sqrt{120}} = 1.4484.$$

Since the target width 1.8979 lies between the widths at $n = 60$ and $n = 120$, we conclude $\boxed{60 \leq n \leq 120.}$

Example 6.14 (Gasoline prices and highway speeds)**.** (a) From a sample of 42 gasoline stations statewide, the average price of a gallon of unleaded gas was found to be \$1.12 and the standard deviation was \$0.04 per gallon. Construct a 99% CI on the mean price of a gallon of unleaded gas.

(b) A random sample of 57 vehicles on a highway is clocked with an average speed of 23.2 mph and a standard deviation of 0.3 mph. Estimate (i) the standard deviation of the population and (ii) the standard error of the mean for this population.

Solution.

(a) $n = 42$, $\bar{x} = 1.12$, $s = 0.04$, $\alpha = 0.01$, $t_{0.01/2,\,41} = t_{0.005,41} = 2.7$:

$$\bar{x} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}} = 1.12 \pm 2.7 \cdot \frac{0.04}{\sqrt{42}} \Rightarrow \boxed{99\%\ \text{CI} = [\$1.10,\ \$1.14].}$$

(b) $n = 57$, $\bar{x} = 23.2$, $s = 0.3$, assume $\alpha = 0.05$. *(i)* A 95% CI for σ^2:

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}},\ \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}\right) = \left(\frac{56 \cdot 0.3^2}{77.4},\ \frac{56 \cdot 0.3^2}{36.5}\right) = (0.065,\ 0.138),$$

so the 95% CI for σ is $(0.255,\ 0.37)$. *(ii)* The standard error of the mean is estimated as

$$\frac{(0.255 + 0.37)/2}{\sqrt{57}} = 0.041.$$

Example 6.15 (Contamination level in food packs)**.** The level of contamination in % was studied in 6 packs of food with the following data:

$$16.8,\ 17.2,\ 17.4,\ 16.9,\ 16.5,\ 17.1.$$

Find a 99% confidence interval on the mean % μ, and provide a practical interpretation of this interval.

Solution.

$$\bar{X} = \frac{101.9}{6} = 16.98,$$

$$s^2 = \frac{0.0324 + 0.0484 + 0.1764 + 0.006 + 0.2304 + 0.0144}{5} = 0.1016, \qquad s = 0.318.$$

99% CI, $t_{0.01/2,\,5} = t_{0.005,5} = 4.032$:

$$\bar{x} \pm t_{0.005,5}\frac{s}{\sqrt{n}} = 16.98 \pm 4.032 \cdot \frac{0.318}{\sqrt{6}} = 16.98 \pm 0.523 \ \Rightarrow\ \boxed{[16.457,\ 17.503].}$$

Example 6.16 (Glass-bottle wall thickness — lower bound)**.** The wall thickness of 25 glass bottles was measured, and the sample mean was 4.05 mm and the sample standard deviation was 0.08 mm. Find a 95% lower confidence bound for the mean wall thickness, and interpret the interval.

Solution. $n = 25$, $\bar{x} = 4.05$, $s = 0.08$, $\alpha = 0.05$, $t_{0.05,24} = 1.711$:

$$\mu \geq \bar{X} - t_{\alpha,\,n-1}\frac{s}{\sqrt{n}} = 4.05 - 1.711 \cdot \frac{0.08}{\sqrt{25}} \ \Rightarrow\ \boxed{\mu \geq 4.02\,\text{mm}.}$$

The minimum population mean wall thickness is 4.02 mm.

6.6 Confidence Interval on the Variance and Standard Deviation

The χ^2 distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2, and let s^2 be the sample variance. Then the random variable

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

has a chi-square (χ^2) distribution with $n-1$ degrees of freedom.

CI on the Variance

If s^2 is the sample variance from a random sample of n observations from a normal distribution with unknown variance σ^2, then a $100(1-\alpha)\%$ confidence interval on σ^2 is

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}},$$

where $\chi^2_{\alpha/2,\,n-1}$ and $\chi^2_{1-\alpha/2,\,n-1}$ are the upper and lower $100\alpha/2\%$ points of the chi-square distribution with $n-1$ degrees of freedom respectively. A confidence interval for σ has lower and upper limits that are the square roots of the corresponding limits for σ^2.

One-Sided Confidence Bounds on the Variance

The $100(1-\alpha)\%$ lower and upper confidence bounds on σ^2 are

$$\frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} \le \sigma^2 \qquad \text{and} \qquad \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,\,n-1}}.$$

6.7 Worked Problems: Variance and Standard Deviation

Example 6.17 (Reading χ^2 percentiles)**.** Determine the values of the following percentiles: $\chi^2_{0.05,10}$, $\chi^2_{0.95,16}$, $\chi^2_{0.005,25}$.

Solution.

$$\chi^2_{0.05,10} = 18.31, \qquad \chi^2_{0.95,16} = 5.14, \qquad \chi^2_{0.005,25} = 46.93.$$

Example 6.18 (χ^2 percentiles for CIs)**.** Determine the χ^2 percentile that is required to construct each of the following CIs.

(a) Confidence level $= 95\%$, dof $= 24$, one-sided (upper).

(b) Confidence level $= 99\%$, dof $= 9$, one-sided (lower).

(c) Confidence level $= 90\%$, dof $= 19$, two-sided.

Solution.

(a) $\chi^2_{0.05,24} = 36.42$.

(b) $\chi^2_{1-0.01,9} = \chi^2_{0.99,9} = 2.09$.

(c) $\chi^2_{0.1/2,\,19} = \chi^2_{0.05,19} = 30.14$.

Example 6.19 (Rivet hole diameter — lower bound on variance)**.** A rivet is to be inserted into a hole. A random sample of $n = 15$ parts is selected, and the hole diameter is measured. The sample standard deviation of the hole diameter measurement is $s = 0.008$ mm. Construct a 99% lower confidence bound for the variance.

Solution. $n = 15$, $s = 0.008$. Lower confidence bound $\dfrac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} \le \sigma^2$, with $\chi^2_{0.01,14} = 29.14$:

$$\sigma^2 \ge \frac{14 \cdot 0.008^2}{29.14} = 3.074 \times 10^{-5} \;\Rightarrow\; \boxed{\text{CI} = [0.00003074,\ \infty).}$$

Example 6.20 (Sugar content of canned peaches)**.** The sugar content of the syrup in canned peaches is normally distributed. A random sample of $n = 10$ cans yields a sample standard deviation of $s = 4.8$ mg. Calculate a 95% two-sided CI for σ.

Solution. $n = 10$, $s = 4.8$ mg, CL $= 95\%$, $\alpha = 0.05$, $\chi^2_{0.025,9} = 19.02$, $\chi^2_{0.975,9} = 2.70$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}} = \frac{9 \cdot 4.8^2}{19.02} \text{ to } \frac{9 \cdot 4.8^2}{2.70}$$

$$\Rightarrow\ 10.90 \le \sigma^2 \le 76.8 \ \Rightarrow\ \sqrt{10.9} \le \sigma \le \sqrt{76.8} \ \Rightarrow\ \boxed{3.30 \le \sigma \le 8.76.}$$

Example 6.21 (CI on population variance — small sample)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, construct a 95% confidence interval on the population variance.

Solution. $s = 1.6583$, $n = 9$, $\chi^2_{0.05,8} = 15.51$, $\chi^2_{0.95,8} = 2.73$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}} = \frac{8 \cdot 1.6583^2}{15.51} \text{ to } \frac{8 \cdot 1.6583^2}{2.73}$$

$$\Rightarrow\ 1.4189 \le \sigma^2 \le 8.058\ \Rightarrow\ \boxed{1.19 \le \sigma \le 2.838.}$$

Example 6.22 (Amino acid concentration in dog feeds)**.** Amino acid concentrations in 8 packs of dog feed were recorded in mg/kg of dog feed as follows:

$$22,\ 24,\ 20,\ 26,\ 27,\ 24,\ 26,\ 23.$$

(i) Perform a test at 5% significance to see if the population variance is greater than 5.5 $(\text{mg/kg})^2$. What is the P-value of the test?

(ii) Construct an upper and a lower 90% confidence bound for the population variance.

Solution. $H_0 : \sigma^2 = 5.5$, $H_1 : \sigma^2 > 5.5$ $(\text{mg/kg})^2$, $\alpha = 0.05$, $n = 8$, $\bar{X} = 24$, $s = 2.3299$.

$$s^2 = \frac{4+0+16+4+9+0+4+1}{7} = \frac{38}{7} = 5.428, \qquad s = \sqrt{5.428} = 2.3299.$$

Test statistic:

$$\chi^2_0 = \frac{(n-1)s^2}{\sigma^2_0} = \frac{7 \cdot 2.3299^2}{5.5} = 6.9091,$$

and $\chi^2_{\text{tab}} = \chi^2_{0.05,7} = 14.07$. Since $\chi^2_0 < \chi^2_{\text{tab}}$, do not reject H_0.

P-value: $P = \text{P}(\chi^2_7 > 6.91)$, which is between 0.4 and 0.5 (since $\text{P}(\chi^2_7 > 6.346) = 0.5$ and $\text{P}(\chi^2_7 > 7.283) = 0.4$).

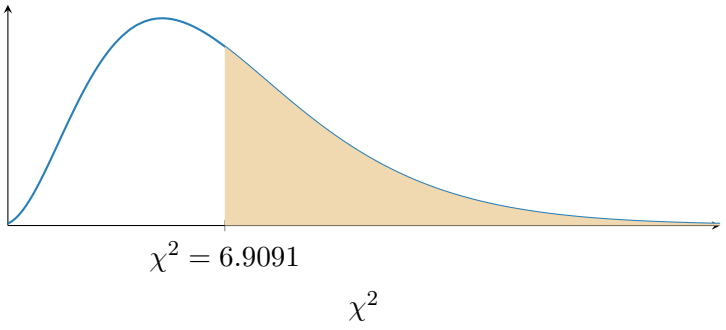


Figure 6.1: Chi-square distribution with 7 degrees of freedom; the shaded upper tail beyond $\chi^2 = 6.9091$ gives the P-value $(0.4 < P < 0.5)$.

(ii) 90% confidence bounds on the variance. Upper bound:

$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,\,n-1}} = \frac{7 \cdot 5.4286}{2.833} = 13.41.$$

Lower bound:

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{7 \cdot 5.4286}{12.02} = 3.16.$$

$$\therefore\ \boxed{3.16 \le \sigma^2 \le 13.41.}$$

Example 6.23 (Population standard deviation — two-sided and one-sided)**.** (Continuation.) For the same sample:

(a) Find a 90% two-sided confidence interval for the population standard deviation.

(b) Find a 95% upper bound and a 95% lower bound for the population standard deviation. How do these bounds compare with the limits of part (a)?

Solution. $n = 10$, $s = 4.8$ (data of the peaches problem).

(a) $\alpha = 0.10$, two-sided, $\alpha/2 = 0.05$, $\chi^2_{0.05,9} = 16.92$, $\chi^2_{0.95,9} = 3.325$:

$$90\%\text{ CI for } \sigma^2: \left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}},\ \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}\right) = \left(\frac{9\cdot 4.8^2}{16.92},\ \frac{9\cdot 4.8^2}{3.325}\right) = (12.26,\ 62.36).$$

Interval for σ: $\boxed{(3.5,\ 7.9).}$

(b) 95% upper bound, $\alpha = 0.05$, $1-\alpha = 0.95$, $\chi^2_{0.95,9} = 3.325$:

$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,\,n-1}} = \frac{9\cdot 4.8^2}{3.325} = 62.36 \ \Rightarrow\ \sigma \le 7.9.$$

95% lower bound, $\chi^2_{0.05,9} = 16.92$:

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{9\cdot 4.8^2}{16.92} = 12.26 \ \Rightarrow\ \sigma \ge 3.5.$$

Both bounds equal the limits of the two-sided CI, since $\alpha/2$ in part (a) equals α in part (b).

Example 6.24 (Hole diameter — 99% lower bound on variance)**.** A hole diameter is measured in a random sample of 15 parts. The sample standard deviation of the hole diameter is 0.008 mm. Construct a 99% lower confidence bound for σ^2.

Solution. $n = 15$, $s = 0.008$, 99% CI $\Rightarrow \alpha = 0.01$, $\chi^2_{0.01,14} = 29.14$:

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{14\cdot 0.008^2}{29.14} = 3.84\times 10^{-3} \ \Rightarrow\ \boxed{\sigma \ge 0.062.}$$

Example 6.25 (Weight measurements — test and lower bound on σ)**.** A set of 15 weight measurements gives sample standard deviation $s = 0.0088$ gm.

(i) Does the population standard deviation σ of all the weight measurements exceed 0.008 gm at 5% significance level? Find the P-value for this test.

(ii) Answer part (i) with the help of a lower confidence interval for σ with $\alpha = 5\%$.

Solution. $n = 15$, $s = 0.0088$ gm, $H_0 : \sigma = 0.008$ gm, $H_1 : \sigma > 0.008$ gm, $\alpha = 0.05$.

(i)

$$\chi^2_0 = \frac{(n-1)s^2}{\sigma^2} = \frac{14\cdot 0.0088^2}{0.008^2} = 16.94.$$

From tables, $\chi^2_{\text{table}} = \chi^2_{0.05,14} = 23.68$. Since $\chi^2_0 < \chi^2_{\text{table}}$, we fail to reject H_0, i.e. the standard deviation does not exceed 0.008 gm at 5% significance. *P-value:* $\chi^2_0 = 16.94$ lies between 0.5 and 0.1 (i.e. $21.06 < \chi^2$-range< 13.34), so $0.1 < P < 0.5$.

(ii) Lower confidence bound, $\alpha = 0.05$, $\chi^2_{0.05,14} = 23.68$:

$$\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{14 \cdot 0.0088^2}{23.68} = 4.578 \times 10^{-5} \;\Rightarrow\; \boxed{\sigma \geq 6.76 \times 10^{-3} \approx 0.0067.}$$

Since the lower bound $0.0067 < 0.008$, the interval is consistent with failing to reject H_0.

Example 6.26 (Titanium percentage in an aerospace alloy)**.** The % of titanium in an alloy used in aerospace casting is measured in 51 randomly selected parts. The sample standard deviation is $s = 0.28$. Construct a 95% two-sided confidence interval for σ.

Solution. $n = 51$, $s = 0.28$, $\alpha = 0.05$, $\chi^2_{0.025,50} = 71.42$, $\chi^2_{0.975,50} = 32.36$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}} = \frac{50 \cdot 0.28^2}{71.42} \text{ to } \frac{50 \cdot 0.28^2}{32.36}$$

$$\Rightarrow\; 0.055 \leq \sigma^2 \leq 0.121.$$

Taking the square root of the endpoints of this interval,

$$\boxed{0.234 \leq \sigma \leq 0.348.}$$

Compiler's Note on Fidelity and Coverage

This textbook was reconstructed directly from the scanned handwritten notes (76 pages). Wherever the handwriting was legible, the content — every definition, formula, worked problem and final numerical answer — was reproduced one-to-one. The original's final answers are treated as authoritative; only intermediate working that was ambiguous in the scan was reconstructed so as to reach those answers. No topics were invented and no problems were fabricated.

Coverage

All six chapters cover the full manuscript: hypothesis-testing fundamentals (Ch. 1); tests on the mean with variance known (Ch. 2); tests on the variance (Ch. 3); tests on the mean with variance unknown, the t-test (Ch. 4); goodness-of-fit tests, including the Poisson, binomial and uniform (fair-die) examples (Ch. 5); and the analysis of statistical intervals — confidence intervals on the mean (variance known and unknown) and on the variance/standard deviation, with sample-size selection and one-sided bounds (Ch. 6).

Points reconstructed, clarified, or carrying an apparent slip in the source

- **All normal / t / χ^2 curves** are schematic TikZ/pgfplots redrawings of the hand-sketched bell curves and rejection regions; shaded areas mark the critical or β regions described in the text. Exact sketch widths are illustrative, but every labelled critical value, tail area, and axis mark matches the notes.
- **Ch. 2, "P-values for a two-sided test", case (b)** $Z_0 = -1.84$. The manuscript records 0.64668, but $2[1 - \Phi(1.84)] = 2(0.032884) = 0.0658$; the notes' value appears to be a decimal slip. Both are shown.
- **Ch. 2, battery-life sample size** evaluates to $n \approx 0.844$, rounded up to $n \approx 1$ in the notes; reproduced exactly as written (physically trivial).
- **Ch. 3, ice-hockey σ problem** momentarily mixes $n = 16$ and $n = 17$ while forming the bound; the tabulated values used ($\chi^2_{0.05,16} = 26.30$) and the boxed final answers ($0.004 \le \sigma^2$, $0.07 \le \sigma$) are kept as in the source.
- **Ch. 5, goodness-of-fit tables** reproduce the notes' pooling of low-expected-frequency classes and their χ^2_{cal} values (2.94, 2.2935, 2.6859, 4.0, 5.8545) and degrees-of-freedom bookkeeping

exactly, including the radar-component example's use of $\chi^2_{0.05,4-1-1} = \chi^2_{0.05,2} = 5.99$.

- **Ch. 6, several CI problems** carry rounding in the source (e.g. $s = 1.449$ vs. $s = 1.658$ for the same 9-point sample in different problems, and the "overfill" variance CI). Each problem's own stated intermediate values and boxed answers are reproduced as written; where the source pools or rounds, the book follows the source rather than silently "correcting" it.
- **Ch. 6, standard-error-of-the-mean estimate** in the gasoline/speed problem follows the notes' own (non-standard) averaging of the σ CI endpoints to obtain 0.041.